

Mathematics Boot Camp

**Mathematical Foundations for Operations Research:
Optimization and Stochastic Processes**

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¹<https://github.com/bradchattergoon/math-boot-camp-notes>

Foreword

Who should use these notes?

These notes are designed as a mathematics boot camp specifically for incoming PhD students to an Operations Research / Industrial Engineering PhD program. The goal of these notes is to provide the mathematical foundations needed to be successful in first-year Optimization and Stochastic Processes sequences.

There are two specific goals for Optimization:

- A basic working knowledge of the theoretical foundation of gradient descent.
- A basic working knowledge of optimization theory as it pertains to primal and dual problems, illustrated through linear programming.

There are two specific goals for Stochastic Processes:

- A basic working knowledge of the Poisson process.
- A basic working knowledge of Brownian motion.

In pursuit of both of these goals we only make an assumption that the reader has a foundational quantitative background as might be obtained from an Industrial Engineering, Applied Mathematics, or related, undergraduate curriculum. However, we treat these goals through a rigorous mathematical lens, building from mathematical fundamentals in Real Analysis, Linear Algebra, and Probability theory with a proof-based approach.

These notes are not intended to be used primarily as reference material, but rather primarily as learning material. With that in mind, the pedagogy employed in the notes focuses on building intuition and motivation for the results before developing the formal theorems and proofs. In particular, the notes are structured for independent reading and learning by a reader, although

they can be paired with active teaching as a supplemental resource.

The pedagogical focus of these notes is on the parts of the foundations specifically needed for the goals outlined above. This focus therefore is not intended to give a thorough treatment of Real Analysis, Linear Algebra, or Probability Theory as might be found in a course specifically dedicated to these areas, and thus these notes are not a substitute for a course dedicated to any of those areas.

The strength of these notes, however, is that they are tailored to what a graduate student would need to know to have a strong understanding of the specified topics as they pertain to the outlined goals, in as straightforward a path as possible. With that in mind, these notes may also be of interest to graduate students or advanced undergraduates in other areas of study which have crossover with Optimization and Stochastic Processes who are looking for a streamlined introduction to these ideas. These might be students in fields like Statistics, Computer Science, Data Science, Economics, or Applied Mathematics.

As of this current version of these notes, the content covers the goals for Optimization but does not yet provide any content for the goals related to Stochastic Processes. The latter will be added in a future version.

If you'd like to be updated when the content on Stochastic Processes is added, or would like to get in touch in general, the author can be reached via [firstname] [lastname]@[institution].edu, where institution is "Berkeley."

If you find these notes useful, or are planning to use them at your institution, that information is especially appreciated!

How these notes are organized

These notes are intended to serve as a means of preparing students for the first-year PhD coursework in an Operations Research or related quantitative department.

With that theme in mind, the pedagogy behind these notes recognizes that there may be a variety of academic backgrounds present in the target population, and we make no assumption about prior preparation other than that an admissions committee has determined that incoming students are sufficiently quantitatively prepared and/or have sufficient quantitative skill and ability to handle the first-year coursework.

As such, the notes are structured to start from absolute fundamentals and progress linearly, making use of earlier work to support subsequent topics, and we give a full treatment of the material with respect to the outlined goals for Optimization and Stochastic Processes. In short, we do not assume that development of a topic can be outsourced to prior coursework. Indeed, the students who would benefit most from a boot camp are the ones who are unable to rely on prior coursework.

Our treatment of the topics covered, therefore, may be repetitive for more prepared students, and these students are encouraged to skip content they already feel very comfortable with. That said, we have aimed to give a perspective that focuses on “how to think” rather than “what to know”, and so even a more prepared student might find some value in looking through familiar topics to see if there is additional intuition to be gained from the review.

The general arc of the notes proceeds as follows. We start with a primer on proofs that introduces what a proof is, the use of the contrapositive, and contradiction in proofs. We begin the content proper with the generation of the real numbers starting from the natural numbers, and use this as motivation to introduce tools like limits and sequences, and concepts like metric spaces. We then move on to topology, e.g. open and closed sets, building up to compactness. After compactness we introduce functions and build to continuity, followed by differentiation. This is followed by vector-valued functions and linear algebra (viewed specifically as linear vector-valued functions). Linear algebra includes discussion of eigenvalues and eigenvectors, as well as determinants (motivated by a need to compute eigenvalues). After the treatment of linear algebra we return to general vector-valued functions and develop the multivariable results corresponding to single-variable differentiation.

After the mathematical fundamentals above, we move into a chapter on optimization. This chapter starts with an introduction to convexity in functions and sets, building into stronger notions of convexity, i.e. strict and strong convexity. These are then used to introduce (vanilla) gradient descent, which is paired with a contraction and fixed-point argument to illustrate convergence of the algorithm. The chapter then concludes with the introduction of linear programming and the development of the dual problem, weak and strong duality, and then feasibility arguments that introduce Farkas’ Lemma.

The table of contents gives finer grained detail about the topics covered. Along the way, there are small exercises that test application of the previously covered content. These may be in the form of an applied problem, an understanding question, or to complete a proof. In all cases the exercises are answerable based on content covered to that point so far, usually relying on content covered immediately prior.

This information will be updated with the discussion for Stochastic Processes content when that is developed.

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Chapter 1

Introduction to Proofs

Mathematics is the science of quantitative logic. When we develop mathematical ideas we do so using the language of logic as a foundation and then localize and augment it for our quantitative setting. As a precursor to understanding and discussing mathematics, then, we must understand the language in which it is discussed. That is the purpose of this chapter. If the reader is already comfortable with the language of mathematics, such as concepts like “for all,” “there exists,” and “assume for contradiction,” then the reader should consider skipping ahead to the next chapter. Otherwise, the reader should start here.

As with any language we must start with definitions so that the “speaker/writer” and the “listener/reader/audience” both share a common understanding of what is being discussed. Definitions are an essential and extremely common feature of mathematical thinking for this reason.

Much like with any language, words/phrases/terms are meant to represent a concept in the speaker’s mind, and when two different speakers use a word to represent a concept, they may be referring to two slightly different versions of that concept without realizing it. This can become a problem because the language of logic is very precise, so a discussion of an idea can get to different conclusions if the originating concepts are not sufficiently identical.

For this reason, in mathematics we *always* seek to present the definition of our starting concept explicitly so everyone engaging in our discussion knows precisely what we mean. Once the definition is clear we can make a “proposition” and support it with a “proof.”

Definition 1.1 (Proposition). A *proposition* is a statement that has a truth value, i.e. True or False. The most typical proposition in math takes the form “if A then B ” where A and B are two concepts.

Definition 1.2 (Proof). A *proof* is a sequence of logical arguments that illustrate the logical connection given in a proposition. For a proposition of the form “if A then B ” a proof starts assuming A is true and then argues rigorously that B must also be true.

A note about the term “proposition.” There are two other common terms used to represent the same idea. These are “lemma” and “theorem.” By convention the use of these terms is selected to distinguish how “important” a result is. “Lemma” is typically used to denote a result that is developed as a helper to prove a bigger or more important result. For instance it might be used to develop part of a proof separately to reduce the size of the proof. “Theorem” is typically used for major results. “Proposition” often sits between a lemma and a theorem. The specific conventions for these terms vary from field to field.

In these notes we will avoid these challenges by simply referring to essentially everything as a Theorem. There is one term we will adopt which is “corollary.” A corollary is a proposition that follows directly from a theorem. Corollaries are therefore almost exclusively presented immediately after a theorem is proved. We will use propositions sometimes as a statement that is not a standard result.

The propositions/theorems we develop will be of the form “if statement 1 then statement 2.” This encodes a very specific “truth value.” It says that if statement 1 is true then statement 2 is true. However, it does not say anything about statement 2 if statement 1 is false. Let us illustrate this idea concretely by an example.

Definition 1.3 (Multiple). Let x and a be non-zero integers. x is said to be a *multiple* of a if there exists another integer y such that x can be written as the product of a and y . Symbolically, $x = a \cdot y$.

Definition 1.4 (Divisible). An integer x is said to be *divisible* by a non-zero integer y if x is a multiple of y .

Definition 1.5 (Even number). An *even number* is an integer which is divisible by 2.

Proposition 1.6. *If x is a multiple of 4, then x is an even number.*

Proof. x is a multiple of 4 so $x = 4 \cdot y$ for some integer y . 4 is also a multiple of 2 since $4 = 2 \cdot 2$. Let $y' = 2 \cdot y$; then y' is also an integer. We can thus write $x = 2 \cdot y'$. Since $2 \cdot y'$ is divisible by 2, x is an even number. $\square$

We use “ $\square$ ” at the end of proofs to visually signal to a reader where the proof ends when scanning the page.

The proposition is made up of two statements:

- (i) x is a multiple of 4;
- (ii) x is an even number.

There are four ways the truth values of these statements can be combined while preserving the structure of the statement. Let $\text{truth}(A)$ be whether statement A is true. Our four combinations can be derived from:

If $\text{truth}(x \text{ is a multiple of } 4)$ then $\text{truth}(x \text{ is an even number})$.

Let us consider the other three versions of the statement.

- **If x is a multiple of 4 then x is not an even number.** Of course this cannot be true since we just proved every multiple of 4 is an even number.
- **If x is not a multiple of 4 then x is an even number.** Consider $x = 3$. This is not a multiple of 4 but it is also not an even number.
- **If x is not a multiple of 4 then x is not an even number.** Consider $x = 2$. This is not a multiple of 4 but it is an even number.

This illustrates our point: an “if A then B ” proposition constrains B only when the “if” statement A is true. When A is false the proposition imposes no restriction on B — B can go either way—so the statement holds regardless of B .

Note we often write $A \Rightarrow B$ to represent “if A then B ” symbolically. We can represent this information in a “truth table” with A being the statement “ x is a multiple of 4” and B being the statement “ x is an even number” as follows.

A	B	$A \Rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

Note we adopt the convention that when A is false the truth value of the statement is “True” rather than “False.” Truth values have to be either True or False, there is no “uncertain,” and assigning it as true preserves the expected interaction with other logical structures.

In fact we would benefit from adopting the tools from logic to help illustrate and prove the ideas in this chapter. The truth table is one such tool. Another such tool is to generalize the relationships between two statements A and B beyond just “if... then...”

It is the case, although we shall not prove it here, that we can build all other relationships between statements from two base relationships plus a bonus individual one.

We use A and B as two statements throughout. The individual one is trivial; it is called “not” and acts on a single statement as “not A ,” written symbolically as $\sim A$. In words this is

$$\text{not}(x \text{ is a multiple of } 4) = x \text{ is not a multiple of } 4.$$

The truth table is simple:

A	$\sim A$
T	F
F	T

The first base relationship is called “AND.” For our two statements symbolically we have “ $A \wedge B$ ” read as “ A and B .” In words, “ x is a multiple of 4 and x is an even number.” The truth table is as follows:

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

which illustrates the name: “ A and B must both be true for the output to be true.”

The second base relationship is called “OR.” For our two statements symbolically we have “ $A \vee B$ ” read as “ A or B .” In words, “ x is a multiple of 4 or x is an even number.” The truth table is as follows:

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

which highlights a nuance of the name: “ A or B , or both, can be true for the output to be true.” Alternatively, “ A and B must both be false for the output to be false.” This suggests some kind of relationship between NOT(AND) and OR. In fact this is an allusion to what we call Boolean Algebra. I will give a small sample here before moving on.

A	B	$A \vee B$	$\sim A$	$\sim B$	$\sim A \wedge \sim B$	$\sim (\sim A \wedge \sim B)$
T	T	T	F	F	F	T
T	F	T	F	T	F	T
F	T	T	T	F	F	T
F	F	F	T	T	T	F

This shows that an A OR B relationship is logically equivalent to NOT(NOT(A) AND NOT(B)). The reader is not expected to know this extremely well and it is only provided as a sample of logical analysis. We now continue with our focus on proofs and propositions.

Given that we have said all relationships can be developed from NOT, OR, and AND, we show the form of $A \Rightarrow B$ in these relationships.

A	B	$A \Rightarrow B$	$\sim A$	$\sim A \vee B$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

so $A \Rightarrow B$ is equivalent to $\sim A$ OR B .

Now we can use this equivalency to consider rigorously whether we can derive any other “if... then...” relationships from two statements A and B after we prove $A \Rightarrow B$.

First we consider whether $\sim A \Rightarrow \sim B$ is logically equivalent to $A \Rightarrow B$. $\sim A \Rightarrow \sim B$ is called the *inverse* of $A \Rightarrow B$. To check for logical equivalence we check if the truth tables align. If they do then they must be equivalent. $\sim A \Rightarrow \sim B$ is equivalent to $\sim (\sim A) \vee \sim B \equiv A \vee \sim B$.

A	B	$\sim B$	$A \vee \sim B$	$\sim A \Rightarrow \sim B$	$A \Rightarrow B$
T	T	F	T	T	T
T	F	T	T	T	F
F	T	F	F	F	T
F	F	T	T	T	T

Since the truth table does not align, $A \Rightarrow B$ does not tell us anything about the inverse!

Let us now assess the *converse* which is $B \Rightarrow A$. $B \Rightarrow A$ is equivalent to $\sim B \vee A$.

A	B	$\sim B$	$\sim B \vee A$	$B \Rightarrow A$	$A \Rightarrow B$
T	T	F	T	T	T
T	F	T	T	T	F
F	T	F	F	F	T
F	F	T	T	T	T

Again we see the truth table does not align so the converse is also not equivalent to the original statement in general.

Now we check one more derived statement called the *contrapositive*. The contrapositive is given as $\sim B \Rightarrow \sim A$. The equivalent statement in OR and NOT is $\sim (\sim B) \vee \sim A$, which is $B \vee \sim A$, but since OR is symmetric this is equivalent to $\sim A \vee B$ which is $A \Rightarrow B$. The proposition $A \Rightarrow B$ is equivalent to $\sim B \Rightarrow \sim A$!

For our two example statements this means “if x is a multiple of 4 then x is an even number” is equivalent to “if x is not an even number then x is not a multiple of 4.” This can be especially useful when we need to prove a proposition. Instead of proving the proposition directly it may be easier to prove its contrapositive.

Proposition 1.7. *If x is a multiple of 4 then x is an even number.*

Proof. We proceed by considering the contrapositive statement “if x is not an even number then x is not a multiple of 4.”

x is not an even number so x cannot be written as $x = 2 \cdot y$ for an integer y . Every multiple of 4 can be written as $4 \cdot b$ where b is an integer. $4 \cdot b$ can be written as $2 \cdot b'$ where $b' = 2 \cdot b$. Since x cannot be written as $2 \cdot y$ for an integer y , but every multiple of 4 can be written in this way, x cannot be a multiple of 4.

Since we have proven the contrapositive of the proposition, the proposition must also be true. $\square$

This illustrates the use of the contrapositive of a proposition as an equivalent proposition that can be used to prove the original target proposition. It is worth clarifying the idea of (logically) “equivalent.” We might be better served by describing what “not equivalent” means. When we look at the truth tables comparing a proposition and its converse or inverse, we see that there are some truth values that align and some that don’t. This means that for a given combination of input truth values we have different outputs for the respective propositions (original and derived).

As we’ve established, a proposition $A \Rightarrow B$ only imposes a restriction on B when A is true. Of course that should tell us that the converse proposition $B \Rightarrow A$, which only imposes a restriction on A when B is true, is not equivalent to a proposition that focuses on when A is true. It does not mean that the converse of the original proposition cannot be proven true, it just means that proving one of the propositions $A \Rightarrow B$ or $B \Rightarrow A$ true does not tell us whether the other proposition is true or not. Equivalence tells us that one proposition being true guarantees the other proposition is true, and if one proposition is false the other must also be false as they are logically equivalent.

We can use this idea to employ another proof strategy inspired by the idea of logical equivalence. We can use the contrapositive as a logically equivalent proposition to prove the original statement, but what if we use a logically “anti-equivalent” proposition. One such proposition is $A \wedge \sim B$. Consider the following truth table.

A	B	$\sim B$	$A \wedge \sim B$	$\sim (A \wedge \sim B)$	$A \Rightarrow B$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	F	T	T
F	F	T	F	T	T

Since $\text{NOT}(A \wedge \sim B)$ is equivalent to $A \Rightarrow B$ it means we can prove $A \Rightarrow B$ by proving $\text{NOT}(A \wedge \sim B)$. For “proposition” $\text{NOT}(A \wedge \sim B)$ to be true we need “proposition” $A \wedge \sim B$ to be false.

We employ this insight via a strategy called *proof by contradiction*. The strategy is as follows. Consider the proposition “if A then B .” To prove it let us assume $\text{not}(B)$ is true so that we assume both A and $\text{not}(B)$ are true. We then use a logical argument to arrive at a result that says $\text{not}(A)$, but we cannot have both A and $\text{not}(A)$ true at the same time so our assumption about $\text{not}(B)$ being true must be false, and thus it must be the case that B is actually true.

Proposition 1.8. *If x is a multiple of 4 then x is an even number.*

Proof. Assume for contradiction that x is not even. Since x is not even we can write it as $x = 2k + 1$ for some integer k since all non-even numbers are odd. k could be even or odd itself.

Case 1: k is even. Since k is even it can be written as $k = 2m$ for some integer m . Then

$$x = 2(2m) + 1 = 4m + 1,$$

which tells us x is not a multiple of 4.

Case 2: k is odd. Since k is odd it can be written as $k = 2m + 1$ for some integer m . Thus

$$x = 2(2m + 1) + 1 = 4m + 2 + 1 = 4m + 3.$$

This also tells us x is not a multiple of 4.

Hence after considering both cases we conclude that if we assume x is not even then x cannot be a multiple of 4, which is a contradiction, and we must conclude that x is in fact an even number if x is a multiple of 4. $\square$

We have developed and illustrated the three main proof strategies: direct, contrapositive, and contradiction. All of these are used throughout math but direct and contradiction are much more typical than contrapositive. Contradiction is especially useful for a specific type of mathematical proposition.

Mathematical statements often involve two main logical quantifiers, “for all” and “there exists.” Examples are:

- If x is an even number there exists an integer y such that $x = 2 \cdot y$.
- If $x = 0$ then for all integers y the product of x and y is 0.

These are not very insightful propositions but they give an example of the use of “for all” and “there exists.” Contradiction can be very useful in these settings because “for all,” when transformed using $\text{NOT}(\text{for all})$, we get “there exists.” For instance in the second proposition we would assume for contradiction “there exists an integer y such that the product of x and y is not 0.” Having just a single specific object to work with can be easier than trying to prove something for infinitely many objects.

We have one more idea we need to cover before we close the chapter on an introduction to proofs. We have established that a proposition $A \Rightarrow B$ does not tell us anything about a proposition $B \Rightarrow A$. However, sometimes both propositions are true and we will see a number of those in these notes. As an example consider two new statements.

A : A triangle has three equal sides.

B : A triangle has three equal angles.

In fact because of the relationship between angles and sides in a triangle we know both $A \Rightarrow B$ and $B \Rightarrow A$ must be true, and such a triangle is an equilateral triangle. We call this a *biconditional* proposition and write it as: a triangle has three equal sides if and only if it has three equal angles. We often shorten “if and only if” to “iff,” and write it symbolically as $A \Leftrightarrow B$. The truth table for this looks like:

A	B	$B \Rightarrow A$	$A \Rightarrow B$	$(A \Rightarrow B) \wedge (B \Rightarrow A)$	$A \Leftrightarrow B$
T	T	T	T	T	T
T	F	T	F	F	F
F	T	F	T	F	F
F	F	T	T	T	T

Note that $(A \Rightarrow B) \wedge (B \Rightarrow A)$ is about the truth value of the AND over the two propositions. This tells us that if A and B satisfy $A \Leftrightarrow B$ then one being true means the other is true, and one being false means the other is false. In this way the statements are equivalent in a similar manner to the way we previously discussed propositions being equivalent. In order to prove $A \Leftrightarrow B$ we almost always prove separately $A \Rightarrow B$ and $B \Rightarrow A$. We typically refer to $A \Rightarrow B$ as the “forward” direction and we typically refer to $B \Rightarrow A$ as the “backward” direction.

Additionally, in a general $A \Rightarrow B$ we typically call statement B “necessary” as in we must necessarily have B if A is true, and we typically call the statement A “sufficient” as in if A is true this is sufficient to conclude B . For $A \Leftrightarrow B$ we can describe the relationship as “ A is necessary and sufficient for B .” This terminology is less straightforward than “if... then...” so we do not use it in these notes but the reader should be aware that it may be used in other materials.

One final note: if the reader is asked to prove a proposition *false* the typical strategy is to find an example that satisfies the “if” statement but does not satisfy the “then” statement.

That concludes our discussion of proofs.

Chapter 2

Mathematical Fundamentals

2.1 Basics of Analysis: From the Natural Numbers to the Reals

Our eventual goal in these notes is to reason rigorously about optimization: about functions defined on spaces of real vectors, about when they attain their minima, and about the algorithms that find those minima. Every one of those ideas rests on the real numbers $\mathbb{R}$. It is tempting to take $\mathbb{R}$ for granted—we have used real numbers since grade school—but the entire discipline of analysis exists because “the real numbers” are subtler than they appear. In this section we build up to $\mathbb{R}$ starting from almost nothing, and along the way we develop the vocabulary (sets, functions, distance, sequences, limits, countability) that the rest of the course depends on.

2.1.1 Spaces, Axioms, and Why Context Matters

Mathematics begins with *axioms*: statements we take to be true without proof. From a small collection of axioms we *derive* everything else. A recurring theme, and a common source of confusion, is that many facts which feel like axioms are actually derived properties, and many assumptions we make without noticing are genuinely axioms.

Remark 2.1 (The water we swim in). There is a well-known parable of two young fish who are asked by an older fish, “How’s the water?”, and who swim on wondering “what is water?” Axioms are the water. They are the assumptions so basic that we operate inside them without noticing they are there. A central skill in this course is learning to notice the water—to ask, at every step, “what space am I in, and what am I allowed to assume?”

To see why the ambient context matters, suppose someone tells you that

$$1 + 1 = 0.$$

You might object that this is wrong because $1 + 1 = 2$. But whether the statement is true depends entirely on the space in which we are working. In the integers modulo 2 (the space with only the elements 0 and 1, where we add and then take the remainder upon division by 2) the statement $1 + 1 = 0$ is perfectly correct. The objection assumed, implicitly, that we were working in something like the ordinary Euclidean numbers. Neither party was wrong about arithmetic; they simply failed to agree on the space. We must *always* identify what space we are in.

A space, at its most basic, is a *set* of elements together with a *set of operations* among those elements that provide structure. We now make the three objects we need—a set, a field, and a vector space—precise.

Definition 2.2 (Set). A *set* is a well-defined collection of distinct objects, called its *elements*. We write $x \in S$ to mean “ x is an element of S ” and $x \notin S$ otherwise. A set is specified either by listing its elements, e.g. $\{0, 1\}$, or by a defining property, e.g. $\{x \mid x \text{ is an even integer}\}$.

A set on its own carries no arithmetic. To add and multiply we need a *field*: a set equipped with two operations that behave the way we expect “numbers” to behave.

Definition 2.3 (Field). A *field* is a set F together with two operations, addition (+) and multiplication ($\cdot$), satisfying the following axioms for all $a, b, c \in F$.

- (F1) **Closure:** $a + b \in F$ and $a \cdot b \in F$.
- (F2) **Associativity:** $(a + b) + c = a + (b + c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- (F3) **Commutativity:** $a + b = b + a$ and $a \cdot b = b \cdot a$.
- (F4) **Identities:** there exist distinct elements $0, 1 \in F$ with $a + 0 = a$ and $a \cdot 1 = a$ for all a .
- (F5) **Additive inverses:** for each $a \in F$ there exists $-a \in F$ with $a + (-a) = 0$.
- (F6) **Multiplicative inverses:** for each $a \in F$ with $a \neq 0$ there exists $a^{-1} \in F$ with $a \cdot a^{-1} = 1$.
- (F7) **Distributivity:** $a \cdot (b + c) = a \cdot b + a \cdot c$.

The rational numbers $\mathbb{Q}$ and the real numbers $\mathbb{R}$ are fields; the integers modulo a prime form a field; the integers $\mathbb{Z}$ are *not* a field (they lack multiplicative inverses). Once we have a field of “scalars” we can build a *vector space* of objects those scalars act on.

Definition 2.4 (Vector space). A *vector space* over a field F is a set V (whose elements are called *vectors*) together with an addition $+: V \times V \rightarrow V$ and a scalar multiplication $\cdot: F \times V \rightarrow V$, satisfying for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all $a, b \in F$:

- (V1) **Additive closure, associativity, commutativity:** $\mathbf{u} + \mathbf{v} \in V$, $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$, and $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
- (V2) **Zero vector:** there exists $\mathbf{0} \in V$ with $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v}$.
- (V3) **Additive inverse:** for each $\mathbf{v} \in V$ there exists $-\mathbf{v} \in V$ with $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.
- (V4) **Compatibility of scalars:** $a \cdot (b \cdot \mathbf{v}) = (a \cdot b) \cdot \mathbf{v}$.
- (V5) **Scalar identity:** $1 \cdot \mathbf{v} = \mathbf{v}$, where 1 is the multiplicative identity of F .
- (V6) **Distributivity:** $a \cdot (\mathbf{u} + \mathbf{v}) = a \cdot \mathbf{u} + a \cdot \mathbf{v}$ and $(a + b) \cdot \mathbf{v} = a \cdot \mathbf{v} + b \cdot \mathbf{v}$.

The prototypical example is $\mathbb{R}^d$: d -tuples of real numbers, added coordinate-wise, with real scalars. The axioms are deliberately spare. Everything else we believe about vectors must be *derived* from them. To illustrate the standard technique for proving that an object is *unique*, we prove the zero vector is unique.

Theorem 2.5 (Uniqueness of the zero vector). *In any vector space V the zero vector is unique.*

Proof. This is the typical strategy for proving uniqueness: assume there are two objects with the defining property and show they must be equal. Suppose $\mathbf{0}$ and $\mathbf{0}'$ are both zero vectors, i.e. each satisfies axiom (V2). Using that $\mathbf{0}'$ is a zero vector,

$$\mathbf{0} = \mathbf{0} + \mathbf{0}'.$$

Using that $\mathbf{0}$ is a zero vector (and commutativity),

$$\mathbf{0} + \mathbf{0}' = \mathbf{0}' + \mathbf{0} = \mathbf{0}'.$$

Chaining the two equalities gives $\mathbf{0} = \mathbf{0}'$. Hence the zero vector is unique. $\square$

Exercise 2.6. Show that additive inverses are unique: for each $\mathbf{v} \in V$ there is exactly one vector $-\mathbf{v}$ with $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$. (*Hint: mimic the proof above.*)

Next we derive a fact that *looks* like an axiom but is not: that the scalar 0 annihilates every vector.

Theorem 2.7. For every vector $\mathbf{v}$ in a vector space V , $0 \cdot \mathbf{v} = \mathbf{0}$, where 0 is the additive identity of the field F .

Proof. Because $0 = 0 + 0$ in the field, scalar-distributivity (V6) gives

$$0 \cdot \mathbf{v} = (0 + 0) \cdot \mathbf{v} = 0 \cdot \mathbf{v} + 0 \cdot \mathbf{v}.$$

By (V3) the vector $0 \cdot \mathbf{v}$ has an additive inverse $-(0 \cdot \mathbf{v})$. Add it to both sides:

$$0 \cdot \mathbf{v} + (-(0 \cdot \mathbf{v})) = (0 \cdot \mathbf{v} + 0 \cdot \mathbf{v}) + (-(0 \cdot \mathbf{v})).$$

The left side is $\mathbf{0}$. Using associativity on the right,

$$\mathbf{0} = 0 \cdot \mathbf{v} + (0 \cdot \mathbf{v} + (-(0 \cdot \mathbf{v}))) = 0 \cdot \mathbf{v} + \mathbf{0} = 0 \cdot \mathbf{v}.$$

Hence $0 \cdot \mathbf{v} = \mathbf{0}$. Note this is a *derived property*, not an assumption. $\square$

Exercise 2.8. Show that any scalar times the zero vector is the zero vector: $a \cdot \mathbf{0} = \mathbf{0}$ for all $a \in F$.

The same axioms let us identify the additive inverse concretely: scaling by -1 produces it.

Theorem 2.9. For every vector $\mathbf{v}$, the vector $(-1) \cdot \mathbf{v}$ is the additive inverse of $\mathbf{v}$; that is, $(-1) \cdot \mathbf{v} = -\mathbf{v}$.

Proof. Using the scalar identity (V5), scalar-distributivity (V6), the field fact $1 + (-1) = 0$, and Theorem 2.7,

$$\mathbf{v} + (-1) \cdot \mathbf{v} = 1 \cdot \mathbf{v} + (-1) \cdot \mathbf{v} = (1 + (-1)) \cdot \mathbf{v} = 0 \cdot \mathbf{v} = \mathbf{0}.$$

Thus $(-1) \cdot \mathbf{v}$ satisfies the defining property of an additive inverse of $\mathbf{v}$. Since additive inverses are unique (Exercise 2.6), $(-1) \cdot \mathbf{v} = -\mathbf{v}$. $\square$

Exercise 2.10. Verify directly, from the axioms, that scalar multiplication by -1 yields the vector inverse. (*This is the statement proved above; write the argument in your own words, citing each axiom you use.*)

Exercise 2.11. Show that if $a \cdot \mathbf{v} = \mathbf{0}$ then either $a = 0$ (in F) or $\mathbf{v} = \mathbf{0}$ (in V). (*Hint: if $a \neq 0$, multiply by a^{-1} .*)

We will usually work with the real numbers $\mathbb{R}$, or with $\mathbb{R}^d$ for some dimension d . The point of the exercises above is not the particular facts—they are unsurprising—but the discipline they instill: we cannot simply *assume* how objects behave, we must argue rigorously from the axioms. With this discipline in hand, our goal for the remainder of the section is concrete: **construct the real numbers**. We build them up, starting from the counting numbers.

2.1.2 The Natural Numbers, Cardinality, and Countability

Definition 2.12 (Natural numbers). The *natural numbers* are $\mathbb{N} = \{1, 2, 3, \dots\}$, named for the “natural” origin of counting objects.

There is a persistent ambiguity about whether $\mathbb{N}$ includes 0: sometimes it does, sometimes it does not. To avoid confusion we introduce a separate symbol for the version with zero,

$$\mathbb{N}^+ = \{0, 1, 2, 3, \dots\},$$

and reserve $\mathbb{N}$ for $\{1, 2, 3, \dots\}$.

How “big” is $\mathbb{N}$? To make “big” precise we need the notion of the *size*, or *cardinality*, of a set.

Definition 2.13 (Cardinality). The *cardinality* of a set S , written $|S|$, is a measure of the number of elements it contains. Two sets S and T have the *same cardinality* if there is a bijection between them (see Definition 2.15); we then write $|S| = |T|$.

Defining equal size *via bijections*—rather than by counting—is what lets us compare infinite sets, where naive counting is impossible. This leads directly to the single most important classification of set sizes in analysis and probability: countable versus uncountable. We give the definition after we have functions in hand, in Section 2.1.3, and we return to build $\mathbb{N}$ ’s comparisons there. First, functions.

2.1.3 Functions

The manuscript motivates functions with a subtle point about *notation versus mapping*. Consider a rule from $\mathbb{N}$ to $\mathbb{N}^+$ given by $f(1) = 0$, $f(2) = 1$, and so on. We can *represent* this compactly as $f(n) = n - 1$, but the equation $f(n) = n - 1$ is not itself the function; the function is the mapping

$$1 \mapsto 0, \quad 2 \mapsto 1, \quad 3 \mapsto 2, \quad \dots$$

and $f(n) = n - 1$ is merely a convenient means of capturing that mapping in notation. Notice too that the input and output here live in *different* spaces: the inputs are drawn from $\mathbb{N}$ and 0 is an output, but $0 \notin \mathbb{N}$, so domain and codomain are genuinely different settings.

Definition 2.14 (Function). A *function* $f : X \rightarrow Y$ from a set X (the *domain*) to a set Y (the *codomain*) is a mapping that assigns to *every* element $x \in X$ exactly *one* element $f(x) \in Y$. The *image* of f is the set of outputs actually attained,

$$\text{im}(f) = \{y \in Y \mid y = f(x) \text{ for some } x \in X\} \subseteq Y.$$

Two requirements are hiding in that definition and are worth stating aloud: a function must assign an output to *everything* in the domain, and it must assign only *one* output to each input. The codomain Y is the space the outputs are declared to live in; the image is the (possibly smaller) set of outputs that are actually hit. For $f(n) = n$ from $\mathbb{N}$ to $\mathbb{N}^+$ the codomain is $\mathbb{N}^+$ but the image is $\mathbb{N}$, so codomain and image differ; for $f(n) = n - 1$ the image is all of $\mathbb{N}^+$, so codomain and image coincide.

Two properties describe how tightly a function fills and separates its codomain.

Definition 2.15 (Injection, surjection, bijection). Let $f : X \rightarrow Y$ be a function.

- f is *injective* (one-to-one) if distinct inputs give distinct outputs: $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
- f is *surjective* (onto) if every element of the codomain is attained: $\text{im}(f) = Y$.
- f is a *bijection* if it is both injective and surjective.

For example, take $f : \mathbb{N} \rightarrow \mathbb{N}^+$. If we define $f(n) = n$ then f is *not surjective*, because nothing maps onto 0. If instead we set $1 \mapsto 0$ and $2 \mapsto 0$, then $3 \mapsto 1$, $4 \mapsto 2$, and so on, the map is *not injective*, because 0 is mapped onto by two different inputs. The map $f(n) = n - 1$ is a bijection. These three cases are shown in Figure 2.1.

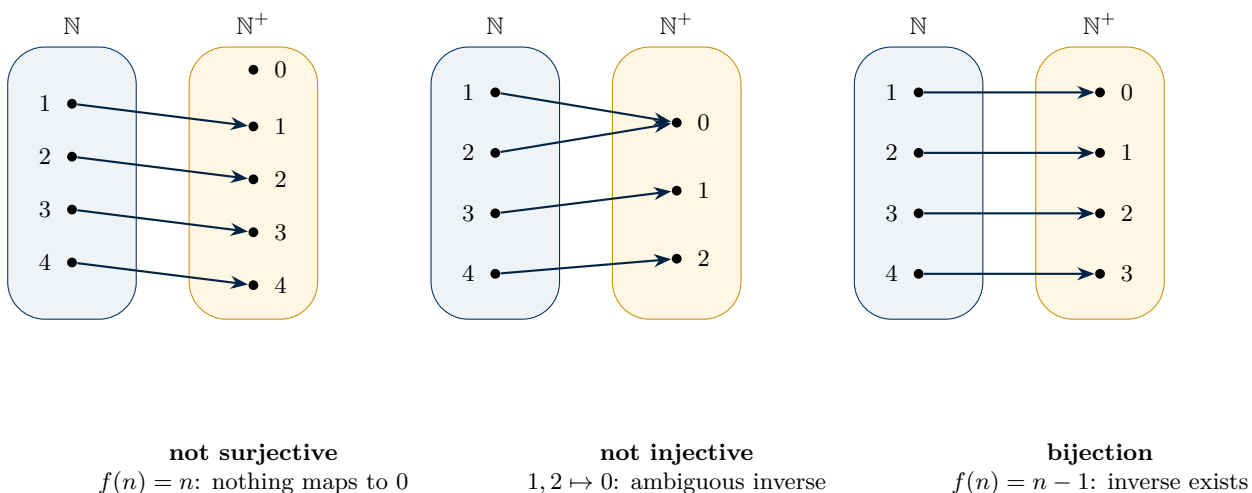


Figure 2.1: Three maps from $\mathbb{N}$ to $\mathbb{N}^+$. A function has an inverse if and only if it is a bijection: it must hit every element of the codomain (surjective) and never send two inputs to the same output (injective).

We often want to reverse a function: to *map back* from the codomain to the domain. This reverse map is the *inverse*. If f fails to be surjective, then some element of the codomain (e.g. 0 under the identity-style map) has nothing mapping to it, so the reverse map is undefined there. If f fails to be injective, then some element (e.g. 0 under $1, 2 \mapsto 0$) is the image of two inputs, so the reverse assignment is ambiguous—it would send one element to two things, which is not a function at all. Hence an inverse exists exactly when neither failure occurs.

Theorem 2.16. *A function $f : X \rightarrow Y$ has an inverse $f^{-1} : Y \rightarrow X$ if and only if f is a bijection.*

Proof. ($\Leftarrow$) Suppose f is a bijection. Define $g : Y \rightarrow X$ as follows: given $y \in Y$, surjectivity guarantees at least one x with $f(x) = y$, and injectivity guarantees at most one such x ; so there is exactly one, and we set $g(y) = x$. Then g assigns exactly one output to every input, so it is a function, and by construction $g(f(x)) = x$ and $f(g(y)) = y$. Thus $g = f^{-1}$ is an inverse.

($\Rightarrow$) Suppose an inverse f^{-1} exists. If f were not surjective, some $y \in Y$ would have no preimage, so $f^{-1}(y)$ would be undefined—contradicting that f^{-1} is a function on Y . If f were not injective, some y would satisfy $f(x_1) = f(x_2) = y$ with $x_1 \neq x_2$, so $f^{-1}(y)$ would have to equal both x_1 and x_2 , contradicting single-valuedness. Hence f is both surjective and injective, i.e. a bijection. $\square$

Bijections are exactly the maps that certify two sets have the same size, which is what we need to compare infinite sets.

Definition 2.17 (Countable set). A set S is *countable* if it is finite, or if there exists a bijection between S and $\mathbb{N}$. A set that is not countable is *uncountable*. A set in bijection with $\mathbb{N}$ is called *countably infinite*.

Clearly $\mathbb{N}$ is countable. It is a good first check of the definition that $\mathbb{N}$ and $\mathbb{N}^+$ have the same cardinality, even though $\mathbb{N}^+$ “has one more element.”

Theorem 2.18. $\mathbb{N}$ and $\mathbb{N}^+$ have the same cardinality.

Proof. The map $f : \mathbb{N} \rightarrow \mathbb{N}^+$, $f(n) = n - 1$, is a bijection (its inverse is $g(m) = m + 1$). Hence $|\mathbb{N}| = |\mathbb{N}^+|$. $\square$

This is the characteristic feature of infinite sets: a proper subset can have the same cardinality as the whole. It also shows why we *must* define “same size” through bijections rather than by counting. We record two more facts we will lean on. First, countability is *transitive* through bijections: if $|S| = |T|$ and T is countable, then S is countable, because a bijection $S \rightarrow T$ composed with a bijection $T \rightarrow \mathbb{N}$ is a bijection $S \rightarrow \mathbb{N}$. Second, we distinguish once more between the two flavors of countable: *finite* sets (in bijection with $\{1, 2, \dots, n\}$ for some n) and *countably infinite* sets (in bijection with all of $\mathbb{N}$).

Exercise 2.19. Define the integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ and show that $\mathbb{Z}$ is countable. (*Hint: interleave the non-negatives and the negatives, e.g. $0, 1, -1, 2, -2, \dots$, and write down the bijection with $\mathbb{N}$ explicitly.*)

Next we introduce the rationals and establish, perhaps surprisingly, that there are “no more” rationals than natural numbers.

Definition 2.20 (Rational numbers). The *rational numbers* are $\mathbb{Q} = \{p/q \mid p \in \mathbb{Z}, q \in \mathbb{N}\}$, the ratios of integers.

Theorem 2.21. $\mathbb{Q}$ is countable.

Proof. Every nonzero rational has a unique representation p/q with $q \in \mathbb{N}$, $p \in \mathbb{Z}$, and $\gcd(|p|, q) = 1$. Group the rationals by the “height” $h = |p| + q \in \mathbb{N}$. For each fixed h there are only finitely many pairs (p, q) with $|p| + q = h$, hence finitely many rationals of height h . List the rationals by increasing height, and within each height in some fixed order (say increasing p):

$$\frac{0}{1}, \frac{1}{1}, \frac{-1}{1}, \frac{1}{2}, \frac{-1}{2}, \frac{2}{1}, \frac{-2}{1}, \dots$$

This procedure visits every rational exactly once and thereby defines a bijection between $\mathbb{N}$ and $\mathbb{Q}$. Hence $\mathbb{Q}$ is countable. (Equivalently, $\mathbb{Q}$ is a countable union of the finite “height” sets, which is countable by Theorem 2.38 below.) Note the striking conclusion: $|\mathbb{Q}| = |\mathbb{N}|$, so the rationals have the same cardinality as the naturals despite being dense on the line. $\square$

We can now motivate the leap to the real numbers. The rationals are rich, but they have *gaps*. The classic witness is $\sqrt{2}$.

Theorem 2.22. There is no rational number x with $x^2 = 2$; that is, $\sqrt{2}$ is irrational.

Proof. Assume for contradiction that $\sqrt{2}$ is rational. Then we may write $\sqrt{2} = p/q$ with $p, q \in \mathbb{N}$ and the fraction in lowest terms, so p and q are not both even. Squaring gives $p^2 = 2q^2$, so p^2 is even, and therefore p is even (if p were odd, p^2 would be odd). Write $p = 2k$. Then $2q^2 = p^2 = 4k^2$, so $q^2 = 2k^2$, whence q^2 is even and q is even. But now p and q are both even, contradicting that the fraction was in lowest terms. Hence no such rational exists. $\square$

We “know” $\sqrt{2}$ exists—we can point to the diagonal of a unit square—yet it is not in $\mathbb{Q}$. How do we get from $\mathbb{Q}$ to a number system that contains it? The idea is to *approximate* $\sqrt{2}$ by rationals and let the approximations define it. We can certainly trap it between rationals: since $1 < 2 < 9$ we have $1 < \sqrt{2} < 3$, and we can do better—from $\frac{25}{16} < 2 < \frac{9}{4}$ we get $\frac{5}{4} < \sqrt{2} < \frac{3}{2}$, and so on, tightening the bracket indefinitely (Figure 2.2). To make “getting closer” mean something precise, though, we first need a notion of *distance*, which we have not yet introduced.

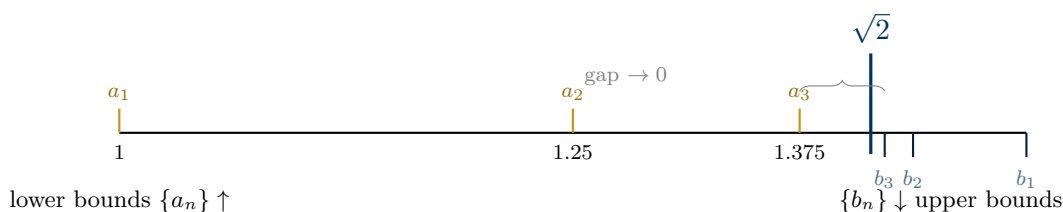


Figure 2.2: Two Cauchy sequences of rationals close in on $\sqrt{2}$ from below ($\{a_n\}$) and above ($\{b_n\}$). Their intersection is empty in $\mathbb{Q}$: the point they squeeze toward is a “gap” that the construction of $\mathbb{R}$ fills.

2.1.4 Metric Spaces and Distance

Definition 2.23 (Metric, metric space). Let S be a set. A *metric* (or *distance*) on S is a function $d : S \times S \rightarrow \mathbb{R}$ satisfying, for all $x, y, z \in S$:

- (M1) **Non-negativity:** $d(x, y) \geq 0$, with $d(x, y) = 0$ iff $x = y$;
- (M2) **Symmetry:** $d(x, y) = d(y, x)$;
- (M3) **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (S, d) is a *metric space*. Note that a distance is, above all, a *function* $d : S \times S \rightarrow \mathbb{R}$.

Exercise 2.24. Show that the Euclidean norm on $\mathbb{R}^2$, $d(x, y) = \|x - y\|_2 = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$, satisfies the metric axioms and hence is a valid distance.

Exercise 2.25. Determine whether $d(x, y) = |x^2 - y^2|$ is a valid distance on $\mathbb{R}$.

Exercise 2.26. Determine whether $d(x, y) = |x - y|^2$ is a valid distance on $\mathbb{R}$.

A metric space need *not* be a vector space. For instance, $\mathbb{Q} \cup \{\sqrt{2}\}$ with the distance $d(x, y) = |x - y|$ (the 1-norm on the line) is a metric space, but it is not a vector space: it fails to be closed under addition, since $\sqrt{2} + \sqrt{2} = 2\sqrt{2}$ is not of the required form. We *can*, however, build a genuine vector space that includes $\sqrt{2}$, namely

$$V = \mathbb{Q} + \mathbb{Q}\sqrt{2} = \left\{ a + b\sqrt{2} \mid a, b \in \mathbb{Q} \right\},$$

which is closed under rational scalar multiplication and addition.

Exercise 2.27. Show that $V = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is a vector space over $\mathbb{Q}$.

When a space is both a vector space and a metric space it is called, in general, a *metric vector space*. Typically the vector spaces we care about carry their distance through a *norm* (a notion of length), so more commonly we speak of *normed vector spaces*. We will make norms precise when we return to linear algebra; for now the important upgrade is that we finally have a way to say two points are “close.”

2.1.5 Sequences, Limits, and Convergence

Return to the metric space $\mathbb{Q} \cup \{\sqrt{2}\}$ with $d = |\cdot|$. Recall the brackets around $\sqrt{2}$: the lower bounds $1, \frac{5}{4}, \dots$ form a list of numbers getting progressively closer to $\sqrt{2}$ from below, and the upper bounds $3, \frac{3}{2}, \dots$ get progressively closer from above. Each such list is a *sequence*. Because we now have a distance, the claim that they get “closer” has content:

$$|1 - \sqrt{2}| > \left|\frac{5}{4} - \sqrt{2}\right| > \dots$$

Sequences are a standard and powerful tool precisely because they let us *start* in a setting we understand (here, the rationals) and *approach* a setting we are trying to reason about (here, $\sqrt{2}$). We now formalize these ideas.

Definition 2.28 (Sequence). A *sequence* in a set S is a function $a : \mathbb{N} \rightarrow S$, written $\{a_n\}_{n=1}^{\infty}$ with $a_n = a(n)$. A sequence is an arbitrary ordered collection: $1, 1, 1, \dots$ is a sequence; $1, -1, 1, -1, \dots$ is a sequence; $1, 2, 10, 7, 2, 5, -1, 0, \dots$ is a sequence even if it has no discernible pattern. Typically we care about *useful* sequences.

Definition 2.29 (Limit and convergence). Let (S, d) be a metric space and $\{a_n\}$ a sequence in S . We say $\{a_n\}$ *converges* to a *limit* $L \in S$, written $a_n \rightarrow L$, if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n > N \Rightarrow d(a_n, L) < \varepsilon.$$

A sequence that does not converge to any $L \in S$ is said to *diverge*.

The definition deserves slow reading. It says: no matter how small a tolerance ε you name, the sequence is eventually (from some index N onward) within ε of L . The choice of N is allowed to depend on ε : smaller tolerances generally require going further out.

A subtlety that foreshadows the construction of $\mathbb{R}$: whether a sequence has a limit depends on the space. In $\mathbb{Q} \cup \{\sqrt{2}\}$ a sequence converging to $\sqrt{2}$ does attain its limit, but a sequence of rationals marching toward, say, π has *no* limit in this space, because π is not in it—even though the sequence is “trying” to converge. We will return to this shortly.

The constant sequence $1, 1, 1, \dots$ converges to 1 trivially.

Exercise 2.30. Does the sequence $1, -1, 1, -1, \dots$ converge? Argue, using the definition, whether it does or does not.

The definition also settles a popular puzzle: does $0.999\dots = 1$? To claim $0.999\dots \neq 1$ we would need some $\varepsilon > 0$ with $|0.999\dots - 1| > \varepsilon$, for if the two numbers are distinct a metric requires a nonzero distance between them (axiom (M1)). But for *every* $\varepsilon > 0$ we have $|0.999\dots - 1| < \varepsilon$, so no such ε exists and therefore $0.999\dots = 1$. Notice this reasoning fails

for a would-be number “1.00...01”: to define 1.00...01 we must say at which place the final 1 occurs, and in doing so we automatically produce an $\varepsilon > 0$ with $|1.00...01 - 1| > \varepsilon$.

There is clearly a relationship between how small ε is and how far into a convergent sequence we must go to find a valid N . Consider the sequence $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$, i.e. $1, \frac{1}{2}, \frac{1}{3}, \dots$, which clearly has $\frac{1}{n} \rightarrow 0$. For $\varepsilon > 0$ choose $N = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1$, where $\lceil \cdot \rceil$ is the *ceiling function* (the smallest integer at least as large as its argument). Then for all $n > N$,

$$n > \left\lceil \frac{1}{\varepsilon} \right\rceil + 1 > \frac{1}{\varepsilon} \implies \varepsilon > \frac{1}{n},$$

so that

$$\left| \frac{1}{n} - 0 \right| = \left| \frac{1}{n} \right| = \frac{1}{n} < \varepsilon.$$

Hence for any $\varepsilon > 0$ the choice $N_{\varepsilon} = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1$ satisfies the definition of convergence, confirming $\frac{1}{n} \rightarrow 0$.

Exercise 2.31. Repeat the analysis for $\left\{\frac{1}{n^2}\right\}_{n=1}^{\infty}$: exhibit an explicit N_{ε} verifying $\frac{1}{n^2} \rightarrow 0$.

2.1.6 Cauchy Sequences and the Construction of the Reals

The convergence definition has an awkward feature: to use it we must already know the limit L . But when we are *constructing* new numbers (like $\sqrt{2}$ or π) the limit is exactly what we do not yet have. The fix is to describe convergence-like behavior *internally*, in terms of the sequence's own terms getting close to *each other*.

Definition 2.32 (Cauchy sequence). A sequence $\{a_n\}$ in a metric space (S, d) is a *Cauchy sequence* if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$m, n > N \implies d(a_m, a_n) < \varepsilon.$$

For example, a sequence of rationals converging (in $\mathbb{R}$) to π is Cauchy—its terms bunch together, their pairwise distances shrinking to zero—yet it does not converge *within* $\mathbb{Q} \cup \{\sqrt{2}\}$, because its would-be limit is absent from that space. It converges in $\mathbb{R}$. The two notions are closely related in one direction.

Theorem 2.33. *Every convergent sequence is Cauchy.*

Exercise 2.34. Prove Theorem 2.33. (*Hint: start from convergence $a_n \rightarrow L$ and, to compare two terms, add and subtract the limit and use the triangle inequality: $d(a_m, a_n) \leq d(a_m, L) + d(L, a_n)$.*)

The converse—every Cauchy sequence converges—is precisely what *fails* in $\mathbb{Q}$ and what $\mathbb{R}$ is built to repair. Here is the idea. Return to the two sequences of rationals bracketing $\sqrt{2}$,

$$\{a_n\}_{n=1}^{\infty} = \left\{1, \frac{5}{4}, \dots\right\} \quad \text{and} \quad \{b_n\}_{n=1}^{\infty} = \left\{3, \frac{3}{2}, \dots\right\},$$

where $\{a_n\}$ approaches from below and $\{b_n\}$ from above. Both are Cauchy, so both are “trying” to converge to $\sqrt{2}$, and their intersection is empty: no rational lies in every bracket. The number $\sqrt{2}$ is thus a point that *partitions* the rationals into those above it and those below it, with no overlap and nothing in between.

Definition 2.35 (Partition). A *partition* of a set S is a collection of nonempty subsets that are pairwise disjoint and whose union is all of S ; that is, every element of S lies in exactly one subset of the collection.

By considering, for *every* possible partition of $\mathbb{Q}$ into a “lower” part and an “upper” part, the pair of rational sequences that close in on the dividing point, we construct the real numbers $\mathbb{R}$ by *filling in the gaps* of $\mathbb{Q}$. This is not a fully rigorous construction, but it is exactly the idea behind the construction via *Dedekind cuts*, which the interested reader can look up. Two remarks: this construction does *not* include $+\infty$ or $-\infty$; and, equivalently, $\mathbb{R}$ can be built as the set of limits of Cauchy sequences of rationals, which is why $\mathbb{R}$ is called *complete*—in $\mathbb{R}$, every Cauchy sequence converges.

2.1.7 Where Countable Crosses into Uncountable

We built $\mathbb{R}$ by filling the gaps in $\mathbb{Q}$. Since $\mathbb{Q}$ is countable, it is natural to ask: does filling the gaps keep us countable? The answer is no, and it is one of the most important facts in the course.

Theorem 2.36. $\mathbb{R}$ is uncountable.

Proof. It suffices to show the subinterval $(0, 1)$ is uncountable, since a subset of a countable set would be countable. Assume for contradiction that the reals in $(0, 1)$ are countable, so they can be listed $x_1, x_2, x_3, \dots$, each written as an infinite decimal

$$x_i = 0.d_{i1}d_{i2}d_{i3}\cdots, \quad d_{ij} \in \{0, 1, \dots, 9\}.$$

Construct a new number $y = 0.e_1e_2e_3\cdots$ by choosing each digit to *differ* from the corresponding diagonal digit: for instance

$$e_j = \begin{cases} 5, & d_{jj} \neq 5, \\ 6, & d_{jj} = 5, \end{cases}$$

which also avoids the ambiguous all-9/all-0 tails. Then $y \in (0, 1)$, but y differs from x_j in the j -th decimal place for every j , so y is not in the list. This contradicts the assumption that the list contained *all* reals in $(0, 1)$. Hence $\mathbb{R}$ is uncountable. This is the “differs in each place” argument, i.e. *Cantor diagonalization*. $\square$

This is remarkable. Since $\mathbb{Q} \subset \mathbb{R}$ is countable, if the gaps we filled had somehow been only “countably many,” then $\mathbb{R}$ would be a union of two countable sets and hence countable (Theorem 2.38). Because $\mathbb{R}$ is uncountable while $\mathbb{Q}$ is countable, it must be the irrationals $\mathbb{R} \setminus \mathbb{Q}$ that are uncountable. The gaps vastly outnumber the rationals.

Where exactly does countable cross into uncountable? Unions are a good place to probe.

Theorem 2.37. A finite union of countable sets is countable.

Proof. We argue by induction on the number of sets—a good first example of *proof by induction*. *Base case* ($n = 1$): a single countable set is countable, trivially. *Inductive step*: suppose the union of any k countable sets is countable, and let $A_1, \dots, A_{k+1}$ be countable. By the inductive hypothesis $B = A_1 \cup \dots \cup A_k$ is countable, so it remains to show $B \cup A_{k+1}$ is countable, i.e. that the union of *two* countable sets is countable. Enumerate $B = \{b_1, b_2, \dots\}$ and $A_{k+1} = \{c_1, c_2, \dots\}$ and interleave them,

$$b_1, c_1, b_2, c_2, b_3, c_3, \dots,$$

skipping any element already listed; this visits every element of $B \cup A_{k+1}$ and gives a bijection with $\mathbb{N}$ (or a finite initial segment). Hence $B \cup A_{k+1}$ is countable, and by induction the claim holds for every finite n . Note carefully that this argument is valid only for *finite* n : induction proves a statement for each natural number n , but it does not, by itself, reach an infinite union. $\square$

Theorem 2.38. *A countable union of countable sets is countable.*

Proof. Because induction cannot reach an infinite union, we argue directly with an explicit pairing. Let $A_1, A_2, A_3, \dots$ be countable sets and enumerate each one as

$$A_i = \{a_{i1}, a_{i2}, a_{i3}, \dots\}.$$

Arrange all the elements in an infinite grid whose i -th row is A_i , and traverse the grid along its finite anti-diagonals:

$$a_{11}; \quad a_{21}, a_{12}; \quad a_{31}, a_{22}, a_{13}; \quad \dots,$$

that is, in order of increasing $i + j$, skipping repeats. Each anti-diagonal $\{a_{ij} : i + j = s\}$ is finite, so this procedure reaches every element a_{ij} after finitely many steps and thus defines a surjection from $\mathbb{N}$ onto $\bigcup_i A_i$; removing repeats yields a bijection. Hence the union is countable. Figure 2.3 illustrates the traversal. $\square$

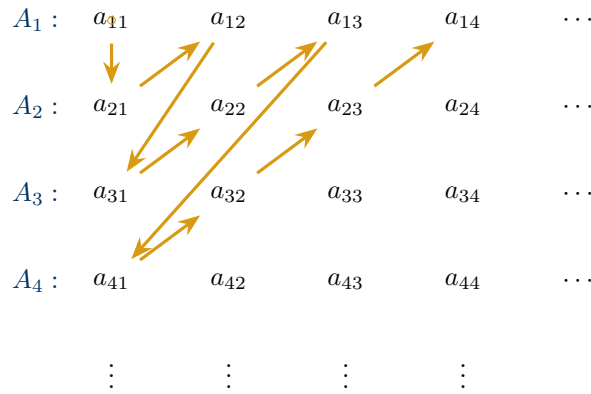


Figure 2.3: Enumerating a countable union $\bigcup_i A_i$. Row i lists the elements of $A_i = \{a_{i1}, a_{i2}, a_{i3}, \dots\}$. Following the finite anti-diagonals (the sets $\{a_{ij} : i + j = s\}$ for $s = 2, 3, 4, \dots$) visits every entry after finitely many steps, giving a surjection $\mathbb{N} \rightarrow \bigcup_i A_i$; deleting repeats yields a bijection.

Together with $\mathbb{Q} = \bigcup_{q \in \mathbb{N}} \{p/q : p \in \mathbb{Z}\}$, this reproves that $\mathbb{Q}$ is countable. Contrast it with the following, where countability finally breaks.

Theorem 2.39. *The set $\{0, 1\}^{\mathbb{N}}$ of all infinite binary sequences is uncountable.*

Exercise 2.40. Prove Theorem 2.39. (*Hint: use Cantor diagonalization, as in the proof that $\mathbb{R}$ is uncountable: assume the sequences can be listed, then build a new binary sequence that flips the j -th bit of the j -th listed sequence.*)

Finally, we localize the countable/uncountable boundary to a single unit interval, which is the version we will use most in probability.

Theorem 2.41. *The rational numbers in $[0, 1]$ form a countable set.*

Proof. For each $q \in \mathbb{N}$ let

$$A_q = \left\{ \frac{p}{q} : p = 0, 1, 2, \dots, q \right\},$$

the fractions in $[0, 1]$ with denominator q . Each A_q is finite (it has $q + 1$ elements), hence countable, and every rational in $[0, 1]$ appears in some A_q . Therefore

$$\mathbb{Q} \cap [0, 1] = \bigcup_{q \in \mathbb{N}} A_q$$

is a countable union of countable (indeed finite) sets, which is countable by Theorem 2.38. $\square$

Theorem 2.42. *The real numbers in $[0, 1]$ form an uncountable set.*

Proof. Apply Cantor diagonalization exactly as in Theorem 2.36, now with the list ranging over all reals in $[0, 1]$: assume they can be enumerated as decimals, then construct $y \in [0, 1]$ differing from the j -th listed number in its j -th decimal digit (choosing digits in $\{4, 5\}$, say, to avoid the $0.999\dots$ ambiguity). The number y is in $[0, 1]$ but not in the list, a contradiction. Hence $[0, 1]$ is uncountable. $\square$

Countable versus uncountable will matter a great deal when we turn to probability. Discrete distributions are defined on *countable* supports (a countable number of possible values), while continuous distributions are defined on *uncountable* supports. The distinction we have drawn here is exactly the one that separates a probability *mass* function from a probability *density* function.

Remark 2.43 (Important takeaways from this section). Three ideas from this section will recur throughout the course:

- the introduction of the ε - N (and $\varepsilon > 0$) proof strategy for making “arbitrarily close” precise;
- the introduction of sequences and limits, and their internal cousin, Cauchy sequences, which build and complete $\mathbb{R}$;
- the distinction between countable and uncountable sets.

2.2 Sets and Topology

In order to eventually establish properties related to more applied settings such as optimization, we need to formalize mathematical objects related to sets—that is, the objects we are optimizing over. Throughout this section, let X be a metric space with distance $d(x, y)$, and assume all elements and sets below are elements of X and subsets of X respectively.

2.2.1 Open Sets, Closed Sets, and Limit Points

Definition 2.44 (Neighborhood / ball). The *neighborhood*, or *ball*, around a point p with radius $r > 0$ is

$$B_r(p) = \{x \in X : d(x, p) < r\}.$$

In $\mathbb{R}$ a ball is an open interval (x, y) ; in $\mathbb{R}^2$ a ball is a disk without its boundary.

Definition 2.45 (Limit point). A point p is a *limit point* of the set S if every neighborhood of p contains a point $q \neq p$ such that $q \in S$.

It is worth comparing the definition of a limit point with the definition of a limit (Definition 2.29): a limit point p of S is precisely a point that some sequence in S can converge to, a correspondence we make exact in Lemma 2.54.

Definition 2.46 (Isolated point). If $p \in S$ and p is not a limit point of S , then p is an *isolated point* of S .

Definition 2.47 (Closed set). A set S is *closed* if every limit point of S is a point of S ; i.e. S contains all of its limit points.

Definition 2.48 (Interior point). A point p is an *interior point* of S if there exists a neighborhood $B_r(p)$ of p that is entirely contained in S , i.e. $B_r(p) \subseteq S$.

Definition 2.49 (Open set). A set S is *open* if every point of S is an interior point of S .

Definition 2.50 (Complement). The *complement* of a set S , denoted S^c , is the set of all points of X that are not in S : $S^c = \{x \in X : x \notin S\}$.

Let us build some intuition for what these objects mean. Let $X = (\mathbb{R}^2, \|\cdot\|_2)$, i.e. the metric space on $\mathbb{R}^2$ with distance $d(x, y) = \|x - y\|_2 = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$. Consider the ball around the origin with radius 1, $B_1(0)$. Visually this is the open unit disk (Figure 2.4): every point $p = \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}$ with $\|p - 0\|_2 = \sqrt{p_1^2 + p_2^2} < 1$, i.e. $p_1^2 + p_2^2 < 1$.

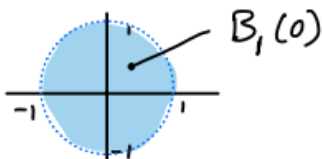


Figure 2.4: The open ball $B_1(0)$ in $\mathbb{R}^2$: all points strictly within distance 1 of the origin. The dashed boundary indicates that the bounding circle is *not* included.

Theorem 2.51. *Every neighborhood is an open set. Equivalently: if $B_r(p)$ is a neighborhood of radius r around a point p , then it is an open set.*

To prove this we always go back to the definitions as a starting point. A neighborhood is a set $B_r(p) = \{x \in X : d(x, p) < r\}$; an open set is one where every point is an interior point; and an interior point of a set is one around which there is a neighborhood contained inside the set.

Proof. Consider an arbitrary point $x \in B_r(p)$. Since $x \in B_r(p)$ we know $d(x, p) < r$. Since $r > 0$ there exists some $\varepsilon > 0$ such that $d(x, p) + \varepsilon = r$, i.e. $d(x, p) = r - \varepsilon$. Consider any point y with $d(x, y) < \varepsilon$, i.e. $y \in B_\varepsilon(x)$. We want to show that every such y satisfies $y \in B_r(p)$, as this would imply $B_\varepsilon(x) \subseteq B_r(p)$, so that every point $x \in B_r(p)$ is an interior point and thus $B_r(p)$ is open. By the triangle inequality,

$$d(y, p) \leq d(y, x) + d(x, p) < \varepsilon + (r - \varepsilon) = r,$$

so $d(y, p) < r$. Since y was chosen arbitrarily in $B_\varepsilon(x)$, this implies $B_\varepsilon(x) \subseteq B_r(p)$, so there exists a neighborhood of x fully contained in $B_r(p)$, and thus x is an interior point of $B_r(p)$. Since x was arbitrary, every $x \in B_r(p)$ is an interior point and thus $B_r(p)$ is an open set. $\square$

We often use balls of this type in reasoning about mathematics and typically call them *open balls* to distinguish them from closed balls.

Definition 2.52 (Closed ball). The *closed ball* around a point p with radius r is the set $\overline{B}_r(p) = \{x \in X : d(x, p) \leq r\}$.

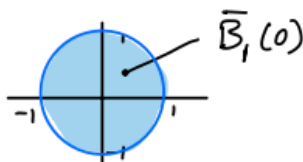


Figure 2.5: The closed ball $\overline{B}_1(0)$ in $\mathbb{R}^2$: all points within distance 1 of the origin, boundary included (solid circle).

Theorem 2.53. *The closed ball is a closed set.*

In order to prove this let us first prove a lemma that lets us work with sequences (rather than open balls) when reasoning about limit points.

Lemma 2.54. *A point $p \in X$ is a limit point of a set $S \subseteq X$ if and only if there exists a sequence $\{p_n\}_{n=1}^\infty$ in S with $p_n \neq p$ such that $\lim_{n \rightarrow \infty} p_n = p$ in X .*

Proof. ($\Rightarrow$) Suppose p is a limit point of S . Every neighborhood of p contains a point of S (distinct from p), so for every $\varepsilon > 0$ the set $B_\varepsilon(p) \cap S$ is non-empty. Let $\varepsilon_n = \frac{1}{n} > 0$, so that $\{\varepsilon_n\}_{n=1}^\infty$ is a sequence of radii with $\varepsilon_n \rightarrow 0$, and let $\tilde{B}_{\varepsilon_n}(p) = B_{\varepsilon_n}(p) \cap S$. Choose the points of the sequence as follows. Because p is a limit point, $\tilde{B}_{\varepsilon_n}(p) = B_{\varepsilon_n}(p) \cap S$ contains a point distinct from p ; select any such point and call it p_n (we do not need a point of maximal distance—such a

point may not exist, since the distances to p can approach but never attain a supremum). Then $p_n \in S$, $p_n \neq p$, and $d(p_n, p) < \varepsilon_n \rightarrow 0$, so the sequence $\{p_n\}$ is in S and $p_n \rightarrow p$ in X .

($\Leftarrow$) Suppose there is a sequence $\{p_n\}$ in S with $p_n \neq p$ and $p_n \rightarrow p$. Let $B_r(p)$ be any neighborhood of p . By convergence there exists N such that $n > N$ implies $d(p_n, p) < r$, i.e. $p_n \in B_r(p)$. Any such p_n is a point of S with $p_n \neq p$ lying in $B_r(p)$. Hence every neighborhood of p contains a point of S distinct from p , so p is a limit point of S . $\square$

This allows us to work with sequences for limit points instead of open balls/neighborhoods. Now we return to the proof of the theorem.

Proof. Assume for contradiction that the closed ball $\overline{B}_r(p)$ is not closed. Then there exists a limit point x of $\overline{B}_r(p)$ with $x \notin \overline{B}_r(p)$. Since x is a limit point of $\overline{B}_r(p)$, by Lemma 2.54 there is a sequence $\{x_n\}_{n=1}^\infty \subseteq \overline{B}_r(p)$ with $x_n \rightarrow x$ in X . Since $x \notin \overline{B}_r(p)$ we have $d(x, p) > r$. Since the sequence $\{x_n\}$ converges to x , choose $\varepsilon > 0$ and then $m > N_\varepsilon$ so that $d(x_m, x) < \varepsilon$. By the triangle inequality,

$$d(x, p) \leq d(x, x_m) + d(x_m, p) \leq d(x, x_m) + r.$$

Now take the limit as $m \rightarrow \infty$ on both sides. The left side is constant in m so it stays the same; the right side has $\lim_{m \rightarrow \infty} d(x, x_m) = 0$, and since it is a weak inequality it is preserved in the limit. Therefore

$$d(x, p) \leq r,$$

which contradicts $d(x, p) > r$. So $\overline{B}_r(p)$ must in fact contain all its limit points and thus be closed. $\square$

2.2.2 Complements, Closure, and the Structure of Open/Closed Sets

Theorem 2.55 (De Morgan's law for sets). *Let $\{S_\alpha\}$ be a (finite or infinite) collection of sets. Then*

$$\left(\bigcup_{\alpha} S_{\alpha}\right)^c = \bigcap_{\alpha} (S_{\alpha}^c).$$

Proof. We show the two sets contain the same points. A point x lies in $(\bigcup_{\alpha} S_{\alpha})^c$ if and only if $x \notin \bigcup_{\alpha} S_{\alpha}$, i.e. x belongs to no S_{α} . This holds if and only if for every α we have $x \notin S_{\alpha}$, i.e. $x \in S_{\alpha}^c$ for every α , which is exactly the statement $x \in \bigcap_{\alpha} S_{\alpha}^c$. Since the two sets have the same elements they are equal. $\square$

Theorem 2.56. *A set S is open if and only if its complement S^c is closed.*

Proof. ($\Rightarrow$) Suppose S is open. Let p be a limit point of S^c ; we show $p \in S^c$. Suppose instead $p \in S$. Since S is open, p is an interior point, so there is a ball $B_r(p) \subseteq S$, and hence $B_r(p) \cap S^c = \emptyset$. But then no neighborhood of p contains a point of S^c , contradicting that p is a limit point of S^c . Therefore $p \in S^c$, so S^c contains all its limit points and is closed.

($\Leftarrow$) Suppose S^c is closed. Let $p \in S$; we show p is an interior point. Suppose not: then every ball $B_r(p)$ contains a point of S^c . Since $p \in S$, that point is $\neq p$, so p is a limit point of S^c . As S^c is closed, $p \in S^c$, contradicting $p \in S$. Hence p is an interior point, and since $p \in S$ was arbitrary, S is open. $\square$

Corollary 2.57. *A set F is closed if and only if its complement F^c is open.*

Proof. Apply Theorem 2.56 to $S = F^c$ and use $(F^c)^c = F$. $\square$

It is useful to note some canonical aspects of open and closed sets. The set X defining the entire space is closed since it of course contains all its limit points; by complementarity the empty set $\emptyset$ is open. However, by a trivial technicality the empty set also contains all its limit points (it has none), so it is closed; by complementarity the whole space X is open. Thus the empty set and X are *clopen* (both open and closed). There are other ways to construct clopen sets but they are not typical sets one runs into except for $\emptyset$ and X . Note this is different from something like $(0, 1]$ on the real line: this interval is not clopen—it is closed on one side and open on the other, so it would be described as neither open nor closed.

Theorem 2.58.

- (a) For any collection $\{E_\alpha\}$ of open sets, $\bigcup_\alpha E_\alpha$ is open.
- (b) For any collection $\{F_\alpha\}$ of closed sets, $\bigcap_\alpha F_\alpha$ is closed.
- (c) For any finite collection $E_1, \dots, E_n$ of open sets, $\bigcap_{i=1}^n E_i$ is open.
- (d) For any finite collection $F_1, \dots, F_n$ of closed sets, $\bigcup_{i=1}^n F_i$ is closed.

Exercise 2.59. Prove Theorem 2.58. (Hint: for (a) an interior point of one E_α is an interior point of the union; obtain (b) from (a) by De Morgan and Corollary 2.57; for (c) intersect the finitely many radii to get a common ball; obtain (d) from (c) by De Morgan.)

Definition 2.60 (Closure). If X is a metric space, $S \subseteq X$, and S' denotes the set of all limit points of S in X , then the *closure* of S is the set $\bar{S} = S \cup S'$.

Theorem 2.61. If X is a metric space and $S \subseteq X$, then

- (a) $\bar{S}$ is closed;
- (b) $S = \bar{S}$ if and only if S is closed;
- (c) $\bar{S} \subseteq F$ for every closed set $F \subseteq X$ such that $S \subseteq F$.

Proof. (a) We show the complement of $\bar{S}$ is open. Let $p \in (\bar{S})^c$, so $p \notin S$ and p is not a limit point of S . Because p is not a limit point, there is a ball $B_r(p)$ containing no point of S (there is a neighborhood missing S except possibly p , and $p \notin S$). Take any $q \in B_r(p)$. Since $B_r(p)$ is open, some ball around q lies inside $B_r(p)$ and hence also contains no point of S ; thus q is neither in S nor a limit point of S , so $q \notin \bar{S}$. Therefore $B_r(p) \subseteq (\bar{S})^c$, so $(\bar{S})^c$ is open and $\bar{S}$ is closed.

(b) S is closed $\iff S' \subseteq S \iff S \cup S' = S \iff \bar{S} = S$.

(c) Let F be closed with $S \subseteq F$. Every limit point of S is a limit point of F (a sequence in S is a sequence in F), so $S' \subseteq F' \subseteq F$, the last inclusion because F is closed. Hence $\bar{S} = S \cup S' \subseteq F$. In other words, $\bar{S}$ is the smallest closed set containing S . $\square$

Remark 2.62. Two ways to view the closure. First, closing a set S amounts to “filling in” the missing limit points to make it closed. Second, part (c) says the closure of S is equivalently the intersection of all closed sets containing S ; this intersection may be uncountably infinite. This characterization is useful in topology and we will see it again when we discuss convexity in optimization.

2.2.3 Boundedness, Suprema, and Infima

Definition 2.63 (Bounded set). A set S is *bounded* if there is a positive real number M and a point $q \in X$ such that $d(p, q) < M$ for all $p \in S$.

The definition of bounded does not depend on a unique reference point q ; we only need *any* fixed q to act as a reference against which to measure how far away the furthest point is. We can operationalize unboundedness with a fixed reference (e.g. the origin 0) by saying: S is unbounded if for every $\alpha > 0$ there exists $x \in S$ with $d(x, 0) > \alpha$.

Definition 2.64 (Supremum). Let $S \subseteq \mathbb{R}$ be bounded above. The *supremum* $\sup S$ is the *least upper bound* of S : the number y such that (i) $x \leq y$ for all $x \in S$, and (ii) if y' is any upper bound of S then $y \leq y'$.

Definition 2.65 (Infimum). Let $S \subseteq \mathbb{R}$ be bounded below. The *infimum* $\inf S$ is the *greatest lower bound* of S : the number z such that (i) $z \leq x$ for all $x \in S$, and (ii) if z' is any lower bound of S then $z' \leq z$.

Definition 2.66 (Diameter). The *diameter* of a set S is $\text{diam}(S) = \sup \{d(x, y) : x, y \in S\}$.

Theorem 2.67. *A set is bounded if and only if its diameter is finite.*

Proof. If S is bounded there is M and q with $d(p, q) < M$ for all $p \in S$. Then for any $x, y \in S$, the triangle inequality gives $d(x, y) \leq d(x, q) + d(q, y) < 2M$, so $\text{diam}(S) \leq 2M$ is finite. Conversely, if $\text{diam}(S) = D < \infty$, fix any $q \in S$; then $d(p, q) \leq D$ for all $p \in S$, so S is bounded (take $M = D + 1$). $\square$

Corollary 2.68. *A set is unbounded if and only if its diameter is not finite.*

Theorem 2.69. *Let S be a nonempty set of real numbers which is bounded above, and let $y = \sup S$. Then $y \in \overline{S}$, and hence $y \in S$ if S is closed.*

Proof. If $y \in S$ then $y \in \overline{S}$ trivially. Suppose $y \notin S$. For every $\varepsilon > 0$, the number $y - \varepsilon$ is not an upper bound of S (else y would not be the *least* upper bound), so there exists $x \in S$ with $y - \varepsilon < x \leq y$. Since $y \notin S$ and $x \in S$ we have $x \neq y$, so $x \in B_\varepsilon(y) \cap S$ with $x \neq y$. Thus every neighborhood of y contains a point of S distinct from y , so y is a limit point of S and $y \in S' \subseteq \overline{S}$. If S is closed then $\overline{S} = S$, so $y \in S$. $\square$

Corollary 2.70. *If S is bounded below and $z = \inf S$, then $z \in \overline{S}$.*

2.2.4 Euclidean Space

So far we have discussed sets generally, but since we will focus on applications involving the real numbers we introduce the standard/default space we will work in, called *Euclidean space* $\mathbb{E}^k$. This is a normed vector space on $\mathbb{R}^k$ for a fixed k with the Euclidean norm. For an element $\mathbf{x} \in \mathbb{R}^k$ we have $\mathbf{x} = (x_1, x_2, \dots, x_k)$ where each $x_i \in \mathbb{R}$.

Remark 2.71 (Notation for vectors). As we did when we introduced vector spaces and scalars (Section 2.1.1), we write **vectors in bold** and scalars in ordinary (non-bold) type. For a bold vector $\mathbf{x}$, the unbolded symbol x_i denotes the i -th entry of $\mathbf{x}$. When writing by hand, or in any setting where boldness is hard to convey, we instead recommend the *arrow* notation to denote a vector, for example $\vec{x} = (x_1, x_2, \dots, x_k)$; the two notations $\mathbf{x}$ and $\vec{x}$ mean the same thing. Finally,

sometimes it is notationally awkward to carry two subscripts at once—for instance when $\mathbf{x}_n$ is itself a term of a *sequence* of vectors. In such cases we write $x_n^{(i)}$ to denote the i -th entry of the vector $\mathbf{x}_n$. We will generally make this clear from context, but we mention it now so the reader is not surprised when encountering the notation.

Definition 2.72 (k -cell). A k -cell with bounds $a_i < b_i$ for $i = 1, \dots, k$ is the set of all points in $\mathbb{R}^k$ such that $a_i \leq x_i \leq b_i$ for $i = 1, \dots, k$:

$$C^k = \left\{ \mathbf{x} \in \mathbb{R}^k : a_i \leq x_i \leq b_i, \ i = 1, \dots, k \right\}.$$

Clearly C^k is a closed set. Moreover C^1 is a closed interval $[a, b]$ on $\mathbb{R}$; we call the corresponding open set (a, b) a *segment* (a 1-D open ball). C^2 is a closed rectangle in $\mathbb{R}^2$. Typically we work with $k = 1$ for simplicity but we highlight when the result is applicable to $\mathbb{R}^k$.

Euclidean space is a normed vector space, but it is a specific type of space called an *inner product space* where an operation called the *inner product* is defined and the norm derives from this inner product. For two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k$, the inner product is

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^k x_i y_i.$$

An additional, equivalent notation for the inner product—previewing a representation from linear algebra—is

$$\mathbf{x}^\top \mathbf{y} = \mathbf{y}^\top \mathbf{x} = \mathbf{x} \cdot \mathbf{y},$$

where $\mathbf{x}^\top$ denotes the *transpose* of $\mathbf{x}$. The norm is

$$\|\mathbf{x}\|_2 = (\mathbf{x} \cdot \mathbf{x})^{1/2} = \left(\sum_{i=1}^k x_i^2 \right)^{1/2}.$$

The following results are given for $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^k$ and $\alpha \in \mathbb{R}$:

- (a) $\|\mathbf{x}\|_2 \geq 0$;
- (b) $\|\mathbf{x}\|_2 = 0$ if and only if $\mathbf{x} = \mathbf{0}$;
- (c) $\|\alpha \mathbf{x}\|_2 = |\alpha| \|\mathbf{x}\|_2$;
- (d) $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$ (Cauchy–Schwarz);
- (e) $\|\mathbf{x} + \mathbf{y}\|_2 \leq \|\mathbf{x}\|_2 + \|\mathbf{y}\|_2$ (triangle inequality);
- (f) $\|\mathbf{x} - \mathbf{z}\|_2 \leq \|\mathbf{x} - \mathbf{y}\|_2 + \|\mathbf{y} - \mathbf{z}\|_2$.

2.2.5 Compactness

Compactness is the property of $\mathbb{R}^k$ that makes the things we want to do in optimization work properly. We will often directly appeal to compactness of subsets of the real numbers as a means to assert a claim about behavior or to support the derivation of a result. The standard definition of compactness is a bit abstract but it is the canonical definition, so we introduce it here before appealing to a more intuitive equivalent definition.

Definition 2.73 (Open cover). An *open cover* of a set S in a metric space X is a collection $\{E_\alpha\}$ of open subsets of X such that $S \subseteq \bigcup_\alpha E_\alpha$.

To understand an example of an open cover, consider the most trivial version, which is to let E_α be an open ball of radius $\varepsilon > 0$ around each point $x \in S$. Then trivially $S \subseteq \bigcup_\alpha B_\varepsilon(x)$. This open cover has cardinality equal to the cardinality of S , which can be uncountably infinite. By cardinality of the open cover we mean the number of open sets which make up the cover. Note that these open sets in the cover can overlap.

Definition 2.74 (Compact set). A subset K of a metric space X is said to be *compact* if every open cover of K contains a *finite* subcover. That is, from every collection $\{E_\alpha\}$ of open sets that covers K we can select a finite subset such that

$$K \subseteq E_{\alpha_1} \cup E_{\alpha_2} \cup \cdots \cup E_{\alpha_n}, \quad n < \infty.$$

We will show a relationship between compactness and closedness plus boundedness later, but for now we show an illustration of a non-compact set to build intuition. Consider the half-open interval $(0, 1]$ in $\mathbb{E}^1$. To show it is not compact we need only exhibit one open cover with no finite subcover. Let $U_m = \left(\frac{1}{m}, 2\right)$, the segment from $\frac{1}{m}$ to 2, and let $\mathcal{U} = \{U_m : m \in \mathbb{N}\}$, so $U_1 = (1, 2)$, $U_2 = \left(\frac{1}{2}, 2\right)$, and so on. Clearly $\lim_{m \rightarrow \infty} U_m = (0, 2)$ and thus $(0, 1] \subseteq \bigcup_{m=1}^\infty U_m$. However, for any finite m , $\frac{1}{m} > 0$, so if we use subindexing $m_1, m_2, \dots, m_n$ to identify a finite n -sized subcover, the union $U_{m_1} \cup U_{m_2} \cup \cdots \cup U_{m_n}$ does not contain $(0, 1]$ for any possible finite selection (it misses points near 0 below $\frac{1}{\max_i m_i}$). Hence $(0, 1]$ is not compact.

Definition 2.75 (Sequential compactness). A set S in a metric space X is *sequentially compact* if every sequence of elements of S contains a convergent subsequence whose limit lies in S .

It is the case that compactness and sequential compactness are equivalent. However, in order to prove it we need to introduce some additional results that build on what we introduced before.

2.2.6 More on Sequences

Theorem 2.76. Let $\{p_n\}$ be a sequence in a metric space X .

- (a) $\{p_n\}$ converges to $p \in X$ if and only if every neighborhood of p contains all but finitely many of the terms $\{p_n\}$.
- (b) If $p, p' \in X$ and $\{p_n\}$ converges to both p and p' , then $p = p'$.
- (c) If $\{p_n\}$ converges then $\{p_n\}$ is bounded.

Proof. (a) $(\Rightarrow)$ Suppose $p_n \rightarrow p$ and let $B_r(p)$ be a neighborhood. Taking $\varepsilon = r$ in the definition of convergence, there is N with $d(p_n, p) < r$ for all $n > N$, i.e. $p_n \in B_r(p)$; only the finitely many terms $p_1, \dots, p_N$ can lie outside. $(\Leftarrow)$ Given $\varepsilon > 0$, the neighborhood $B_\varepsilon(p)$ contains all but finitely many terms, so there is N (the largest excluded index) with $p_n \in B_\varepsilon(p)$, i.e. $d(p_n, p) < \varepsilon$, for all $n > N$; hence $p_n \rightarrow p$.

(c) Suppose $p_n \rightarrow p$. Taking $\varepsilon = 1$, there is N with $d(p_n, p) < 1$ for $n > N$. Let $M' = \max_{1 \leq n \leq N} d(p_n, p)$, the largest distance among the finitely many early terms, and set $M = \max(M', 1) + 1$. Then $d(p_n, p) < M$ for all n , so $\{p_n\}$ is bounded (with reference point $q = p$). $\square$

Exercise 2.77. Prove part (b) of Theorem 2.76. (Hint: try to assert $d(p, p') = 0$ starting from the definitions of convergence.)

The results above, and the proof technique, highlight a useful way to think about converging sequences: they can be separated into the “finite” part up to $n \leq N_\varepsilon$ and then the infinite part made up of terms $n > N_\varepsilon$ which are arbitrarily close to the limit.

Part (a) also motivates an augmentation. It says that if $\{p_n\}$ converges to p then (1) every neighborhood around p contains infinitely many members of the sequence, and (2) there are only finitely many elements in the sequence outside of each neighborhood. Let us relax the second condition. Consider the sequence $1, -1, 1, -1, \dots$. Both 1 and -1 have infinitely many terms in every neighborhood of 1 and -1 respectively (satisfying the first criterion), but the second criterion fails. Let us introduce a term for this.

Definition 2.78 (Cluster point). A point c is a *cluster point* of a sequence $\{p_n\}$ in a metric space X if for every $\varepsilon > 0$ the set $\{n \in \mathbb{N} : x_n \in B_\varepsilon(c)\}$ is an infinite set.

In the sequence $1, -1, 1, -1, \dots$ the points 1 and -1 are both cluster points. Another example is the sequence defined by

$$x_n = \begin{cases} 1 - \frac{1}{n}, & n \text{ even,} \\ \frac{1}{n}, & n \text{ odd.} \end{cases}$$

For n even the sequence clusters at 1; for n odd it clusters at 0. This sequence does not converge but clearly there is a subsequence that converges to 1 and a subsequence that converges to 0.

Definition 2.79 (Subsequence). Given a sequence $\{p_n\}$, consider a strictly increasing sequence of positive integers indexed by $k \in \mathbb{N}$, written $\{n_k\}$ with $n_1 < n_2 < n_3 < \dots$. Note this is not restricted to a consecutive ordering: $n_1 = 1, n_2 = 2, n_3 = 5, \dots$ is valid, but so is $n_1 = 1, n_2 = 5, n_3 = 11, \dots$. The index k identifies the k -th element of the subsequence. The sequence $\{p_{n_k}\}_{k=1}^\infty$ is a *subsequence* of $\{p_n\}$. If $\{p_{n_k}\}$ converges, its limit is called a *subsequential limit* or cluster point of the sequence $\{p_n\}_{n=1}^\infty$.

For the sequence $\{x_n\}$ above we can define two subsequences, $\{x_{n_i}\}_{i=1}^\infty$ with $n_1 = 1, n_2 = 3, n_3 = 5, \dots$ (the odd indices) and $\{x_{n_j}\}_{j=1}^\infty$ with $n_1 = 2, n_2 = 4, \dots$ (the even indices), and then

$$x_{n_i} \xrightarrow{i \rightarrow \infty} 0, \quad x_{n_j} \xrightarrow{j \rightarrow \infty} 1.$$

2.2.7 Completeness

We will return to subsequences later. For now we revisit Cauchy sequences.

Definition 2.80 (Complete metric space). A metric space in which every Cauchy sequence converges is said to be *complete*. Note completeness is about metric spaces, not subsets.

By our construction of $\mathbb{R}$ as “filling in” the gaps of $\mathbb{Q}$ we showed that every Cauchy sequence in $\mathbb{R}$ converges to a point in $\mathbb{R}$: $\mathbb{R}$ was generated as the “completion” of $\mathbb{Q}$, so $\mathbb{R}$ is a complete metric space. For the purpose of these bootcamp notes this is sufficient but would be treated more rigorously in an analysis-focused course.

Theorem 2.81. $\mathbb{E}^k$ is a complete metric space.

Proof. Let $\{\mathbf{x}_n\}$ be a Cauchy sequence in $\mathbb{R}^k$. For each coordinate i , $|x_m^{(i)} - x_n^{(i)}| \leq \|\mathbf{x}_m - \mathbf{x}_n\|_2$, so each coordinate sequence $\{x_n^{(i)}\}$ is Cauchy in $\mathbb{R}$. Since $\mathbb{R}$ is complete, each coordinate converges, say $x_n^{(i)} \rightarrow L_i$; let $\mathbf{L} = (L_1, \dots, L_k)$. Then

$$\|\mathbf{x}_n - \mathbf{L}\|_2 = \left(\sum_{i=1}^k (x_n^{(i)} - L_i)^2 \right)^{1/2} \rightarrow 0,$$

so $\mathbf{x}_n \rightarrow \mathbf{L}$ in $\mathbb{R}^k$. Thus the chain is: Cauchy in $\mathbb{R}^k \Rightarrow$ Cauchy coordinate-wise in $\mathbb{R} \Rightarrow$ coordinate-wise convergence $\Rightarrow$ Euclidean convergence. $\square$

If we want to talk about completeness of a subset we need to consider that subset as its own metric space.

Theorem 2.82. *Suppose (X, d) is a complete metric space. If Y is a subset of (X, d) , then $(Y, d|_Y)$ is complete if and only if Y is closed.*

Proof. ($\Leftarrow$) Suppose Y is closed. Let $\{y_n\}$ be a Cauchy sequence in Y ; it is also Cauchy in X , and since X is complete it converges to some $p \in X$. If $p \in Y$ we are done. Otherwise p is a limit of a sequence in Y with terms distinct from p , so p is a limit point of Y ; as Y is closed, $p \in Y$. Hence the sequence converges in Y , and Y is complete.

($\Rightarrow$) Suppose $(Y, d|_Y)$ is complete. Let p be a limit point of Y . By Lemma 2.54 there is a sequence in Y converging to p in X ; being convergent it is Cauchy. By completeness of Y it converges to a point of Y , and since limits are unique (Theorem 2.76(b)) that point is p , so $p \in Y$. Thus Y contains its limit points and is closed. $\square$

2.3 Compactness in $\mathbb{E}^k$, Sequences, and Series

The previous section introduced compactness through open covers and, separately, sequential compactness. We now prove that—together with completeness and total boundedness—these notions coincide, specialize the result to $\mathbb{E}^k$ via the Heine–Borel theorem, and then develop the sequences-and-series machinery we will need for differentiation and Taylor’s theorem.

2.3.1 The Compactness Characterization

Definition 2.83 (Totally bounded). A set S in a metric space X is *totally bounded* if for every $\varepsilon > 0$ there exist finitely many points $x_1, \dots, x_n \in S$ such that $S \subseteq \bigcup_{i=1}^n B_\varepsilon(x_i)$.

For example, $[0, 1] \subseteq \mathbb{E}^1$ is totally bounded: given $\varepsilon > 0$ choose $n > 1/\varepsilon$ and the points $x_i = \frac{i-1}{n}$ for $i = 1, \dots, n+1$ (so $x_1 = 0, \dots, x_{n+1} = 1$). Consecutive points are at distance $\frac{1}{n} < \varepsilon$, so the balls $B_\varepsilon(x_i)$ overlap and cover $[0, 1]$. We can already see a connection here to a finite open cover.

Theorem 2.84. *For a set S in a metric space X , the following are equivalent:*

- (1) S is compact;
- (2) S is complete and totally bounded;
- (3) S is sequentially compact.

Proof. We prove the cycle $(1) \Rightarrow (3) \Rightarrow (2) \Rightarrow (1)$, which establishes all the equivalences (in particular $(1) \Leftrightarrow (2)$ and $(2) \Leftrightarrow (3)$).

$(1) \Rightarrow (3)$. Let $\{x_n\}$ be a sequence in a compact S and suppose, for contradiction, that it has no subsequence converging to a point of S . Then no $p \in S$ is a cluster point, so each p has a radius $\varepsilon_p > 0$ with $B_{\varepsilon_p}(p)$ containing x_n for only finitely many indices n . The balls $\{B_{\varepsilon_p}(p) : p \in S\}$ form an open cover of S ; by compactness finitely many of them cover S , but then only finitely many indices n occur in total—contradicting that the sequence has infinitely many terms. Hence some subsequence converges in S .

$(3) \Rightarrow (2)$. Assume S is sequentially compact. *Complete*: a Cauchy sequence in S has, by (3), a subsequence converging to some $p \in S$; a Cauchy sequence with a convergent subsequence converges to the same limit, so p is its limit and S is complete. *Totally bounded*: if not, there is $\varepsilon > 0$ with no finite ε -net. Pick $x_1 \in S$; having chosen $x_1, \dots, x_k$, the balls $B_\varepsilon(x_1), \dots, B_\varepsilon(x_k)$ do not cover S , so choose $x_{k+1} \in S$ outside them. The resulting sequence has $d(x_i, x_j) \geq \varepsilon$ for all $i \neq j$, so no subsequence is Cauchy and none converges, contradicting (3).

$(2) \Rightarrow (1)$. Assume S is complete and totally bounded, and suppose $\{E_\alpha\}$ is an open cover of S with no finite subcover. Since S is totally bounded it is covered by finitely many balls of radius $\frac{1}{2}$; at least one of them, intersected with S , has no finite subcover—call this piece S_1 , contained in a ball of radius $\frac{1}{2}$. The set S_1 is itself totally bounded, so repeating the argument with radius $\frac{1}{4}$ produces $S_2 \subseteq S_1$ inside a ball of radius $\frac{1}{4}$ with no finite subcover, and inductively a nested sequence $S_1 \supseteq S_2 \supseteq \dots$ with S_k inside a ball of radius 2^{-k} and no finite subcover. Choose $p_k \in S_k$; then $d(p_j, p_k) \leq 2^{-\min(j,k)+1} \rightarrow 0$, so $\{p_k\}$ is Cauchy and, by completeness, converges to some $p \in S$. Now $p \in E_\alpha$ for some α , and since E_α is open, $B_r(p) \subseteq E_\alpha$ for some $r > 0$. For k large enough, $S_k \subseteq B_r(p) \subseteq E_\alpha$, so S_k is covered by the *single* set E_α —contradicting that S_k has no finite subcover. Hence a finite subcover exists and S is compact. $\square$

2.3.2 Monotone Sequences and Bolzano–Weierstrass

We now specialize to sequences of real numbers, where order gives us extra tools.

Definition 2.85 (Monotone sequence). A sequence $\{s_n\}$ of real numbers is *monotonically increasing* if $s_n \leq s_{n+1}$ for all n , and *monotonically decreasing* if $s_n \geq s_{n+1}$ for all n . A sequence that is one or the other is called *monotone*.

Theorem 2.86 (Monotone Convergence). *Suppose $\{s_n\}$ is monotone. Then $\{s_n\}$ converges if and only if it is bounded.*

Proof. If $\{s_n\}$ converges then it is bounded by Theorem 2.76(c). Conversely, suppose $\{s_n\}$ is bounded and (without loss of generality) increasing. The set $\{s_n : n \in \mathbb{N}\}$ is bounded above, so by completeness of $\mathbb{R}$ it has a supremum $L = \sup s_n$. Given $\varepsilon > 0$, $L - \varepsilon$ is not an upper bound, so there is N with $s_N > L - \varepsilon$; by monotonicity $L - \varepsilon < s_N \leq s_n \leq L$ for all $n > N$, whence $|s_n - L| < \varepsilon$. Thus $s_n \rightarrow L$. The decreasing case is identical with the infimum. $\square$

Lemma 2.87 (Monotone subsequence / “Rising Sun”). *Every sequence of real numbers contains a monotone subsequence (increasing, decreasing, or both).*

Proof. Call an index n a *peak* of $\{s_n\}$ if $s_n \geq s_m$ for every $m > n$ (no later term exceeds it). Consider two cases.

If there are infinitely many peaks $n_1 < n_2 < n_3 < \dots$, then by the definition of a peak $s_{n_1} \geq s_{n_2} \geq s_{n_3} \geq \dots$, a decreasing subsequence.

If there are only finitely many peaks, let N be larger than the last peak. Then $n_1 := N + 1$ is not a peak, so there is $n_2 > n_1$ with $s_{n_2} > s_{n_1}$; n_2 is not a peak either, so there is $n_3 > n_2$ with $s_{n_3} > s_{n_2}$; continuing gives an increasing subsequence. $\square$

Theorem 2.88 (Bolzano–Weierstrass). *Every bounded sequence of real numbers contains a convergent subsequence.*

Proof. By Lemma 2.87 the sequence has a monotone subsequence, which is bounded because the original sequence is bounded. By the Monotone Convergence Theorem 2.86 that subsequence converges. $\square$

Remark 2.89. Bolzano–Weierstrass extends to $\mathbb{E}^k$ coordinate by coordinate: a bounded sequence in $\mathbb{R}^k$ has each coordinate bounded, so one extracts a subsequence along which the first coordinate converges, then a further subsequence for the second coordinate, and so on through all k coordinates; the final subsequence converges in every coordinate, hence in $\mathbb{E}^k$. Combining this with completeness, any *closed and bounded* subset of $\mathbb{E}^k$ is therefore sequentially compact—every sequence in it has a subsequence converging to a point of the set—and thus, by the characterization of Theorem 2.84, compact. This is exactly the Heine–Borel theorem proved next.

2.3.3 Heine–Borel and Attainment of Extrema

Theorem 2.90 (Heine–Borel). *Let S be a subset of $\mathbb{E}^k$. Then S is compact if and only if it is closed and bounded.*

Proof. ($\Rightarrow$) Suppose S is compact. By Theorem 2.84 it is totally bounded, hence bounded. It is also closed: if p is a limit point of S , take a sequence in S converging to p (Lemma 2.54); by sequential compactness a subsequence converges to a point of S , and since limits are unique that point is p , so $p \in S$.

($\Leftarrow$) Suppose S is closed and bounded. Let $\{x_n\}$ be any sequence in S . Boundedness and the coordinate-wise Bolzano–Weierstrass theorem give a subsequence converging to some $\mathbf{p} \in \mathbb{E}^k$; since S is closed, $\mathbf{p} \in S$. Thus S is sequentially compact, hence compact by Theorem 2.84. $\square$

This gets us to the core foundational result we want for optimization.

Theorem 2.91. *If a nonempty set $S \subseteq \mathbb{R}$ is compact then its supremum and infimum are members of S . Writing $\max S$ for the largest element and $\min S$ for the smallest, we have in the compact setting $\max S = \sup S$ and $\min S = \inf S$.*

Proof. By Heine–Borel, S is closed and bounded. Boundedness guarantees $\sup S$ and $\inf S$ exist. By Theorem 2.69, $\sup S \in \overline{S} = S$ (as S is closed), and likewise $\inf S \in S$ by Corollary 2.70. Being the least upper bound and greatest lower bound that lie in S , they are the maximum and minimum. $\square$

This is an important result because optimization algorithms often rely on generating a sequence of points that iteratively get closer to the optimal value. Compactness guarantees that if there is such an optimal value—the supremum (maximization) or infimum (minimization)—then the sequence can actually converge to a point that attains it, rather than iterating indefinitely with no convergence.

2.3.4 Equivalence of Norms

We have worked with the Euclidean norm thus far, but it is not uncommon to see other norms such as the 1-norm $\|\mathbf{x}\|_1 = |x_1| + |x_2| + \cdots + |x_k|$ or the infinity norm $\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq k} |x_i|$. Do the results we have shown fail to hold for metrics in other norms? Fortunately for us, they do not.

Definition 2.92 (Lipschitz-equivalent norms). Two norms $\|\cdot\|$ and $\|\cdot\|^*$ on the same vector space X are *Lipschitz-equivalent* if there exist $m, M > 0$ such that

$$m \|\mathbf{x}\| \leq \|\mathbf{x}\|^* \leq M \|\mathbf{x}\| \quad \text{for all } \mathbf{x} \in X.$$

Theorem 2.93. *All norms on $\mathbb{R}^k$ are equivalent.*

Proof. It suffices to show an arbitrary norm $\|\cdot\|$ is equivalent to the Euclidean norm $\|\cdot\|_2$; equivalence is transitive, so this links any two norms. Write $\mathbf{x} = \sum_{i=1}^k x_i \mathbf{e}_i$ in the standard basis. By the triangle inequality and Cauchy–Schwarz,

$$\|\mathbf{x}\| \leq \sum_{i=1}^k |x_i| \|\mathbf{e}_i\| \leq \left(\sum_{i=1}^k \|\mathbf{e}_i\|^2 \right)^{1/2} \left(\sum_{i=1}^k x_i^2 \right)^{1/2} = M \|\mathbf{x}\|_2,$$

with $M = (\sum_i \|\mathbf{e}_i\|^2)^{1/2}$. This upper bound also shows $\|\cdot\|$ is continuous with respect to $\|\cdot\|_2$, since the reverse triangle inequality gives $|\|\mathbf{x}\| - \|\mathbf{y}\|| \leq \|\mathbf{x} - \mathbf{y}\| \leq M \|\mathbf{x} - \mathbf{y}\|_2$.

For the lower bound, consider the Euclidean unit sphere $\mathcal{S} = \{\mathbf{x} : \|\mathbf{x}\|_2 = 1\}$, which is closed and bounded, hence compact by Heine–Borel. Let $m = \inf_{\mathbf{x} \in \mathcal{S}} \|\mathbf{x}\|$. Take a sequence $\{\mathbf{x}_n\} \subseteq \mathcal{S}$ with

$\|\mathbf{x}_n\| \rightarrow m$; by compactness a subsequence converges (in $\|\cdot\|_2$) to some $\mathbf{x}^* \in \mathcal{S}$, and by continuity of $\|\cdot\|$, $\|\mathbf{x}^*\| = m$. Since $\mathbf{x}^* \in \mathcal{S}$ we have $\mathbf{x}^* \neq \mathbf{0}$, so $m = \|\mathbf{x}^*\| > 0$. For any $\mathbf{x} \neq \mathbf{0}$, applying this to $\mathbf{x}/\|\mathbf{x}\|_2 \in \mathcal{S}$ gives $\|\mathbf{x}\| \geq m \|\mathbf{x}\|_2$. Combining, $m \|\mathbf{x}\|_2 \leq \|\mathbf{x}\| \leq M \|\mathbf{x}\|_2$, so the two norms are equivalent. $\square$

Exercise 2.94. Find the constants c_1 and c_2 that provide the equivalence between the infinity norm and the Euclidean norm, $c_1 \|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq c_2 \|\mathbf{x}\|_\infty$.

2.3.5 Some Important Limits

The following limits recur throughout the course.

Proposition 2.95.

(a) If $p > 0$ then $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$.

(b) If $p > 0$ then $\lim_{n \rightarrow \infty} p^{1/n} = 1$.

(c) $\lim_{n \rightarrow \infty} n^{1/n} = 1$.

(d) If $|x| < 1$ then $\lim_{n \rightarrow \infty} x^n = 0$.

Proof. (a) Given $\varepsilon > 0$, take $N > \varepsilon^{-1/p}$; then $n > N$ implies $n^p > 1/\varepsilon$, i.e. $1/n^p < \varepsilon$.

(b) Assume first $p \geq 1$ and write $p^{1/n} = 1 + d_n$ with $d_n \geq 0$. By the binomial inequality $p = (1 + d_n)^n \geq 1 + nd_n$, so $0 \leq d_n \leq (p - 1)/n \rightarrow 0$, giving $p^{1/n} \rightarrow 1$. If $0 < p < 1$, apply the result to $1/p > 1$ and take reciprocals.

(c) Write $n^{1/n} = 1 + d_n$ with $d_n \geq 0$. For $n \geq 2$ the binomial theorem gives $n = (1 + d_n)^n \geq \binom{n}{2} d_n^2 = \frac{n(n-1)}{2} d_n^2$, so $d_n^2 \leq \frac{2}{n-1} \rightarrow 0$ and $n^{1/n} \rightarrow 1$.

(d) If $x = 0$ this is clear. Otherwise $0 < |x| < 1$, so $|x| = 1/(1 + t)$ with $t > 0$, and $|x|^n = (1 + t)^{-n} \leq \frac{1}{1 + nt} \rightarrow 0$; hence $x^n \rightarrow 0$. $\square$

2.3.6 Limit Superior and Limit Inferior

The limit of a sequence need not exist—recall $1, -1, 1, -1, \dots$. At the very least we would like to describe the long-run bounds of a sequence. To this end we track the supremum and infimum of the “tail.”

Definition 2.96 (lim sup and lim inf). For a sequence $\{s_n\}$ in $\mathbb{R}$,

$$\limsup_{n \rightarrow \infty} s_n = \lim_{N \rightarrow \infty} \sup \{s_n : n > N\}, \quad \liminf_{n \rightarrow \infty} s_n = \lim_{N \rightarrow \infty} \inf \{s_n : n > N\}.$$

These are sometimes written $\overline{\lim} s_n$ and $\underline{\lim} s_n$. Unlike the ordinary limit, lim sup and lim inf *always* exist in the *extended reals*

$$\mathbb{R}^* = \mathbb{R} \cup \{-\infty, +\infty\},$$

(notation we reuse later), because the tail supremum is nonincreasing in N and the tail infimum is nondecreasing in N (each is therefore a monotone—hence convergent, in the extended sense—sequence).

Theorem 2.97. *Let $\{s_n\}$ be any sequence in $\mathbb{R}$. There exists a subsequence converging to $\limsup_n s_n$ and a subsequence converging to $\liminf_n s_n$.*

Proof. Let $L = \limsup_n s_n$ and write $\sigma_N = \sup\{s_n : n > N\}$. As noted in Definition 2.96, the sequence $\{\sigma_N\}$ is nonincreasing in N and converges to L ; we abbreviate this by writing $\sigma_N \downarrow L$ (“ σ_N decreases to L ”). Assume L is finite (the cases $L = \pm\infty$ are analogous). Construct the subsequence inductively. Given n_{k-1} , choose $N \geq n_{k-1}$ with $\sigma_N < L + \frac{1}{k}$; since σ_N is a supremum, there is $n_k > N$ with $s_{n_k} > \sigma_N - \frac{1}{k} \geq L - \frac{1}{k}$. Then $L - \frac{1}{k} < s_{n_k} \leq \sigma_N < L + \frac{1}{k}$, so $s_{n_k} \rightarrow L$. The $\liminf$ statement is symmetric. $\square$

Theorem 2.98. *Let $\{s_n\}$ be a sequence in $\mathbb{R}$ and let S^* denote the set of subsequential limits (cluster points) of $\{s_n\}$ together with any of $\{+\infty, -\infty\}$ that arise. Then*

- (i) S^* is non-empty;
- (ii) $\sup S^* = \limsup_n s_n$ and $\inf S^* = \liminf_n s_n$;
- (iii) $\lim_n s_n$ exists if and only if S^* has exactly one element, namely $\lim_n s_n$.

Proof. (i) Theorem 2.97 exhibits a subsequence converging to $\limsup_n s_n \in S^*$.

(ii) By Theorem 2.97, $\limsup_n s_n \in S^*$, so $\sup S^* \geq \limsup_n s_n$. Conversely, if $t \in S^*$ is the limit of a subsequence s_{n_k} , then for each N eventually $n_k > N$, so $t \leq \sup\{s_n : n > N\} = \sigma_N$; letting $N \rightarrow \infty$ gives $t \leq \limsup_n s_n$. Hence $\sup S^* = \limsup_n s_n$. The infimum statement is symmetric.

(iii) If $\lim_n s_n = s$ exists, every subsequence also converges to s , so $S^* = \{s\}$. Conversely, if $S^* = \{s\}$ then by (ii) $\limsup_n s_n = \liminf_n s_n = s$; since $\inf\{s_n : n > N\} \leq s_n \leq \sup\{s_n : n > N\}$ for $n > N$ and both bounds tend to s , the squeeze gives $s_n \rightarrow s$. $\square$

Corollary 2.99. *$\lim_n s_n$ exists if and only if $\limsup_n s_n = \liminf_n s_n$, in which case $\lim_n s_n = \limsup_n s_n = \liminf_n s_n$.*

2.3.7 Series

We now consider sums of sequences, referred to as *series*. Given a sequence $\{a_n\}$ we write $\sum_{n=p}^q a_n$ for the finite sum $a_p + a_{p+1} + \cdots + a_q$, and we consider the sums starting from the first term to be a sequence $\{S_n\}$ of *partial sums*,

$$S_n = \sum_{k=1}^n a_k.$$

If $\{S_n\}$ converges to S we say the *series converges* and write $\sum_{k=1}^{\infty} a_k = S$; if $\{S_n\}$ does not converge the series *diverges*. (Series are sometimes indexed from $k = 0$ when appropriate.)

Theorem 2.100 (Cauchy criterion for series). *$\sum a_n$ converges if and only if for every $\varepsilon > 0$ there is an integer N such that*

$$\left| \sum_{k=n}^m a_k \right| \leq \varepsilon \quad \text{whenever } m \geq n \geq N.$$

Proof. The partial sums satisfy $\sum_{k=n}^m a_k = S_m - S_{n-1}$. The series converges iff $\{S_n\}$ converges iff (since $\mathbb{R}$ is complete) $\{S_n\}$ is Cauchy, which is exactly the stated condition. $\square$

Taking $m = n$ shows $|a_n| \leq \varepsilon$ for $n \geq N$; that is, if a series converges then $\lim_n a_n = 0$. The converse fails: the terms going to 0 does not imply the series converges—they must go to zero “fast enough.”

Theorem 2.101 (*p*-series). $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

Proof. For nonnegative terms the partial sums are increasing, so by Monotone Convergence the series converges iff the partial sums are bounded. Group the terms in blocks of length $1, 2, 4, \dots, 2^j, \dots$ (Cauchy condensation). The block starting at $n = 2^j$ has 2^j terms each at most $1/(2^j)^p = 2^{-jp}$ and each at least $1/(2^{j+1})^p = 2^{-(j+1)p}$, so the block sum lies between $2^{-p} 2^{j(1-p)}$ and $2^{j(1-p)}$. Thus the series converges iff $\sum_j 2^{j(1-p)}$ converges, a geometric series with ratio 2^{1-p} , which converges iff $2^{1-p} < 1$, i.e. $p > 1$. For $p \leq 1$ the ratio is ≥ 1 and the partial sums are unbounded, so the series diverges. $\square$

Remark 2.102. For any $p > 0$ the individual terms $1/n^p \rightarrow 0$ (Proposition 2.95(a)), yet the series converges only for $p > 1$. Convergence of the terms to zero is necessary but not sufficient.

2.3.8 Power Series and the Radius of Convergence

Definition 2.103 (Power series). Given a sequence $\{a_n\}_{n=0}^{\infty}$ of real numbers, the series $\sum_{n=0}^{\infty} a_n x^n$ is called a *power series*.

The name indicates that the series is made up of “powers” of x . The most important power series we will cover later is the *Taylor series*, which gives a polynomial representation for infinitely differentiable functions (with some caveats we discuss when we cover Taylor series). There are other common series representations, such as Fourier series, but they are outside the scope of these notes. **The results we develop in this subsection are groundwork we will draw on when we introduce Taylor series later** (Section 2.4 handled continuity; the Taylor material comes with differentiation). The following theorem gives the defining characteristic of a power series: the *radius of convergence*, which tells us where the series has a finite value.

Theorem 2.104 (Cauchy–Hadamard). For the power series $\sum a_n x^n$, let

$$\beta = \limsup_{n \rightarrow \infty} |a_n|^{1/n}, \quad R = \frac{1}{\beta}$$

(with $R = +\infty$ if $\beta = 0$ and $R = 0$ if $\beta = +\infty$). Then the power series

- (i) converges for $|x| < R$, and
- (ii) diverges for $|x| > R$.

Proof. Apply the root test to the terms $c_n = a_n x^n$, for which $\limsup_n |c_n|^{1/n} = |x| \limsup_n |a_n|^{1/n} = |x| \beta = |x|/R$.

(i) If $|x| < R$ then $|x| \beta < 1$; pick ρ with $|x| \beta < \rho < 1$. By definition of $\limsup$ there is N with $|a_n|^{1/n} |x| < \rho$, i.e. $|c_n| < \rho^n$, for all $n > N$. Comparison with the convergent geometric series $\sum \rho^n$ shows $\sum c_n$ converges absolutely.

(ii) If $|x| > R$ then $|x| \beta > 1$, so $|a_n|^{1/n} |x| > 1$, i.e. $|c_n| > 1$, for infinitely many n . The terms do not tend to 0, so the series diverges. $\square$

Example 2.105. Consider $\sum_{n=0}^{\infty} \frac{x^n}{n!}$, so $a_n = \frac{1}{n!}$. Since $(n!)^{1/n} \rightarrow \infty$, we have $|a_n|^{1/n} = (n!)^{-1/n} \rightarrow 0$, hence $\beta = 0$ and $R = +\infty$. This power series converges for every $x \in \mathbb{R}$. (It is the series for e^x .)

Example 2.106. Consider $\sum_{n=0}^{\infty} x^n$, so $a_n = 1$. Then $|a_n|^{1/n} = 1$ for all n , so $\beta = 1$ and $R = 1$: the series converges for $|x| < 1$ and diverges for $|x| > 1$ (it is the geometric series, summing to $\frac{1}{1-x}$ on $(-1, 1)$). This second example illustrates well that the radius of convergence is centered as a neighborhood around 0.

The radius of convergence is centered as a neighborhood around 0. To consider convergence around some other location x_0 , we recenter.

Definition 2.107 (Generalized power series). A *generalized power series*, or power series expanded around x_0 , has the form

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n.$$

Substituting $y = x - x_0$ turns this into the standard form $\sum_n a_n y^n$, so it has the same radius-of-convergence analysis, now centered at x_0 . For example, $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (x-1)^n$ is a power series expanded around $x_0 = 1$.

2.4 Functions and Continuity

As we discussed when introducing countability, a function is a mapping from one set to another—a description of how we connect one space to another space. For any metric space X with distance d , the distance itself is a “function” $d : X \times X \rightarrow \mathbb{R}$. It is a way to bridge information about two elements in X into a setting on $\mathbb{R}$, and $\mathbb{R}$ comes equipped with the usual distance, the absolute value.

This mapping usually encodes some information in the codomain. In the case of distance, consider $d_1 = d(x_1, y)$ and $d_2 = d(x_2, y)$, with $d_1, d_2 \in \mathbb{R}$. The relationship between d_1 and d_2 in $\mathbb{R}$ tells us something about the relationship between x_1 and y and between x_2 and y in X . In particular, it also tells us something about x_1 and x_2 *relative to* y : for example, x_1 and x_2 may be equidistant from y (when $d_1 = d_2$). This might seem trivial, but the introduction of functions means we need to be able to think about *two* different spaces at the same time—the domain and the codomain.

Our primary goal in this section is a rigorous definition of continuity for a function, and then to show how functions which are continuous (or pseudo-continuous) let us carry the useful results we proved for sets over into statements about functions.

2.4.1 Limits of Functions

Since functions are mappings we must consider behavior in the domain space simultaneously with behavior in the codomain. Typically we prove a result by examining an $\varepsilon > 0$ in the codomain given some distance $\delta > 0$ in the domain.

Definition 2.108 (Limit of a function). Let (X, d_X) and (Y, d_Y) be metric spaces, $S \subseteq X$, $f : S \rightarrow Y$, and let p be a limit point of S . We say the limit of $f(x)$ as $x \rightarrow p$ is $q \in Y$, written $\lim_{x \rightarrow p} f(x) = q$, if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all points $x \in S$ with $0 < d_X(x, p) < \delta$ we have $d_Y(f(x), q) < \varepsilon$.

This canonical definition operationalizes the concept of distance in Y being connected to distance in X for the element x in the domain. Sequences also provide a powerful way of working with functional limits, since convergent sequences implicitly handle the “ $\delta > 0$ in the domain.”

Theorem 2.109. With X, Y, S, f, p as above, $\lim_{x \rightarrow p} f(x) = q$ if and only if $\lim_{n \rightarrow \infty} f(p_n) = q$ for every sequence $\{p_n\}$ in S with $p_n \neq p$ and $\lim_{n \rightarrow \infty} p_n = p$.

Proof. ($\Rightarrow$) Suppose $\lim_{x \rightarrow p} f(x) = q$ and let $p_n \rightarrow p$ with $p_n \in S$, $p_n \neq p$. Given $\varepsilon > 0$, take δ from the definition; since $p_n \rightarrow p$ there is N with $0 < d_X(p_n, p) < \delta$ for $n > N$, hence $d_Y(f(p_n), q) < \varepsilon$. Thus $f(p_n) \rightarrow q$.

($\Leftarrow$) We prove the contrapositive. If $\lim_{x \rightarrow p} f(x) \neq q$ (or fails to exist), there is $\varepsilon > 0$ such that for every $\delta > 0$ some $x \in S$ has $0 < d_X(x, p) < \delta$ yet $d_Y(f(x), q) \geq \varepsilon$. Taking $\delta = \frac{1}{n}$ produces a point $p_n \in S$ with $0 < d_X(p_n, p) < \frac{1}{n}$ and $d_Y(f(p_n), q) \geq \varepsilon$. Then $p_n \rightarrow p$, $p_n \neq p$, but $f(p_n) \not\rightarrow q$, so the sequential condition fails. $\square$

We colloquially call a sequence satisfying the theorem—i.e. $p_n \neq p$ and $p_n \rightarrow p$ implies $f(p_n) \rightarrow q$ with $q = f(p)$ —a *good* sequence. Otherwise, if $p_n \rightarrow p$ but $f(p_n) \not\rightarrow f(p)$ (or the limit does not exist), it is a *bad* sequence.

2.4.2 Continuity

Continuity is one of the most important ideas mathematical analysis is based upon, because pretty much everything we want to evaluate can be encoded as a function: we take in observed data, transform it via a function map, and output some information or decision value. (A simple example is a binary prediction problem, where functional forms such as logistic regression or decision trees take in data and output a probability that an observation should be labelled 1.) Continuity gives valuable guarantees relating properties of the domain to properties of the codomain.

Definition 2.110 (Continuity at a point). Suppose X and Y are metric spaces, $S \subseteq X$, $p \in S$, and $f : S \rightarrow Y$. Then f is *continuous at p* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that whenever $d_X(x, p) < \delta$ (with $x \in S$) we have $d_Y(f(x), f(p)) < \varepsilon$. If f is continuous at every point of S , then f is *continuous on S* .

Note that continuity is always relative to a specific set in a specific metric space, mapping to a (possibly different) metric space with a (possibly different) metric. Usually we work in settings without odd behavior, but it is important to be aware of what we are talking about.

Theorem 2.111. *With f, S, p as above and p a limit point of S , the following are equivalent:*

- (a) f is continuous at p ;
- (b) $\lim_{x \rightarrow p} f(x) = f(p)$;
- (c) every sequence $\{p_n\}$ in S with $p_n \rightarrow p$ is a good sequence, i.e. $f(p_n) \rightarrow f(p)$.

Proof. (a) $\Leftrightarrow$ (b): the continuity condition “ $d_X(x, p) < \delta \Rightarrow d_Y(f(x), f(p)) < \varepsilon$ ” is exactly Definition 2.108 with $q = f(p)$ (the case $x = p$ is automatic since $d_Y(f(p), f(p)) = 0$). (b) $\Leftrightarrow$ (c): this is Theorem 2.109 with $q = f(p)$, noting that a sequence with $p_n = p$ for some n contributes the term $f(p)$ and does not affect convergence to $f(p)$. $\square$

The “every good sequence” formulation is especially helpful when arguing for *discontinuity*, because we need only exhibit *one* bad sequence to show discontinuity.

Theorem 2.112 (Continuity of compositions). Suppose X, Y, Z are metric spaces, $S \subseteq X$, f maps S into Y , g maps $f(S)$ into Z , and $h : S \rightarrow Z$ is defined by $h(x) = g(f(x))$. If f is continuous at $p \in S$ and g is continuous at $f(p)$, then h is continuous at p .

Proof. Let $p_n \rightarrow p$ in S . Since f is continuous at p , Theorem 2.111 gives $f(p_n) \rightarrow f(p)$. Since g is continuous at $f(p)$, applying the same theorem to the sequence $\{f(p_n)\}$ gives $g(f(p_n)) \rightarrow g(f(p))$, i.e. $h(p_n) \rightarrow h(p)$. As this holds for every sequence $p_n \rightarrow p$, h is continuous at p . $\square$

Theorem 2.113 (Algebra of continuous functions). Let f and g be real continuous functions on a metric space X . Then $f + g$, fg , and f/g (where $g \neq 0$) are continuous on X .

Proof. Let $p \in X$ and $p_n \rightarrow p$. Continuity gives $f(p_n) \rightarrow f(p)$ and $g(p_n) \rightarrow g(p)$. The limit laws for convergent real sequences then give $f(p_n) + g(p_n) \rightarrow f(p) + g(p)$, $f(p_n)g(p_n) \rightarrow f(p)g(p)$, and, when $g(p) \neq 0$, $f(p_n)/g(p_n) \rightarrow f(p)/g(p)$. Hence $f + g$, fg , and f/g are continuous at p , and p was arbitrary. (At isolated points of X continuity is automatic.) $\square$

2.4.3 Topological Characterization of Continuity

An important nuance about functions is that their properties are really properties in the *codomain*, because the map lives in the codomain: $f(p) \in Y$ for $p \in X$ and $f : X \rightarrow Y$. To phrase continuity in terms of the sets we have studied, we need a way to pull a set $V \subseteq Y$ back to the domain.

Definition 2.114 (Preimage). For $f : X \rightarrow Y$ and $V \subseteq Y$, the *preimage* of V under f is

$$f^{-1}(V) = \{x \in X : f(x) \in V\}.$$

Note that this is *not* an “inverse”: f need not be a bijection, and $f^{-1}(V)$ is defined for every f and every V .

Theorem 2.115. *A mapping f of a metric space X into a metric space Y is continuous on X if and only if $f^{-1}(V)$ is open in X for every open set V in Y .*

Proof. ($\Rightarrow$) Suppose f is continuous and $V \subseteq Y$ is open. Let $p \in f^{-1}(V)$, so $f(p) \in V$; since V is open there is $\varepsilon > 0$ with $B_\varepsilon(f(p)) \subseteq V$. By continuity there is $\delta > 0$ with $f(B_\delta(p)) \subseteq B_\varepsilon(f(p)) \subseteq V$, so $B_\delta(p) \subseteq f^{-1}(V)$. Hence every point of $f^{-1}(V)$ is interior and $f^{-1}(V)$ is open.

($\Leftarrow$) Suppose preimages of open sets are open. Fix $p \in X$ and $\varepsilon > 0$. The set $V = B_\varepsilon(f(p))$ is open, so $f^{-1}(V)$ is open and contains p ; thus there is $\delta > 0$ with $B_\delta(p) \subseteq f^{-1}(V)$, i.e. $f(B_\delta(p)) \subseteq B_\varepsilon(f(p))$. Hence f is continuous at p , and p was arbitrary. $\square$

Corollary 2.116. *A mapping f of a metric space X into a metric space Y is continuous if and only if $f^{-1}(C)$ is closed in X for every closed set C in Y .*

Proof. For any $C \subseteq Y$, $f^{-1}(C^c) = f^{-1}(C)^c$. If C is closed then C^c is open, so continuity gives $f^{-1}(C^c) = f^{-1}(C)^c$ open, i.e. $f^{-1}(C)$ closed; the converse is identical using Theorem 2.115. $\square$

Again note the statement is grounded in the set in Y . Starting with a set in X does *not* imply the analogous topology in Y : for $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 0$, the open set $A = (0, 1)$ has image $f(A) = \{0\}$, which is not open; and for $f(x) = x^2$ the open set $A = (-1, 1)$ has image $f(A) = [0, 1)$, which is not open.

2.4.4 Compactness, Continuity, and the Extreme Value Theorem

Theorem 2.117. *Suppose f is a continuous mapping of a compact metric space X into a metric space Y . Then $f(X)$ is compact.*

Proof. Let $\{V_\alpha\}$ be an open cover of $f(X)$. By Theorem 2.115 each $f^{-1}(V_\alpha)$ is open, and these cover X (every x has $f(x) \in$ some V_α). By compactness of X finitely many, $f^{-1}(V_{\alpha_1}), \dots, f^{-1}(V_{\alpha_n})$, cover X . Then $V_{\alpha_1}, \dots, V_{\alpha_n}$ cover $f(X)$: any $y = f(x)$ has $x \in f^{-1}(V_{\alpha_i})$ for some i , so $y \in V_{\alpha_i}$. Hence $f(X)$ is compact. $\square$

Theorem 2.118 (Extreme Value Theorem). *Suppose f is a continuous real function on a nonempty compact metric space X , and let $M = \sup_{p \in X} f(p)$ and $m = \inf_{p \in X} f(p)$. Then there exist points $p, q \in X$ such that $f(p) = M$ and $f(q) = m$.*

Proof. By Theorem 2.117, $f(X) \subseteq \mathbb{R}$ is compact, hence closed and bounded (Heine–Borel). Boundedness makes $M = \sup f(X)$ and $m = \inf f(X)$ finite, and by Theorem 2.91 both belong to $f(X)$. Thus $M = f(p)$ for some $p \in X$ and $m = f(q)$ for some $q \in X$. $\square$

This is our headline result for functions and continuity: when we want to solve an optimization such as $\min_{x \in X} f(x)$ where X is compact and f is continuous, we have a guarantee that a solution *exists*. (This does not mean we can necessarily *compute* the optimal solution—that comes later—but the conditions necessary for a solution to exist are met.) The result also gives us a statement about the inverse function, when it exists.

Theorem 2.119. *Suppose f is a continuous bijective map from a compact metric space X onto a metric space Y . Then the inverse map $f^{-1} : Y \rightarrow X$, defined by $f^{-1}(f(x)) = x$, is a continuous mapping of Y onto X .*

Proof. Write $g = f^{-1}$. By Corollary 2.116, g is continuous iff $g^{-1}(C)$ is closed for every closed $C \subseteq X$. But $g^{-1}(C) = f(C)$ (since $g = f^{-1}$). If $C \subseteq X$ is closed, then C is a closed subset of the compact X , hence compact; by Theorem 2.117, $f(C)$ is compact, hence closed in Y . Thus $g^{-1}(C)$ is closed for every closed C , so g is continuous. $\square$

2.4.5 Uniform Continuity

We provide one more general continuity result before exploring discontinuity. Ordinary continuity allows the δ to depend on the point p ; uniform continuity requires a single δ that works everywhere.

Definition 2.120 (Uniform continuity). A function $f : X \rightarrow Y$ is *uniformly continuous* on X if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in X$, $d_X(x, y) < \delta$ implies $d_Y(f(x), f(y)) < \varepsilon$.

Uniform continuity clearly implies continuity, but the reverse is not true.

Example 2.121. (a) *A uniformly continuous function.* Let $f(x) = 2x$ on $\mathbb{R}$. Given $\varepsilon > 0$, choose $\delta = \varepsilon/2$. Then for any $x, y \in \mathbb{R}$ with $|x - y| < \delta$ we have $|f(x) - f(y)| = 2|x - y| < 2\delta = \varepsilon$. The same δ works at every point, so f is uniformly continuous.

(b) *A continuous function that is not uniformly continuous.* Let $f(x) = x^2$ on $\mathbb{R}$, which is continuous everywhere by Theorem 2.113. It is *not* uniformly continuous: fix any $\delta > 0$ and take $x_n = n$, $y_n = n + \frac{\delta}{2}$. Then $|x_n - y_n| = \frac{\delta}{2} < \delta$, yet

$$|f(x_n) - f(y_n)| = \left| n^2 - \left(n + \frac{\delta}{2}\right)^2 \right| = n\delta + \frac{\delta^2}{4} \xrightarrow{n \rightarrow \infty} \infty,$$

so for (say) $\varepsilon = 1$ no single δ can succeed at every point. The function $f(x) = 1/x$ on $(0, 1)$ fails to be uniformly continuous for the same reason as $x \rightarrow 0$.

Theorem 2.122. *If f is a continuous mapping of a compact metric space X into a metric space Y , then f is uniformly continuous on X .*

Proof. Suppose not. Then there is $\varepsilon > 0$ such that for every $\delta > 0$ some pair violates the condition; taking $\delta = \frac{1}{n}$ produces sequences $\{x_n\}, \{y_n\}$ in X with $d_X(x_n, y_n) < \frac{1}{n}$ but $d_Y(f(x_n), f(y_n)) \geq \varepsilon$. By compactness (sequential) a subsequence $x_{n_k} \rightarrow x^* \in X$; since $d_X(x_{n_k}, y_{n_k}) \rightarrow 0$ we also have $y_{n_k} \rightarrow x^*$. Continuity gives $f(x_{n_k}) \rightarrow f(x^*)$ and $f(y_{n_k}) \rightarrow f(x^*)$, so $d_Y(f(x_{n_k}), f(y_{n_k})) \rightarrow 0$, contradicting $d_Y(f(x_{n_k}), f(y_{n_k})) \geq \varepsilon$. Hence f is uniformly continuous. $\square$

2.4.6 What Compactness Buys: Failure on Noncompact Domains

In a setting where the domain set is not compact in $\mathbb{R}$, what do we lose?

Theorem 2.123. *Let S be a noncompact set in $\mathbb{R}$. Then*

- (a) *there exists a continuous function on S which is not bounded;*
- (b) *there exists a continuous and bounded function on S which has no maximum;*
- (c) *if in addition S is bounded, there exists a continuous function on S which is not uniformly continuous.*

Proof. By Heine–Borel, S noncompact means S is unbounded or not closed.

(a) If S is unbounded, $f(x) = x$ is continuous and unbounded on S . If S is bounded but not closed, it has a limit point $p \notin S$; then $f(x) = \frac{1}{x-p}$ is continuous on S (as $x \neq p$ there) and unbounded, since $|f(x)| \rightarrow \infty$ as $x \rightarrow p$ along S .

(b) If S is unbounded, $f(x) = \frac{|x|}{1+|x|}$ is continuous, bounded (values in $[0, 1)$), and has supremum 1 approached as $|x| \rightarrow \infty$ along S but never attained, so it has no maximum. (We use $|x|$ rather than x so that this works for any unbounded S , even one unbounded in only one direction.) If S is bounded but not closed with limit point $p \notin S$, then $f(x) = \frac{1}{1+|x-p|}$ is continuous and bounded on S with supremum 1 approached as $x \rightarrow p$ but never attained (since $p \notin S$), so it has no maximum.

(c) If S is bounded and noncompact it is not closed, so it has a limit point $p \notin S$. Then $f(x) = \frac{1}{x-p}$ is continuous on S but not uniformly continuous: near p , points arbitrarily close together have f -values arbitrarily far apart. $\square$

These results build intuition about what can still be asserted when we lack compactness. Compactness is useful for optimization because it gives us, via the Extreme Value Theorem, a guarantee that an optimum exists. Continuity, however, is not always *strictly* required for an optimal point to exist. For example, the function

$$f(x) = \begin{cases} 0, & x = 0, \\ 1, & 0 < x \leq 1, \end{cases}$$

is discontinuous at $x = 0$, yet $\min_{x \in [0,1]} f(x) = 0$ is attained at $x = 0$: f attains a minimum despite the discontinuity. We explore discontinuity more carefully next.

2.5 Discontinuities

Continuity is typically required to iteratively find an optimal point via a method like gradient descent, which generates a sequence of points that converges to the optimal point in the limit under conditions weaker than compactness. We will introduce gradient descent later. First we look more carefully at how continuity can fail.

Definition 2.124 (Discontinuity). If x is a point in the domain of f and f is not continuous at x , we say f is *discontinuous at x* , or has a *discontinuity at x* .

To classify discontinuities we need one-sided limits.

Definition 2.125 (One-sided limits). Let f be defined on (a, b) and consider a point x with $a \leq x < b$. The *right-hand limit* is the limit approaching from the right side of x : we consider sequences $\{t_n\} \subseteq (x, b)$ with $t_n \rightarrow x$. If $f(t_n) \rightarrow q$ for all such sequences, we write

$$f(x+) = \lim_{t \rightarrow x^+} f(t) = q.$$

Analogously, the *left-hand limit* $f(x-) = \lim_{t \rightarrow x^-} f(t)$ is defined using sequences $\{t_n\} \subseteq (a, x)$ with $t_n \rightarrow x$.

Theorem 2.126. For any point $x \in (a, b)$, $\lim_{t \rightarrow x} f(t)$ exists if and only if

$$f(x+) = f(x-) = \lim_{t \rightarrow x} f(t).$$

Exercise 2.127. Prove Theorem 2.126. (Hint: a full-limit sequence can be split into its terms above x and its terms below x ; use the sequential characterization of the limit.)

Definition 2.128 (Simple discontinuity). Let f be defined on (a, b) . f is said to have a *simple discontinuity* at x if it is discontinuous at x but $f(x+)$ and $f(x-)$ both exist. In such a case either $f(x+) \neq f(x-)$ (a “jump”), or $f(x+) = f(x-) \neq f(x)$ (a “removable” discontinuity).

Example 2.129. The step function

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0, \end{cases}$$

has a simple discontinuity at $x = 0$: the one-sided limits $f(0-) = 0$ and $f(0+) = 1$ both exist but differ, so it is a jump discontinuity.

Definition 2.130 (Discontinuity of the second kind). If f is discontinuous at x and either $f(x+)$, $f(x-)$, or both fail to exist, then f is said to have a *discontinuity of the second kind* at x .

Example 2.131 (Dirichlet function). Let

$$f(x) = \begin{cases} 1, & x \text{ rational}, \\ 0, & x \text{ irrational}. \end{cases}$$

At every point x , both one-sided limits fail to exist (every neighborhood contains both rationals and irrationals, so f takes both values 0 and 1 arbitrarily close to x). Hence f has a discontinuity of the second kind at every point.

Example 2.132. Let

$$f(x) = \begin{cases} x, & x \text{ rational,} \\ 0, & x \text{ irrational.} \end{cases}$$

This f is continuous *only* at $x = 0$ (there the two cases agree in the limit) and has a discontinuity of the second kind at every other point.

Exercise 2.133 (Rudin 4.18). Every rational x can be written in the form $x = m/n$ where $n > 0$ and $m, n \in \mathbb{Z}$ with $\gcd(m, n) = 1$; when $x = 0$ take $n = 1$. Consider the function f defined on $\mathbb{R}$ by

$$f(x) = \begin{cases} 0, & x \text{ irrational,} \\ \frac{1}{n}, & x = m/n. \end{cases}$$

Prove that f is continuous at every irrational point and that f has a simple discontinuity at every rational point.

2.6 Differentiation

The typical motivation for the derivative originates in physics. Given the location of a particle along an axis indexed by time, we might be interested in the rate of change of location with respect to time—the velocity. Let $f(t)$ be the location on the axis at time t , as in Figure 2.6.

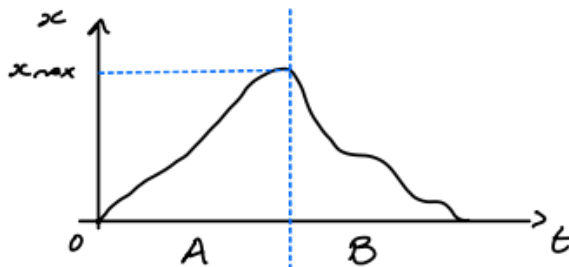


Figure 2.6: A particle’s position $f(t)$ along an axis over time. During period A the particle moves away from the origin; during period B it returns. The peak $x_{\max}$ is where the instantaneous velocity is zero.

If we consider the average velocity over the whole period $A+B$, we compute total displacement over total time,

$$\frac{0 - 0}{A + B} = 0,$$

since the particle starts and ends in the same place. This is not a meaningful metric. We can do better by splitting the trip into the period A moving away and the period B returning: the average velocity leaving is $\frac{x_{\max} - 0}{A} = \frac{x_{\max}}{A}$ and the average velocity returning is $\frac{-x_{\max}}{B}$. To find the *maximum* velocity we repeat this over ever-smaller intervals, computing the average velocity $\frac{f(t_{\text{end}}) - f(t_{\text{start}})}{t_{\text{end}} - t_{\text{start}}}$ over each. Taking the limit as the interval size goes to 0 isolates the *instantaneous velocity* at each point.

This instantaneous velocity also encodes *direction*: during period A it is positive (moving away from the origin) and during period B it is negative (moving back). At the switch between them the particle is momentarily stationary, with instantaneous velocity 0. This feature of the instantaneous velocity, called the *gradient*, is very important in optimization.

2.6.1 The Derivative

Definition 2.134 (Derivative). Let f be defined (and real-valued) on $[a, b]$. For any $x \in [a, b]$ define

$$\phi(t) = \frac{f(t) - f(x)}{t - x} \quad (a < t < b, t \neq x),$$

and define

$$f'(x) = \lim_{t \rightarrow x} \phi(t),$$

provided the limit exists. We call f' the *derivative* of f .

An equivalent formulation writes $t = x + \Delta t$ and lets the increment $\Delta t \rightarrow 0$:

$$f'(x) = \lim_{\Delta t \rightarrow 0} \frac{f(x + \Delta t) - f(x)}{\Delta t}.$$

Since $f'(x)$ measures the change in f near x over a minuscule interval, this inspires the notation $\frac{df}{dx}$, where $\Delta f = f(x + \Delta x) - f(x)$ and

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f(x)}{\Delta x}.$$

The $\frac{df}{dx}$ notation is more common because it records which variable the derivative is taken with respect to; the notation $f'(x)$ is used when f depends on a single variable.

Example 2.135. For $f(x) = x^2$,

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} = \frac{(x + \Delta x)^2 - x^2}{\Delta x} = \frac{2x \Delta x + (\Delta x)^2}{\Delta x} = 2x + \Delta x \xrightarrow{\Delta x \rightarrow 0} 2x,$$

so $f'(x) = 2x$.

Exercise 2.136. Determine the derivative of $f(x) = x^3$ from the definition.

Exercise 2.137. Determine the derivative of $f(x) = x$ from the definition.

If f' is defined at a point x , we say f is *differentiable at x* ; if f' is defined at every point of a set $S \subseteq X$, then f is *differentiable on S* . Because the definition involves a limit, it is natural to ask whether differentiability relates to continuity. It does, but only in one direction.

Theorem 2.138. *Let f be defined on $[a, b]$. If f is differentiable at a point $x \in (a, b)$, then f is continuous at x .*

Proof. For $t \neq x$ we have $f(t) - f(x) = \phi(t)(t - x)$ with $\phi(t) \rightarrow f'(x)$ as $t \rightarrow x$. Therefore

$$\lim_{t \rightarrow x} (f(t) - f(x)) = \left(\lim_{t \rightarrow x} \phi(t) \right) \left(\lim_{t \rightarrow x} (t - x) \right) = f'(x) \cdot 0 = 0,$$

so $\lim_{t \rightarrow x} f(t) = f(x)$, i.e. f is continuous at x . □

The converse is not true.

Example 2.139. The function $f(x) = |x|$ is continuous at 0, but not differentiable there: the difference quotient $\frac{|t| - 0}{t - 0}$ equals +1 for $t > 0$ and -1 for $t < 0$, so the one-sided limits disagree and $f'(0)$ does not exist.

Remark 2.140. If f is differentiable on (a, b) then it is continuous on (a, b) ; indeed, differentiability at a point implies continuity there, since $f(x + \Delta x) - f(x) = \frac{f(x + \Delta x) - f(x)}{\Delta x} \cdot \Delta x \rightarrow f'(x) \cdot 0 = 0$ as $\Delta x \rightarrow 0$.

2.6.2 Chain Rule and the Algebra of Derivatives

As with continuity, it is convenient to have chained results over compositions rather than differentiating each composite from the definition.

Theorem 2.141 (Chain rule). *Let f be differentiable at t and g differentiable at $f(t)$, and define $h(t) = g(f(t))$. Then h is differentiable at t and*

$$h'(t) = g'(f(t)) f'(t).$$

Proof. Let $y = f(t)$. Because g is differentiable at y , we may write

$$g(z) - g(y) = [g'(y) + v(z)](z - y),$$

where $v(z) \rightarrow 0$ as $z \rightarrow y$ (set $v(y) = 0$). Substituting $z = f(s)$,

$$\frac{h(s) - h(t)}{s - t} = [g'(y) + v(f(s))] \frac{f(s) - f(t)}{s - t}.$$

As $s \rightarrow t$, continuity of f (Theorem 2.138) gives $f(s) \rightarrow y$, hence $v(f(s)) \rightarrow 0$, while $\frac{f(s) - f(t)}{s - t} \rightarrow f'(t)$. Therefore $h'(t) = g'(y)f'(t) = g'(f(t))f'(t)$. $\square$

Theorem 2.142 (Algebra of derivatives). *Suppose f and g are defined on $[a, b]$ and differentiable at a point $x \in [a, b]$. Then $f + g$, fg , and f/g are differentiable at x , and*

- (a) $(f + g)'(x) = f'(x) + g'(x)$;
- (b) $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$;
- (c) $\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - g'(x)f(x)}{g^2(x)}$, provided $g(x) \neq 0$.

Proof. (a) The difference quotient of $f + g$ is the sum of the difference quotients of f and g ; taking limits gives $f'(x) + g'(x)$.

(b) Write, with $h(x) = f(x)g(x)$,

$$\frac{h(x+t) - h(x)}{t} = f(x+t) \frac{g(x+t) - g(x)}{t} + g(x) \frac{f(x+t) - f(x)}{t}.$$

As $t \rightarrow 0$, $f(x+t) \rightarrow f(x)$ (continuity), and the quotients tend to $g'(x)$ and $f'(x)$, giving $f(x)g'(x) + g(x)f'(x)$.

(c) First, applying the chain rule to $g^{-1}(x) = 1/g(x)$ (the composition of g with $z \mapsto 1/z$, whose derivative is $-1/z^2$) gives $(1/g)'(x) = -g'(x)/g^2(x)$. Writing $f/g = f \cdot (1/g)$ and applying the product rule (b),

$$\left(\frac{f}{g}\right)'(x) = f'(x) \frac{1}{g(x)} + f(x) \left(-\frac{g'(x)}{g^2(x)}\right) = \frac{g(x)f'(x) - g'(x)f(x)}{g^2(x)}. \quad \square$$

2.6.3 Stationary Points and the Interior Extremum Theorem

We now develop foundational results underlying a number of critical optimization results.

Recall our discussion of instantaneous velocity from the particle path in Figure 2.6. During period A the particle is consistently moving away from the start/origin, so its instantaneous velocity is positive, while during period B it is consistently moving back toward the origin, so its instantaneous velocity is negative. At the point where the trip switches from period A to period B —for this particular graph—there must be an instantaneous velocity of 0, where the particle is momentarily neither moving away from nor toward the origin. In the path shown, this switch point is also the maximum distance from the origin that the particle attains; restricting the domain to the interval shown, it is a global maximum. This is precisely the link we now formalize: a point where the derivative (the “velocity”) is zero is a candidate for an extremum.

Definition 2.143 (Local extremum). Let f be a real function defined on a metric space X . f has a *local maximum* at $p \in X$ if there exists $\delta > 0$ such that $f(q) \leq f(p)$ for all $q \in X$ with $d(p, q) < \delta$. A *local minimum* is defined analogously. These are local, not necessarily *global*, extrema.

Definition 2.144 (Global extremum). $f : X \rightarrow \mathbb{R}$ has a *global maximum* if there is a point $p \in X$ with $f(q) \leq f(p)$ for all $q \in X$; a *global minimum* is defined analogously. A global maximum is automatically a local maximum (any $\delta > 0$ works, since $f(q) \leq f(p)$ for *all* $q \in X$); the converse is not true in general.

Theorem 2.145 (Interior extremum / Fermat). Let $f : [a, b] \rightarrow \mathbb{R}$. If f has a local maximum (or minimum) at a point $x \in (a, b)$ and $f'(x)$ exists, then $f'(x) = 0$.

Note a few subtleties before proving this. First, f is defined on the compact set $[a, b]$, so $f([a, b])$ is also compact and f attains its minimum and maximum on this set. However, those extreme values could occur at the boundaries, i.e. at a and/or b . Because the derivative relies on evaluating a limit, it is not actually defined at the edges of an interval: for the (two-sided) limit to exist, both the left-hand and right-hand limits must exist (recall Theorem 2.126). Therefore the theorem here is stated on the *open* interval (a, b) , and standard differentiation gives no result at the endpoints. There is a notion of a *one-sided derivative* that considers only a one-sided limit, but the theorem does *not* hold for one-sided derivatives. As a simple example consider $f(x) = x$ on $[0, 1]$: the maximum is clearly at $x = 1$, and the one-sided derivative there can be computed to be $f'(1) = 1$, so we have a local maximum with one-sided derivative not equal to 0. The standard (two-sided) derivative is simply not defined at $x = 1$.

Proof. Suppose f has a local maximum at x , so $f(t) \leq f(x)$ for t near x . For $t > x$ near x , $\frac{f(t)-f(x)}{t-x} \leq 0$, and letting $t \rightarrow x^+$ gives $f'(x) \leq 0$. For $t < x$ near x , $\frac{f(t)-f(x)}{t-x} \geq 0$, and letting $t \rightarrow x^-$ gives $f'(x) \geq 0$. Hence $f'(x) = 0$. The minimum case is symmetric. $\square$

Another important subtlety is that while $f'(x) = 0$ is *necessary* to identify an interior local minimum or maximum, it is not *sufficient*: there are other points which also exhibit this behavior.

Definition 2.146 (Stationary point). A point x is a *stationary point* of f if $f'(x) = 0$ (equivalently, the instantaneous rate of change—the “velocity”—is zero there).

Definition 2.147 (Saddle point). A stationary point that is neither a local maximum nor a local minimum is a *saddle point*. In one dimension, $f(x) = x^3$ has a saddle point at $x = 0$: $f'(0) = 0$, but 0 is neither a maximum nor a minimum.

We will typically identify stationary points first and then determine whether they are local minima or maxima. Now we develop the results that support this.

2.6.4 Rolle’s Theorem and the Mean Value Theorem

Theorem 2.148 (Rolle). Let f be continuous on $[a, b]$, differentiable on (a, b) , with $f(a) = f(b)$. Then there exists (at least one) $x \in (a, b)$ with $f'(x) = 0$.

Proof. Since $[a, b]$ is compact and f continuous, the Extreme Value Theorem 2.118 gives points where f attains its maximum M and minimum m . If both are attained only at the endpoints, then $M = m$ (as $f(a) = f(b)$), so f is constant and $f'(x) = 0$ for every $x \in (a, b)$. Otherwise an extremum is attained at some interior point $x \in (a, b)$, and by Fermat’s Theorem 2.145, $f'(x) = 0$. $\square$

Theorem 2.149 (Mean Value Theorem). *Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists (at least one) $x \in (a, b)$ with*

$$f'(x) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Define $g(t) = f(t) - f(a) - \frac{f(b) - f(a)}{b - a}(t - a)$, which is continuous on $[a, b]$ and differentiable on (a, b) . Then $g(a) = 0$ and $g(b) = f(b) - f(a) - (f(b) - f(a)) = 0$, so $g(a) = g(b)$. By Rolle's Theorem there is $x \in (a, b)$ with $g'(x) = 0$, i.e. $f'(x) - \frac{f(b) - f(a)}{b - a} = 0$, which rearranges to the claim. $\square$

This is very useful, particularly written as

$$f(b) = f(a) + f'(x)(b - a) \quad \text{for some } x \in (a, b).$$

Recall the limit definition of the derivative was motivated by the gradient of the shortest line at t as a way of estimating instantaneous velocity; the Mean Value Theorem gives us that same idea *without* dealing with the limit—an *exact* value of the slope somewhere in the interval.

2.6.5 Local Linear Approximation and Little- o Notation

Recall the limit definition of the derivative, $f'(t) = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t}$. In developing this definition we were trying to obtain the gradient of the shortest line at t as a way of estimating instantaneous velocity.

The Mean Value Theorem lets us connect the derivative to a description of it as the best local linear approximation for a function f at a point t . Consider a very small $\Delta t > 0$ and the open interval $(t, t + \Delta t)$. By the Mean Value Theorem there is some point $z \in (t, t + \Delta t)$ such that

$$f(t + \Delta t) = f(t) + f'(z)((t + \Delta t) - t) = f(t) + f'(z) \Delta t.$$

As $\Delta t \rightarrow 0$, the point z —trapped between t and $t + \Delta t$ —satisfies $z \rightarrow t$, so if f' is continuous we can write

$$f(t + \Delta t) = f(t) + f'(t) \Delta t + r(t),$$

where $r(t) = (f'(z) - f'(t))\Delta t$ is the remainder, or error, of the approximation. If $r(t)$ goes to 0 *faster* than Δt goes to 0, we consider this a good approximation, since the error becomes very small in proportion to the approximation itself. We denote this idea of an error smaller than the approximation by *little- o* .

Definition 2.150 (Little- o). We write $r(t) = o(\Delta t)$ to mean $\frac{r(t)}{\Delta t} \rightarrow 0$ as $\Delta t \rightarrow 0$; that is, r vanishes faster than Δt .

We will show, via the Taylor series (Section 2.7), that indeed $r(t) = o(\Delta t)$. Granting this, and writing $x = t + \Delta t$,

$$f(x) = f(t) + f'(t)(x - t) + o(x - t),$$

so $f(t) + f'(t)(x - t)$ is the *best local linear approximation* for f at t : its error is $o(x - t)$, i.e. it vanishes faster than the displacement $x - t$. Note the subtle but important point that we could *not* simply assert the approximation error was small; the Mean Value Theorem supplies an *exact* intermediate slope $f'(z)$, and continuity of f' is what lets us replace $f'(z)$ by $f'(t)$ in the limit. This distinction becomes clearer and more useful when we describe the Taylor series next.

2.6.6 Monotonicity from the Derivative

Theorem 2.151. *Suppose f is differentiable on (a, b) . Then:*

- (i) *if $f'(x) \geq 0$ for all $x \in (a, b)$, then f is monotonically increasing;*
- (ii) *if $f'(x) = 0$ for all $x \in (a, b)$, then f is constant;*
- (iii) *if $f'(x) \leq 0$ for all $x \in (a, b)$, then f is monotonically decreasing.*

Proof. Take any $x_1 < x_2$ in (a, b) . By the Mean Value Theorem there is $\xi \in (x_1, x_2)$ with $f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1)$. Since $x_2 - x_1 > 0$, the sign of $f(x_2) - f(x_1)$ matches the sign of $f'(\xi)$: (i) $f'(\xi) \geq 0$ gives $f(x_2) \geq f(x_1)$; (ii) $f'(\xi) = 0$ gives $f(x_2) = f(x_1)$, so f is constant; (iii) $f'(\xi) \leq 0$ gives $f(x_2) \leq f(x_1)$. $\square$

Intuitively, to the left of an interior minimum the gradient is negative (the function is decreasing) and to the right it is positive (the function is increasing), consistent with Theorem 2.151.

2.7 Taylor's Theorem

For notation, we consider higher-order derivatives such as f'' for the derivative of f' (the second derivative), and denote the n -th derivative by $f^{(n)}(x)$.

Before diving in, it is worth asking *why* we should expect to be able to write a function as an infinite polynomial $f(x) = \sum_{k=0}^{\infty} a_k x^k$ at all. The motivation comes from the *Weierstrass approximation theorem*: every continuous function on a closed, bounded interval $[a, b]$ can be *uniformly approximated* as closely as we like by polynomials. That is, for any $\varepsilon > 0$ there is a polynomial p with $\sup_{x \in [a, b]} |f(x) - p(x)| < \varepsilon$. In words, on a compact interval continuous functions “look like” polynomials to arbitrary precision. If a function is not merely continuous but *differentiable*—indeed infinitely differentiable—it is reasonable to hope we can do better than approximate: we can represent it *exactly* as the limit of a sequence of polynomials, i.e. as a power series $\sum_k a_k x^k$. Taylor's theorem makes this precise by identifying what the coefficients a_k must be and by controlling the error of the finite-degree approximations. (A caveat for the careful reader: the polynomials in Weierstrass's theorem need not be the partial sums of a single power series, and not every infinitely differentiable function equals its Taylor series; nonetheless this is the right intuition for why the power-series representation is natural.)

To introduce the Taylor series, consider a power series with some radius of convergence $R > 0$, $f(x) = \sum_{k=0}^{\infty} a_k x^k$. We first develop the tools to differentiate such a series term by term.

2.7.1 Differentiating and Integrating Power Series

Lemma 2.152. *If the power series $\sum_{n=0}^{\infty} a_n x^n$ has radius of convergence R , then the series*

$$\sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{and} \quad \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$$

also have radius of convergence R .

Proof. By Cauchy–Hadamard (Theorem 2.104) the radius is governed by $\limsup_n |\cdot|^{1/n}$ of the coefficients. Since $n^{1/n} \rightarrow 1$ (Proposition 2.95(c)),

$$\limsup_n |n a_n|^{1/n} = \limsup_n n^{1/n} |a_n|^{1/n} = \limsup_n |a_n|^{1/n} = \beta,$$

so the differentiated series has the same β and hence the same radius $R = 1/\beta$. Likewise $(n+1)^{1/n} \rightarrow 1$, so $\limsup_n |a_n/(n+1)|^{1/n} = \beta$, giving radius R for the integrated series. $\square$

Theorem 2.153. *Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ have radius of convergence $R > 0$. Then f is differentiable on $(-R, R)$ and*

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{for } |x| < R.$$

Proof. Fix $|x| < R$ and choose ρ with $|x| < \rho < R$; restrict h so that $|x+h| \leq \rho$. Let $g(x) = \sum_{n \geq 1} n a_n x^{n-1}$, which converges for $|x| < R$ by Lemma 2.152. Using the algebraic identity $(x+h)^n - x^n - n x^{n-1} h = h^2 \sum_{k=2}^n \binom{n}{k} x^{n-k} h^{k-2}$ and $|x|, |x+h| \leq \rho$, one gets the bound

$$\left| \frac{(x+h)^n - x^n}{h} - n x^{n-1} \right| \leq |h| \frac{n(n-1)}{2} \rho^{n-2}.$$

Summing against $|a_n|$,

$$\left| \frac{f(x+h) - f(x)}{h} - g(x) \right| \leq |h| \sum_{n \geq 2} |a_n| \frac{n(n-1)}{2} \rho^{n-2}.$$

The series $\sum_n |a_n| n(n-1) \rho^{n-2}$ converges, because applying Lemma 2.152 twice shows the second-derivative series has radius $R > \rho$. Hence the right-hand side is a finite constant times $|h|$ and tends to 0 as $h \rightarrow 0$, so $f'(x) = g(x)$. $\square$

Theorem 2.154. *Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ have radius of convergence $R > 0$. Then, granting the Fundamental Theorem of Calculus (developed with integration later),*

$$\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1} \quad \text{for } |x| < R.$$

Proof. Let $F(x) = \sum_{n \geq 0} \frac{a_n}{n+1} x^{n+1}$, which has radius R by Lemma 2.152. By Theorem 2.153, differentiating F term by term gives $F'(x) = \sum_{n \geq 0} a_n x^n = f(x)$, and clearly $F(0) = 0$. By the Fundamental Theorem of Calculus, $\int_0^x f(t) dt = F(x) - F(0) = F(x)$. (We introduce integration formally later; the point here is that differentiation and integration act term by term inside the radius of convergence.) $\square$

2.7.2 The Taylor Series

Applying Theorem 2.153 repeatedly to $f(x) = \sum_{k \geq 0} a_k x^k$,

$$f'(x) = \sum_{k \geq 1} k a_k x^{k-1}, \quad f''(x) = \sum_{k \geq 2} k(k-1) a_k x^{k-2},$$

and, recursively, the n -th derivative is

$$f^{(n)}(x) = \sum_{k \geq n} k(k-1) \cdots (k-n+1) a_k x^{k-n} \quad (|x| < R).$$

Setting $x = 0$ kills every term except $k = n$, leaving

$$f^{(n)}(0) = n(n-1) \cdots 1 \cdot a_n = n! a_n \quad \implies \quad a_n = \frac{f^{(n)}(0)}{n!}.$$

Substituting back, we find

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k \quad (|x| < R),$$

the *Taylor series* of f about $x = 0$. As with power series generally, we can recenter the expansion at any location.

Definition 2.155 (Taylor series). Let f be a function defined on an open interval containing c . If f possesses derivatives of all orders at c , the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!} (x-c)^k$$

is called the *Taylor series* for f about c .

For $n \geq 1$, the *remainder* $R_n(x)$ measures how far the degree $(n - 1)$ Taylor polynomial (the $(n - 1)$ -th partial sum) is from f :

$$R_n(x) = f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!} (x - c)^k.$$

Theorem 2.156 (Taylor's Theorem, Lagrange remainder). *Let f be defined on (a, b) with $a < c < b$ (allowing $a = -\infty$ or $b = +\infty$), and suppose the n -th derivative $f^{(n)}$ exists on (a, b) . Then for each $x \neq c$ in (a, b) there is some y between c and x such that*

$$R_n(x) = \frac{f^{(n)}(y)}{n!} (x - c)^n,$$

equivalently

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!} (x - c)^k + \frac{f^{(n)}(y)}{n!} (x - c)^n.$$

Proof. Fix $x \neq c$ and define the number M by

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!} (x - c)^k + M(x - c)^n,$$

i.e. $M = R_n(x)/(x - c)^n$. Consider

$$g(t) = f(t) - \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!} (t - c)^k - M(t - c)^n.$$

By construction $g(x) = 0$, and differentiating the Taylor polynomial shows $g(c) = g'(c) = \dots = g^{(n-1)}(c) = 0$. Applying Rolle's Theorem 2.148 to g on the interval between c and x yields a point ξ_1 with $g'(\xi_1) = 0$; applying Rolle to g' (which vanishes at c and ξ_1) yields ξ_2 with $g''(\xi_2) = 0$; iterating n times produces a point y between c and x with $g^{(n)}(y) = 0$. Since $g^{(n)}(t) = f^{(n)}(t) - n! M$, we get $f^{(n)}(y) = n! M$, hence $M = f^{(n)}(y)/n!$ and $R_n(x) = \frac{f^{(n)}(y)}{n!} (x - c)^n$. $\square$

This y functions as the point x does in the Mean Value Theorem; indeed for $n = 1$ Taylor's Theorem is exactly the MVT, $f(x) = f(c) + f'(y)(x - c)$. Taylor's Theorem is important because if $f^{(n)}$ is continuous we can, by continuity, get arbitrarily close in the approximation with a finite number of terms; it also says there is a mean-value-type equality for every order of derivative. The Taylor series appears throughout mathematics and especially in optimization and stochastic calculus. Its most useful form, expanding in an increment Δt about a point t , is

$$f(t + \Delta t) = f(t) + f'(t) \Delta t + \frac{f''(t)}{2!} \Delta t^2 + \frac{f'''(t)}{3!} \Delta t^3 + \dots.$$

This form settles the claim deferred in Section 2.6.5. Comparing it with the local linear approximation $f(t + \Delta t) = f(t) + f'(t)\Delta t + r(t)$, the remainder is

$$r(t) = \frac{f''(t)}{2!} \Delta t^2 + \frac{f'''(t)}{3!} \Delta t^3 + \dots,$$

and every term carries a factor of Δt^2 or higher. Hence $\frac{r(t)}{\Delta t} \rightarrow 0$ as $\Delta t \rightarrow 0$, i.e. $r(t) = o(\Delta t)$: the tangent line's error vanishes faster than the displacement, exactly as promised.

Before we continue we make a note about the Taylor Series approximation for a function. Most of the functions that we work with in optimization coursework will be well approximated by its Taylor Series representation, but this is not true for functions in general. An infinitely differentiable function may have a Taylor Series but it does not mean that it is well represented by that Taylor Series at a point. We specify a term for functions which can be well represented by a power series at a point.

Definition 2.157 (Analytic function). Let $U \subseteq \mathbb{R}$ be open and let $f : U \rightarrow \mathbb{R}$. We say f is *analytic at a point* $c \in U$ if there exists $r > 0$ such that

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!} (x - c)^k$$

for every $x \in (c - r, c + r) \subseteq U$; that is, f agrees with its Taylor series on some open neighborhood of c . We say f is *analytic on* U if it is analytic at every point $c \in U$.

Intuitively, an analytic function is one that can be reconstructed locally from all of its derivatives at a single point: its Taylor series is not merely an approximation but an exact local representation of the function.

The essential distinction is that analyticity at c implies f is infinitely differentiable (C^∞) near c , but the converse is false. Smoothness only guarantees that the Taylor coefficients $f^{(k)}(c)/k!$ exist; analyticity additionally requires the resulting Taylor series to converge to f on a neighborhood of c . Put another way, two logically distinct requirements are bundled together: (i) the Taylor series has a positive radius of convergence, and (ii) within that radius its sum equals the original function.

Example 2.158. Consider

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

This function is infinitely differentiable at 0, with $f^{(k)}(0) = 0$ for every k , so its Taylor series at 0 is identically zero. That series converges everywhere—to the zero function—but it does not equal $f(x)$ for any $x \neq 0$. Therefore f is not analytic at 0, even though it is C^∞ .

2.7.3 Second-Order Conditions for Extrema

The Taylor series helps us assess whether stationary points are local minima or maxima. For a function that is at least twice differentiable, the Taylor expansion at a stationary point t (where $f'(t) = 0$) is

$$f(t + \Delta t) = f(t) + f'(t) \Delta t + \frac{f''(t)}{2} \Delta t^2 + o(\Delta t^2) = f(t) + \frac{f''(t)}{2} \Delta t^2 + o(\Delta t^2).$$

Since $\Delta t^2 > 0$, the dominant term shows: if $f''(t) > 0$ the function value increases under a small perturbation, so t is a local *minimum*; if $f''(t) < 0$, t is a local *maximum*; and if $f''(t) = 0$ we cannot conclude without additional examination.

Example 2.159. The “additional examination” matters. Consider $f(x) = x^4$, which has a stationary point at $x = 0$. Here $f''(x) = 12x^2$, so $f''(0) = 0$ and the second-order test is inconclusive. Continuing, $f^{(3)}(x) = 24x$ so $f^{(3)}(0) = 0$, while $f^{(4)}(x) = 24$ so $f^{(4)}(0) = 24 > 0$; the first nonvanishing derivative is of even order and positive, which tells us $x = 0$ is a minimum.

When a function has quadratic behavior in a neighborhood around the stationary point this makes optimization easier, and we will return to this when we discuss convexity.

2.8 Lipschitz Continuity and a Preview of Gradient Descent

Before we wrap up this section on derivatives we introduce a useful concept that sits between derivatives and continuity.

Definition 2.160 (Lipschitz continuity). Let X and Y be metric spaces and $f : X \rightarrow Y$. Then f is *Lipschitz continuous* if there exists a constant $L < \infty$ such that

$$d_Y(f(x), f(y)) \leq L d_X(x, y) \quad \text{for all } x, y \in X.$$

For $f : \mathbb{R} \rightarrow \mathbb{R}$ this reads $|f(x) - f(y)| \leq L|x - y|$, which can be rearranged (for $x \neq y$) as

$$\frac{|f(x) - f(y)|}{|x - y|} \leq L.$$

This form illustrates the intuition: the rate of change of the function is bounded—it cannot change arbitrarily fast—so Lipschitz continuity guarantees a level of smoothness. The constant L is called a *Lipschitz constant*.

Definition 2.161 (C^k functions). We denote by C^k the set of functions that are k -times continuously differentiable, i.e. that have continuous derivatives up to order k .

Theorem 2.162. If $f : I \rightarrow \mathbb{R}$ on an interval I is C^1 with bounded derivative, then f is Lipschitz continuous on I .

Proof. Let $L = \sup_t |f'(t)| < \infty$. For any $x < y$ in I , the Mean Value Theorem 2.149 gives $\xi \in (x, y)$ with $f(y) - f(x) = f'(\xi)(y - x)$, so

$$|f(y) - f(x)| = |f'(\xi)| |y - x| \leq L |y - x|.$$

Hence f is Lipschitz continuous with constant L . □

Theorem 2.163. If f is Lipschitz continuous then f is continuous.

Proof. Let L be a Lipschitz constant (if $L = 0$, f is constant and continuous; assume $L > 0$). Given $\varepsilon > 0$, choose $\delta = \varepsilon/L$. Then $d_X(x, y) < \delta$ implies $d_Y(f(x), f(y)) \leq L d_X(x, y) < L\delta = \varepsilon$. Since the same δ works everywhere, f is in fact *uniformly* continuous, and in particular continuous. □

In developing guarantees for optimization algorithms we often assume f is a function with *Lipschitz gradient*, i.e.

$$|f'(x) - f'(y)| \leq L |x - y|.$$

This guarantees that the gradient does not change too quickly, which helps when using a gradient-descent-type algorithm to find a minimum point.

2.8.1 A Preview of Gradient Descent

Suppose we want to find the minimum point of a continuous, differentiable function f . Often this is some loss function that measures the distance between a model's parameters and a set of data, and minimizing it is called “fitting the model to the data.” Typically we are taught to set the gradient equal to zero and solve, which is supported by the results we have shown (Fermat's Theorem 2.145). However, this analytic approach only works for certain simple models.

To remove that restriction we explore computational methods that iteratively find the optimal point; the most common is *gradient descent*, and we now have the tools to introduce it.

Start with $f : X \rightarrow Y$, where $Y = \mathbb{R}$ and, for simplicity, $X = \mathbb{R}$. Our goal is a point x^* that is a local minimum of f . We do not know where that is, so we randomly initialize a point x_0 (perhaps $x_0 = 0$), assuming we can compute derivatives. We want to know which direction to move in to decrease f . Let this direction be Δx , so $x_0 + \Delta x$ is the point we move to. Look at the Taylor series expanded around x_0 ,

$$f(x_0 + \Delta x) = f(x_0) + f'(x_0)\Delta x + \frac{f''(x_0)}{2}\Delta x^2 + o(\Delta x^2),$$

and distill it to the first-derivative term, the best linear approximation,

$$f(x_0 + \Delta x) \approx f(x_0) + f'(x_0) \Delta x.$$

If we pick a direction with $f(x_0 + \Delta x) - f(x_0) < 0$ we have a *descent direction*. An easy way to achieve this is to choose $\Delta x = -f'(x_0)$: then, for $x_1 = x_0 + \Delta x$,

$$f(x_1) - f(x_0) \approx f'(x_0)\Delta x = -[f'(x_0)]^2 \leq 0,$$

and since $[f'(x_0)]^2 > 0$ whenever $f'(x_0) \neq 0$, we obtain a strict decrease. Applying this update in X gives

$$x_1 = x_0 - f'(x_0).$$

So far we have implicitly taken a “step” of one unit of the gradient, which may be suboptimal; for instance since the gradient is a linear approximation one full gradient step may overshoot along the line defined by the gradient and get to a point where the function value is actually higher. To see this clearly consider $f : \mathbb{R} \rightarrow \mathbb{R}$ given as $f(x) = x^4$ with $f'(x) = 4x^3$. If we are at $x = 1$ then $f'(1) = 4$ and a full gradient step gives $x_1 = 1 - 4 = -3$. However, $f(1) = 1 < 81 = f(-3)$ so we have not decreased in function value. We therefore introduce a *step size* α ,

$$x_1 = x_0 - \alpha f'(x_0),$$

or more generally, for a sequence of iterates $\{x_k\}_{k=0}^\infty$,

$$\boxed{x_{k+1} = x_k - \alpha_k f'(x_k).}$$

This introduces the idea, but we still need to discuss selection of the step size and whether the method converges—especially in higher-dimensional spaces. We return to this in the optimization chapter.

Exercise 2.164. If we wanted to extend the approximation from just the first derivative to the second derivative, what is a good idea for a direction Δx ? (Feel free to ignore the step size. *Hint: we need a condition on $f''(x_0)$.*)

Chapter 3

Linear Algebra and Vector Spaces

In our earlier results we generally restricted the analysis either to general sets or, especially for functions, to analysis on $\mathbb{R}$. We also mentioned some specific results involving $\mathbb{R}^k$ but did not fully develop the idea of the vector space $\mathbb{R}^k$. We do so now, building from vector-valued functions up through linear maps, matrices, eigenvalues, and determinants.

3.1 Vector-Valued Functions and Projections

Recall the definition of a normed vector space and its properties (Section 2.2.4). We also record the notion of a linear combination, which we will use constantly.

Definition 3.1 (Linear combination). A *linear combination* of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ in a vector space V over a field F is any vector of the form $\sum_{i=1}^k \alpha_i \mathbf{v}_i$ with scalars $\alpha_i \in F$.

Fortunately, the work we have done for functions $f : \mathbb{R} \rightarrow \mathbb{R}$ (in particular, on continuity) extends in a natural way to functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The bridge is the family of *coordinate mappings*.

Definition 3.2 (Coordinate mapping / projection). For each i , let $P_i : \mathbb{R}^n \rightarrow \mathbb{R}$ be the function

$$P_i(\mathbf{x}) = x_i,$$

where x_i is the i -th entry of $\mathbf{x}$. We call P_i the i -th *coordinate mapping*. More generally, a *projection* is a linear map that returns the component of its input along a chosen coordinate or subspace; the coordinate mapping P_i projects a vector onto its i -th coordinate. (When a projection P maps a space to itself it is characterized by idempotence, $P \circ P = P$.)

Projections are a very common tool when working across different spaces; we will return to them.

Proposition 3.3. *Coordinate mappings are Lipschitz continuous.*

Proof. For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$|P_i(\mathbf{x}) - P_i(\mathbf{y})| = |x_i - y_i| \leq \left(\sum_{j=1}^n (x_j - y_j)^2 \right)^{1/2} = \|\mathbf{x} - \mathbf{y}\|_2,$$

so P_i is Lipschitz with constant $L = 1$. In particular, coordinate mappings are continuous (Theorem 2.163). $\square$

Proposition 3.4. *A function $f : S \rightarrow \mathbb{R}^n$ (with $S \subseteq \mathbb{R}^m$) is continuous if and only if each of the coordinate functions $P_1 \circ f, \dots, P_n \circ f$ from S into $\mathbb{R}$ is continuous.*

Proof. ($\Rightarrow$) If f is continuous then each $P_i \circ f$ is a composition of continuous maps (Proposition 3.3 and Theorem 2.112), hence continuous.

($\Leftarrow$) Suppose each $P_i \circ f$ is continuous and let $\mathbf{p}_k \rightarrow \mathbf{p}$ in S . Then each coordinate satisfies $(P_i \circ f)(\mathbf{p}_k) \rightarrow (P_i \circ f)(\mathbf{p})$, i.e. f converges coordinate-wise, and therefore $\|f(\mathbf{p}_k) - f(\mathbf{p})\|_2 = (\sum_i |(P_i \circ f)(\mathbf{p}_k) - (P_i \circ f)(\mathbf{p})|^2)^{1/2} \rightarrow 0$. Hence $f(\mathbf{p}_k) \rightarrow f(\mathbf{p})$ and f is continuous. $\square$

This is powerful connective tissue to our existing results. Consider $\mathbf{x} \in \mathbb{R}^2$ and $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$,

$$f(\mathbf{x}) = f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_1 x_2 \\ e^{x_1 + x_2} \end{bmatrix}.$$

Since each entry of $f(\mathbf{x}) \in \mathbb{R}^3$ is continuous with respect to the domain, we know the entire function is continuous. Note that this representation, as a vertical stack of individual “entries” corresponding to the dimension of the space, is (in a sense) completely arbitrary, but it is the agreed-upon standard notation. Another possible representation is the row

$$f(\mathbf{x}) = \begin{bmatrix} x_1 + x_2 & x_1 x_2 & e^{x_1 + x_2} \end{bmatrix},$$

or even the three entries placed at the vertices of a triangle (Figure 3.1)—or, more generally, at the vertices of a polygon of side k . The point is that all of these are possible representations of the same vector.

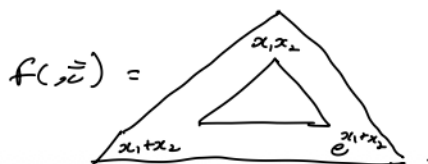


Figure 3.1: A “triangle” representation of the vector-valued function $f(\mathbf{x})$, placing the three entries $x_1 + x_2$, $x_1 x_2$, and $e^{x_1 + x_2}$ at the vertices of a triangle. This is just as valid a representation of the vector as the standard column or row.

This is a small but important point. We use representations, or notation, as a means to write down a mathematical object or operation, and we often select notation or representations that seem most “natural” or most convenient. This can lead to a reliance on notation as a guide to how an object or operation behaves, but the notation is not a substitute for understanding the objects or operations themselves. As a benchmark for reference, you should understand what is going on well enough that if you were dropped into an alternate reality where vectors are represented as k -sided polygons, you could quickly identify what is happening once someone tells you “this is a vector.” Now we return to projections.

3.2 Spans, Linear Independence, Bases, and Dimension

To introduce a notion of “size” for a vector space, first consider a rather silly—but rigorous—construction of a make-shift inverse. Consider $P : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $P(\mathbf{x}) = \begin{bmatrix} P_1(\mathbf{x}) \\ P_2(\mathbf{x}) \end{bmatrix}$, and define $P^{-1} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$P^{-1}\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{bmatrix} a \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Clearly $P^{-1}P(\mathbf{x}) = \mathbf{x}$, so P^{-1} is a valid inverse. Even though this example is silly, it illustrates an equivalency between an arbitrary vector $\mathbf{x}$ and two other vectors in $\mathbb{R}^2$. Let

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix};$$

we have illustrated that any vector $\mathbf{x} \in \mathbb{R}^2$ can be written as $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2$, and more generally, for $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{y} = \sum_{i=1}^n y_i\mathbf{e}_i$. Moreover $\mathbf{x} = \mathbf{0}$ if and only if $x_1 = x_2 = 0$.

Definition 3.5 (Span). Let $\{\mathbf{v}_i\}_{i=1}^k$ be a collection of vectors with each $\mathbf{v}_i \in V$. The *span* of $\{\mathbf{v}_i\}_{i=1}^k$ is the set of all vectors that can be generated from a linear combination of the $\mathbf{v}_i$.

Example 3.6. If $k = 1$ and $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, then $\text{span}\{\mathbf{v}_1\} = \{\mathbf{u} : \mathbf{u} = \begin{bmatrix} \alpha \\ 2\alpha \end{bmatrix}, \alpha \in \mathbb{R}\}$. Clearly $\mathbb{R}^2 = \text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$. A spanning set is not unique: we also have $\mathbb{R}^2 = \text{span}\{\mathbf{e}_1, \mathbf{e}_2, \begin{bmatrix} 1 \\ 2 \end{bmatrix}\}$.

Definition 3.7 (Linear independence). A set of vectors $\{\mathbf{v}_i\}_{i=1}^k$ is *linearly independent* if $\sum_{i=1}^k \alpha_i \mathbf{v}_i = \mathbf{0}$ implies $\alpha_i = 0$ for all $i \in \{1, \dots, k\}$. Otherwise the set is *linearly dependent*.

There is an equivalent definition which is perhaps more illuminating.

Theorem 3.8. A set of vectors $\{\mathbf{v}_i\}_{i=1}^k$ is linearly dependent if and only if one of them is $\mathbf{0}$ or a linear combination of the others.

Exercise 3.9. Prove Theorem 3.8.

Definition 3.10 (Basis). A set $\{\mathbf{v}_i\}_{i=1}^k$ of vectors from a vector space V is a *basis* for V if $\text{span}\{\mathbf{v}_i\}_{i=1}^k = V$ and $\{\mathbf{v}_i\}_{i=1}^k$ is linearly independent.

A basis is the fewest number of vectors needed to generate the space.

Exercise 3.11. In the examples given after the definition of span, which are bases for $\mathbb{R}^2$, which are not, and why?

Definition 3.12 (Dimension). The *dimension* of a vector space is the number of linearly independent vectors needed to span the space.

Theorem 3.13. Let r be a positive integer. If a vector space V is spanned by a set of r vectors, then $\dim V \leq r$.

Proof. It suffices to show that any linearly independent set in V has at most r elements (the dimension is the size of a largest such set). Let $\{\mathbf{s}_1, \dots, \mathbf{s}_r\}$ span V and suppose $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly independent with $m > r$; we derive a contradiction by an exchange argument. Since the $\mathbf{s}_j$ span V , write $\mathbf{v}_1 = \sum_j c_j \mathbf{s}_j$; as $\mathbf{v}_1 \neq \mathbf{0}$, some $c_j \neq 0$, and we may solve for that $\mathbf{s}_j$ and replace it by $\mathbf{v}_1$, so $\{\mathbf{v}_1, \mathbf{s}_2, \dots, \mathbf{s}_r\}$ (after relabeling) still spans V . Repeating, we swap in $\mathbf{v}_2, \dots, \mathbf{v}_r$

one at a time, at each step retaining a spanning set of size r ; independence of the $\mathbf{v}_i$ guarantees a nonzero coefficient on a remaining $\mathbf{s}_j$ at each step. After r steps $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ spans V , so $\mathbf{v}_{r+1}$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_r$, contradicting linear independence. Hence $m \leq r$ and $\dim V \leq r$. $\square$

Corollary 3.14. $\dim \mathbb{R}^n = n$.

Clearly the vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ form a basis for $\mathbb{R}^2$, and we refer to basis vectors of this form as the *standard basis vectors*. Are these the only basis vectors for $\mathbb{R}^2$? No.

Exercise 3.15. Show that $\mathbf{v}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$ are basis vectors for $\mathbb{R}^2$. (*Hint: see if you can appeal to the standard basis vectors.*)

Theorem 3.16. Let V be a vector space with dimension n .

- (a) A set $\{\mathbf{v}_i\}_{i=1}^n$ of n vectors in V spans V if and only if $\{\mathbf{v}_i\}_{i=1}^n$ is linearly independent.
- (b) V has a basis, and every basis consists of n vectors.
- (c) If $1 \leq r \leq n$ and $\{\mathbf{u}_i\}_{i=1}^r$ is a linearly independent set in V , then V has a basis containing $\{\mathbf{u}_i\}_{i=1}^r$.

Exercise 3.17. Prove Theorem 3.16.

3.3 Subspaces and Direct Sums

Now that we have a notion of size through dimension, we can continue our discussion of inverses. Consider two finite-dimensional vector spaces X and Y , with dimensions m and n , respectively, and a function

$$f : X \rightarrow Y.$$

For completely arbitrary functions, dimension alone does not determine whether f can be injective or surjective. We will instead be interested in functions that preserve enough geometric structure for dimension to remain meaningful, such as linear maps and, later, differentiable maps. For functions of this kind, the following provides a useful way to think about mappings between spaces of different dimensions:

- **($m < n$) (mapping to a higher-dimensional space):** the domain has fewer independent directions than the codomain. The map may be injective, so that different inputs remain distinguishable, but its image will generally occupy only a lower-dimensional portion of (Y) . We should therefore not expect the map to be surjective.
- **($m > n$) (mapping to a lower-dimensional space):** the codomain has fewer independent directions than the domain. The map must therefore compress or discard some of the information in (X) , so we should not expect it to be injective. Distinct inputs will generally be mapped to the same output.
- **($m = n$):** an inverse is possible, but not guaranteed. The function must still preserve enough "information" that distinct inputs produce distinct outputs and every element of the codomain is reached.

These statements should initially be understood as geometric intuition. In the next section we introduce linear maps, for which the corresponding dimension statements are exact: a linear map from a lower-dimensional space to a higher-dimensional space cannot be surjective, while a linear map from a higher-dimensional space to a lower-dimensional space cannot be injective.

These are illustrated even by nonlinear examples. For $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = \begin{bmatrix} x_1 \\ x_1^2 + x_2 \end{bmatrix}$ is nonlinear yet invertible, with $f^{-1}\left(\begin{smallmatrix} y_1 \\ y_2 \end{smallmatrix}\right) = \begin{bmatrix} y_1 \\ y_2 - y_1^2 \end{bmatrix}$. In contrast, $f\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = \begin{bmatrix} x_1 x_2 \\ e^{x_1 + x_2} \end{bmatrix}$ has no inverse because of the symmetry $f\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = f\left(\begin{smallmatrix} x_2 \\ x_1 \end{smallmatrix}\right)$ (no injectivity). And $f : \mathbb{R} \rightarrow \mathbb{R}^2$, $f(x) = \begin{bmatrix} x \\ 2x \end{bmatrix}$, has an inverse $f^{-1}\left(\begin{smallmatrix} y_1 \\ y_2 \end{smallmatrix}\right) = y_1$ despite mapping to a higher-dimensional space, because its image is really one-dimensional; strictly speaking it is really that for this function $f : \mathbb{R} \rightarrow \text{Im}(f)$ (with $\text{Im}(f) \subset \mathbb{R}^2$) has an inverse and we say this function if considering as $f : \mathbb{R} \rightarrow \mathbb{R}^2$ has a left inverse. This motivates the concept of a subspace which is a term we use to describe a usually smaller space that lives within a vector space.

Definition 3.18 (Subspace). A subset U of V is called a *subspace* of V if U is also a vector space with the same additive identity, addition, and scalar multiplication as on V .

Theorem 3.19. A subset U of V is a subspace of V if and only if U satisfies:

- (i) **Additive identity:** $\mathbf{0} \in U$;
- (ii) **Closed under addition:** $\mathbf{u}, \mathbf{w} \in U$ implies $\mathbf{u} + \mathbf{w} \in U$;
- (iii) **Closed under scalar multiplication:** if $\alpha \in F$ and $\mathbf{u} \in U$, then $\alpha \mathbf{u} \in U$.

Exercise 3.20. Prove Theorem 3.19.

A subspace is therefore a vector space in its own right; if U is a finite subspace it has a dimension and basis. Returning to the example, $f(x) = \begin{bmatrix} x \\ 2x \end{bmatrix}$ lives in the subspace of $\mathbb{R}^2$ spanned by $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, which has dimension 1; we are really mapping from a space of dimension 1 to a space of dimension 1. The dimension of a vector space captures, in a sense, the amount of “information” in the space.

Definition 3.21 (Sum of subspaces). If $V_1, \dots, V_m$ are subspaces of a vector space V , the *sum* $V_1 + \dots + V_m$ is the set of all possible sums of elements of the V_i :

$$V_1 + \dots + V_m = \{ \mathbf{v}_1 + \dots + \mathbf{v}_m : \mathbf{v}_i \in V_i \}.$$

Definition 3.22 (Direct sum). Suppose $V_1, \dots, V_m$ are subspaces of V . If each element of $V_1 + \dots + V_m$ can be written in only one way as a sum $\mathbf{v}_1 + \dots + \mathbf{v}_m$ with each $\mathbf{v}_k \in V_k$, then the sum is called a *direct sum*, denoted $V_1 \oplus \dots \oplus V_m$.

Example 3.23. In $\mathbb{R}^3$, consider the four subspaces

$$V_1 = \{ \alpha_1 \mathbf{e}_1 \}, \quad V_2 = \{ \alpha_2 \mathbf{e}_2 \}, \quad V_3 = \{ \alpha_3 \mathbf{e}_3 \}, \quad V_4 = \left\{ \alpha_4 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}.$$

Consider the sum $V_1 + V_3 + V_4$ and the vector $\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$. Within this subspace the vector can be written in more than one way. Placing the four subspaces as columns, each row of the table below is a different decomposition of $\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$ as a sum $\alpha_1 \mathbf{e}_1 + \alpha_2 \mathbf{e}_2 + \alpha_3 \mathbf{e}_3 + \alpha_4 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$:

	$V_1: \alpha_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$V_2: \alpha_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$	$V_3: \alpha_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	$V_4: \alpha_4 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$
$\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} =$	$1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$	0	$0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	$1 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$
$\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} =$	$2 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$	0	$2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	$0 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$
$\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} =$	$0 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$	0	$-2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	$2 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$

Since the sum here is $V_1 + V_3 + V_4$, the V_2 column contributes 0 in every row; the point is that the V_1, V_3, V_4 contributions are *not* the same across rows, so the representation is not unique and the sum is *not* a direct sum. By contrast, because $\mathbf{e}_1$ and $\mathbf{e}_2$ are linearly independent, each vector in $V_1 + V_2$ has a unique representation, so $V_1 + V_2 = V_1 \oplus V_2$.

Direct sums have a clean relationship to linear independence.

Theorem 3.24. *Suppose $V_1, \dots, V_m$ are subspaces of V . Then $V_1 + \dots + V_m$ is a direct sum if and only if the only way to write $\mathbf{0}$ as a sum $\mathbf{v}_1 + \dots + \mathbf{v}_m$ with $\mathbf{v}_k \in V_k$ is to take every $\mathbf{v}_k = \mathbf{0}$.*

Proof. ($\Rightarrow$) In a direct sum every element, including $\mathbf{0}$, has a unique representation; since $\mathbf{0} = \mathbf{0} + \dots + \mathbf{0}$ is one such representation, it is the only one.

($\Leftarrow$) Suppose $\mathbf{0}$ is written uniquely (all-zero). If some $\mathbf{v} \in V_1 + \dots + V_m$ had two representations $\mathbf{v} = \sum_k \mathbf{u}_k = \sum_k \mathbf{u}'_k$ with $\mathbf{u}_k, \mathbf{u}'_k \in V_k$, then subtracting gives $\mathbf{0} = \sum_k (\mathbf{u}_k - \mathbf{u}'_k)$ with $\mathbf{u}_k - \mathbf{u}'_k \in V_k$. By hypothesis each $\mathbf{u}_k - \mathbf{u}'_k = \mathbf{0}$, i.e. $\mathbf{u}_k = \mathbf{u}'_k$. Hence representations are unique and the sum is direct. $\square$

Theorem 3.25. *Suppose U and W are subspaces of V . Then $U + W$ is a direct sum if and only if $U \cap W = \{\mathbf{0}\}$.*

Proof. ($\Rightarrow$) Suppose $U + W$ is direct and let $\mathbf{v} \in U \cap W$. Then $\mathbf{0} = \mathbf{v} + (-\mathbf{v})$ with $\mathbf{v} \in U$ and $-\mathbf{v} \in W$; by Theorem 3.24 both terms are $\mathbf{0}$, so $\mathbf{v} = \mathbf{0}$.

($\Leftarrow$) Suppose $U \cap W = \{\mathbf{0}\}$ and $\mathbf{0} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$, $\mathbf{w} \in W$. Then $\mathbf{u} = -\mathbf{w} \in U \cap W = \{\mathbf{0}\}$, so $\mathbf{u} = \mathbf{w} = \mathbf{0}$. By Theorem 3.24, $U + W$ is a direct sum. $\square$

Theorem 3.26. *Every linearly independent list of vectors in a finite-dimensional vector space can be extended to a basis of the vector space.*

Proof. Let $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ be linearly independent in the n -dimensional space V . If it spans V it is already a basis. Otherwise there is a vector $\mathbf{w} \in V$ not in its span; then $\{\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{w}\}$ is still linearly independent (a dependence would express $\mathbf{w}$ in the span of the $\mathbf{u}_i$). Repeat this process; because $\dim V = n$ is finite and each step increases the size of an independent set by one, it terminates after at most $n - r$ steps with a spanning independent set, i.e. a basis containing the original list. $\square$

Theorem 3.27. *Suppose V is a finite-dimensional vector space and U is a subspace of V . Then there is a subspace W of V such that $V = U \oplus W$.*

Exercise 3.28. Prove Theorem 3.27. (Hint: extend a basis of U to a basis of V and let W be the span of the added vectors; use Theorem 3.25.)

3.4 Linear Maps and Matrices

We began this chapter connecting the results for single-variable functions to functions that take in a vector variable, which required developing infrastructure about the vector space containing the domain. Now we discuss the most common and well-studied type of vector map: the *linear map*.

Definition 3.29 (Linear map). Let X and Y be vector spaces and $f : X \rightarrow Y$. Then f is a *linear map* (or *linear transformation*) if for all scalars $\alpha_1, \alpha_2 \in F$ and all $\mathbf{u}, \mathbf{v} \in X$,

$$f(\alpha_1 \mathbf{u} + \alpha_2 \mathbf{v}) = \alpha_1 f(\mathbf{u}) + \alpha_2 f(\mathbf{v}).$$

For example, $f\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = x_1 + x_2$, $g\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = x_1 - x_2$, and $h : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $h\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \end{bmatrix}$ are all linear; e.g.

$$h(\mathbf{u} + \mathbf{v}) = h\left(\begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}\right) = \begin{bmatrix} (u_1 + v_1) + (u_2 + v_2) \\ (u_1 + v_1) - (u_2 + v_2) \end{bmatrix} = \begin{bmatrix} u_1 + u_2 \\ u_1 - u_2 \end{bmatrix} + \begin{bmatrix} v_1 + v_2 \\ v_1 - v_2 \end{bmatrix} = h(\mathbf{u}) + h(\mathbf{v}).$$

Importantly, for a function to be a linear map, all of the output coordinates must be generated as a linear combination of the input coordinates. This means every linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$ is uniquely defined by the coefficients that make up each output coordinate.

3.4.1 Matrix Notation

It is natural to represent these linear maps more explicitly by listing the coefficients of the linear combination corresponding to each output coordinate in a grid. We represent $h(\mathbf{x})$ as

$$h(\mathbf{x}) = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \mathbf{x}, \quad \text{and similarly} \quad f(\mathbf{x}) = \begin{bmatrix} 1 & 1 \end{bmatrix} \mathbf{x}.$$

These representations are called *matrices*. For completeness, the matrix representation proceeds as

$$h(\mathbf{x}) = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \end{bmatrix},$$

as desired. We adopt matrix notation for linear maps going forward. For a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the number of columns corresponds to n (from the domain) and the number of rows corresponds to m (from the codomain). We denote the set of linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$ by $L(\mathbb{R}^n, \mathbb{R}^m)$, and shorten $L(\mathbb{R}^n, \mathbb{R}^n)$ to $L(\mathbb{R}^n)$.

We can define operations of addition and scalar multiplication on $L(\mathbb{R}^n, \mathbb{R}^m)$: for $S, T \in L(\mathbb{R}^n, \mathbb{R}^m)$,

$$(S + T)\mathbf{v} = S\mathbf{v} + T\mathbf{v}, \quad (\lambda T)\mathbf{v} = \lambda(T\mathbf{v}).$$

Theorem 3.30. $L(\mathbb{R}^n, \mathbb{R}^m)$ is a vector space with addition and scalar multiplication as defined.

Proof. Each axiom is inherited from $\mathbb{R}^m$ pointwise: e.g. the zero map is the additive identity, $(-T)$ is the additive inverse of T , and associativity, commutativity, and distributivity hold because they hold for the outputs in $\mathbb{R}^m$ at every input $\mathbf{v}$. (The usual properties hold; the one that fails for the related operation of *composition* of maps is commutativity of multiplication.) $\square$

3.4.2 Linear Maps and Bases

Adopting matrix notation lets us relate linear maps and bases. Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be a basis for $\mathbb{R}^n$ and $A \in L(\mathbb{R}^n, \mathbb{R}^m)$. For a vector $\mathbf{v} \in \mathbb{R}^n$ we can express it as a linear combination $\mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{v}_i$. Since A is linear,

$$A\mathbf{v} = A\left(\sum_{i=1}^n \alpha_i \mathbf{v}_i\right) = \sum_{i=1}^n \alpha_i A\mathbf{v}_i = \sum_{i=1}^n \alpha_i \mathbf{u}_i, \quad \mathbf{u}_i = A\mathbf{v}_i.$$

So a linear map is completely determined by what it does to a basis. The vectors $\mathbf{u}_i \in \mathbb{R}^m$ span the image space $\subseteq \mathbb{R}^m$, but they are not guaranteed to be linearly independent, so they need not form a basis. If $n = m$ we can make a stronger claim.

Theorem 3.31. *Suppose V and W are vector spaces with $\dim V = \dim W = n$ and bases $\mathbf{v}_1, \dots, \mathbf{v}_n$ and $\mathbf{w}_1, \dots, \mathbf{w}_n$ respectively. There exists a unique linear map $A : V \rightarrow W$ such that $A\mathbf{v}_i = \mathbf{w}_i$ for $i = 1, \dots, n$.*

Proof. Existence. Every $\mathbf{v} \in V$ has a unique representation $\mathbf{v} = \sum_i \alpha_i \mathbf{v}_i$; define $A\mathbf{v} = \sum_i \alpha_i \mathbf{w}_i$. This is well-defined (by uniqueness of the representation) and linear (representations add and scale coordinate-wise), and $A\mathbf{v}_i = \mathbf{w}_i$. *Uniqueness.* If B is another linear map with $B\mathbf{v}_i = \mathbf{w}_i$, then for any $\mathbf{v} = \sum_i \alpha_i \mathbf{v}_i$, $B\mathbf{v} = \sum_i \alpha_i B\mathbf{v}_i = \sum_i \alpha_i \mathbf{w}_i = A\mathbf{v}$, so $B = A$. $\square$

Theorem 3.32. *Linear maps take $\mathbf{0}_n$ to $\mathbf{0}_m$.*

Proof. $A\mathbf{0} = A(0 \cdot \mathbf{0}) = 0 \cdot A\mathbf{0} = \mathbf{0}$, using linearity and Proposition 2.7. $\square$

3.4.3 Null Space, Image, and Rank–Nullity

We previously discussed issues around dimension when mapping between spaces of different sizes. For $L(V, W)$ we can now formalize those ideas.

Definition 3.33 (Null space / kernel). For $A \in L(V, W)$, the *null space* of A , denoted $\text{Null } A$, is the subset of V of vectors $\mathbf{v} \in V$ with $A\mathbf{v} = \mathbf{0}$:

$$\text{Null } A = \{ \mathbf{v} \in V : A\mathbf{v} = \mathbf{0} \}.$$

The null space is also called the *kernel*, $\text{Ker } A$.

Theorem 3.34. *Let $A \in L(V, W)$. Then $\text{Null } A$ is a subspace of V .*

Proof. By Theorem 3.32, $A\mathbf{0} = \mathbf{0}$, so $\mathbf{0} \in \text{Null } A$. If $\mathbf{u}, \mathbf{w} \in \text{Null } A$ then $A(\mathbf{u} + \mathbf{w}) = A\mathbf{u} + A\mathbf{w} = \mathbf{0}$, and $A(\alpha\mathbf{u}) = \alpha A\mathbf{u} = \mathbf{0}$. So $\text{Null } A$ satisfies the subspace test (Theorem 3.19). $\square$

The null space corresponds to injectivity of the linear map. If the null space has dimension greater than 0, i.e. it contains at least one non-zero vector, then A maps more than one vector to $\mathbf{0}$. Recall our definition of injectivity is about each vector in the image of a map having a unique/single vector in the domain mapping to it. The next theorem formalizes this idea.

Theorem 3.35. *Let $A \in L(V, W)$. Then A is injective if and only if $\text{Null } A = \{\mathbf{0}\}$.*

Proof. $(\Rightarrow)$ If A is injective and $A\mathbf{v} = \mathbf{0} = A\mathbf{0}$, then $\mathbf{v} = \mathbf{0}$. $(\Leftarrow)$ If $\text{Null } A = \{\mathbf{0}\}$ and $A\mathbf{u} = A\mathbf{w}$, then $A(\mathbf{u} - \mathbf{w}) = \mathbf{0}$, so $\mathbf{u} - \mathbf{w} \in \text{Null } A = \{\mathbf{0}\}$ and $\mathbf{u} = \mathbf{w}$. $\square$

Definition 3.36 (Image / range). For $A \in L(V, W)$, the *range* or *image* of A , denoted $\text{Im } A$, is the set of vectors in W equal to $A\mathbf{v}$ for some $\mathbf{v} \in V$:

$$\text{Im } A = \{ A\mathbf{v} \in W : \mathbf{v} \in V \}.$$

Theorem 3.37. *The image of A is a subspace of W .*

Exercise 3.38. Prove Theorem 3.37.

If $\text{Im } A = W$ then A is surjective. We now reach a headline result tying together the earlier discussion of maps between different dimensions.

Theorem 3.39 (Fundamental Theorem of Linear Maps / Rank–Nullity). *Suppose V is a finite-dimensional vector space and $A \in L(V, W)$. Then $\text{Im } A$ is finite-dimensional and*

$$\dim V = \dim \text{Null } A + \dim \text{Im } A.$$

Proof. Let $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be a basis of $\text{Null } A$ and extend it (Theorem 3.26) to a basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{s}_1, \dots, \mathbf{s}_r\}$ of V , so $\dim V = k + r$. We claim $\{A\mathbf{s}_1, \dots, A\mathbf{s}_r\}$ is a basis of $\text{Im } A$.

Spanning: any $\mathbf{v} = \sum_i a_i \mathbf{u}_i + \sum_j b_j \mathbf{s}_j$ has $A\mathbf{v} = \sum_j b_j A\mathbf{s}_j$, since $A\mathbf{u}_i = \mathbf{0}$; so the $A\mathbf{s}_j$ span $\text{Im } A$. *Independence:* if $\sum_j c_j A\mathbf{s}_j = \mathbf{0}$ then $A(\sum_j c_j \mathbf{s}_j) = \mathbf{0}$, so $\sum_j c_j \mathbf{s}_j \in \text{Null } A = \text{span } \{\mathbf{u}_i\}$; but $\{\mathbf{u}_i, \mathbf{s}_j\}$ is independent, forcing all $c_j = 0$. Hence $\dim \text{Im } A = r$ and $\dim \text{Null } A = k$, giving $\dim V = k + r = \dim \text{Null } A + \dim \text{Im } A$. $\square$

The Fundamental Theorem of Linear Maps is also known as the *Rank–Nullity Theorem*, where $\text{rank}(A) = \dim \text{Im}(A)$. It is powerful for understanding how dimension—which relates to linear independence, and to the “amount of information” carried in a map—behaves.

Exercise 3.40. Using Rank–Nullity, argue that a linear map to a lower-dimensional space is not injective.

Exercise 3.41. Using Rank–Nullity, argue that a linear map to a higher-dimensional space is not surjective.

So far we have discussed linear maps between spaces of possibly different dimension. There are special properties when we look at linear maps within the same space, i.e. $A \in L(V)$. Linear maps within the same space are an extremely common operation in applied mathematics, and for this reason we give them a unique name: they are called *operators*. We take up operators, eigenvalues, and diagonalization in the next section.

3.5 Operators and Eigenvalues

When we discussed general linear maps we briefly explored decomposing the action of a linear map $A \in L(V, W)$ on a vector into its action on the linear combination of basis vectors that generates the vector, for a chosen basis of the domain. Symbolically, if $\{\mathbf{v}_i\}_{i=1}^n$ is a basis for V and $\mathbf{v} \in V$, then

$$\mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{v}_i \quad \text{and} \quad A\mathbf{v} = \sum_{i=1}^n \alpha_i A\mathbf{v}_i.$$

We noted that the mapped basis vectors $A\mathbf{v}_i \in W$ need not form a linearly independent set in W , and by Rank–Nullity (Theorem 3.39) we may not even have enough of them to span W , even if $\dim V \geq \dim W$. However, we also saw (Theorem 3.31) that if $\dim V = \dim W$ there exists a linear map that carries a basis for V to a basis for W .

Now we ask a slightly different question. Given a linear map within the same space, $A \in L(V)$ —i.e. an *operator*—is there a basis such that A maps the basis to itself? Note two things. First, this is a question we can only ask of operators: it only makes sense to ask whether we can map a basis to itself if we are mapping into the same space, since the basis defines the space. Second, if we have such an “invariant” basis, we do not need to keep track of what the new basis is after the map, because it will be the same before and after the map.

3.5.1 A Motivating Example

To make this clearer, consider the following operator on the standard basis vectors in $\mathbb{R}^2$, with matrix $A = \begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix}$:

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \quad \begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

The vectors $\{\begin{bmatrix} 5 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix}\}$ are linearly independent and form a basis for $\mathbb{R}^2$. However, with the standard basis vectors I could define axes to describe or visualize my space in a typical x_1 – x_2 way, positioning a vector by scaling just the first coordinate along the x_1 axis and scaling just the second coordinate along the x_2 axis. After the map, my axes now run along the lines $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, so I have shifted the orientation of my axes—my basis. When I apply the map again, I have again shifted my axes (Figure 3.2). This makes the current active “orientation” of directions—really the basis—tedious to track.

If there were a basis $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^2$ such that

$$A\mathbf{v}_1 = \mathbf{v}_1 \quad \text{and} \quad A\mathbf{v}_2 = \mathbf{v}_2,$$

then I could fix my orientation once into $\mathbf{v}_1$ and $\mathbf{v}_2$ and preserve it regardless of how many times I apply the map. This intuition is good but is actually too restrictive, as the only operator which maps every vector to itself is the identity map, and this is unique. Fortunately we can relax it and still get our desired behavior. If $\mathbf{v}_1$ and $\mathbf{v}_2$ are a basis, then $\lambda_1 \mathbf{v}_1$ and $\lambda_2 \mathbf{v}_2$ are also a basis, where $\lambda_1, \lambda_2 \in \mathbb{R} \setminus \{0\}$ are nonzero scalar multipliers (a zero multiplier would collapse a basis vector to $\mathbf{0}$). This means we can relax the condition to

$$A\mathbf{v}_1 = \lambda_1 \mathbf{v}_1 \quad \text{and} \quad A\mathbf{v}_2 = \lambda_2 \mathbf{v}_2,$$

which allows λ_i to capture the action of A on scaling $\mathbf{v}_i$.

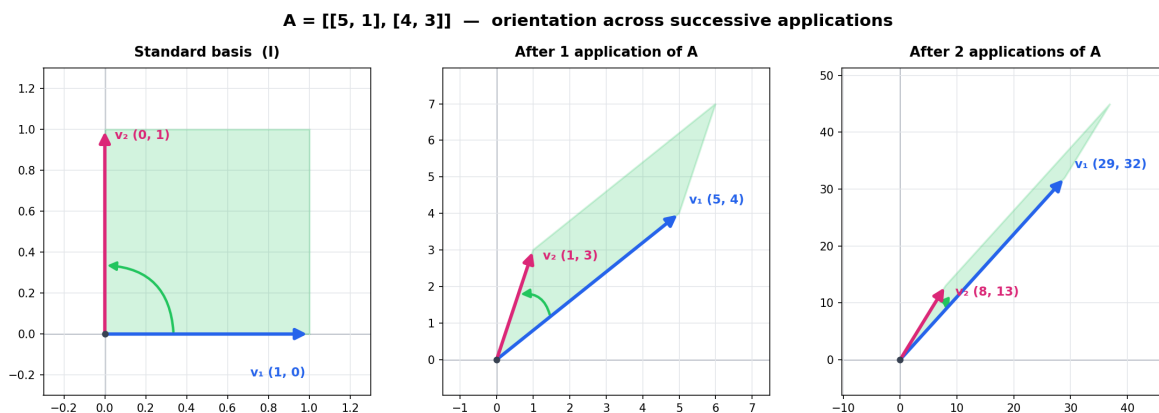


Figure 3.2: The unit square spanned by e_1, e_2 (left) and its image after one and two applications of $A = \begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix}$. Each application shifts the orientation of the basis, so tracking the active basis becomes tedious.

3.5.2 Eigenvalues, Eigenvectors, and Invariant Subspaces

Definition 3.42 (Eigenvalue / eigenvector). We call a $\lambda \in \mathbb{R}$ and a $v \neq 0 \in V$ pair such that

$$Av = \lambda v$$

an *eigenvalue–eigenvector* pair, where λ is an *eigenvalue* and v is a corresponding *eigenvector*.

If a basis of eigenvectors exists, then the application of our operator to a vector becomes

$$Av = \sum_{i=1}^n \alpha_i Av_i = \sum_{i=1}^n \alpha_i \lambda_i v_i = \sum_{i=1}^n \lambda_i (\alpha_i v_i).$$

Denote the subspace formed (spanned) by each such basis vector v_i as V_i . Then $Av \in V_1 + \dots + V_n$ for a space of dimension n . Clearly, for each such subspace the action of the operator A on a vector in the subspace results in a vector also in the subspace, and we give this behavior a specific name.

Definition 3.43 (Invariant subspace). Let $A \in L(V)$ and let U be a subspace of V . U is called an *invariant subspace* under A if for every $u \in U$ we also have $Au \in U$; i.e. the map on the subspace does not go out of the subspace. We denote this as $A|_U$ is an operator on U .

Example 3.44. Two trivial examples of invariant subspaces are V and $\{0\}$. Recall every vector space is a subspace of itself.

Exercise 3.45. Determine whether the kernel and image of a linear operator are invariant subspaces.

3.5.3 Existence and Properties of Eigenvalues

Now we develop an understanding of the existence and properties of eigenvalues and eigenvectors.

Lemma 3.46. *Let V and W be finite-dimensional vector spaces with $\dim V = \dim W$ and let $A \in L(V, W)$. We have the following equivalence:*

$$A \text{ is invertible} \Leftrightarrow A \text{ is injective} \Leftrightarrow A \text{ is surjective.}$$

Exercise 3.47. Prove Lemma 3.46. *Hint:* Use Rank–Nullity.

Theorem 3.48. *Let V be a finite-dimensional vector space and let $A \in L(V)$, $\lambda \in \mathbb{R}$. The following are equivalent:*

- (a) λ is an eigenvalue of A ;
- (b) $A - \lambda I$ is not injective;
- (c) $A - \lambda I$ is not surjective;
- (d) $A - \lambda I$ is not invertible.

Exercise 3.49. Prove Theorem 3.48. *Hint:* $Av = \lambda v \Leftrightarrow Av = \lambda Iv$, and we have a useful theorem about injectivity from earlier (Theorem 3.35, Lemma 3.46).

Remark 3.50. Theorem 3.48 tells us A is not invertible if and only if 0 is an eigenvalue (take $\lambda = 0$).

Theorem 3.51. *Let $A \in L(V)$. Every list of eigenvectors of A corresponding to distinct eigenvalues of A is linearly independent.*

Remark 3.52. There can be multiple eigenvalues that are equal, which we describe as having *multiplicity* greater than 1. In this theorem we are selecting only one eigenvector for each distinct eigenvalue.

Proof. Let $\mathbf{v}_1, \dots, \mathbf{v}_m$ be eigenvectors with distinct eigenvalues $\lambda_1, \dots, \lambda_m$. Suppose, for contradiction, the list is dependent. Let k be the smallest index such that $\mathbf{v}_k \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{k-1}\}$; such a k exists and $k \geq 2$ since a single nonzero eigenvector is independent. Write

$$\mathbf{v}_k = c_1 \mathbf{v}_1 + \dots + c_{k-1} \mathbf{v}_{k-1}. \quad (*)$$

Apply A to both sides, using $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$:

$$\lambda_k \mathbf{v}_k = c_1 \lambda_1 \mathbf{v}_1 + \dots + c_{k-1} \lambda_{k-1} \mathbf{v}_{k-1}.$$

Now multiply $(*)$ by λ_k and subtract:

$$\mathbf{0} = c_1(\lambda_1 - \lambda_k) \mathbf{v}_1 + \dots + c_{k-1}(\lambda_{k-1} - \lambda_k) \mathbf{v}_{k-1}.$$

By minimality of k , the vectors $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$ are linearly independent, so every coefficient vanishes: $c_i(\lambda_i - \lambda_k) = 0$. The eigenvalues are distinct, so $\lambda_i - \lambda_k \neq 0$, forcing $c_i = 0$ for all i . But then $(*)$ gives $\mathbf{v}_k = \mathbf{0}$, contradicting that eigenvectors are nonzero. Hence the list is linearly independent. $\square$

Theorem 3.53. *Let V be a finite-dimensional vector space. Every operator on V , $A \in L(V)$, has at most $\dim V$ distinct eigenvalues.*

Proof. By Theorem 3.51, eigenvectors corresponding to distinct eigenvalues form a linearly independent list. A linearly independent list in V has at most $\dim V$ elements (Theorem 3.13 and Definition 3.12). Choosing one eigenvector per distinct eigenvalue therefore gives at most $\dim V$ distinct eigenvalues. $\square$

These two results tell us that we are guaranteed to have no more linearly independent eigenvectors (or eigen-directions, see Remark below) than the dimension of the space, and that when the number of distinct eigenvalues equals the dimension of the space we will have an eigenbasis for the space. This means we have some conditions under which we can obtain a basis of eigenvectors for the space.

However, it does not tell us whether we can guarantee the existence of an eigenvalue for $\mathbb{R}^n$. It turns out we cannot guarantee existence of an eigenvalue in general, but we can under some fairly common conditions. Before we continue to develop these, we formalize the invariant-subspace idea we used to introduce eigenvalues and eigenvectors.

Remark 3.54. When discussing eigenvectors it is worth noting that eigenvectors are not unique. Consider an eigenvector $\mathbf{v}$ corresponding to a linear map $A \in L(V)$. For any $\alpha \in \mathbb{R}$, $\alpha\mathbf{v}$ is also an eigenvector. It may therefore be useful to think of the concept of eigenvectors more as “eigen-directions” as the “direction” idea more naturally aligns with the way we think about eigenvalues, i.e. a finite dimensional vector space having at most $\dim V$ distinct eigenvalues corresponds to the idea that the vector space also has at most $\dim V$ distinct eigen-directions, but the vector space has infinite eigenvectors if there exists at least one eigenvector. When discussing eigenvectors, specifying that they have unit norm is an equivalent way to getting to the eigen-direction idea.

Definition 3.55 (Eigenspace). Let $A \in L(V)$ for a vector space V and $\lambda \in \mathbb{R}$. The *eigenspace* of A corresponding to λ is the subspace $E(\lambda, A)$ of V defined by

$$E(\lambda, A) = \{ \mathbf{v} \in V : A\mathbf{v} = \lambda\mathbf{v} \} = \text{Ker}(A - \lambda I).$$

Remark 3.56. There may be more than one linearly independent eigenvector corresponding to this space.

Theorem 3.57. Let $A \in L(V)$ and let $\lambda_1, \dots, \lambda_m$ be distinct eigenvalues of A . Then

$$E(\lambda_1, A) + \dots + E(\lambda_m, A)$$

is a direct sum, and if V is finite-dimensional then

$$\dim E(\lambda_1, A) + \dots + \dim E(\lambda_m, A) \leq \dim V.$$

Proof. To show the sum is direct, suppose

$$\mathbf{u}_1 + \dots + \mathbf{u}_m = \mathbf{0}, \quad \mathbf{u}_i \in E(\lambda_i, A).$$

Each nonzero $\mathbf{u}_i$ is an eigenvector for the distinct eigenvalue λ_i . If some $\mathbf{u}_i$ were nonzero, the nonzero terms would form a linearly dependent list of eigenvectors with distinct eigenvalues, contradicting Theorem 3.51. Hence every $\mathbf{u}_i = \mathbf{0}$, and by the subspace characterization of direct sums (Theorem 3.24) the sum is direct.

For the dimension bound, pick a basis of each $E(\lambda_i, A)$. Their union is linearly independent: any dependence, grouped by eigenspace, yields vectors $\mathbf{u}_i \in E(\lambda_i, A)$ summing to $\mathbf{0}$, which we just showed forces each group—and hence each coefficient—to vanish. A linearly independent list in V has at most $\dim V$ elements, so the total dimension $\sum_i \dim E(\lambda_i, A) \leq \dim V$. $\square$

3.6 Polynomials and the Existence of Eigenvalues

The previous section established that eigenvectors corresponding to distinct eigenvalues are independent and that there are at most $\dim V$ of them, but it did not tell us whether an eigenvalue exists at all. To settle existence we make use of the originating definition. Since an eigenvalue–eigenvector pair satisfies

$$Av = \lambda v \Leftrightarrow (A - \lambda I)v = \mathbf{0}$$

for a linear operator $A \in L(V)$, and we know $v \neq \mathbf{0}$, the “zeroness” must “live” in the $(A - \lambda I)$ term somehow. In fact we know eigenvectors correspond to $\text{Ker}(A - \lambda I)$.

Suppose there are m distinct eigenvalues for A . We then have m terms of the form $(A - \lambda_i I)$ for $i = 1, \dots, m$. If we consider a “polynomial” of these linear operators we get

$$(A - \lambda_1 I)(A - \lambda_2 I) \cdots (A - \lambda_m I)v_i = \mathbf{0}$$

for any eigenvector v_i corresponding to some eigenvalue λ_i (the factor $(A - \lambda_i I)$ annihilates v_i , and the factors commute). There is some sense in which the zeroness “lives” in the product, so that we might identify eigenvalues by looking at these terms, and it turns out we can. We first develop some results about polynomials in order to develop the analysis.

3.6.1 Polynomials

In this subsection we will make use of the complex field $\mathbb{C}$ to develop results, but we will then use these results to develop the corresponding results for the real field $\mathbb{R}$. These notes are targeted at a graduate audience, so we assume familiarity with the complex numbers and will not develop the complex field here; many resources are available for an introduction to complex numbers.

Much like we have been using V as a general vector space over $\mathbb{R}$, for this subsection we will have $\mathbb{F}$ as a general field which is either $\mathbb{R}$ or $\mathbb{C}$, and when we are developing a result specifically for $\mathbb{C}$ we will highlight this.

Definition 3.58 (Polynomial). A *polynomial* is a function over a field $\mathbb{F}$ such that $p : \mathbb{F} \rightarrow \mathbb{F}$. Polynomials are described by their *degree*. A polynomial p of degree $m \in \mathbb{N}$ is given as

$$p(z) = a_0 + a_1 z + \cdots + a_m z^m$$

for $z \in \mathbb{F}$ and $a_0, \dots, a_m \in \mathbb{F}$ with $a_m \neq 0$. We denote the set of polynomials over a field $\mathbb{F}$ by $\mathcal{P}(\mathbb{F})$.

Definition 3.59 (Zero / root). A number $\lambda \in \mathbb{F}$ is called a *zero* (or *root*) of a polynomial $p \in \mathcal{P}(\mathbb{F})$ if $p(\lambda) = 0$.

Theorem 3.60 (Factor theorem). Let $p \in \mathcal{P}(\mathbb{F})$ be a polynomial of degree m and $\lambda \in \mathbb{F}$. Then $p(\lambda) = 0$ if and only if there exists a polynomial $q \in \mathcal{P}(\mathbb{F})$ of degree $m - 1$ such that

$$p(z) = (z - \lambda)q(z)$$

for $z \in \mathbb{F}$.

Proof. ($\Leftarrow$) If $p(z) = (z - \lambda)q(z)$, then $p(\lambda) = (\lambda - \lambda)q(\lambda) = 0$.

($\Rightarrow$) Suppose $p(\lambda) = 0$. Perform polynomial division of p by $(z - \lambda)$: there exist $q \in \mathcal{P}(\mathbb{F})$ of degree $m - 1$ and a constant remainder $r \in \mathbb{F}$ with

$$p(z) = (z - \lambda)q(z) + r.$$

The remainder is constant because the divisor has degree 1. Evaluating at $z = \lambda$ gives $0 = p(\lambda) = r$, so $r = 0$ and $p(z) = (z - \lambda)q(z)$. $\square$

Theorem 3.61 (Fundamental Theorem of Algebra). *Every nonconstant polynomial with complex coefficients has a zero in $\mathbb{C}$.*

Proof. We give the classical analytic argument, which uses the compactness tools from Chapter 2. Let $p \in \mathcal{P}(\mathbb{C})$ be nonconstant of degree $m \geq 1$. Writing $p(z) = a_m z^m (1 + a_{m-1}/(a_m z) + \cdots)$, we see $|p(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$. Hence there is a radius R with $|p(z)| > |p(0)|$ for all $|z| > R$. On the closed disk $\{|z| \leq R\}$, which is closed and bounded and therefore compact (Theorem 2.90), the continuous function $|p|$ attains a minimum (Extreme Value Theorem, Theorem 2.118) at some point z_0 ; by the choice of R this z_0 is a global minimizer of $|p|$ on $\mathbb{C}$.

We claim $p(z_0) = 0$. Suppose not. Expand p about z_0 : writing $w = z - z_0$,

$$p(z_0 + w) = p(z_0) + c_k w^k + c_{k+1} w^{k+1} + \cdots + c_m w^m,$$

where c_k is the first nonzero coefficient after the constant term ($1 \leq k \leq m$). Choose a direction $w = t\delta$, $t > 0$ small, with $\delta^k = -p(z_0)/c_k$ (a k -th root exists in $\mathbb{C}$). Then

$$p(z_0 + t\delta) = p(z_0)(1 - t^k) + O(t^{k+1}),$$

so for sufficiently small $t > 0$, $|p(z_0 + t\delta)| < |p(z_0)|$, contradicting minimality. Therefore $p(z_0) = 0$. $\square$

Theorem 3.62 (Fundamental Theorem of Algebra, second version). *If $p \in \mathcal{P}(\mathbb{C})$ is a nonconstant polynomial, then p has a unique factorization (except for the order of the factors) of the form*

$$p(z) = c(z - \lambda_1) \cdots (z - \lambda_m),$$

where $c, \lambda_1, \dots, \lambda_m \in \mathbb{C}$ and m is the degree of p .

Proof. Existence is by induction on m using Theorems 3.61 and 3.60. If $m = 1$, $p(z) = a_1 z + a_0 = a_1(z + a_0/a_1)$. For $m > 1$, the Fundamental Theorem of Algebra (Theorem 3.61) gives a root λ_m , and the Factor Theorem (Theorem 3.60) gives $p(z) = (z - \lambda_m)q(z)$ with $\deg q = m - 1$; apply the inductive hypothesis to q . Uniqueness follows because any two such factorizations must share the same multiset of roots: each factor $(z - \lambda_i)$ contributes a root, and the roots of p are determined by p ; matching leading coefficients gives the same c . $\square$

We now generate results for $\mathcal{P}(\mathbb{R})$, starting with two results that should already be familiar and are presented without detailed proof.

Theorem 3.63 (Conjugate roots). *Let $p \in \mathcal{P}(\mathbb{C})$ be a polynomial with real coefficients. If $\lambda \in \mathbb{C}$ is a zero of p , then so is its conjugate $\bar{\lambda}$.*

Proof. Since the coefficients a_j are real, $\bar{a_j} = a_j$, and conjugation is multiplicative and additive, so $\overline{p(\lambda)} = \sum_j \bar{a_j} \bar{\lambda}^j = \sum_j a_j \bar{\lambda}^j = p(\bar{\lambda})$. If $p(\lambda) = 0$ then $p(\bar{\lambda}) = \overline{p(\lambda)} = 0$. $\square$

Theorem 3.64 (Real quadratic factorization). *Consider a polynomial of degree 2 with $b, c \in \mathbb{R}$, given by $x^2 + bx + c$. This polynomial factorizes as*

$$x^2 + bx + c = (x - \lambda_1)(x - \lambda_2) \quad \text{with } \lambda_1, \lambda_2 \in \mathbb{R}$$

if and only if $b^2 \geq 4c$.

Proof. This comes directly from the quadratic formula: the roots are $\lambda_{1,2} = \frac{1}{2}(-b \pm \sqrt{b^2 - 4c})$, which are real if and only if the discriminant $b^2 - 4c \geq 0$. $\square$

Theorem 3.65 (Real factorization). *Let $p \in \mathcal{P}(\mathbb{R})$ be a nonconstant polynomial. Then p has a unique factorization of the form*

$$p(x) = c(x - \lambda_1) \cdots (x - \lambda_m)(x^2 + b_1x + c_1) \cdots (x^2 + b_Mx + c_M),$$

where $c, \lambda_i, b_j, c_j \in \mathbb{R}$ for $i = 1, \dots, m, j = 1, \dots, M$, with $b_j^2 < 4c_j$.

Proof. View p as a polynomial with complex coefficients and apply Theorem 3.62 to factor it completely over $\mathbb{C}$. Because the coefficients of p are real, Theorem 3.63 implies its non-real roots occur in conjugate pairs $\lambda, \bar{\lambda}$. Each such pair multiplies to a real quadratic,

$$(x - \lambda)(x - \bar{\lambda}) = x^2 - 2\operatorname{Re}(\lambda)x + |\lambda|^2 = x^2 + b_jx + c_j,$$

whose discriminant is $b_j^2 - 4c_j = 4(\operatorname{Re} \lambda)^2 - 4|\lambda|^2 = -4(\operatorname{Im} \lambda)^2 < 0$, i.e. $b_j^2 < 4c_j$ (consistent with Theorem 3.64). The real roots give the linear factors $(x - \lambda_i)$. Uniqueness is inherited from the uniqueness of the complex factorization. $\square$

3.6.2 Existence of Eigenvalues

Now we can develop results about the existence of eigenvalues. We start with some notation. We previously discussed, as motivation, the repeated application of a linear map, and we give some notation for that idea.

Let $A \in L(V)$ and $m \in \mathbb{N}^+$. Denote:

- $A^m = A \cdots A$, i.e. the product (composition) of the map m times;
- $A^0 = I$;
- if A is invertible, the inverse map A^{-1} , and $A^{-m} = (A^{-1})^m$;
- the usual exponent rules apply: $A^m A^n = A^{m+n}$ and $(A^m)^n = A^{mn}$.

If we have a polynomial $p \in \mathcal{P}(\mathbb{R})$ given as $p(x) = a_0 + a_1x + \cdots + a_mx^m$ for all $x \in \mathbb{R}$, we also allow a linear operator to be an argument for the polynomial, as

$$p(A) = a_0I + a_1A + \cdots + a_mA^m.$$

For the results we want to develop we want to consider polynomials of degree m with $a_m = 1$.

Definition 3.66 (Monic polynomial). A *monic polynomial* is a polynomial whose highest-degree coefficient is 1.

Definition 3.67 (Minimal polynomial). Let V be a finite-dimensional vector space and $A \in L(V)$. The *minimal polynomial* of A is the unique (in coefficients) monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that $p(A) = 0$.

Proposition 3.68. *Let V be a finite-dimensional vector space and $A \in L(V)$. There exists a minimal polynomial $p \in \mathcal{P}(\mathbb{F})$ with $\deg p \leq \dim V$.*

Proof. First we show a monic annihilating polynomial exists. The space $L(V)$ is finite-dimensional, so the list $I, A, A^2, \dots$ cannot be linearly independent forever; some finite linear combination vanishes, giving a nonzero polynomial q with $q(A) = 0$. Dividing by its leading coefficient makes it monic, so the set of monic polynomials annihilating A is nonempty. Its degrees form a nonempty subset of $\mathbb{N}$, which has a least element by well-ordering; a monic polynomial of that least degree is a minimal polynomial. Uniqueness holds because the difference of two distinct monic minimal polynomials would be a nonzero polynomial of smaller degree annihilating A , after normalization contradicting minimality.

The degree bound $\deg p \leq \dim V$ is the sharper statement; it can be proved by an inductive argument on a generating vector. We take it as given here and refer the reader to Axler's *Linear Algebra Done Right* for the details. $\square$

Now we finally get to the headline result.

Theorem 3.69. *Let V be a finite-dimensional vector space and $A \in L(V)$.*

- (i) *The zeros of the minimal polynomial of A are the eigenvalues of A .*
- (ii) *If V is a complex vector space, then the minimal polynomial of A has the form*

$$(z - \lambda_1) \cdots (z - \lambda_m),$$

where $\lambda_1, \dots, \lambda_m$ are the eigenvalues of A , possibly with repetition.

- (iii) *If V is a real vector space, then the minimal polynomial of A factorizes as*

$$(x - \lambda_1) \cdots (x - \lambda_m)(x^2 + b_1x + c_1) \cdots (x^2 + b_Mx + c_M),$$

and if $b_j^2 < 4c_j$ for any $j = 1, \dots, M$ then we do not obtain a full set of eigenvalues for A .

Proof. (i) Suppose λ is a zero of the minimal polynomial p . By the Factor Theorem (Theorem 3.60), $p(z) = (z - \lambda)q(z)$ with $\deg q < \deg p$. By minimality $q(A) \neq 0$, so there is $\mathbf{u}$ with $\mathbf{w} = q(A)\mathbf{u} \neq \mathbf{0}$. Then $(A - \lambda I)\mathbf{w} = p(A)\mathbf{u} = \mathbf{0}$, so $\mathbf{w}$ is an eigenvector with eigenvalue λ . Conversely, if λ is an eigenvalue with eigenvector $\mathbf{v}$, then $p(A)\mathbf{v} = p(\lambda)\mathbf{v}$ (since $A^k\mathbf{v} = \lambda^k\mathbf{v}$), and $p(A) = 0$ forces $p(\lambda)\mathbf{v} = \mathbf{0}$; as $\mathbf{v} \neq \mathbf{0}$, $p(\lambda) = 0$.

- (ii)–(iii) Apply Theorems 3.62 and 3.65 to the minimal polynomial, together with part (i) identifying the real roots as the eigenvalues. $\square$

This finally gives us the answer we need for existence of an eigenvalue. If the vector space is complex then an eigenvalue always exists, and in fact a full set of eigenvalues always exists. If the vector space is real we could end up in a situation where we can only reduce to complex-root quadratics and thus have no real eigenvalues. If the real vector space, particularly $\mathbb{R}^n$, has odd dimension, then the factors of the minimal polynomial must include at least one term of the form $(x - \lambda)$ —so a real eigenvalue exists. This can be proved via induction but we omit it here; see Sheldon Axler's *Linear Algebra Done Right*, result 5.34, for details.

As we stated before, we will give common conditions under which eigenvalues are guaranteed to exist for real matrices later. Additionally, we have omitted an actual example of computing a minimal polynomial, because this is not a practical way to find eigenvalues when they exist, even though they are defined in this way. We return to this later as well.

3.7 Diagonalization and the Spectral Theorem

Now that we have established results about the existence of eigenvalues, we motivate the actual execution of applying the eigenvalue–eigenvector structure. If the eigenspaces defined by the eigenvalues of the operator produce a direct sum of dimension equal to the space, then the eigenvectors produce a full set of linearly independent vectors that form a basis for the vector space. We call this an *eigenbasis*.

If we can “rotate” the basis for a vector into the eigenbasis, then we can apply the operator directly by scaling each eigenbasis vector by its corresponding eigenvalue, and then “unrotate” the basis back to the original basis to get the mapped-to vector in the original basis. We formalize this now.

3.7.1 Deriving the Diagonalization

Let V be a finite-dimensional vector space and $A \in L(V)$ with eigenvalues $\{\lambda_i\}_{i=1}^n$ and eigenvectors $\{\mathbf{v}_i\}_{i=1}^n$ that form an eigenbasis. Let $\mathbf{v} \in V$ and $\{\alpha_i\}_{i=1}^n$ be the coefficients defining the linear combination of the eigenbasis that generates $\mathbf{v}$. What we want to do is

$$A\mathbf{v} = \sum_{i=1}^n \alpha_i A\mathbf{v}_i = \sum_{i=1}^n \alpha_i \lambda_i \mathbf{v}_i = \mathbf{w}.$$

In order to represent a vector in a vector space we always need a basis. In $\mathbb{R}^n$ this is implicitly the standard basis vectors $\{\mathbf{e}_i\}_{i=1}^n$. If this is not obvious, take a minute to think about how we have been describing vectors this entire time: we talk about the i -th coordinate of a vector, but to have a coordinate we must have a chosen coordinate system, which we represent via axes. The default coordinate system for $\mathbb{R}^n$ is defined by the standard basis vectors.

Recall an earlier theorem (Theorem 3.31) that says if two vector spaces have the same dimension—which is trivially true for the same space—and two bases $\{\mathbf{v}_i\}_{i=1}^n$ and $\{\mathbf{w}_i\}_{i=1}^n$, one for each space (the same space in this case), then there exists a unique linear map, call it U , such that $U\mathbf{v}_i = \mathbf{w}_i$ for each $i = 1, \dots, n$. Taking the columns of U to be the eigenvectors, U itself sends the standard basis vectors to the eigenbasis; consequently, to express a vector’s *standard-basis coordinates* in terms of its *eigenbasis coordinates* we apply the inverse U^{-1} . We illustrate the idea in $\mathbb{R}^2$:

$$U = \begin{bmatrix} v_{11} & v_{21} \\ v_{12} & v_{22} \end{bmatrix}, \quad \text{where the eigenbasis is } \mathbf{v}_1 = \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix}.$$

Consider a vector $\mathbf{x} \in \mathbb{R}^2$ with $A\mathbf{x} = \mathbf{w}$. In the standard basis

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

We want to write $\mathbf{x}$ in the eigenbasis, so denote by subscript ε the eigenbasis representation:

$$\mathbf{x}_\varepsilon = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 = \alpha_1 \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} + \alpha_2 \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix},$$

where the $\mathbf{v}_1$ and $\mathbf{v}_2$ here are given in the standard basis. Reading off each coordinate gives

$$x_1 = \alpha_1 v_{11} + \alpha_2 v_{21}, \quad x_2 = \alpha_1 v_{12} + \alpha_2 v_{22}.$$

Equivalently, in matrix form,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} v_{11} & v_{21} \\ v_{12} & v_{22} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}.$$

Therefore, to get the vector $\mathbf{x}$ in eigenbasis coordinates, i.e. $\mathbf{x}_\varepsilon$, we need to use the inverse map U^{-1} :

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} v_{11} & v_{21} \\ v_{12} & v_{22} \end{bmatrix}^{-1} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \text{i.e.} \quad \mathbf{x}_\varepsilon = U^{-1}\mathbf{x}.$$

(We will discuss how to actually calculate an inverse later.) Now we can scale each coordinate by the eigenvalue. Let $\Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$; then

$$\mathbf{w}_\varepsilon = \Lambda \mathbf{x}_\varepsilon = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} \lambda_1 \alpha_1 \\ \lambda_2 \alpha_2 \end{bmatrix} = \Lambda U^{-1}\mathbf{x}.$$

But now we are still in eigenbasis space, so we need to return to standard basis vector space to continue to work as usual. This is basically the same process as above but reversing the bases. The result is that we find

$$A\mathbf{x} = \mathbf{w} = U\mathbf{w}_\varepsilon = U\Lambda U^{-1}\mathbf{x}.$$

Exercise 3.70. Show that mapping a vector from its eigenbasis representation to its standard basis representation is given by $\mathbf{w} = U\mathbf{w}_\varepsilon$ as above. Feel free to use $\mathbb{R}^2$.

By the equation above it should be clear that

$$\boxed{A = U\Lambda U^{-1}}$$

where Λ is a diagonal matrix of eigenvalues and U is a matrix of eigenvectors as columns.

We derived this relationship by illustrating the application of using a linearly independent set of eigenvectors as a basis, but this relationship can also be developed directly from the eigenvalue–eigenvector definition. Stacking $A\mathbf{v}_i = \lambda_i\mathbf{v}_i$ across the columns,

$$A \begin{bmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix},$$

i.e. $AU = U\Lambda$, and therefore $A = U\Lambda U^{-1}$.

The process of generating a diagonal matrix that represents a linear map is called *diagonalization* and is an extremely commonly used technique in applied mathematics. Eigenvalues are one way to perform diagonalization that works for square matrices/operators if an eigenbasis exists. Another very common diagonalization is the *singular value decomposition*, which has a relationship with eigenvalues but is different in origin and more general. Singular values and other diagonalization strategies are worth looking into but are outside the scope of these notes.

3.7.2 Conditions for Diagonalization

Now we present a theorem that gives structure for this target behavior.

Theorem 3.71. Let V be a finite-dimensional vector space and $A \in L(V)$. Let λ_i , $i = 1, \dots, m$, be the distinct eigenvalues of A . The following are equivalent:

- (i) A is diagonalizable;
- (ii) V has an eigenbasis, i.e. A has a set of $n = \dim V$ linearly independent eigenvectors;
- (iii) $V = E(\lambda_1, A) \oplus \cdots \oplus E(\lambda_m, A)$;
- (iv) $\dim V = \dim E(\lambda_1, A) + \cdots + \dim E(\lambda_m, A)$.

Proof. We prove the chain of equivalences.

- $(i) \Leftrightarrow (ii)$: If $A = U\Lambda U^{-1}$ with Λ diagonal, then $AU = U\Lambda$ says each column $\mathbf{u}_i$ of U satisfies $A\mathbf{u}_i = \lambda_i \mathbf{u}_i$; U invertible means its columns are a basis of eigenvectors. Conversely, a basis of eigenvectors placed as the columns of U gives $AU = U\Lambda$, hence $A = U\Lambda U^{-1}$.
- $(ii) \Rightarrow (iv)$: A basis of eigenvectors, grouped by eigenvalue, gives $\dim E(\lambda_i, A)$ vectors in each eigenspace summing to n vectors, so $\sum_i \dim E(\lambda_i, A) \geq n$; the reverse inequality is Theorem 3.57, giving equality.
- $(iv) \Leftrightarrow (iii)$: By Theorem 3.57 the sum $E(\lambda_1, A) + \cdots + E(\lambda_m, A)$ is always direct; it equals V exactly when its dimension $\sum_i \dim E(\lambda_i, A)$ equals $\dim V$.
- $(iii) \Rightarrow (ii)$: Concatenating bases of the eigenspaces of a direct sum equal to V yields a basis of V consisting of eigenvectors.

□

Theorem 3.72. *Let V be a finite-dimensional vector space and $A \in L(V)$ that has $\dim V$ distinct eigenvalues. Then A is diagonalizable.*

Proof. Choose one eigenvector for each of the $\dim V$ distinct eigenvalues. By Theorem 3.51 these are linearly independent, and a linearly independent list of $\dim V$ vectors in V is a basis. This is an eigenbasis, so by Theorem 3.71 A is diagonalizable. □

3.7.3 The Spectral Theorem

The next theorem is a fundamental theorem in linear algebra that gives us a powerful result for applying diagonalization.

Definition 3.73 (Symmetric matrix). Let $A \in \mathbb{R}^{n \times n}$ be a square $n \times n$ matrix such that $A = A^\top$. Then A is said to be a *symmetric matrix*.

Theorem 3.74 (Spectral Theorem, real matrices). *If $A \in \mathbb{R}^{n \times n}$ is symmetric, then A has an orthonormal basis of eigenvectors and all real eigenvalues.*

Proof. We sketch the standard induction on n . First, the eigenvalues are real. Viewing A as a complex matrix it has an eigenvalue $\lambda \in \mathbb{C}$ with eigenvector $\mathbf{z} \neq \mathbf{0}$ (Theorem 3.69 over $\mathbb{C}$). Using the Hermitian inner product and $A = A^\top$ real,

$$\lambda \langle \mathbf{z}, \mathbf{z} \rangle = \langle A\mathbf{z}, \mathbf{z} \rangle = \langle \mathbf{z}, A\mathbf{z} \rangle = \bar{\lambda} \langle \mathbf{z}, \mathbf{z} \rangle,$$

and $\langle \mathbf{z}, \mathbf{z} \rangle > 0$ forces $\lambda = \bar{\lambda}$, so $\lambda \in \mathbb{R}$; a real eigenvalue then has a real eigenvector (a nonzero real or imaginary part of $\mathbf{z}$ works). Normalize it to a unit vector $\mathbf{u}_1$.

Now induct. The orthogonal complement $W = \{\mathbf{x} : \langle \mathbf{x}, \mathbf{u}_1 \rangle = 0\}$ is invariant under A : if $\langle \mathbf{x}, \mathbf{u}_1 \rangle = 0$ then $\langle A\mathbf{x}, \mathbf{u}_1 \rangle = \langle \mathbf{x}, A\mathbf{u}_1 \rangle = \lambda_1 \langle \mathbf{x}, \mathbf{u}_1 \rangle = 0$, using symmetry. The restriction $A|_W$ is symmetric on the $(n - 1)$ -dimensional space W , so by the inductive hypothesis W has an orthonormal eigenbasis. Adjoining $\mathbf{u}_1$ gives an orthonormal eigenbasis of $\mathbb{R}^n$, all with real eigenvalues. $\square$

This is very powerful, as it gives us a somewhat common condition for diagonalization, which is that if a matrix is symmetric we can diagonalize it.

We are now almost done with our treatment of linear algebra, except for one more topic: determinants.

3.8 Determinants

Determinants arise naturally from our discussion of eigenbases and invariant subspaces. As usual we will start with a simple motivating example in $\mathbb{R}^2$.

3.8.1 A Motivating Example

As we have discussed, the default basis for $\mathbb{R}^n$ is the set of standard basis vectors. Let us consider the unit standard basis vectors e_1 and e_2 . Using the standard two-dimensional axes these are shown as in Figure 3.3.

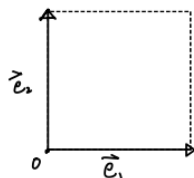


Figure 3.3: The unit square spanned by the standard basis vectors e_1 and e_2 . Its area is 1 unit, and $\det I = 1$.

The area generated by the square here is 1 unit. For a linear operator $A \in L(\mathbb{R}^2)$, we can use the effect of the operator on the “area”—really “volume” more generally—generated by the 1-unit-norm basis vectors as a measure of how “big” the operator is in a norm sense. We have to be very clear that the operation we are about to describe is not a norm. For instance, the operation which we call “determinant” does not satisfy $\det A \geq 0$. However, it can be helpful for building intuition to think of it semantically as a norm, in the sense of “size” of some sort, but which can have “negative size,” and we will explain what this means.

Consider the identity operator $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ in $\mathbb{R}^2$. Clearly $Ie_1 = e_1$ and $Ie_2 = e_2$, so I maps unit volume to unit volume. We say $\det I = 1$. Now let us consider an augmented matrix $J = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$:

$$Je_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad Je_2 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

So J maps unit volume to volume 2 (Figure 3.4), and $\det J = 2$.

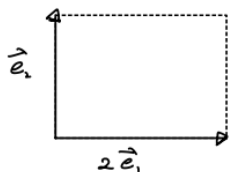


Figure 3.4: The image of the unit square under $J = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$: a 2×1 rectangle of area 2, so $\det J = 2$.

To build more intuition consider $2I$:

$$2Ie_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad 2Ie_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

The volume from this map is 4 (Figure 3.5). Therefore $\det(2I) = 4$, so the map takes 1 unit volume to 4.

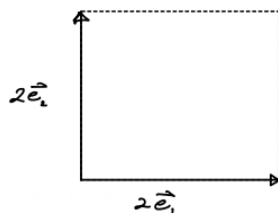


Figure 3.5: The image of the unit square under $2I$: a 2×2 square of area 4, so $\det(2I) = 4$.

Before continuing, we might wonder how this map works if we are spreading out 1 unit volume of points to 4 units area of the same set of points. We recall our discussion of set cardinality and note that both $[0, 1]$ and $\mathbb{R}$ have the same cardinality; similarly the cardinality is preserved during the map. Back to determinants.

3.8.2 The Determinant of a 2×2 Matrix

It should start to be clear that for a diagonal matrix the determinant is the product of the entries on the diagonal. For a diagonalizable operator $A = U\Lambda U^{-1}$ we have Λ diagonal, but it is not clear how we isolate Λ from the outer matrices U and U^{-1} . This also introduces the question of the determinant of an inverse map. We develop some more intuition via $A \in L(\mathbb{R}^2)$.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then

$$Ae_1 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ c \end{bmatrix}, \quad Ae_2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} b \\ d \end{bmatrix}.$$

So the mapped basis vectors span the parallelogram shown in Figure 3.6, and we can calculate its area directly.

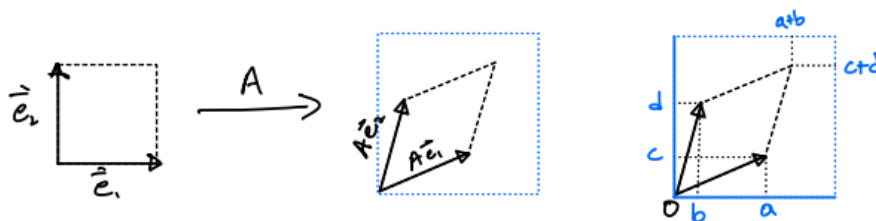


Figure 3.6: The unit square (left) is mapped by $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ to the parallelogram spanned by $Ae_1 = \begin{bmatrix} a \\ c \end{bmatrix}$ and $Ae_2 = \begin{bmatrix} b \\ d \end{bmatrix}$. The signed area of this parallelogram is the determinant.

Recall that for a parallelogram the area is given by height $\times$ base $= pq \sin \theta$ (Figure 3.7). We can identify p and q for our parallelogram by comparing Figures 3.6 and 3.7: the side p is the mapped vector $Ae_1 = \begin{bmatrix} a \\ c \end{bmatrix}$, the side q is $Ae_2 = \begin{bmatrix} b \\ d \end{bmatrix}$, and the diagonal r joins their tips. So $p = \sqrt{a^2 + c^2}$ by Pythagoras (or just the 2-norm), $q = \left\| \begin{bmatrix} b \\ d \end{bmatrix} \right\|_2 = \sqrt{b^2 + d^2}$, and $r = \sqrt{(a-b)^2 + (c-d)^2}$.

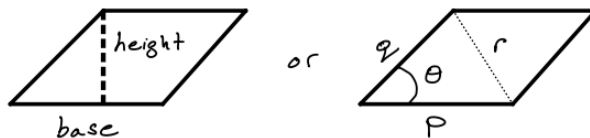


Figure 3.7: Area of a parallelogram as base $\times$ height, or equivalently $pq \sin \theta$, with the diagonal r used in the cosine rule.

We can use the cosine rule to get

$$r^2 = p^2 + q^2 - 2pq \cos \theta \Rightarrow \cos \theta = \frac{p^2 + q^2 - r^2}{2pq},$$

and $\sin^2 \theta + \cos^2 \theta = 1 \Rightarrow \sin \theta = \sqrt{1 - \cos^2 \theta}$. Putting it all together we obtain

$$\begin{aligned} \det A &= pq \sqrt{1 - \left(\frac{p^2 + q^2 - r^2}{2pq} \right)^2} = pq \frac{\sqrt{4p^2q^2 - (p^2 + q^2 - r^2)^2}}{2pq} \\ &= \frac{1}{2} \sqrt{4(a^2 + c^2)(b^2 + d^2) - (a^2 + c^2 + b^2 + d^2 - (a - b)^2 - (c - d)^2)^2} \\ &= \frac{1}{2} \sqrt{4(a^2 + c^2)(b^2 + d^2) - (2ab + 2cd)^2} \\ &= \frac{1}{2} \cdot 2 \sqrt{a^2d^2 - 2abcd + b^2c^2} = \sqrt{(ad - bc)^2} \\ &= ad - bc. \end{aligned}$$

Remark 3.75. Technically, we actually have $\sqrt{(ad - bc)^2} = |ad - bc|$ since area cannot be negative in general, but if the orientation of the axes switches we represent this by a sign change for the area and simply letting $\det A = ad - bc$ incorporates this sign convention. The idea is described next and shown in Figure 3.8.

This is the determinant for any square 2×2 matrix, but it does not give any intuition for how determinants are calculated in general, and we will not develop that here; it comes about from a different (equivalent) definition for the determinant. We have derived the determinant for $A \in L(\mathbb{R}^2)$ in order to illustrate, from basic principles, general properties of the determinant.

This also lets us return to the “signed volume” aspect of the determinant that we mentioned when introducing it—the idea that the “size” can be negative. Recall the operator from our motivating example, $A = \begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix}$, which has $\det A = (5)(3) - (1)(4) = 11 > 0$. Now define B by swapping the columns of A ,

$$B = \begin{bmatrix} 1 & 5 \\ 3 & 4 \end{bmatrix}, \quad \det B = (1)(4) - (5)(3) = -11 < 0.$$

The determinant is now negative. Applying B to the standard basis vectors,

$$Be_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad Be_2 = \begin{bmatrix} 5 \\ 4 \end{bmatrix},$$

the relative orientation of Be_1 and Be_2 is *flipped* compared to that of e_1 and e_2 : the ordered pair goes from right-handed to left-handed, which is exactly what the sign of the determinant records. On a second application of B the orientation flips back, so the sign of $\det(B^2) = (\det B)^2 = 121$ is positive again. Figure 3.8 illustrates this behavior.

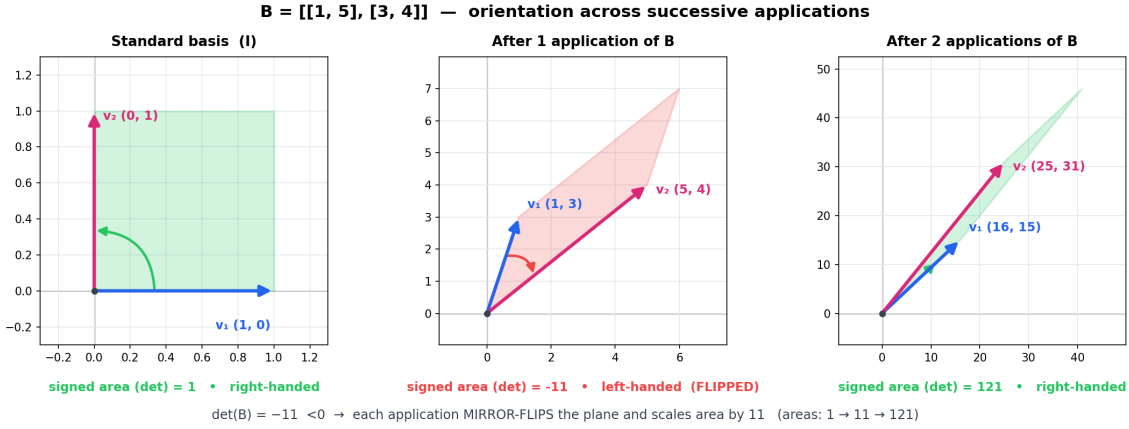


Figure 3.8: The column-swapped matrix $B = \begin{bmatrix} 1 & 5 \\ 3 & 4 \end{bmatrix}$ has $\det B = -11 < 0$. Each application mirror-flips the plane (right-handed $\rightarrow$ left-handed $\rightarrow$ right-handed) and scales area by 11, so the signed areas run $1 \rightarrow -11 \rightarrow 121$.

3.8.3 The Determinant and the Inverse

Now we calculate from first principles A^{-1} . Let $A^{-1} = \begin{bmatrix} t & u \\ v & w \end{bmatrix}$; then we need

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} t & u \\ v & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \Rightarrow \begin{cases} at + bv = 1, \\ au + bw = 0, \\ ct + dv = 0, \\ cu + dw = 1. \end{cases}$$

Since this is for motivation, let us assume $a, b, c, d \neq 0$ so we do not have to consider many different cases. From the first and third equations, $v = (1 - at)/b$ and $ct + dv = 0$ give $t = \frac{d}{ad - bc}$ provided $ad - bc \neq 0$, and then $v = \frac{-c}{ad - bc}$. From the second and fourth equations, $w = -au/b$ and $cu + dw = 1$ give $u = \frac{-b}{ad - bc}$ and $w = \frac{a}{ad - bc}$. Thus we find

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

We establish two observations here:

- (i) For a matrix $A \in L(\mathbb{R}^2)$, the reciprocal of the determinant shows up as a factor in the inverse A^{-1} .
- (ii) For a matrix $A \in L(\mathbb{R}^2)$, the inverse does not exist if the determinant of A is 0.

We will prove that both of these observations actually generalize.

3.8.4 The Determinant Across Diagonalization

Before we get to formal results we finally can illustrate the behavior of the determinant across diagonalization. Let $\mathbf{v}_1 = \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix}$ be the eigenbasis for A , so

$$U = \begin{bmatrix} v_{11} & v_{21} \\ v_{12} & v_{22} \end{bmatrix} \Rightarrow U^{-1} = \frac{1}{v_{11}v_{22} - v_{12}v_{21}} \begin{bmatrix} v_{22} & -v_{21} \\ -v_{12} & v_{11} \end{bmatrix}.$$

The application of the map $U\Lambda U^{-1}$ on the area generated by the standard basis vectors can be tracked across the individual maps, as in Figure 3.9.

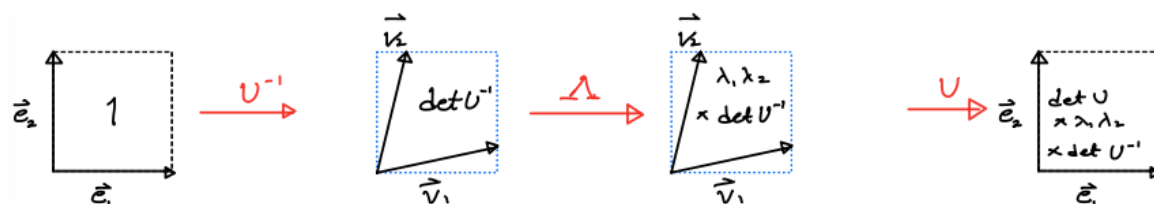


Figure 3.9: Tracking the unit volume through $A = U\Lambda U^{-1}$. Applying U^{-1} scales volume by $\det U^{-1}$; applying Λ scales by $\lambda_1\lambda_2$; applying U scales by $\det U$. The net scaling is $\det U \cdot \lambda_1\lambda_2 \cdot \det U^{-1} = \lambda_1\lambda_2$.

Note that the determinant's volume definition is about a 1-unit volume in the current basis. We have been using the standard basis prior to this illustration because we are usually in the standard basis. By starting off with the map U^{-1} we change the basis, so multiplying by $\det U$ to obtain the final area does not need additional justification. We know what each term evaluates to:

$$\det U = v_{11}v_{22} - v_{12}v_{21}, \quad \det U^{-1} = \frac{v_{11}v_{22} - v_{12}v_{21}}{(v_{11}v_{22} - v_{12}v_{21})^2} = \frac{1}{v_{11}v_{22} - v_{12}v_{21}}.$$

Therefore

$$\det A = (v_{11}v_{22} - v_{12}v_{21}) \lambda_1\lambda_2 \frac{1}{v_{11}v_{22} - v_{12}v_{21}} = \lambda_1\lambda_2.$$

This tells us a couple of important things:

- (i) The determinant of a linear operator is governed by the “diagonal/scaling” part of the matrix.
- (ii) The determinant of a product of linear operators for the same space can be considered as the product of the individual determinants.

3.8.5 Formal Results

Now we formalize these ideas.

Definition 3.76 (Determinant). Let $A \in L(V)$ for a finite-dimensional real vector space V of dimension n . The *determinant* $\det A$ is the signed n -dimensional volume of the image under A of the unit cube spanned by a positively oriented orthonormal basis; equivalently, it is the factor by which A scales signed n -dimensional volume. In particular $\det I = 1$, and for a diagonal matrix the determinant is the product of the diagonal entries.

Remark 3.77. This is the volume-based definition. It suffices for our purposes, but there is a more rigorous definition and related results that can be found in Sheldon Axler's *Linear Algebra Done Right* if a more rigorous treatment is desired. The volume idea is equivalent and is especially useful when we get to probability. Typically students will not be asked to work in matrix settings where the dimension is greater than 3; there are a couple of standard approaches for generating the formula for the determinant of a 3×3 that can be found in standard references, and the higher dimensional formulas are developed via the more rigorous definition such as in Axler.

Theorem 3.78 (Multiplicativity). *If $A, B \in L(V)$ then $\det(AB) = \det A \cdot \det B$.*

Proof. Using the volume-scaling interpretation (Definition 3.76), applying the composition AB to the unit cube first applies B , scaling signed volume by $\det B$, and then applies A , scaling the resulting signed volume by $\det A$. The net scaling factor is $\det A \cdot \det B$, which is $\det(AB)$. $\square$

Theorem 3.79 (Determinant and invertibility). *Let $A \in L(V)$. Then A is invertible if and only if $\det A \neq 0$. If A is invertible then $\det(A^{-1}) = \frac{1}{\det A}$.*

Proof. If A is invertible, then $AA^{-1} = I$, so by Theorem 3.78, $\det A \cdot \det(A^{-1}) = \det I = 1$. This requires $\det A \neq 0$ and gives $\det(A^{-1}) = 1/\det A$. Conversely, if A is not invertible then it is not injective (Lemma 3.46), so it collapses a positive-volume region onto a lower-dimensional set of zero n -dimensional volume; hence the volume scaling factor is $\det A = 0$. Contrapositively, $\det A \neq 0$ implies A is invertible. $\square$

Theorem 3.80 (Conjugation invariance). *Let $A \in L(V)$ and let $T : W \rightarrow V$ be an invertible map. Then*

$$\det(T^{-1}AT) = \det A.$$

Proof. By Theorem 3.78 and Theorem 3.79,

$$\det(T^{-1}AT) = \det(T^{-1})\det(A)\det(T) = \frac{1}{\det T}\det(A)\det(T) = \det A. \quad \square$$

Theorem 3.81 (Determinant as product of eigenvalues). *If a linear operator $A \in L(V)$ has a complete eigenbasis for V , then its determinant is the product of its eigenvalues.*

Proof. A complete eigenbasis means $A = U\Lambda U^{-1}$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ (Theorem 3.71). By Theorems 3.78 and 3.79,

$$\det A = \det U \cdot \det \Lambda \cdot \det(U^{-1}) = \det U \cdot \det \Lambda \cdot \frac{1}{\det U} = \det \Lambda = \prod_{i=1}^n \lambda_i,$$

since the determinant of a diagonal matrix is the product of its diagonal entries (Definition 3.76). $\square$

Now we get to the most immediately useful result that encourages our development of determinants in these notes, as a way to find eigenvalues.

Theorem 3.82. *Let $A \in L(V)$ and $\lambda \in \mathbb{R}$. Then λ is an eigenvalue of A if and only if $\det(\lambda I - A) = 0$.*

Proof. By Theorem 3.48, λ is an eigenvalue of A if and only if $A - \lambda I$ is not invertible, equivalently $\lambda I - A$ is not invertible (since $\det(\lambda I - A) = (-1)^n \det(A - \lambda I)$, the two determinants vanish together). By Theorem 3.79, $\lambda I - A$ is not invertible if and only if $\det(\lambda I - A) = 0$. $\square$

By using this result we obtain another, more common, polynomial associated with eigenvalues.

Definition 3.83 (Characteristic polynomial). Let $A \in L(V)$ for a finite-dimensional vector space V . The *characteristic polynomial* of A is

$$\chi_A(\lambda) = \det(\lambda I - A),$$

a monic polynomial of degree $n = \dim V$ in λ whose roots are the eigenvalues of A .

The characteristic polynomial approach is the typical method by which we find eigenvalues analytically for smaller dimensions. For large dimensions computational methods are employed.

Chapter 4

Vector and Matrix Calculus

We began the preceding chapter by connecting continuity for vector-valued functions back to continuity for each coordinate-wise output. This motivated our discussion of linear maps, inverses, and invariant subspaces. We now return to general maps, i.e. general functions, for vector-valued inputs and/or outputs. We focus specifically on functions of real variables $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Having already established the result for continuity, we now turn to differentiation, as this will connect us to optimization.

4.1 The Derivative as a Linear Map

We start by introducing an equivalent but slightly different definition for differentiation than the one we used in the single-variable case. This equivalent definition will allow us to work in the multivariable case a bit more easily.

Recall that for a function $f : \mathbb{R} \rightarrow \mathbb{R}$, it is differentiable at a point $x \in \mathbb{R}$ if the limit in the definition exists (Definition 2.134):

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

Recall also our discussion of $f'(x)$ as the best linear approximation for f at x , and our discussion of the error resulting from the Taylor series as $o(\Delta x)$. We can consider the following algebraic reorganization:

$$\begin{aligned} f'(x) &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \\ 0 &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} - f'(x) \\ 0 &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x) - f'(x)\Delta x}{\Delta x}. \end{aligned}$$

We note two things. The first is that the term $f'(x)\Delta x$ is linear in Δx . The second is a bit circular, but bear with it for now: this limit exists if and only if $f'(x)$ exists. We can generalize this a bit by defining $\lambda : \mathbb{R} \rightarrow \mathbb{R}$ as a linear function of Δx , so $\lambda(\Delta x) = a \Delta x$ for some $a \in \mathbb{R}$, and we argue that

$$\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \text{ exists if and only if there exists a } \lambda(\Delta x) \text{ such that}$$

$$\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x) - \lambda(\Delta x)}{\Delta x} = 0.$$

In other words, $f'(x)$ exists if and only if such a $\lambda(\Delta x)$ exists, and in fact we would have $\lambda(\Delta x) = f'(x)\Delta x$ in the single-variable setting.

We can use this to give an equivalent definition of the derivative.

Definition 4.1 (Differentiable). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *differentiable* at a point $\mathbf{x} \in \mathbb{R}^n$ if there is a linear transformation $\lambda : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$\lim_{\Delta \mathbf{x} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{x} + \Delta \mathbf{x}) - f(\mathbf{x}) - \lambda(\Delta \mathbf{x})\|}{\|\Delta \mathbf{x}\|} = 0.$$

Note that this does not explicitly give us a reasonable way to find a derivative in general. In the single-variable setting the definition we used also gives us a means of finding the derivative, but in higher dimensions it is not as straightforward. We will show that in this setting, much like continuity, we need only consider coordinate-wise outputs to actually find the derivative.

Also note that we rely on norms in order to map from the higher-dimensional space to $\mathbb{R}$ for comparison of sizes, i.e. we compare the “size” of the numerator to the “size” of the denominator in the limit. Without applying a norm, the concept of “division” does not occur across spaces nor in higher dimension. There is no definition of $\mathbf{x}/\mathbf{y}$ as whole objects; if trying to denote element-wise division, this must be made explicit, as such notation is not standard.

The linear function $\lambda(\Delta \mathbf{x})$ maps $\lambda : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and as we have previously established, we can use a matrix representation for this linear map. With $\Delta \mathbf{x} = [\Delta x_1, \Delta x_2, \dots, \Delta x_n]^\top$ we can write $\lambda(\Delta \mathbf{x}) = A\Delta \mathbf{x}$ with $A \in \mathbb{R}^{m \times n}$. We define the *derivative* of f as the matrix A . We can mirror the notation from the single-variable setting as $f'(\mathbf{x}) = A$, but more typically the notation uses the symbol D , and $Df(\mathbf{x})$ is the “derivative of f at $\mathbf{x}$.”

The matrix A is itself often referred to as the *Jacobian*. Finding the actual derivative involves finding the matrix A . Most commonly, in optimization settings, $f : \mathbb{R}^n \rightarrow \mathbb{R}$, so $Df(\mathbf{x}) \in \mathbb{R}^{1 \times n}$. In fact, functions with a single output, called *scalar-valued functions* $f : \mathbb{R}^n \rightarrow \mathbb{R}$, are given a unique name for their derivative, called the *gradient*, with notation $\vec{\nabla} f(\mathbf{x}) = Df(\mathbf{x})^\top$, a column vector rather than a $1 \times n$ row matrix. We will give a proper definition for the gradient later (Section 4.4), but we introduce it now as a preview.

We establish some results we can obtain just from the definition before we develop the typical way we find Df . First we propose that λ is unique.

Theorem 4.2. *If a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at $\mathbf{x} \in \mathbb{R}^n$, there is a unique linear transformation $\lambda : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that*

$$\lim_{\Delta \mathbf{x} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{x} + \Delta \mathbf{x}) - f(\mathbf{x}) - \lambda(\Delta \mathbf{x})\|}{\|\Delta \mathbf{x}\|} = 0.$$

Proof. Suppose two linear maps λ and μ both satisfy the limit. Let $\eta = \lambda - \mu$, which is linear. Subtracting the two limit expressions and using the triangle inequality,

$$\frac{\|\eta(\Delta \mathbf{x})\|}{\|\Delta \mathbf{x}\|} \leq \frac{\|f(\mathbf{x} + \Delta \mathbf{x}) - f(\mathbf{x}) - \lambda(\Delta \mathbf{x})\|}{\|\Delta \mathbf{x}\|} + \frac{\|f(\mathbf{x} + \Delta \mathbf{x}) - f(\mathbf{x}) - \mu(\Delta \mathbf{x})\|}{\|\Delta \mathbf{x}\|} \rightarrow 0$$

as $\Delta \mathbf{x} \rightarrow \mathbf{0}$. Fix any direction $\mathbf{v} \neq \mathbf{0}$ and take $\Delta \mathbf{x} = t\mathbf{v}$ with $t \rightarrow 0^+$. By linearity $\eta(t\mathbf{v}) = t\eta(\mathbf{v})$, so

$$\frac{\|\eta(t\mathbf{v})\|}{\|t\mathbf{v}\|} = \frac{\|\eta(\mathbf{v})\|}{\|\mathbf{v}\|},$$

which is independent of t . Since the left side tends to 0, we must have $\eta(\mathbf{v}) = \mathbf{0}$ for every $\mathbf{v}$, i.e. $\lambda = \mu$. $\square$

The chain rule also generalizes.

Theorem 4.3 (Chain Rule). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}^p$ be two functions differentiable at $\mathbf{x}$ and $f(\mathbf{x})$ respectively. The composition $g \circ f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is differentiable at $\mathbf{x}$ and*

$$D(g \circ f)(\mathbf{x}) = Dg(f(\mathbf{x})) \cdot Df(\mathbf{x}).$$

Proof. Write $\mathbf{y} = f(\mathbf{x})$, $A = Df(\mathbf{x})$, $B = Dg(\mathbf{y})$. By differentiability, $f(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{y} + A\Delta\mathbf{x} + \mathbf{r}(\Delta\mathbf{x})$ with $\|\mathbf{r}(\Delta\mathbf{x})\| / \|\Delta\mathbf{x}\| \rightarrow 0$, and $g(\mathbf{y} + \Delta\mathbf{y}) = g(\mathbf{y}) + B\Delta\mathbf{y} + \mathbf{s}(\Delta\mathbf{y})$ with $\|\mathbf{s}(\Delta\mathbf{y})\| / \|\Delta\mathbf{y}\| \rightarrow 0$. Substituting $\Delta\mathbf{y} = A\Delta\mathbf{x} + \mathbf{r}(\Delta\mathbf{x})$,

$$g(f(\mathbf{x} + \Delta\mathbf{x})) = g(\mathbf{y}) + BA\Delta\mathbf{x} + B\mathbf{r}(\Delta\mathbf{x}) + \mathbf{s}(\Delta\mathbf{y}).$$

The remainder $B\mathbf{r}(\Delta\mathbf{x}) + \mathbf{s}(\Delta\mathbf{y})$ is $o(\|\Delta\mathbf{x}\|)$: the first term because $\mathbf{r}$ is, and the second because $\|\Delta\mathbf{y}\| \leq \|A\| \|\Delta\mathbf{x}\| + \|\mathbf{r}(\Delta\mathbf{x})\| = O(\|\Delta\mathbf{x}\|)$ while $\mathbf{s}(\Delta\mathbf{y}) / \|\Delta\mathbf{y}\| \rightarrow 0$. Hence $g \circ f$ is differentiable at $\mathbf{x}$ with derivative $BA = Dg(f(\mathbf{x})) Df(\mathbf{x})$. $\square$

4.1.1 Coordinate Functions and Results from the Definition

Now recall our coordinate projection function $P_i : \mathbb{R}^m \rightarrow \mathbb{R}$ (Definition 3.2). Let $f^i = P_i(f)$, so $f^i(\mathbf{x}) = P_i(f(\mathbf{x}))$ is the i -th output coordinate of the function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$; thus $f^i : \mathbb{R}^n \rightarrow \mathbb{R}$. We can then write, in vector notation,

$$f(\mathbf{x}) = \begin{bmatrix} f^1(\mathbf{x}) \\ \vdots \\ f^m(\mathbf{x}) \end{bmatrix}.$$

We now present several useful results by simply applying the definition.

Theorem 4.4. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$.*

- (i) *If f is constant, so $f(\mathbf{x}) = \mathbf{k}$ for some $\mathbf{k} \in \mathbb{R}^m$ and all $\mathbf{x} \in \mathbb{R}^n$, then $\lambda(\Delta\mathbf{x}) = \mathbf{0}$, so $Df = 0$.*
- (ii) *If f is a linear map, so $f(\mathbf{x}) = A\mathbf{x}$, then $\lambda(\Delta\mathbf{x}) = f(\Delta\mathbf{x})$, so $Df = A$.*
- (iii) *f is differentiable at $\mathbf{x} \in \mathbb{R}^n$ if and only if each f^i is differentiable at $\mathbf{x} \in \mathbb{R}^n$. Additionally, with $\lambda^i = P_i(\lambda)$ and $\lambda = (\lambda^1, \dots, \lambda^m)$, for each λ^i we have an $A_i \in \mathbb{R}^n$ such that $\lambda^i(\Delta\mathbf{x}) = A_i\Delta\mathbf{x} = Df^i\Delta\mathbf{x} = \vec{\nabla} f^i{}^\top \Delta\mathbf{x}$. Recall that $\vec{\nabla} f^i \in \mathbb{R}^n$, so each $(f^i)'(\mathbf{x}) = Df^i(\mathbf{x}) = \vec{\nabla} f^i(\mathbf{x})^\top$ is a $1 \times n$ row. Stacking these rows, $Df(\mathbf{x})$ is the $m \times n$ matrix whose i -th row is $\vec{\nabla} f^i(\mathbf{x})^\top$:*

$$Df(\mathbf{x}) = \begin{bmatrix} - & A_1 & - \\ & \vdots & \\ - & A_m & - \end{bmatrix} = \begin{bmatrix} - & \vec{\nabla} f^1{}^\top & - \\ & \vdots & \\ - & \vec{\nabla} f^m{}^\top & - \end{bmatrix}.$$

- (iv) *If $s : \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined by $s\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = x + y$, then $\lambda\left(\begin{smallmatrix} \Delta x \\ \Delta y \end{smallmatrix}\right) = \Delta x + \Delta y = s\left(\begin{smallmatrix} \Delta x \\ \Delta y \end{smallmatrix}\right)$, so $\lambda = s$ and $Ds\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \begin{bmatrix} 1 & 1 \end{bmatrix}$.*

(v) If $p : \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined by $p\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = xy$, then $\lambda\left(\begin{smallmatrix} \Delta x \\ \Delta y \end{smallmatrix}\right) = y \Delta x + x \Delta y$, so $Dp\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \begin{bmatrix} y & x \end{bmatrix}$.

Proof. (i) $f(\mathbf{x} + \Delta \mathbf{x}) - f(\mathbf{x}) = \mathbf{k} - \mathbf{k} = \mathbf{0}$, so $\lambda = 0$ makes the limit vanish; by uniqueness $Df = 0$.

(ii) $f(\mathbf{x} + \Delta \mathbf{x}) - f(\mathbf{x}) = A(\mathbf{x} + \Delta \mathbf{x}) - A\mathbf{x} = A\Delta \mathbf{x}$, so $\lambda(\Delta \mathbf{x}) = A\Delta \mathbf{x}$ exactly and the remainder is zero; $Df = A$.

(iii) The norm on $\mathbb{R}^m$ controls each coordinate ($|P_i(\mathbf{v})| \leq \|\mathbf{v}\|$) and is controlled by them ($\|\mathbf{v}\| \leq \sum_i |P_i(\mathbf{v})|$), so the vector limit vanishes if and only if each coordinate limit does; the matrix of the λ^i stacks as claimed.

(iv) $s(\mathbf{x} + \Delta \mathbf{x}) - s(\mathbf{x}) = (x + \Delta x) + (y + \Delta y) - x - y = \Delta x + \Delta y$, which is linear in $\Delta \mathbf{x}$, so it equals λ exactly.

(v) $p(\mathbf{x} + \Delta \mathbf{x}) - p(\mathbf{x}) = (x + \Delta x)(y + \Delta y) - xy = y \Delta x + x \Delta y + \Delta x \Delta y$; the term $\Delta x \Delta y = o(\|\Delta \mathbf{x}\|)$, leaving the linear part $y \Delta x + x \Delta y$.

□

Corollary 4.5. Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at $\mathbf{x}$.

(i) Let $h = f + g$. Then $\lambda_h(\Delta \mathbf{x}) = \lambda_f(\Delta \mathbf{x}) + \lambda_g(\Delta \mathbf{x})$, so

$$Dh(\mathbf{x}) = D(f + g)(\mathbf{x}) = Df(\mathbf{x}) + Dg(\mathbf{x}).$$

(ii) Let $h = f \cdot g$. Then $\lambda_h(\Delta \mathbf{x}) = g(\mathbf{x})\lambda_f(\Delta \mathbf{x}) + f(\mathbf{x})\lambda_g(\Delta \mathbf{x})$, so

$$Dh(\mathbf{x}) = g(\mathbf{x})Df(\mathbf{x}) + f(\mathbf{x})Dg(\mathbf{x}).$$

Proof. Write $\mathbf{F}(\mathbf{x}) = \begin{pmatrix} f(\mathbf{x}) \\ g(\mathbf{x}) \end{pmatrix} : \mathbb{R}^n \rightarrow \mathbb{R}^2$, which has $D\mathbf{F}(\mathbf{x}) = \begin{pmatrix} Df(\mathbf{x}) \\ Dg(\mathbf{x}) \end{pmatrix}$ by Theorem 4.4(iii).

(i) $h = s \circ \mathbf{F}$ with $s\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right) = u + v$. By the Chain Rule (Theorem 4.3) and Theorem 4.4(iv),

$$Dh(\mathbf{x}) = Ds D\mathbf{F}(\mathbf{x}) = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} Df(\mathbf{x}) \\ Dg(\mathbf{x}) \end{bmatrix} = Df(\mathbf{x}) + Dg(\mathbf{x}).$$

(ii) $h = p \circ \mathbf{F}$ with $p\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right) = uv$. By the Chain Rule and Theorem 4.4(v), $Dp\left(\begin{smallmatrix} f(\mathbf{x}) \\ g(\mathbf{x}) \end{smallmatrix}\right) = \begin{bmatrix} g(\mathbf{x}) & f(\mathbf{x}) \end{bmatrix}$, so

$$Dh(\mathbf{x}) = \begin{bmatrix} g(\mathbf{x}) & f(\mathbf{x}) \end{bmatrix} \begin{bmatrix} Df(\mathbf{x}) \\ Dg(\mathbf{x}) \end{bmatrix} = g(\mathbf{x})Df(\mathbf{x}) + f(\mathbf{x})Dg(\mathbf{x}).$$

□

If the reader has already had some exposure to multivariable calculus, the results above might seem obvious based on knowing the actual way to find the derivatives we cover next; but what is noteworthy about the results above is that we can obtain them directly from the definition without appealing to the exact calculation of such derivatives. Now we develop a general way to calculate the derivative, via partial derivatives.

4.2 Partial Derivatives

The definition of the derivative focuses on the difference between the value of a function at a point and at a small perturbation from that point, i.e. $f(\mathbf{x} + \Delta\mathbf{x}) - f(\mathbf{x})$, as the perturbation goes to magnitude 0 in the limit. We have not put any restrictions on $\Delta\mathbf{x}$, so the perturbation can be in any or all coordinates; e.g. if $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ then $\Delta\mathbf{x} = \begin{pmatrix} \Delta x_1 \\ \Delta x_2 \end{pmatrix}$, and $f(\mathbf{x} + \Delta\mathbf{x})$ is evaluated at a perturbation both in the first and second coordinate.

However, we can consider a restriction on the perturbation. In particular we want to consider restricting the perturbation to a single coordinate. For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$ we use the notation $\frac{\partial f}{\partial x_i}$ to represent the i -th *partial derivative*:

Definition 4.6 (Partial derivative). For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$, the i -th partial derivative is

$$\frac{\partial f}{\partial x_i} = \lim_{\Delta x_i \rightarrow 0} \frac{f([x_1, \dots, x_i + \Delta x_i, \dots, x_n]^\top) - f([x_1, \dots, x_i, \dots, x_n]^\top)}{\Delta x_i}.$$

This looks quite similar to the derivative in single-variable calculus, and it is. In fact, if we consider each x_j with $j \neq i$ as a constant value, we can define a single-variable function that has an equivalent derivative. Let $g(y) = f(x_1, \dots, y, \dots, x_n)$. Then

$$\frac{dg}{dy} = \lim_{\Delta y \rightarrow 0} \frac{g(y + \Delta y) - g(y)}{\Delta y}.$$

This means that we already know how to find partial derivatives, because

$$\frac{\partial f}{\partial x_i} = \frac{dg}{dy} \quad \text{for } y = x_i.$$

Example 4.7. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as $f\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 e^{x_2}$. Then

$$\frac{\partial f}{\partial x_1} = \frac{d}{dx_1}(x_1 e^{x_2}) = e^{x_2}, \quad \frac{\partial f}{\partial x_2} = \frac{d}{dx_2}(x_1 e^{x_2}) = x_1 e^{x_2}.$$

Note we distinguish between partials, written $\frac{\partial f}{\partial x_i}$, and the single-variable derivative operation, written $\frac{d}{dx_i}$.

Partial derivatives extract out the rate of change of the function at a point in the specific coordinate/direction that the partial is taken with respect to. A reasonable conjecture is that partial derivatives in all coordinates/directions should correspond to the overall derivative of a function. This conjecture is correct.

Theorem 4.8. If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at a point $\mathbf{x} \in \mathbb{R}^n$, then $\frac{\partial f^i}{\partial x_j}$ exists for all $i = 1, \dots, m$ and $j = 1, \dots, n$. Additionally, $Df(\mathbf{x})$ is the $m \times n$ matrix with entry in row i and column j given as $\frac{\partial f^i}{\partial x_j}$. This $m \times n$ matrix is the Jacobian.

Proof. By Theorem 4.4(iii) it suffices to treat each scalar component f^i , so assume $m = 1$ and write $A = Df(\mathbf{x}) \in \mathbb{R}^{1 \times n}$ with entries A_j . Restrict the perturbation to the j -th coordinate, $\Delta\mathbf{x} = \Delta x_j \mathbf{e}_j$. Differentiability gives

$$\frac{f(\mathbf{x} + \Delta x_j \mathbf{e}_j) - f(\mathbf{x}) - A_j \Delta x_j}{|\Delta x_j|} \rightarrow 0,$$

which rearranges to $\frac{f(\mathbf{x} + \Delta x_j \mathbf{e}_j) - f(\mathbf{x})}{\Delta x_j} \rightarrow A_j$. The left side is exactly the difference quotient defining $\frac{\partial f}{\partial x_j}$, so the partial exists and equals A_j . Hence the entries of $Df(\mathbf{x})$ are the partials, and stacking over i gives the stated $m \times n$ matrix. $\square$

Note that this says differentiability—based on the definition over a limit defined approaching from any direction—implies existence of the partials, and the derivative itself can be represented as the specific matrix. However, the existence of the partials does not itself imply overall differentiability. Consider the following example.

Example 4.9. Consider

$$f(x, y) = f\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Along either axis f is identically 0, so both partials exist at the origin:

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - 0}{\Delta x} = 0, \quad \frac{\partial f}{\partial y}(0, 0) = 0.$$

But approaching along the line $y = x$, for $x > 0$,

$$f(x, x) = \frac{x^2}{\sqrt{2x^2}} = \frac{x}{\sqrt{2}}, \quad \text{so} \quad \frac{f(x, x) - f(0, 0)}{\|(x, x)\|} = \frac{x/\sqrt{2}}{x\sqrt{2}} = \frac{1}{2} \not\rightarrow 0.$$

So the linear approximation by the (zero) partials fails along $y = x$, and f is not differentiable at the origin even though both partials exist there.

If we put an additional restriction on the partial derivatives of the function we can get equivalence, i.e. an if-and-only-if statement.

Theorem 4.10. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. If all partials $\frac{\partial f^i}{\partial x_j}$ exist in an open set containing a point $\mathbf{x} \in \mathbb{R}^n$ and each partial is continuous at $\mathbf{x}$, then f is differentiable at $\mathbf{x}$.*

Proof. Again treat one scalar component, so $m = 1$. Write the increment as a sum of one-coordinate steps: with $\mathbf{z}_0 = \mathbf{x}$ and $\mathbf{z}_k = \mathbf{x} + \sum_{j \leq k} \Delta x_j \mathbf{e}_j$,

$$f(\mathbf{x} + \Delta \mathbf{x}) - f(\mathbf{x}) = \sum_{k=1}^n (f(\mathbf{z}_k) - f(\mathbf{z}_{k-1})).$$

Each term changes only in coordinate k , so by the single-variable Mean Value Theorem there is a point $\boldsymbol{\xi}_k$ on that segment with $f(\mathbf{z}_k) - f(\mathbf{z}_{k-1}) = \frac{\partial f}{\partial x_k}(\boldsymbol{\xi}_k) \Delta x_k$. Then

$$f(\mathbf{x} + \Delta \mathbf{x}) - f(\mathbf{x}) - \sum_k \frac{\partial f}{\partial x_k}(\mathbf{x}) \Delta x_k = \sum_k \left(\frac{\partial f}{\partial x_k}(\boldsymbol{\xi}_k) - \frac{\partial f}{\partial x_k}(\mathbf{x}) \right) \Delta x_k.$$

As $\Delta \mathbf{x} \rightarrow \mathbf{0}$ each $\boldsymbol{\xi}_k \rightarrow \mathbf{x}$, and continuity of the partials makes each bracket $\rightarrow 0$; dividing by $\|\Delta \mathbf{x}\|$ and using $|\Delta x_k| \leq \|\Delta \mathbf{x}\|$ shows the remainder is $o(\|\Delta \mathbf{x}\|)$. Hence f is differentiable at $\mathbf{x}$ with $Df(\mathbf{x}) = \left[\frac{\partial f}{\partial x_1}(\mathbf{x}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}) \right]$. $\square$

This is a very common condition that we apply when working in optimization, so we give it a specific name.

Definition 4.11 (Continuously differentiable). A function is called *continuously differentiable* at a point if all the partial derivatives are continuous at the point. A function is continuously differentiable in the general sense if its partials are continuous at every point.

We sometimes also want to talk not just about whether the function is continuously differentiable, but also about whether the derivative itself is also continuously differentiable. In order to do this we use a prefix to capture the information, such as “twice continuously differentiable” to indicate that the derivative of the function is also continuously differentiable. We also use “infinitely continuously differentiable” to communicate that derivatives of any order (e.g. 2nd, 10th, etc.) are all continuously differentiable. Recall we denote the set of functions of i -th order of continuous differentiability as C^i (Definition 2.161). Of course it should be clear that

$$C^\infty \subset \dots \subset C^k \subset \dots \subset C^2 \subset C^1.$$

This naturally motivates discussion of higher-order derivatives, but before we continue on to higher-order derivatives we present one more topic.

4.3 Directional Derivatives

When we introduced partial derivatives at a point we appealed to the representation of a multivariable function as a single-variable function, i.e. for $f : \mathbb{R}^n \rightarrow \mathbb{R}$ as $f(\mathbf{x})$ we considered $g : \mathbb{R} \rightarrow \mathbb{R}$ as $g(x_i)$ at a given point $\mathbf{x}$, where we fixed all other coordinates x_j for $j \neq i$ in order to get the instantaneous change of f in coordinate x_i at $\mathbf{x}$.

We can generalize this idea to considering the partial derivative as being with respect to the standard basis vectors in a very natural way. Consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$:

$$\frac{df}{dx_i} = \lim_{\Delta x \rightarrow 0} \frac{f(\mathbf{x} + \Delta x \mathbf{e}_i) - f(\mathbf{x})}{\Delta x},$$

which is equivalent to our previous definition for a partial derivative with respect to coordinate x_i . Moreover, given that we find the derivative, i.e. Jacobian, for $f : \mathbb{R}^n \rightarrow \mathbb{R}$ as

$$Df(\mathbf{x}) = \left[\frac{\partial f}{\partial x_1}(\mathbf{x}), \quad \dots, \quad \frac{\partial f}{\partial x_n}(\mathbf{x}) \right],$$

we can extract out the derivative corresponding to the direction x_i by considering the Jacobian map on $\mathbf{e}_i$. We omit the argument $(\mathbf{x})$ for simpler notation, but it should always be understood we are evaluating the derivative at a point:

$$Df \cdot \mathbf{e}_i = \left[\frac{\partial f}{\partial x_1}, \quad \dots, \quad \frac{\partial f}{\partial x_i}, \quad \dots, \quad \frac{\partial f}{\partial x_n} \right] \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} = \frac{\partial f}{\partial x_i}.$$

In this way we have found the instantaneous rate of change in the coordinate x_i , or equivalently we have found the “directional derivative” in the direction $\mathbf{e}_i$.

Note that the equivalency of the partial derivative with respect to coordinate x_i and the directional derivative in the direction $\mathbf{e}_i$ depends on $\|\mathbf{e}_i\| = 1$. Specifically, if we instead considered the direction given by the vector $\mathbf{e}'_i = 2\mathbf{e}_i$, then

$$Df \cdot \mathbf{e}'_i = 2 \frac{\partial f}{\partial x_i},$$

which would not have exact equivalence with the partial derivative. For this reason the convention is typically to consider the normalized vector. We will return to this shortly. This generalizes in two ways we will want to consider.

(1) Vector-valued functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$

In this case we again illustrate the idea using $\mathbf{e}_i$:

$$Df \cdot \mathbf{e}_i = \begin{bmatrix} \frac{\partial f^1}{\partial x_1} & \dots & \frac{\partial f^1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f^m}{\partial x_1} & \dots & \frac{\partial f^m}{\partial x_n} \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{\partial f^1}{\partial x_i} \\ \frac{\partial f^2}{\partial x_i} \\ \vdots \\ \frac{\partial f^m}{\partial x_i} \end{bmatrix} = D_{\mathbf{e}_i} f,$$

where we introduce the notation $D_{\mathbf{e}_i}f$ to indicate the derivative of f in direction $\mathbf{e}_i$, compared with Df , which is the derivative across the entire space $\mathbb{R}^m$. As we develop the directional derivative, it may be useful to reflect on the linear algebra ideas we had around bases and dimension.

When $m = 1$ the directional derivative (with respect to $\mathbf{e}_i$) tells us if we increase or decrease in function value, i.e. do we move “up” or “down” with respect to a change in input coordinate x_i at the target point. When $m > 1$ the directional derivative tells us more generally what direction the function moves towards from a point $f(\mathbf{x})$ when we move in the direction $\mathbf{e}_i$ from the point $\mathbf{x}$ in the input space.

(2) General vectors other than the standard basis vectors

Using the example of the equivalence between a partial derivative and the representation of a perturbation in a standard basis vector, we can develop a definition for a directional derivative in an arbitrary direction $\mathbf{u} \in \mathbb{R}^n$.

Definition 4.12 (Directional derivative). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a continuously differentiable function. The *directional derivative* of f in the direction $\mathbf{u} \in \mathbb{R}^n$ evaluated at a point $\mathbf{x} \in \mathbb{R}^n$ is given as

$$D_{\mathbf{u}}f(\mathbf{x}) = \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{u}) - f(\mathbf{x})}{t}.$$

Similar to the definition for a derivative based around the existence of a linear function $\lambda_f(\Delta\mathbf{x})$, this definition does not immediately present a method to find such a $D_{\mathbf{u}}f(\mathbf{x})$.

Thinking back to our ideas about a basis and the ability to represent any vector based on its decomposition into a linear combination of the basis vectors, this suggests we might be able to develop a way to find the directional derivative by appealing to some kind of basis decomposition. As usual, we use the standard basis vectors. Ideally we would like to take the idea that $\mathbf{u} = \sum_{i=1}^n u_i \mathbf{e}_i$ and apply it in the form $D_{\mathbf{u}}f(\mathbf{x}) = \sum_{i=1}^n u_i D_{\mathbf{e}_i}f(\mathbf{x})$. This gives

$$D_{\mathbf{u}}f(\mathbf{x}) = \sum_{i=1}^n u_i D_{\mathbf{e}_i}f(\mathbf{x}) = \sum_{i=1}^n u_i Df(\mathbf{x})\mathbf{e}_i = Df(\mathbf{x}) \sum_{i=1}^n u_i \mathbf{e}_i = Df(\mathbf{x}) \cdot \mathbf{u},$$

where we take $Df(\mathbf{x})$ outside the sum by linearity. It turns out we can find this equivalence.

Theorem 4.13. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a continuously differentiable function. Let $\mathbf{u} \in \mathbb{R}^n$ be a direction in $\mathbb{R}^n$ and $\mathbf{x} \in \mathbb{R}^n$ be a point. Then

$$D_{\mathbf{u}}f(\mathbf{x}) = Df(\mathbf{x}) \cdot \mathbf{u}.$$

Proof. We proceed by reducing the multivariable context to a single-variable representation. Let $\gamma : \mathbb{R} \rightarrow \mathbb{R}^n$ be the function $\gamma(t) = \mathbf{x} + t\mathbf{u}$. Let $g(t) = f(\gamma(t))$, so by the Chain Rule (Theorem 4.3) we obtain

$$g'(t) = f'(\gamma(t)) \gamma'(t).$$

Clearly at $t = 0$ we obtain

$$g'(0) = f'(\gamma(0)) \cdot \gamma'(0) = f'(\mathbf{x}) \cdot \mathbf{u} = Df(\mathbf{x}) \cdot \mathbf{u}.$$

Further, $g(t) - g(0) = f(\gamma(t)) - f(\gamma(0)) = f(\mathbf{x} + t\mathbf{u}) - f(\mathbf{x})$, so

$$g'(0) = \lim_{t \rightarrow 0} \frac{g(t) - g(0)}{t} = \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{u}) - f(\mathbf{x})}{t} = D_{\mathbf{u}}f(\mathbf{x}).$$

Therefore $D_{\mathbf{u}}f(\mathbf{x}) = Df(\mathbf{x}) \cdot \mathbf{u}$. Observe that the definition of the limit for the derivative here does not require norms, because we are dividing by a scalar so the operation is well defined; and if $m > 1$ the directional derivative is an m -dimensional vector, and if $m = 1$ the directional derivative is a scalar value, as desired. $\square$

4.4 The Gradient

The results above are given for general functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, but in optimization settings we are almost always in a setting with $m = 1$, i.e. $f : \mathbb{R}^n \rightarrow \mathbb{R}$. We therefore now restrict further discussion to *scalar-valued* functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$. These are also called *real-valued* functions.

With this restriction we can now revisit the term “gradient” with a more useful definition that does not frame it as a “special case of the derivative/Jacobian.”

Definition 4.14 (Gradient). The *gradient* of a scalar-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at a point $\mathbf{x} \in \mathbb{R}^n$ is the vector whose directional derivative with f gives the greatest magnitude; i.e. it is the direction of greatest instantaneous/linear increase of the function at the point $\mathbf{x}$. We denote this vector by $\vec{\nabla} f(\mathbf{x})$. More formally, the gradient is the vector whose dot product with any vector $\mathbf{v}$ at each point $\mathbf{x}$ is the directional derivative of f along $\mathbf{v}$:

$$D_{\mathbf{v}} f(\mathbf{x}) = \vec{\nabla} f(\mathbf{x}) \cdot \mathbf{v}$$

In Euclidean space the dot product of two vectors $\mathbf{u}, \mathbf{v}$ is given as

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^\top \mathbf{v} = \cos \theta \|\mathbf{u}\| \|\mathbf{v}\|.$$

We also know by Cauchy–Schwarz that

$$\mathbf{u} \cdot \mathbf{v} \leq |\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

So we only obtain a maximum dot product between $\mathbf{u}$ and $\mathbf{v}$ if $\cos \theta = 1 \Rightarrow \theta = 0^\circ$, so the two vectors must be pointing in the same direction.

For the directional derivative to be maximized we can consider $\mathbf{u}^\top = Df(\mathbf{x})$, so $\mathbf{u} \in \mathbb{R}^n$ since $Df(\mathbf{x}) \in \mathbb{R}^{1 \times n}$. If we want to pick a direction $\vec{\nabla} f(\mathbf{x})$ so that $D_{\vec{\nabla} f} f(\mathbf{x})$ is maximized (recall $D_{\mathbf{u}} f(\mathbf{x}) = Df(\mathbf{x}) \cdot \mathbf{u}$), we need to set

$$\vec{\nabla} f(\mathbf{x}) = Df(\mathbf{x})^\top = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}.$$

It may seem like we are playing word games rather than defining something of substance. However, in order to be rigorous in our development of the mathematical infrastructure we are building, we need to be very specific in the definitions we employ. In the single-variable setting the terms *derivative* and *gradient* may be used interchangeably because they evaluate to the same result, i.e. if $f : \mathbb{R} \rightarrow \mathbb{R}$ then the derivative at a point is $f'(x)$ and the direction of greatest increase is given as $\text{sign}[f'(x)]$.

However, the derivative and the gradient are conceptually distinct but related ideas. In the multivariable setting $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $f'(\mathbf{x}) \neq \vec{\nabla} f$, because $f'(\mathbf{x}) \in \mathbb{R}^{1 \times n}$ while $\vec{\nabla} f \in \mathbb{R}^n$. We have to be careful to respect the structure of the spaces and recognize $f'(\mathbf{x})$ as a linear map acting on a vector in $\mathbb{R}^n$, usually $\Delta \mathbf{x}$, while $\vec{\nabla} f$ is a vector/direction in $\mathbb{R}^n$.

4.5 Higher-Order Derivatives and the Hessian

Higher-order derivatives still follow the same definitions as the first-order derivative, as we are simply applying the same definition to the function obtained from the first-order derivative. It is worthwhile to illustrate how this works. First we establish some notation.

As we saw, partial derivatives largely determine the derivative function at a point, so we introduce higher-order partial derivative notation, starting with comparison to the single-variable case. In the single-variable setting for a function $f : \mathbb{R} \rightarrow \mathbb{R}$ the second-order derivative notation is

$$f''(x) = \frac{d}{dx} \left(\frac{df}{dx} \right) = \frac{d^2 f}{dx^2} \equiv \frac{d^2 f}{(dx)^2}.$$

Note the little notational nuance in the final term; the typical notation should really be thought of as applying the squared function to dx , not just to x in the notation. When we get to integration we will see why this distinction is important for capturing the correct intuition with the notation. Back to our partial derivative notation.

We already established that $f'(\mathbf{x}) = Df(\mathbf{x})$, so we would reserve $f''(\mathbf{x})$ for $D(Df)(\mathbf{x}) = D^2 f(\mathbf{x})$. We therefore use the partial notation

$$\frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right) = \frac{\partial^2 f}{\partial x_j \partial x_i} \quad \text{for } f : \mathbb{R}^n \rightarrow \mathbb{R}.$$

We now state without proof a very useful result for partials.

Theorem 4.15 (Symmetry of second partials). *If $\frac{\partial^2 f}{\partial x_j \partial x_i}$ and $\frac{\partial^2 f}{\partial x_i \partial x_j}$ are continuous in an open set containing $\mathbf{x}$, then*

$$\frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}) = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}).$$

This is very useful because it tells us these “cross partials” must be equal, i.e. taking one partial first and then the other is not a different operation when the order is exchanged. This gives us a useful property we highlight later.

4.5.1 Evaluating the Second Derivative

Now we illustrate how to evaluate the second derivative. Recall that for linear functions $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ we represent the linear function as an $m \times n$ matrix, since each coordinate of the output is a linear combination of the input coordinates, i.e. $g^i(\mathbf{x}) = \sum_{j=1}^n \alpha_j x_j$ for $i = 1, \dots, m$. For a concrete illustration consider a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$:

$$Df(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix},$$

but this really represents the linear function

$$\lambda(\Delta \mathbf{x}) = Df(\mathbf{x})\Delta \mathbf{x} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix} = \frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2.$$

The focus of this function is really about the point $\mathbf{x}$. We will therefore slightly abuse notation this one time and consider this function as $\lambda(\mathbf{x}) = \frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2$ for use in the definition of the derivative for this function. The input argument $\mathbf{x}$ refers to $\frac{\partial f}{\partial x_1}(\mathbf{x})$ and $\frac{\partial f}{\partial x_2}(\mathbf{x})$.

The derivative of this function is defined as a linear function $\gamma(\Delta \mathbf{x})$ satisfying

$$\lim_{\Delta \mathbf{x} \rightarrow \mathbf{0}} \frac{\|\lambda(\mathbf{x} + \Delta \mathbf{x}) - \lambda(\mathbf{x}) - \gamma(\Delta \mathbf{x})\|}{\|\Delta \mathbf{x}\|}.$$

Remark 4.16. Technically, this isn't correct in general since we should really consider an independent perturbation $\Delta \mathbf{y}$ and therefore a $\gamma(\Delta \mathbf{y})$. However, we will limit our focus to the case where $\Delta \mathbf{y} = \Delta \mathbf{x}$ so we continue accordingly, but the reader should be aware of this choice.

Then applying the derivative function to $\lambda(\mathbf{x})$ is equivalent to

$$\gamma(\Delta \mathbf{x}) = D\lambda(\mathbf{x}) \cdot \Delta \mathbf{x} = D(Df)(\mathbf{x}) \cdot \Delta \mathbf{x} = D^2f(\mathbf{x}) \Delta \mathbf{x}.$$

Computing directly,

$$\begin{aligned} \gamma(\Delta \mathbf{x}) &= D\lambda(\mathbf{x}) \cdot \Delta \mathbf{x} = \begin{bmatrix} \frac{\partial \lambda}{\partial x_1} & \frac{\partial \lambda}{\partial x_2} \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2 \right) & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2 \right) \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} \Delta x_1 + \frac{\partial^2 f}{\partial x_1 \partial x_2} \Delta x_2 & \frac{\partial^2 f}{\partial x_2 \partial x_1} \Delta x_1 + \frac{\partial^2 f}{\partial x_2^2} \Delta x_2 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix} \\ &= \frac{\partial^2 f}{\partial x_1^2} \Delta x_1^2 + \frac{\partial^2 f}{\partial x_1 \partial x_2} \Delta x_2 \Delta x_1 + \frac{\partial^2 f}{\partial x_2 \partial x_1} \Delta x_1 \Delta x_2 + \frac{\partial^2 f}{\partial x_2^2} \Delta x_2^2. \end{aligned}$$

We can write this function in matrix notation as

$$\gamma(\Delta \mathbf{x}) = \begin{bmatrix} \Delta x_1 & \Delta x_2 \end{bmatrix} \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix}.$$

Definition 4.17 (Hessian). Consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. The matrix of second-order partial derivatives is called the *Hessian* and denoted symbolically as

$$H(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

This tells us that $D^2f(\mathbf{x}) = \Delta \mathbf{x}^\top H(\mathbf{x})$. While this is the definition of the second-order derivative, since the term $\Delta \mathbf{x}^\top$ is simply the transpose of the corresponding input variable $\Delta \mathbf{x}$, when we consider the second-order derivative we typically only focus on the Hessian, $H(\mathbf{x}) \in \mathbb{R}^{n \times n}$.

Remark 4.18. By the symmetry of second partials (Theorem 4.15), whenever the cross partials are continuous the Hessian is a symmetric matrix. This is the useful property alluded to above, and it lets us apply the Spectral Theorem (Theorem 3.74) to the Hessian when we study optimization.

As the reader might already be able to infer, while further higher-order derivatives can exist and be calculated, the higher-order derivatives become quite complex. In most typical contexts we usually limit analysis to the Hessian as the highest-order derivative.

Now we connect these results to optimization.

4.6 Stationary Points in $\mathbb{R}^n$

We again can expect that the single-variable results should inform behavior in the multivariable setting, and they do. We first reintroduce some definitions updated to the $\mathbb{R}^n$ setting. For the following definitions, let U be an open set in $\mathbb{R}^n$ and $f : U \rightarrow \mathbb{R}$.

Definition 4.19 (Critical point). A point $\mathbf{x} \in \mathbb{R}^n$ is a *critical point* of f if $\vec{\nabla} f(\mathbf{x}) = \mathbf{0}$ or f is not differentiable at $\mathbf{x}$.

Critical points indicate a location where a possible maximum or minimum might exist. It does not guarantee that the point is a maximum or minimum, either local or global, but it identifies candidates for a maximum or minimum. The derivative being equal to $\mathbf{0}$ is the typical condition we employ, but f not being differentiable does two things. The more important one is that it identifies the boundary of the domain as a possible location for a maximum or minimum. The less relevant one is that it generalizes to functions which are not continuous or not differentiable over the domain. The second is only “less relevant” in the sense that we will typically work with continuously differentiable functions, but there are some functions which we might want to work with such as the L^1 norm $\|\cdot\|_1$, which is not differentiable at $\mathbf{0}$ because it is not smooth at $\mathbf{0}$.

Definition 4.20 (Singular point). A point $\mathbf{x} \in \mathbb{R}^n$ is a *singular point* of f if the derivative of f is not defined there.

Definition 4.21 (Stationary point). A point $\mathbf{x} \in \mathbb{R}^n$ is a *stationary point* of f if $\vec{\nabla} f(\mathbf{x}) = \mathbf{0}$.

Let $B_\varepsilon^U(\mathbf{x}) = U \cap B_\varepsilon(\mathbf{x})$, where we recall $B_\varepsilon(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^n : \|\mathbf{y} - \mathbf{x}\|_2 < \varepsilon\}$. The set $B_\varepsilon^U(\mathbf{x})$ gives us the right concept for the following definitions, especially at “borders” of U .

Definition 4.22 (Local maximum). A point $\mathbf{x} \in \mathbb{R}^n$ is a *local maximum* of f if there exists $\varepsilon > 0$ such that $f(\mathbf{y}) \leq f(\mathbf{x})$ for all $\mathbf{y} \in B_\varepsilon^U(\mathbf{x})$.

Definition 4.23 (Local minimum). A point $\mathbf{x} \in \mathbb{R}^n$ is a *local minimum* of f if there exists $\varepsilon > 0$ such that $f(\mathbf{x}) \leq f(\mathbf{y})$ for all $\mathbf{y} \in B_\varepsilon^U(\mathbf{x})$.

Definition 4.24 (Saddle point). A point $\mathbf{x} \in \mathbb{R}^n$ which is a stationary point but not a local minimum or a local maximum is called a *saddle point*.

Saddle points are an important concept and are the generalization of a stationary point that is a point of inflection in the single-variable setting. Note that points of inflection in the single-variable setting can also occur at non-stationary points (where the concavity changes but the derivative is nonzero; we discuss convexity and concavity later and it is not critical to understand this idea right now), which is why we specifically refer to a *stationary* point of inflection. If thinking about a saddle point by starting via the single-variable setting, the part to pay attention to is direction. Consider the function $f(x) = x^3$, which has a stationary point of inflection, i.e. a saddle point, at $x = 0$. If we move in the direction of Δx for $\Delta x > 0$ from $x = 0$, $f(x)$ increases, while if we move in the direction of $\Delta x < 0$ from $x = 0$ then $f(x)$ decreases. This is the part that generalizes: at a saddle point there is a direction of movement $\Delta \mathbf{x}_1$ where $f(\mathbf{x} + \Delta \mathbf{x}_1) < f(\mathbf{x})$, and some other direction $\Delta \mathbf{x}_2$ where $f(\mathbf{x} + \Delta \mathbf{x}_2) > f(\mathbf{x})$. In the local-min setting $f(\mathbf{x} + \Delta \mathbf{x}) \geq f(\mathbf{x})$ for all $\Delta \mathbf{x} \in \mathbb{R}^n$ (sufficiently small), and analogously for the local-max setting.

Now we can talk about how to find a stationary point.

Theorem 4.25. *Let $U \subset \mathbb{R}^n$ be an open subset of $\mathbb{R}^n$. If a point $\mathbf{x} \in U$ is a stationary point of a function $f : U \rightarrow \mathbb{R}$, then*

$$\frac{df}{dx_i} = 0 \quad \text{for any } i = 1, \dots, n \text{ such that } \frac{df}{dx_i} \text{ exists at } \mathbf{x}.$$

Proof. By definition a stationary point satisfies $\vec{\nabla} f(\mathbf{x}) = \mathbf{0}$. Recall $\vec{\nabla} f(\mathbf{x}) = \left[\frac{\partial f}{\partial x_1}(\mathbf{x}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}) \right]^\top$ (Definition 4.14), so each component $\frac{\partial f}{\partial x_i}(\mathbf{x})$ is an entry of the zero vector and therefore equals 0 wherever it exists. $\square$

Corollary 4.26. *If $f : U \rightarrow \mathbb{R}$ is differentiable and f has a stationary point at $\mathbf{x} \in U$, then $Df(\mathbf{x}) = 0$, i.e. $\vec{\nabla} f(\mathbf{x}) = \mathbf{0}$.*

Proof. When f is differentiable, $Df(\mathbf{x}) = \vec{\nabla} f(\mathbf{x})^\top$ (Section 4.4), and $\mathbf{x}$ being a stationary point means $\vec{\nabla} f(\mathbf{x}) = \mathbf{0}$, so $Df(\mathbf{x}) = 0$. $\square$

4.7 The Multivariate Taylor Series

Now we develop the multivariate Taylor series and Taylor's theorem. We use a similar approach to the one we took for developing the directional derivative, by leveraging the single-variable result.

We again consider an open set $U \subseteq \mathbb{R}^n$ and a function $f : U \rightarrow \mathbb{R}$. We assume f is infinitely continuously differentiable on U . Recall that we expand a Taylor series at a specific point and in a specific direction. Let us expand around $\mathbf{x} \in U$ in direction $\Delta\mathbf{x} \in \mathbb{R}^n$ and define the point of interest as $\mathbf{y} = \mathbf{x} + \Delta\mathbf{x}$. We assume the segment from $\mathbf{x}$ to $\mathbf{y}$ lies in U (for instance when U is convex); since an open set need not contain the segment between two of its points, this is required for the construction below to make sense.

Define $\gamma : \mathbb{R} \rightarrow \mathbb{R}^n$ as $\gamma(t) = \mathbf{x} + t(\mathbf{y} - \mathbf{x})$. Here γ is basically the line connecting $\mathbf{x}$ and $\mathbf{y}$, or the line passing through $\mathbf{x}$ in the direction $\Delta\mathbf{x}$.

Let $g(t) = f(\gamma(t)) = f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$; then $g : \mathbb{R} \rightarrow \mathbb{R}$, so g is a single-variable function which is differentiable at $t = 0$ since $g(0) = f(\gamma(0)) = f(\mathbf{x})$ and f is differentiable at $\mathbf{x}$. Now let us expand g at 0 in a direction Δt :

$$g(0 + \Delta t) = g(0) + \frac{g'(0)}{1!} \Delta t + \frac{g''(0)}{2!} \Delta t^2 + \dots$$

We can use the chain rule to calculate the derivatives of g . For the first derivative,

$$g'(t) = \frac{d}{dt}(f(\gamma(t))) = f'(\gamma(t)) \gamma'(t) \Rightarrow g'(0) = f'(\gamma(0)) \gamma'(0) = f'(\mathbf{x})(\mathbf{y} - \mathbf{x}) = Df(\mathbf{x})\Delta\mathbf{x} = \vec{\nabla} f(\mathbf{x}) \cdot \Delta\mathbf{x}.$$

For the second derivative, since γ is linear, $\gamma''(t) = \mathbf{0}$, so

$$g''(t) = \frac{d}{dt}[\vec{\nabla} f(\gamma(t))^\top \gamma'(t)] = \gamma'(t)^\top \vec{\nabla}^2 f(\gamma(t)) \gamma'(t) + \vec{\nabla} f(\gamma(t))^\top \gamma''(t) = \gamma'(t)^\top H(\gamma(t)) \gamma'(t),$$

since $\gamma''(t) = \mathbf{0}$. With $\gamma'(t) = \mathbf{y} - \mathbf{x} = \Delta\mathbf{x}$, this gives

$$g''(0) = (\mathbf{y} - \mathbf{x})^\top H(\mathbf{x})(\mathbf{y} - \mathbf{x}) = \Delta\mathbf{x}^\top H(\mathbf{x})\Delta\mathbf{x}.$$

This can be continued for higher-order derivatives, but we stop here. We now want to evaluate g at $\Delta t = 1$, since $g(1) = f(\gamma(1)) = f(\mathbf{y}) = f(\mathbf{x} + \Delta\mathbf{x})$. We should be careful here: an infinite Taylor *series* equals g only if g is analytic and the series converges, which is a stronger condition than g being infinitely differentiable, and in particular there is no guarantee it holds at $\Delta t = 1$. Instead we invoke Taylor's Theorem (Theorem 2.156) for the single-variable function g on the interval between 0 and Δt , which gives an *exact* second-order expansion with a remainder evaluated at an intermediate point: for some ξ between 0 and Δt ,

$$g(\Delta t) = g(0) + g'(0) \Delta t + \frac{1}{2} g''(\xi) (\Delta t)^2.$$

Because this identity is exact, we may set $\Delta t = 1$. Recalling $g'(0) = \vec{\nabla} f(\mathbf{x}) \cdot \Delta\mathbf{x}$ and $g''(\xi) = \Delta\mathbf{x}^\top H(\gamma(\xi))\Delta\mathbf{x} = \Delta\mathbf{x}^\top H(\mathbf{x} + \xi\Delta\mathbf{x})\Delta\mathbf{x}$, we obtain, for some $\xi \in (0, 1)$,

$$f(\mathbf{x} + \Delta\mathbf{x}) = f(\mathbf{x}) + \vec{\nabla} f(\mathbf{x}) \cdot \Delta\mathbf{x} + \frac{1}{2} \Delta\mathbf{x}^\top H(\mathbf{x} + \xi\Delta\mathbf{x})\Delta\mathbf{x}.$$

Equivalently, using continuity of the second partials as $\Delta\mathbf{x} \rightarrow \mathbf{0}$ (so that $H(\mathbf{x} + \xi\Delta\mathbf{x}) = H(\mathbf{x}) + o(1)$), we can write the expansion in little-o form. This gives the natural extension from the single-variable Taylor series to the multivariable case:

$$f(\mathbf{x} + \Delta\mathbf{x}) = f(\mathbf{x}) + \vec{\nabla} f(\mathbf{x}) \cdot \Delta\mathbf{x} + \frac{1}{2} \Delta\mathbf{x}^\top H(\mathbf{x})\Delta\mathbf{x} + o(\|\Delta\mathbf{x}\|^2).$$

4.8 Classifying Stationary Points

Now that we have established the multivariate Taylor series, we can discuss how we can identify whether a stationary point is a local minimum, local maximum, or saddle point.

Let $\mathbf{x}_0 \in U \subset \mathbb{R}^n$ be a stationary point of a twice differentiable function $f : U \rightarrow \mathbb{R}$. Consider an arbitrary direction of movement $\Delta \mathbf{x}$ from $\mathbf{x}_0$. The Taylor series expansion in this direction is

$$f(\mathbf{x}_0 + \Delta \mathbf{x}) = f(\mathbf{x}_0) + \vec{\nabla} f(\mathbf{x}_0) \cdot \Delta \mathbf{x} + \frac{1}{2} \Delta \mathbf{x}^\top H(\mathbf{x}_0) \Delta \mathbf{x} + o(\|\Delta \mathbf{x}\|^2),$$

where we only assumed second-order differentiability, so we can only expand up to the second derivative. Since $\mathbf{x}_0$ is a stationary point, $\vec{\nabla} f(\mathbf{x}_0) = \mathbf{0}$, so the Taylor expansion becomes

$$f(\mathbf{x}_0 + \Delta \mathbf{x}) = f(\mathbf{x}_0) + \frac{1}{2} \Delta \mathbf{x}^\top H(\mathbf{x}_0) \Delta \mathbf{x} + o(\|\Delta \mathbf{x}\|^2).$$

Therefore, if we can say something about $\Delta \mathbf{x}^\top H(\mathbf{x}_0) \Delta \mathbf{x}$, then we can say something about the nature of the stationary point. Since $H(\mathbf{x}_0)$ is a matrix, we can appeal to linear algebra results. Specifically, we want to be able to make a statement like $\Delta \mathbf{x}^\top H(\mathbf{x}_0) \Delta \mathbf{x} > 0$ for all $\Delta \mathbf{x} \neq \mathbf{0}$ in order to conclude the stationary point is a local minimum, or $\Delta \mathbf{x}^\top H(\mathbf{x}_0) \Delta \mathbf{x} < 0$ for a local maximum. Fortunately we have such results.

Definition 4.27 (Definiteness). Let $A \in L(\mathbb{R}^n)$ be symmetric. A is said to be *positive definite* if for all $\mathbf{x} \in \mathbb{R}^n$ with $\mathbf{x} \neq \mathbf{0}$,

$$\mathbf{x}^\top A \mathbf{x} > 0.$$

This is also written $A \succ 0$. Analogously, A is *negative definite* if $\mathbf{x}^\top A \mathbf{x} < 0$ for all $\mathbf{x} \neq \mathbf{0}$, written $A \prec 0$. A is *positive semi-definite* if $\mathbf{x}^\top A \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$, written $A \succeq 0$, and *negative semi-definite* if $\mathbf{x}^\top A \mathbf{x} \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n$, written $A \preceq 0$.

The term has two parts, “positive” and “definite.” We need definiteness to classify the stationary point as a max or a min. Semi-definiteness allows for $\mathbf{x}^\top A \mathbf{x} = 0$ for some $\mathbf{x} \in \mathbb{R}^n$ with $\mathbf{x} \neq \mathbf{0}$. In such a case we would need to do additional analysis to determine the nature of the stationary point.

Remark 4.28. Note that lack of positive definiteness does not mean negative definiteness. It means there exists at least one direction $\mathbf{x} \in \mathbb{R}^n$ such that $\mathbf{x}^\top A \mathbf{x} \leq 0$. Importantly, indefiniteness means there exists both positive and negative directions.

Since cross-partials are equal, i.e. for a twice differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ we have $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$ (Theorem 4.15), the Hessian is actually a real symmetric matrix. By the Spectral Theorem for real symmetric matrices (Theorem 3.74), A has a full orthonormal basis of eigenvectors and a full set of real eigenvalues. There is a very useful relationship between eigenvalues and positive definiteness.

Theorem 4.29. Let $A \in L(\mathbb{R}^n)$ be a real symmetric matrix. A is positive definite if and only if all eigenvalues of A are strictly positive.

Proof. By the Spectral Theorem (Theorem 3.74), $A = Q \Lambda Q^\top$ with Q orthogonal ($Q^\top Q = I$) and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ real. For any $\mathbf{x}$, set $\mathbf{z} = Q^\top \mathbf{x}$; since Q is invertible, $\mathbf{x} \neq \mathbf{0}$ if and only if $\mathbf{z} \neq \mathbf{0}$, and $\|\mathbf{z}\| = \|\mathbf{x}\|$. Then

$$\mathbf{x}^\top A \mathbf{x} = \mathbf{x}^\top Q \Lambda Q^\top \mathbf{x} = \mathbf{z}^\top \Lambda \mathbf{z} = \sum_{i=1}^n \lambda_i z_i^2.$$

($\Leftarrow$) If every $\lambda_i > 0$, then for $\mathbf{x} \neq \mathbf{0}$ we have $\mathbf{z} \neq \mathbf{0}$, so $\sum_i \lambda_i z_i^2 > 0$; hence A is positive definite.
 ($\Rightarrow$) Conversely, taking $\mathbf{x} = Q\mathbf{e}_i$ gives $\mathbf{z} = \mathbf{e}_i$ and $\mathbf{x}^\top A\mathbf{x} = \lambda_i$; positive definiteness forces $\lambda_i > 0$ for each i . $\square$

The relationship for positive semi-definiteness is that all eigenvalues are non-negative.

Exercise 4.30. What is the relationship between eigenvalues and negative definiteness and semi-definiteness?

These results give us a set of steps for assessing stationary points of a twice differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. First, find all stationary points by solving $\vec{\nabla} f(\mathbf{x}) = \mathbf{0}$. Next, determine the Hessian at each point found. If the Hessian is positive definite at the stationary point, then it is a local minimum, and analogously a local maximum if negative definite. If the Hessian is positive semi-definite, it may be a minimum point but it is not guaranteed. Similarly, if it is negative semi-definite, it may be a maximum point, but it is not guaranteed. If it is not definite at all, then we know it is a saddle point.

This is a very powerful idea in how we develop methods to solve optimization problems in practice. In the next chapter we introduce optimization problems and develop useful building blocks for thinking about how to solve them. Before that, we round out this chapter with some odds and ends related to vector calculus.

4.9 Applications of Vector Calculus

The discussion above develops rigorously the mathematical objects and ideas we want to use in analysis. While these are important for understanding what they are and how they work, the discussion does not provide much illustration for how to actually apply them, particularly for matrix and vector derivatives. As a chapter end, we explore some very common functions to which we apply the ideas we developed.

4.9.1 The Quadratic Form

Consider $f(\mathbf{x}) = \mathbf{x}^\top Q \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c$, of course $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Let us assume $n = 2$ for illustration:

$$f(\mathbf{x}) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1 & b_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + c.$$

Expanding,

$$f(\mathbf{x}) = q_{11}x_1^2 + q_{21}x_1x_2 + q_{12}x_1x_2 + q_{22}x_2^2 + b_1x_1 + b_2x_2 + c.$$

Taking partial derivatives and assembling the gradient,

$$\begin{aligned} \vec{\nabla} f(\mathbf{x}) &= \begin{bmatrix} 2q_{11}x_1 + (q_{12} + q_{21})x_2 + b_1 \\ (q_{12} + q_{21})x_1 + 2q_{22}x_2 + b_2 \end{bmatrix} = \begin{bmatrix} 2q_{11} & q_{12} + q_{21} \\ q_{12} + q_{21} & 2q_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \mathbf{b} \\ &= (Q + Q^\top) \mathbf{x} + \mathbf{b}. \end{aligned}$$

If Q is symmetric then $Q = Q^\top$, so

$$\vec{\nabla} f(\mathbf{x}) = 2Q\mathbf{x} + \mathbf{b},$$

which is the multivariate analog of $g(x) = ax^2 + bx + c$ with $g'(x) = 2ax + b$.

Remark 4.31. We find that $\frac{d}{d\mathbf{x}}(\mathbf{b}^\top \mathbf{x}) = \frac{d}{d\mathbf{x}}(\mathbf{x}^\top \mathbf{b}) = \mathbf{b}$.

This is a commonly used function as a sample function to introduce an optimization concept or topic when Q is a real symmetric matrix, because it has a full set of eigenvalues and can be used to illustrate how eigenvalues inform properties of optimization algorithms, and it is easy to specify a definiteness property of Q to contextualize if the function has a minimum, maximum, or saddle point. This is because the second derivative is given as

$$\vec{\nabla}^2 f(\mathbf{x}) = H(\mathbf{x}) = \begin{bmatrix} 2q_{11} & 2q_{12} \\ 2q_{21} & 2q_{22} \end{bmatrix} = 2Q$$

since $q_{12} = q_{21}$.

4.9.2 Least Squares

Consider $f(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|_2^2$, so $f : \mathbb{R}^n \rightarrow \mathbb{R}$, and again $n = 2$ with $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, and we choose $m = 2$. We present three methods for finding the derivative.

Method 1 (component-wise). We have

$$A\mathbf{x} - \mathbf{b} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 - b_1 \\ a_{21}x_1 + a_{22}x_2 - b_2 \end{bmatrix},$$

so that

$$\|A\mathbf{x} - \mathbf{b}\|_2 = \sqrt{(a_{11}x_1 + a_{12}x_2 - b_1)^2 + (a_{21}x_1 + a_{22}x_2 - b_2)^2},$$

and

$$\|A\mathbf{x} - \mathbf{b}\|_2^2 = (a_{11}x_1 + a_{12}x_2 - b_1)^2 + (a_{21}x_1 + a_{22}x_2 - b_2)^2.$$

Taking the gradient term by term,

$$\begin{aligned} \vec{\nabla} f(\mathbf{x}) &= \begin{bmatrix} 2a_{11}(a_{11}x_1 + a_{12}x_2 - b_1) + 2a_{21}(a_{21}x_1 + a_{22}x_2 - b_2) \\ 2a_{12}(a_{11}x_1 + a_{12}x_2 - b_1) + 2a_{22}(a_{21}x_1 + a_{22}x_2 - b_2) \end{bmatrix} \\ &= 2 \begin{bmatrix} a_{11}^2x_1 + a_{11}a_{12}x_2 - a_{11}b_1 + a_{21}^2x_1 + a_{21}a_{22}x_2 - a_{21}b_2 \\ a_{11}a_{12}x_1 + a_{12}^2x_2 - a_{12}b_1 + a_{21}a_{22}x_1 + a_{22}^2x_2 - a_{22}b_2 \end{bmatrix} \\ &= 2 \begin{bmatrix} (a_{11}^2 + a_{21}^2)x_1 + (a_{11}a_{12} + a_{21}a_{22})x_2 \\ (a_{11}a_{12} + a_{21}a_{22})x_1 + (a_{12}^2 + a_{22}^2)x_2 \end{bmatrix} - 2 \begin{bmatrix} a_{11}b_1 + a_{21}b_2 \\ a_{12}b_1 + a_{22}b_2 \end{bmatrix} \\ &= 2 \begin{bmatrix} a_{11}^2 + a_{21}^2 & a_{11}a_{12} + a_{21}a_{22} \\ a_{11}a_{12} + a_{21}a_{22} & a_{12}^2 + a_{22}^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - 2A^\top \mathbf{b} \\ &= 2 \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - 2A^\top \mathbf{b}. \end{aligned}$$

Therefore

$$\vec{\nabla} f(\mathbf{x}) = 2(A^\top A\mathbf{x} - A^\top \mathbf{b}).$$

Method 2 (expanding the squared norm). This works, but it is a quite long process. We can use another approach. The squared norm is equivalent to the dot product; for $\mathbf{y} \in \mathbb{R}^n$, $\|\mathbf{y}\|_2^2 = \mathbf{y}^\top \mathbf{y}$. Therefore we can rewrite the function as

$$\begin{aligned} f(\mathbf{x}) &= \|A\mathbf{x} - \mathbf{b}\|_2^2 = (A\mathbf{x} - \mathbf{b})^\top (A\mathbf{x} - \mathbf{b}) = (\mathbf{x}^\top A^\top - \mathbf{b}^\top)(A\mathbf{x} - \mathbf{b}) \\ &= \mathbf{x}^\top A^\top A\mathbf{x} - \mathbf{b}^\top A\mathbf{x} - \mathbf{x}^\top A^\top \mathbf{b} + \mathbf{b}^\top \mathbf{b} = \mathbf{x}^\top A^\top A\mathbf{x} - 2\mathbf{x}^\top A^\top \mathbf{b} + \mathbf{b}^\top \mathbf{b}, \end{aligned}$$

where we used that $\mathbf{x}^\top A^\top \mathbf{b} \in \mathbb{R}$, so $\mathbf{x}^\top A^\top \mathbf{b} = \mathbf{b}^\top A\mathbf{x}$. We know how to take the derivative of this function from our first example (the quadratic form, with $Q = A^\top A$ symmetric), which gives

$$\vec{\nabla} f(\mathbf{x}) = 2A^\top A\mathbf{x} - 2A^\top \mathbf{b} = 2(A^\top A\mathbf{x} - A^\top \mathbf{b}).$$

Method 3 (chain rule). We can also use the chain rule to find the derivative. Let $g(\mathbf{y}) = \|\mathbf{y}\|_2^2 = y_1^2 + y_2^2$, so $\vec{\nabla} g(\mathbf{y}) = \begin{bmatrix} 2y_1 \\ 2y_2 \end{bmatrix} = 2\mathbf{y}$. Let $h(\mathbf{x}) = A\mathbf{x} - \mathbf{b}$. Then $f(\mathbf{x}) = g(h(\mathbf{x}))$, and

$$\vec{\nabla} f(\mathbf{x}) = Dh(\mathbf{x})^\top \vec{\nabla} g(h(\mathbf{x})) = A^\top [2(A\mathbf{x} - \mathbf{b})] = 2(A^\top A\mathbf{x} - A^\top \mathbf{b}).$$

It can sometimes be difficult to assess the right orientation when using the chain rule, or in general to assess if a vector or matrix should be transposed. If it is not possible to get a reference to check, one way to sanity-check the orientation is to ensure the expected orientation is obtained after the derivative is evaluated, and all elements have the correct orientation for the multiplication to align based on dimensions.

Remark 4.32. It is important to understand how matrix and vector derivatives work, but rather than needing to derive them from scratch, a great free resource to keep available for reference is *The Matrix Cookbook* by Kaare Brandt Petersen and Michael Syskind Pedersen.

Chapter 5

Optimization

Optimization is the invisible backbone supporting almost all quantitative decision-making algorithms driving algorithmic choices applied to the real world. These include how Amazon and other delivery-focused companies determine how to most efficiently schedule delivery routes, how credit card companies automate fraud detection for incoming transactions, and even more abstract decision processes such as where a company should locate its warehouses or manufacturing plants.

These algorithms running on optimization have major impacts on the world we live in. Generally speaking, if these algorithms work well, they can significantly reduce the cost of operations, thus either improving affordability or making certain operations economically feasible at all. Consider, for instance, a fraud-detection algorithm. In any given day a credit card issuer may process millions or billions of credit card transactions, some portion of which may be fraudulent in some way, such as a stolen credit card. Without an algorithmic/computational process, the credit card company either chooses to review transactions with human review or to simply accept that fraud will occur and either pass the cost of fraud onto the customer or accept the cost themselves without charging the customer, or some combination. If the company chooses the human-review route, they can choose to review every transaction, which means hiring many, many people to reduce the length of time of review, or they can sample some number of transactions and only review those. The first case does not scale well to millions or billions of transactions without costs exploding, while the second case puts the company back in the situation of potentially missing some fraudulent activity before it gets processed. Through machine learning which runs on optimization, the credit card company has access to an algorithmic solution instead.

The above example illustrates the importance of connecting algorithmic/computational decision-making to the real world, and optimization is the bridge that connects the two.

In this chapter we develop fundamental results for thinking about and solving optimization problems that build on the foundation provided by the work we have done in analysis and linear algebra.

5.1 Basics of Optimization

The absolute basics of optimization consist of a domain in which we are seeking to make a decision; an objective function that maps a choice of point in the domain into a codomain that represents the outcome of the choice when evaluated by the objective function; the codomain as described; and an optimization/objective choice for the problem, i.e. minimize or maximize.

Typically our codomain is $\mathbb{R}$, but it may be different depending on the context. For instance, another codomain could be $\{0, 1\}$ where the output is a binary decision. Our domain is typically $\mathbb{R}^n$, but it may also be $\{0, 1\}$ or $\mathbb{Z}^n$, or $\mathbb{R}^n \times \mathbb{Z}^m$ for a mixture of real numbers and integers. The algorithmic details will usually vary as the domain changes due to changes in the analytic structure; e.g. real numbers are continuous whereas integers are not, which means something like a gradient, which is defined on $\mathbb{R}^n$, may not have meaning in $\mathbb{Z}^n$.

In these notes we will assume $f : \mathbb{R}^n \rightarrow \mathbb{R}$ as our setting and f is twice continuously differentiable on $\mathbb{R}^n$. Additionally our objective choice will be minimization. What we develop for minimization will typically hold for maximization as well by simply considering a maximization objective as the negative of the objective function for the minimization problem.

Putting these together, the basic optimization problem is written as

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \quad \left(= \max_{\mathbf{x} \in \mathbb{R}^n} [-f(\mathbf{x})] \right).$$

This notation is used to focus on the optimal value of the function. The optimal point is usually denoted $\mathbf{x}^*$. Then the notation we use is

$$y^* = f(\mathbf{x}^*) = \min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}), \quad \mathbf{x}^* = \arg \min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}),$$

to denote the optimal value of the function y^* (or $f(\mathbf{x}^*)$), and $\arg \min$ communicates returning or finding the argument of f , i.e. the point at which the minimum occurs.

In the optimization problem we are interested in finding a global minimum, not a local minimum. Since we are looking for a global minimum, if it exists, $y^* = f(\mathbf{x}^*)$ will be unique, but $\mathbf{x}^*$ may not be. As an example, the function

$$f(x) = \begin{cases} 0 & -1 \leq x \leq 1, \\ x - 1 & x \geq 1, \\ -x - 1 & x \leq -1, \end{cases} \quad f : \mathbb{R} \rightarrow \mathbb{R},$$

has global minimum $f(\mathbf{x}^*) = 0$, but the function attains this value at any $\mathbf{x}^* \in [-1, 1]$ (Figure 5.1).

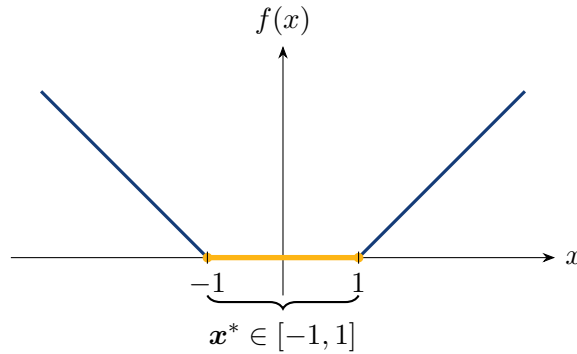


Figure 5.1: The function attains its global minimum value $f(\mathbf{x}^*) = 0$ along the entire flat segment $[-1, 1]$ (gold), so the optimal value $y^* = 0$ is unique but the minimizer $\mathbf{x}^*$ is not.

This means in general we really might have multiple optimal points, if a global minimum exists, and so more correctly we specify:

$$X^* = \arg \min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

where $X^* = \{\mathbf{x} \in X : f(\mathbf{x}) = y^*\}$.

In our discussion of derivatives and the gradient we showed that at local minima the gradient is $\mathbf{0}$, and we will properly introduce the core algorithm that leverages this fact, i.e. gradient descent, later in this chapter. Unfortunately, gradient descent does not provide a simple algorithmic way to find a global minimum, if it exists, and faces a non-trivial risk of instead finding a local, but not global, minimum in general.

In order to manage this challenge we typically restrict our choice of objective function to ones which are guaranteed to have the same function value at every local minimum, so that finding a local minimum is in fact finding a global minimum. Such functions are called *convex functions*.

5.2 Convex Sets

Before introducing the definition of a convex function, note that $\mathbb{R}$ does not include ∞ as an element, so let $\mathbb{R}^* = \mathbb{R} \cup \{-\infty, +\infty\}$ denote the extended reals (as introduced in our discussion of limits superior and inferior, Section 2.3.6). We start with a discussion of convex sets because, as we will see, convex functions must be defined on a convex domain.

Definition 5.1 (Convex set). A set S is *convex* if, given any two points $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$, we have $\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in S$.

This says that any “line” connecting two points in a space must be completely contained within the set (Figure 5.2).



Figure 5.2: A convex set (left) contains the entire segment between any two of its points; a non-convex set (right) does not.

Examples of convex sets are:

- (i) $\mathbb{R}^n$, and $\emptyset$. The empty set trivially fits the definition, because the line between any two points is in fact in the set, as the non-existent point is “in” the empty set.
- (ii) Euclidean balls, both open and closed, are convex, i.e. $B_\varepsilon(\mathbf{x}_0) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\|_2 < \varepsilon\}$ and the closed version $\overline{B}_\varepsilon(\mathbf{x}_0)$ are convex.
- (iii) A line in $\mathbb{R}^n$ defined as $\mathbf{a}^\top \mathbf{x} = b$ for $\mathbf{a}, \mathbf{x} \in \mathbb{R}^n$ and $b \in \mathbb{R}$ is by definition convex. If this isn’t obvious, take some time to think about it until it becomes obvious because this idea will be reused throughout this chapter.
- (iv) Another useful convex subset of $\mathbb{R}^n$ is called a *half-space*, defined as $\mathbf{a}^\top \mathbf{x} \leq b$ for $\mathbf{a}, \mathbf{x} \in \mathbb{R}^n$, $b \in \mathbb{R}$.

An example of a half-space in $\mathbb{R}^2$ is $x_2 - x_1 \leq 0 \Leftrightarrow x_2 \leq x_1$. On a two-dimensional x - y plot, if $x = x_1$ and $y = x_2$, this is the region on and below the line $x = y$ (Figure 5.3).

Another example is $x_1 \geq 0$, and we can combine this with the first half-space as

$$\begin{aligned} x_2 - x_1 \leq 0 \\ -x_1 \leq 0 \end{aligned} \quad \Leftrightarrow \quad \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Each half-space is convex, i.e. $H_1 = \{\mathbf{x} \in \mathbb{R}^2 : x_2 - x_1 \leq 0\}$ and $H_2 = \{\mathbf{x} \in \mathbb{R}^2 : x_1 \geq 0\}$ are each convex (Figure 5.4). The intersection

$$H_1 \cap H_2 = \left\{ \mathbf{x} \in \mathbb{R}^2 : \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$$

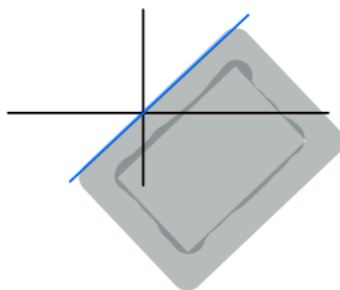


Figure 5.3: The half-space $x_2 - x_1 \leq 0$ in $\mathbb{R}^2$: the shaded region lies on and below the boundary line $x_1 = x_2$.

appears to be convex as well. This motivates our first result about convex sets.

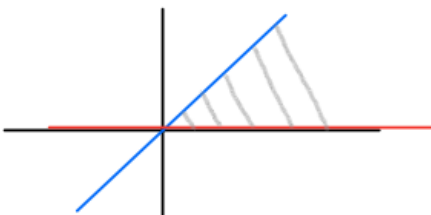


Figure 5.4: The intersection of the half-spaces $x_2 \leq x_1$ (below the blue line) and $x_1 \geq 0$ (above the red line): the shaded wedge is convex.

Theorem 5.2. *Let $\mathcal{H}$ be a collection of convex sets, i.e. every set in $\mathcal{H}$ is a convex set, of possibly uncountably infinite cardinality. The intersection of an arbitrary collection of convex sets,*

$$\bigcap_{H \in \mathcal{H}} H = \{ \mathbf{x} : \forall H \in \mathcal{H}, \mathbf{x} \in H \},$$

is convex.

Exercise 5.3. Prove Theorem 5.2.

Now that we know the arbitrary intersection of convex sets is convex, we can introduce an equivalent definition for convex sets. In order to motivate it, consider the example of a non-convex set that we gave earlier; let us call it S . We can make an augmented version of this set S' such that S' is convex and $S \subseteq S'$ by “filling in” the minimal amount of additional points that S is missing in order to satisfy convexity (Figure 5.5). S' is called the *convex hull* of S .

Definition 5.4 (Convex hull). Let S be an arbitrary set. The *convex hull* of S , $\text{co}(S)$, is the smallest convex set containing S . Equivalently, if $\mathcal{H}$ is the collection of all convex sets containing S , then $\text{co}(S) = \bigcap_{H \in \mathcal{H}} H$.

Theorem 5.5. *A set S is convex if and only if $S = \text{co}(S)$.*

Exercise 5.6. Prove Theorem 5.5.

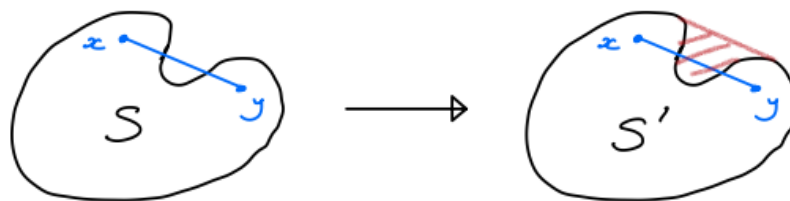


Figure 5.5: The non-convex set S (left) is augmented to its convex hull S' (right) by filling in the minimal set of points needed to make it convex.

We typically wouldn't want to ever use this equivalency as the base definition, even though it is equivalent, because it would circularly depend on defining what a convex set is. However, it is a useful result for thinking about convex sets.

The reader may have noticed that in our definition of a convex set for two points, i.e. $\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in S$, we are equivalently just saying that a linear combination $\alpha\mathbf{x} + \beta\mathbf{y}$ with $0 \leq \alpha, \beta \leq 1$ and $\alpha + \beta = 1$ must be in S . Since a linear combination generalizes over any finite number of vectors, does our concept of the linear combination in this convex setting also generalize over any finite number of vectors? The answer is yes.

Definition 5.7 (Convex combination). Let $\{\mathbf{x}_i\}_{i=1}^m$ be a collection of vectors, and let $\{\alpha_i\}_{i=1}^m$ be a collection of scalars, i.e. $\alpha_i \in \mathbb{R}$, with $0 \leq \alpha_i \leq 1$ and $\sum_{i=1}^m \alpha_i = 1$. The sum $\sum_{i=1}^m \alpha_i \mathbf{x}_i$ is called a *convex combination* of the vectors $\{\mathbf{x}_i\}_{i=1}^m$.

This greatly simplifies our language, as in the definition of a convex set we can instead say the convex combination of any two elements in the set must also be in the set. This language also immediately allows us to generalize our definition.

Theorem 5.8. Let S be an arbitrary set, and let K be the set of vectors consisting of all convex combinations from vectors in S . We have that $K = \text{co}(S)$.

Exercise 5.9. Prove Theorem 5.8. *Hint:* For two sets A and B , $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Since we have already shown a set is convex if and only if it equals its convex hull, we can also say a set is convex if and only if it contains all of its convex combinations, which generalizes the definition beyond just two points.

5.3 Convex Functions

Before introducing definitions, it may be useful to identify a canonical reference function to help the reader conceptualize the results and definitions. The best single-variable function for visualizing convexity is $f(x) = ax^2 + bx + c$ with $a > 0$, i.e. a quadratic function that forms a parabola when plotted. The shape of the parabola illustrates several of the core ideas we will highlight regarding convex functions.

First, the quadratic function clearly has a single minimum point which is a global minimum. Second, if we draw a line between any two points on the parabola—say these points are $(x_1, f(x_1))$ and $(x_2, f(x_2))$ —then the function value for any point $x_1 \leq x \leq x_2$ is on or below this line. Third is less obvious, but the tangent line to any point on the parabola is below the parabola.

The second characteristic is the definition we typically use, because it mirrors the definition of convex sets, which we also introduced.

Definition 5.10 (Convex function). Let $S \subseteq \mathbb{R}^n$ be a convex subset of $\mathbb{R}^n$ with $\mathbf{x}, \mathbf{y} \in S$, $\lambda \in [0, 1]$, and $f : S \rightarrow \mathbb{R}^*$. f is a *convex function* if

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

In other words, f is convex if the value of the function applied to the convex combination of two points in the domain is less than or equal to the convex combination of the function values of the points in $\mathbb{R}^*$ (Figure 5.6).

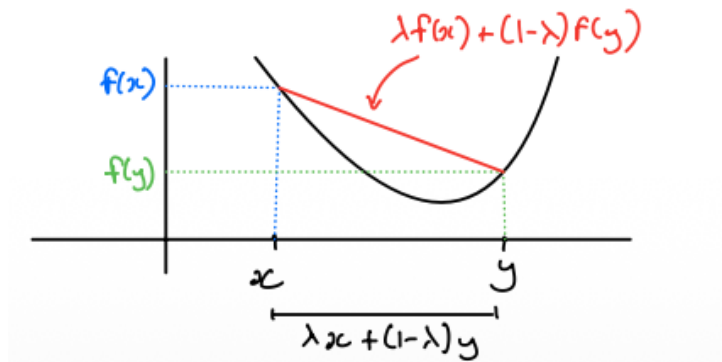


Figure 5.6: For a convex function, the chord between $(x, f(x))$ and $(y, f(y))$ lies on or above the graph: $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$.

It is worth noting two things about this definition. The first is that it mirrors the definition of a convex set by focusing on the convex combination of two points. Since S is a convex set we guarantee that $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in S$, and therefore the definition makes sense because the point in the definition is in the domain. If S were not convex, the definition would not be complete, as there would be some convex combination of two points not in the domain, and the function would not be defined in this case.

The second thing to note is that this definition formalizes the idea we had when discussing the parabola. This relationship—between the function value of the convex combination of two points in the domain being less than the convex combination of the function values at each point—is saying that the function value between any two points is less than or equal to the line connecting the points. This intuition captured by the simple parabola generalizes to higher dimensions.

5.3.1 The Epigraph

Now we show an equivalency between convex functions and convex sets. First we generate intuition using the parabola. In short, since the parabola is a convex function, the set defined by all points above the parabola, i.e.

$$P = \{ (x, y) \in \mathbb{R} \times \mathbb{R} : y \geq f(x), x \in \text{dom}(f) \},$$

is a convex set. P is shown as the highlighted area in Figure 5.7.

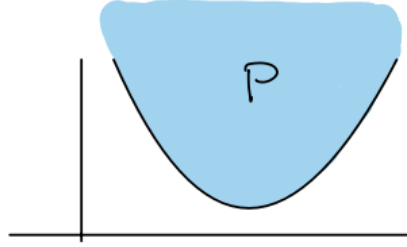


Figure 5.7: The set P of points on or above the parabola is convex.

Definition 5.11 (Epigraph). Let $f : S \rightarrow \mathbb{R}^*$. The *epigraph* of f is the set of all pairs $(\mathbf{x}, y) \in S \times \mathbb{R}$ such that $f(\mathbf{x}) \leq y$.

In the case of the parabola example it is easy to see from the diagram that P is convex. It turns out that if the epigraph of a function is convex, that also tells us the function is convex.

Theorem 5.12. A function $f : S \rightarrow \mathbb{R}^*$ is convex if and only if its epigraph is a convex subset of $S \times \mathbb{R}$.

Proof. ($\Rightarrow$) Suppose f is convex and let $(\mathbf{x}, s), (\mathbf{y}, t)$ be in the epigraph, so $f(\mathbf{x}) \leq s$ and $f(\mathbf{y}) \leq t$. For $\lambda \in [0, 1]$, convexity gives

$$f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}) \leq \lambda s + (1 - \lambda)t,$$

so $(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}, \lambda s + (1 - \lambda)t)$ is in the epigraph; hence the epigraph is convex.

($\Leftarrow$) Suppose the epigraph is convex. The points $(\mathbf{x}, f(\mathbf{x}))$ and $(\mathbf{y}, f(\mathbf{y}))$ are in the epigraph, so their convex combination $(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}, \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}))$ is too. By the definition of the epigraph this means $f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y})$, i.e. f is convex. $\square$

This relationship between a convex function and its epigraph will be helpful with some stronger notions of convexity we discuss later.

Using just the definition we can assert our claim that relates local minima and the global minimum for convex functions.

Theorem 5.13. Let f be a convex function, $f : S \rightarrow \mathbb{R}^*$. Every local minimum point of f is a global minimum point.

Exercise 5.14. Prove Theorem 5.13. *Hint:* Consider an argument by contradiction, i.e. there exists a local minimum that is not a global minimum.

Thus we have established that convex functions give us our guarantee that using a gradient-based approach—i.e. an approach which relies on the result that at a stationary point the gradient is $\mathbf{0}$ —will find a global minimum. Technically, we haven't actually quite shown this yet, because we haven't shown that every stationary point of a convex function is a minimum. It should be fairly obvious just from the definition that a stationary point will not be a (local) maximum in all cases, but it is not as immediately obvious that it will also never be a saddle point. Consider a function in one dimension as in Figure 5.8.

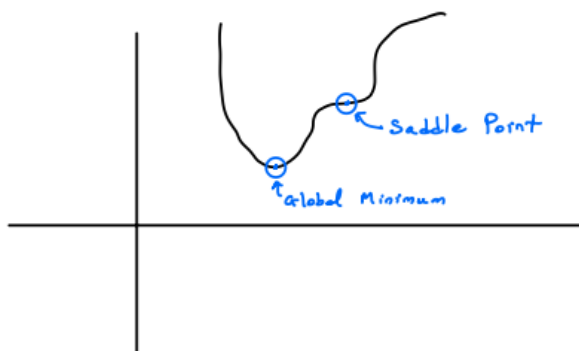


Figure 5.8: A function with a saddle point (a stationary inflection) and a separate global minimum. The chord joining the saddle point and the global minimum drops below the graph, so those two epigraph points have a convex combination outside the epigraph—the function is not convex.

In this case we can use the epigraph result to note that the line joining the saddle point and the global minimum is not part of the epigraph, despite the saddle point and the global minimum being part of the epigraph. This means this function is not convex, even though it might possibly appear to be at first glance.

This gives us intuition that a function with a saddle point is not convex, but a single case is not a proof and our intuition may or may not scale to higher dimensions. We want to be able to claim that if f is a convex function then every stationary point is a global minimum. Additionally, as this example shows, we also want an easy way to check if a function is convex. We tackle the challenge of finding ways to characterize convex functions first, and in doing so we will find a very easy way to prove the claim about stationary points.

5.4 Characterizing Convexity

There are three primary ways to characterize a convex function depending on the differentiability of the function: non-differentiable, once differentiable, or twice differentiable (or higher).

5.4.1 Zeroth-Order Convexity

Theorem 5.15 (Zeroth-Order Convexity). *A function $f : S \rightarrow \mathbb{R}^*$ is convex if and only if it is convex when restricted to any line that intersects its domain. Formally, consider any $\mathbf{x} \in S \subseteq \mathbb{R}^n$, let $\Delta\mathbf{x} \in \mathbb{R}^n$ be any direction, and define the set $S' = S \cap \{\mathbf{x} + t\Delta\mathbf{x} : t \in \mathbb{R}\}$. The function $g : S' \rightarrow \mathbb{R}^*$ given as $g(t) = f(\mathbf{x} + t\Delta\mathbf{x})$ is convex on S' if and only if f is convex on S .*

Before the proof, note that we previously already established that a line is convex, so zeroth-order convexity is effectively saying that if we consider every possible line in the convex set of the domain, and f is convex on every such line, then f is equivalently convex across the union over all such lines. This isn't a rigorous statement but it may be helpful for building intuition. If visualizing a convex set in two dimensions, as an illustration consider $f : S \rightarrow \mathbb{R}^*$, $S \subseteq \mathbb{R}^2$, with $f(\mathbf{x}) = x_1^2 + x_2^2$. Along any line in S , $f(\mathbf{x})$ will look like a parabola when restricted to that line (Figure 5.9).

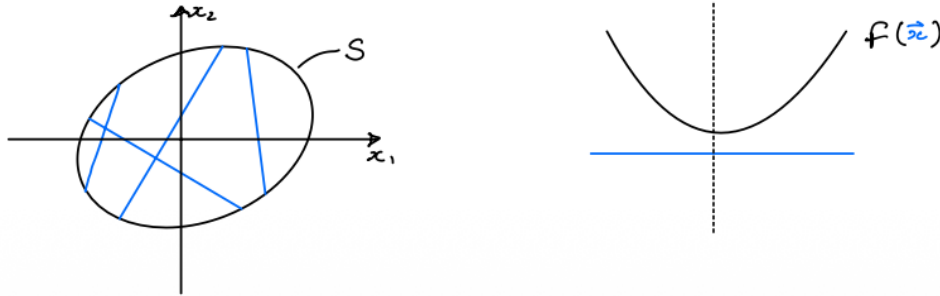


Figure 5.9: Restricting $f(\mathbf{x}) = x_1^2 + x_2^2$ to any line through its domain S (left) produces a parabola (right), which is convex.

Proof. ($\Rightarrow$) If f is convex on S , then for a fixed $\mathbf{x}$ and direction $\Delta\mathbf{x}$ the restriction $g(t) = f(\mathbf{x} + t\Delta\mathbf{x})$ is the composition of f with an affine map. For $t_1, t_2 \in S'$ and $\lambda \in [0, 1]$, the point $\lambda(\mathbf{x} + t_1\Delta\mathbf{x}) + (1 - \lambda)(\mathbf{x} + t_2\Delta\mathbf{x}) = \mathbf{x} + (\lambda t_1 + (1 - \lambda)t_2)\Delta\mathbf{x}$, so

$$g(\lambda t_1 + (1 - \lambda)t_2) = f(\lambda(\mathbf{x} + t_1\Delta\mathbf{x}) + (1 - \lambda)(\mathbf{x} + t_2\Delta\mathbf{x})) \leq \lambda g(t_1) + (1 - \lambda)g(t_2),$$

using convexity of f ; hence g is convex.

($\Leftarrow$) Suppose f is convex on every such line. Take any $\mathbf{x}, \mathbf{y} \in S$ and set $\Delta\mathbf{x} = \mathbf{y} - \mathbf{x}$, so that $\mathbf{x}$ corresponds to $t = 0$ and $\mathbf{y}$ to $t = 1$ on the line through them. By hypothesis $g(t) = f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$ is convex, so for $\lambda \in [0, 1]$,

$$f(\lambda\mathbf{x} + (1 - \lambda)\mathbf{y}) = g(1 - \lambda) = g(\lambda \cdot 0 + (1 - \lambda) \cdot 1) \leq \lambda g(0) + (1 - \lambda)g(1) = \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}),$$

so f is convex on S . □

5.4.2 First-Order Convexity

Theorem 5.16 (First-Order Convexity). *Let $f : S \rightarrow \mathbb{R}^*$ be a differentiable function on $S \subseteq \mathbb{R}^n$ as an open convex set. f is convex if and only if for any two points $\mathbf{x}, \mathbf{y} \in S$,*

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \vec{\nabla} f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}).$$

Before the proof, the general form of the right-hand side of the inequality should remind the reader of the Taylor series. We can define $\Delta \mathbf{x} = \mathbf{y} - \mathbf{x}$ so that $\Delta \mathbf{x}$ is the direction to get to $\mathbf{y}$ from $\mathbf{x}$ in $\mathbb{R}^n$. Additionally, the gradient is evaluated at $\mathbf{x}$, so the first-order Taylor approximation is at $\mathbf{x}$, as the tangent line at $\mathbf{x}$. Again using the parabola for intuition we illustrate the theorem in Figure 5.10.

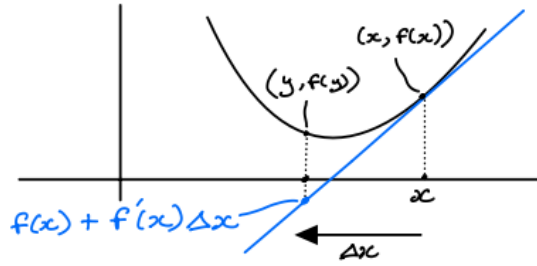


Figure 5.10: The first-order (tangent) approximation $f(x) + f'(x)\Delta x$ at $\mathbf{x}$ lies on or below the graph everywhere: the tangent surface is an underestimator of the function.

The First-Order Theorem is saying that the tangent surface (line in a single variable, plane in two variables, etc.) is always an underestimator for the function. Another way to say this is that the tangent surface at every point is a lower bound on the function value at every point. This has a lot of uses in optimization.

Proof. ($\Rightarrow$) Suppose f is convex. For $\lambda \in (0, 1]$, convexity between $\mathbf{x}$ and $\mathbf{y}$ gives $f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) \leq (1 - \lambda)f(\mathbf{x}) + \lambda f(\mathbf{y})$. Rearranging,

$$\frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda} \leq f(\mathbf{y}) - f(\mathbf{x}).$$

As $\lambda \rightarrow 0^+$, the left side tends to the directional derivative $D_{\mathbf{y}-\mathbf{x}}f(\mathbf{x}) = \vec{\nabla} f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$ (Theorem 4.13), giving $f(\mathbf{y}) \geq f(\mathbf{x}) + \vec{\nabla} f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$.

($\Leftarrow$) Suppose the inequality holds for all pairs. Fix $\mathbf{x}, \mathbf{y} \in S$, $\lambda \in [0, 1]$, and let $\mathbf{z} = \lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$. Applying the inequality at $\mathbf{z}$ toward $\mathbf{x}$ and toward $\mathbf{y}$,

$$f(\mathbf{x}) \geq f(\mathbf{z}) + \vec{\nabla} f(\mathbf{z})^\top (\mathbf{x} - \mathbf{z}), \quad f(\mathbf{y}) \geq f(\mathbf{z}) + \vec{\nabla} f(\mathbf{z})^\top (\mathbf{y} - \mathbf{z}).$$

Multiplying the first by λ , the second by $(1 - \lambda)$, and adding, the gradient terms combine to $\vec{\nabla} f(\mathbf{z})^\top (\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} - \mathbf{z}) = \mathbf{0}$, leaving $\lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}) \geq f(\mathbf{z})$, i.e. f is convex. $\square$

5.4.3 Second-Order Convexity

Theorem 5.17 (Second-Order Convexity). *Let $f : S \rightarrow \mathbb{R}^*$ be a twice continuously differentiable function with $S \subseteq \mathbb{R}^n$ as an open and convex set. f is convex if and only if its Hessian is positive semi-definite, i.e.*

$$H(\mathbf{x}) = \vec{\nabla}^2 f(\mathbf{x}) \succeq \mathbf{0} \quad \forall \mathbf{x} \in S.$$

Proof. By Zeroth-Order Convexity (Theorem 5.15), f is convex if and only if $g(t) = f(\mathbf{x} + t\mathbf{v})$ is convex for every $\mathbf{x} \in S$ and direction $\mathbf{v}$. In the single-variable setting, g is convex if and only if $g''(t) \geq 0$ everywhere. As computed for the multivariate Taylor series, $g''(t) = \mathbf{v}^\top H(\mathbf{x} + t\mathbf{v})\mathbf{v}$. Thus f is convex if and only if $\mathbf{v}^\top H(\mathbf{w})\mathbf{v} \geq 0$ for every point $\mathbf{w} \in S$ and every direction $\mathbf{v}$, which is exactly the statement that $H(\mathbf{w})$ is positive semi-definite for all $\mathbf{w} \in S$. $\square$

We can also connect the Second-Order Theorem to the Taylor series, and in particular we can see that the classification of a stationary point required the Hessian of the function at that point to be positive semi-definite, and convex functions have a positive semi-definite Hessian everywhere.

Unfortunately, this does not guarantee that every stationary point is a minimum point for convex functions. However, the First-Order Theorem does.

Proposition 5.18. *Let $f : S \rightarrow \mathbb{R}^*$ be a differentiable convex function. Every stationary point of f is a global minimum point.*

Proof. Let $\mathbf{x} \in S$ be a stationary point of f . Then $\vec{\nabla}f(\mathbf{x}) = \mathbf{0}$, so the first-order convexity property (Theorem 5.16) is

$$f(\mathbf{y}) \geq f(\mathbf{x}) \quad \forall \mathbf{y} \in S,$$

including $\mathbf{y} \in B_\varepsilon(\mathbf{x}) \cap S$ for any $\varepsilon > 0$. So $\mathbf{x}$ must be a local minimum, and we have already established every local minimum is a global minimum (Theorem 5.13). $\square$

The result that for a convex function every stationary point is a global minimum is very powerful as we seek to solve a decision problem via representation as an optimization problem. If we can find a way to represent the map from decision choice (i.e. domain) to decision output (i.e. codomain) via a convex function, we obtain a guarantee to find a globally optimal solution rather than just a locally optimal one. Additionally, due to properties of gradient descent—which we will cover shortly—a convex function also tends to have better computational properties. If the objective function has to be non-convex to model the decision correctly, a convex relaxation of the objective function can be preferred even if there is a gap between the convex relaxation solution and the true optimal solution.

Note that all of the results we developed for convex functions associated with minimization apply to *concave* functions associated with maximization. This can be developed by either the relationship that the max over a concave function is the min over the negative of the concave function, or using the definition that a concave function g satisfies

$$g(\lambda\mathbf{x} + (1 - \lambda)\mathbf{y}) \geq \lambda g(\mathbf{x}) + (1 - \lambda)g(\mathbf{y}),$$

which says that the function value on the line between any two points in the domain is greater than the line joining the function values of the two points in the codomain. Note that this still requires the domain be a convex set; there is no such thing as a concave set.

5.5 Stronger Notions of Convexity

One of the properties that the basic definition of convexity allows is the existence of multiple different points at which we obtain the global minimum value. For example, the following function is convex and has multiple possible points at which the global minimum of 0 is obtained (Figure 5.11).

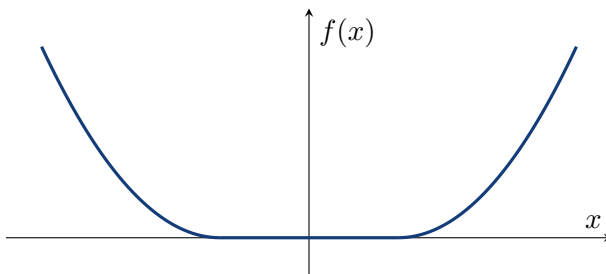


Figure 5.11: A convex function with a flat bottom: the global minimum value 0 is attained at every point of the segment, so the minimizer is not unique.

We may want to communicate explicitly that a certain convex function in fact explicitly has a unique point at which the global minimum is obtained, if a global minimum exists. We call such a convex function *strictly convex*.

Remark 5.19. Note that there are convex, and strictly convex, functions which do not obtain their global minimum. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ given as $f(x) = e^x$ is (strictly) convex and has an infimum of 0, but since this is obtained at $-\infty$ it doesn't have a global minimum (that is obtained on $\mathbb{R}$).

5.5.1 Strict Convexity

Definition 5.20 (Strictly convex function). Let $f : S \rightarrow \mathbb{R}^*$ where $S \subseteq \mathbb{R}^n$ is convex, let $\mathbf{x}, \mathbf{y} \in S$, $\mathbf{x} \neq \mathbf{y}$, and $\lambda \in (0, 1)$ (note the open interval). f is *strictly convex* if

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) < \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

This captures our new notion because the function evaluated at any point on a line from a minimum point to any other point in the domain must be strictly under the line between the minimum of the function and the function value of the other point. In other words, it is not possible to connect a minimum point to any other point in the domain such that the convex combination of the mappings of these two points in the codomain generates a line in the codomain that is “flat”; there has to be an increase in function value relative to the minimum at every other point.

There is a bit of nuance to this definition, however. A strictly convex function has a unique global minimum point, but a convex function with a unique global minimum is not necessarily strictly convex. Consider $f : \mathbb{R} \rightarrow \mathbb{R}^*$ with $f(x) = |x|$ (Figure 5.12).

This function has a unique global minimum point at $x = 0$. However, consider two points $x = 1$, $y = 2$ and $\lambda = \frac{1}{2}$:

$$f(\lambda x + (1 - \lambda)y) = \left| \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 2 \right| = \frac{3}{2} = \frac{1}{2} |1| + \frac{1}{2} |2| = \lambda f(x) + (1 - \lambda) f(y),$$

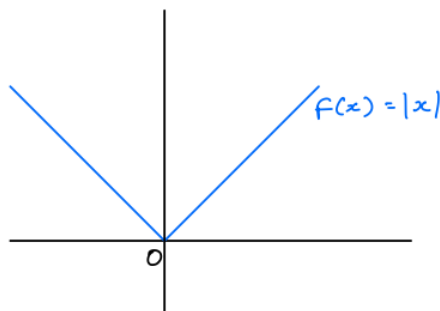


Figure 5.12: $f(x) = |x|$ has a unique global minimum at $x = 0$ but is not strictly convex.

so $f(\lambda x + (1 - \lambda)y) \not\leq \lambda f(x) + (1 - \lambda)f(y)$. Thus $f(x) = |x|$ is not strictly convex. This is important to note because some resources use an absolute value function like this as an example of a strictly convex function. The reader should be able to conclude that any convex function with a straight line segment in the codomain is not strictly convex.

Now we give an actual example of a strictly convex function. An obvious one is again the parabola, but for reasons that will become clear shortly let us use $f : \mathbb{R} \rightarrow \mathbb{R}^*$ as $f(x) = x^4$ (Figure 5.13).

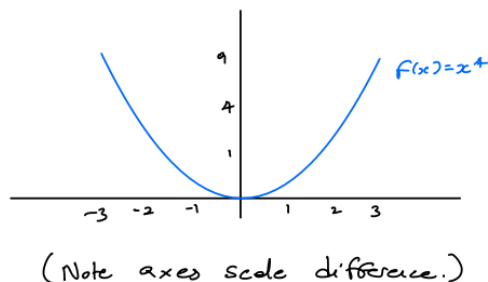


Figure 5.13: $f(x) = x^4$ is strictly convex (note the axis scale difference).

While we have asserted $f(x) = x^4$ is a strictly convex function, it would be helpful if we had a way to characterize functions as strictly convex more generally. We know that strictly convex functions must have a unique global minimum, and more generally only one stationary point. In the single-variable setting this means we can divide a strictly convex function into the part left of the minimum and the part right of the minimum. On the left side the function has to be increasing as we continue moving left, and on the right side it must be increasing as we continue moving right. Recall that the gradient is the direction in the domain in which the function is increasing fastest. This tells us on the left side the gradient must be negative and on the right side the gradient must be positive.

Additionally, since the line connecting the function values of any two points must be strictly above the function values between those two points, the gradient also has to be increasing in magnitude as we move away from the minimum. We can use these observations about gradients and points to derive a potential classification criterion (Figure 5.14).

For points on the same side, $(y_1 - x_1)$ should have the same sign as $(\vec{\nabla} f(y_1) - \vec{\nabla} f(x_1))$, because the relative positions of the two points will be mirrored in the relative directions of the

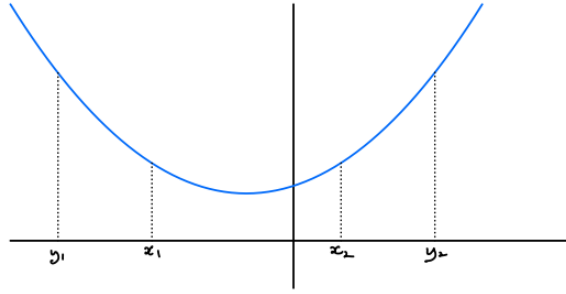


Figure 5.14: For points on the same side of the minimum, the differences in the points and in the gradients share a sign; across the minimum both flip. In all cases the product $(\vec{\nabla}f(x) - \vec{\nabla}f(y))(x - y)$ is positive.

gradients. For points across the minimum point the signs of the gradients will be opposite: e.g. $x_2 > x_1$ and $\vec{\nabla}f(x_1) < 0$, so $(x_2 - x_1) > 0$ and $\vec{\nabla}f(x_2) - \vec{\nabla}f(x_1) > 0$. Therefore, considering two arbitrary points $x, y \in \mathbb{R}$ with $x \neq y$, the test that the two terms generated by the differences of the gradients and the points themselves are pointing in the same direction is if their product is strictly positive:

$$(\vec{\nabla}f(x) - \vec{\nabla}f(y))(x - y) > 0.$$

The relative positions of x and y do not matter, as

$$(\vec{\nabla}f(x) - \vec{\nabla}f(y))(x - y) = (-1)(\vec{\nabla}f(y) - \vec{\nabla}f(x))(-1)(y - x) = (\vec{\nabla}f(y) - \vec{\nabla}f(x))(y - x).$$

This generalizes to $\mathbb{R}^n$ by using the dot product,

$$(\vec{\nabla}f(\mathbf{x}) - \vec{\nabla}f(\mathbf{y})) \cdot (\mathbf{x} - \mathbf{y}) > 0.$$

Recall that the dot product for two vectors decomposes as $\mathbf{v} \cdot \mathbf{u} = \cos \theta_{\mathbf{uv}} \|\mathbf{v}\|_2 \|\mathbf{u}\|_2$, so if $\mathbf{v}$ and $\mathbf{u}$ are pointing in the “same direction” in higher-dimensional space, the cosine of the angle between them is positive.

Theorem 5.21 (Strict Gradient Monotonicity). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^*$ be a differentiable convex function. f is strictly convex if and only if for any two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ with $\mathbf{x} \neq \mathbf{y}$,*

$$(\vec{\nabla}f(\mathbf{y}) - \vec{\nabla}f(\mathbf{x})) \cdot (\mathbf{y} - \mathbf{x}) > 0.$$

Before the proof, we make a couple of notes relating strict convexity to the zeroth-order and first-order convexity results. Both of these transition naturally to strict convexity. The first-order convexity condition simply becomes a strict inequality, and the zeroth-order convexity condition uses strict convexity on any line in the domain rather than regular convexity. The second-order convexity condition cannot differentiate regular convexity and strict convexity.

Exercise 5.22. Prove the strict convexity equivalents for the first-order and zeroth-order convexity properties.

Exercise 5.23. Explain why the second-order convexity property cannot be used to differentiate strict convexity. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^4$ may be useful for intuition.

Proof. We use the strict zeroth-order characterization (Exercise 5.22): f is strictly convex if and only if $g(t) = f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$ is strictly convex on every line, and in the single-variable setting g is strictly convex if and only if g' is strictly increasing. Since $g'(t) = \vec{\nabla} f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) \cdot (\mathbf{y} - \mathbf{x})$, we have $g'(1) > g'(0)$, i.e.

$$\vec{\nabla} f(\mathbf{y}) \cdot (\mathbf{y} - \mathbf{x}) > \vec{\nabla} f(\mathbf{x}) \cdot (\mathbf{y} - \mathbf{x}) \Leftrightarrow (\vec{\nabla} f(\mathbf{y}) - \vec{\nabla} f(\mathbf{x})) \cdot (\mathbf{y} - \mathbf{x}) > 0.$$

Conversely, if this inequality holds for all $\mathbf{x} \neq \mathbf{y}$, then for a fixed line g' is strictly increasing (apply it to nearby parameter values), so g is strictly convex, and by the strict zeroth-order characterization f is strictly convex. $\square$

The reasoning we used to develop the theorem in the strict convexity setting also works in the regular convexity setting but with a weak inequality. This can also be used to characterize general convexity.

Theorem 5.24 (Gradient Monotonicity). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function. f is convex if and only if for any two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,*

$$(\vec{\nabla} f(\mathbf{y}) - \vec{\nabla} f(\mathbf{x})) \cdot (\mathbf{y} - \mathbf{x}) \geq 0.$$

Exercise 5.25. Prove Theorem 5.24. It follows pretty much exactly the same as for Strict Gradient Monotonicity.

Now we can easily check that $f : \mathbb{R} \rightarrow \mathbb{R}^*$ with $f(x) = x^4$ is strictly convex. We have $f'(x) = 4x^3$, so for any two points $x, y \in \mathbb{R}$, $x \neq y$,

$$(4y^3 - 4x^3)(y - x) = 4(y - x)(x^2 + xy + y^2)(y - x) = 4(y - x)^2(x^2 + xy + y^2).$$

Note that for $x \neq y$, at least one of $x \neq 0$ or $y \neq 0$. Consider the factorization:

$$x^2 + xy + y^2 = \left(x + \frac{y}{2}\right)^2 + \frac{3}{4}y^2 \geq 0$$

which is zero only when $x = y = 0$ so we conclude:

$$4(y - x)^2(x^2 + xy + y^2) > 0$$

for $x \neq y$, and thus $f(x) = x^4$ is strictly convex.

Strictly Convex Sets

When we discussed regular (or “vanilla”) convexity we were able to relate convex functions and convex sets via the epigraph of the convex function. A natural extension of that idea is whether the epigraph of a strictly convex function is related to some concept of a “strictly convex set” that is similarly a stronger notion of convexity on sets as strictly convex is for functions. The answer is yes. We first state the theorem, then give the definition it relies on, and then prove it.

Theorem 5.26. *A function $f : S \rightarrow \mathbb{R}$ (no infinity) is strictly convex if and only if its epigraph forms a strictly convex set on an open domain $S \subseteq \mathbb{R}^n$.*

Definition 5.27 (Strictly convex set). A set S is *strictly convex* if the line segment connecting any two points $\mathbf{x}, \mathbf{y} \in S$, excluding the points themselves, is fully in the interior of S . Formally, a set S is strictly convex if for all $\mathbf{x}, \mathbf{y} \in S$ with $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0, 1)$,

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in \text{int}(S).$$

Proof of Theorem 5.26. Recall $\text{int}(\text{epi } f) = \{(\mathbf{w}, v) : \mathbf{w} \in \text{int}(S), v > f(\mathbf{w})\}$ when S is open.

($\Rightarrow$) Suppose f is strictly convex. Take two distinct points $(\mathbf{x}, s), (\mathbf{y}, t) \in \text{epi } f$ (so $s \geq f(\mathbf{x})$, $t \geq f(\mathbf{y})$) and $\lambda \in (0, 1)$, and set $\mathbf{w} = \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$, $v = \lambda s + (1 - \lambda)t$. Since S is open and convex, $\mathbf{w} \in \text{int}(S)$. If $\mathbf{x} \neq \mathbf{y}$, strict convexity gives $f(\mathbf{w}) < \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}) \leq \lambda s + (1 - \lambda)t = v$. If $\mathbf{x} = \mathbf{y}$ then $s \neq t$ (the points are distinct), and $v = \lambda s + (1 - \lambda)t > \min(s, t) \geq f(\mathbf{x}) = f(\mathbf{w})$. In both cases $v > f(\mathbf{w})$, so $(\mathbf{w}, v) \in \text{int}(\text{epi } f)$; the epigraph is a strictly convex set.

($\Leftarrow$) Suppose the epigraph is a strictly convex set. Take $\mathbf{x} \neq \mathbf{y}$ in S and $\lambda \in (0, 1)$. The boundary points $(\mathbf{x}, f(\mathbf{x}))$ and $(\mathbf{y}, f(\mathbf{y}))$ lie in $\text{epi } f$, so their open segment is in the interior:

$$(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}, \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y})) \in \text{int}(\text{epi } f),$$

which means $\lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}) > f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y})$. This is exactly strict convexity of f . $\square$

Note that this theorem requires that a strictly convex function's epigraph be evaluated on an open subset of $\mathbb{R}^n$ as domain. If we use a closed subset then the boundary becomes a flat surface and the epigraph is no longer strictly convex. Let us illustrate in a single variable using $f(x) = (x - 1)^4$, so we shift the minimum point to $x = 1$, and use the closed interval $[1, 2]$ as domain (Figure 5.15).

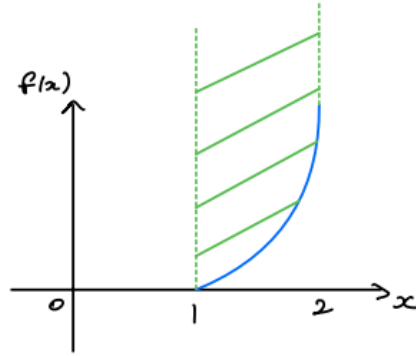


Figure 5.15: The epigraph of $f(x) = (x - 1)^4$ on the closed interval $[1, 2]$. The vertical boundary at $x = 1$ is flat, so the segment joining $(1, 0)$ and $(1, 1)$ lies on the boundary rather than the interior, and the epigraph is not a strictly convex set.

The epigraph of f on $[1, 2]$ is everything above the graph including the domain, i.e.

$$\{(x, y) : x \in [1, 2], y \geq f(x)\}.$$

This includes the line $x = 1$ that extends vertically on $[0, +\infty)$ in the codomain. To be more explicit, it includes the points $(1, 0)$ and $(1, 1)$. Clearly the line segment joining $(1, 0)$ and $(1, 1)$ is on the boundary of the epigraph, not the interior, so on this domain the epigraph of $f(x) = (x - 1)^4$ does not meet the definition for a strictly convex set which is why the statement in the theorem focusing on an open domain is important.

Exercise 5.28. Use the epigraph equivalence to argue that $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = |x|$ is not strictly convex.

Strict convexity provides a very useful strengthening of vanilla convexity by introducing a unique global minimum point. When we use an iterative method of finding the minimum point we generate a sequence of points in the domain and codomain corresponding to iteration 0, iteration 1, etc. In order to analyze the performance of such an iterative algorithm we would like to be able to model how the “error” of such an algorithm changes with each iteration.

For a general function that is not convex, we have no guarantee that every stationary point is a global minimum. When using a gradient-based algorithm, the best “error” measurement we can use is $\|\vec{\nabla} f\|$; if the algorithm converges to a stationary point, all we know is the gradient becomes $\mathbf{0}$, so $\|\vec{\nabla} f\| \rightarrow 0$. Importantly we can’t use the values $f_i = f(\mathbf{x}_i)$ generated by the sequence, or the points $\mathbf{x}_i$, to determine error because we have no guarantees about the uniqueness of these to use as a reference.

For a convex function we know every stationary point is a global minimum, so we know the sequence $\{f_i\}$, if it converges, is converging to a single value that is the global minimum $f^* = f(\mathbf{x}^*)$. Since we know there is an equivalence between f^* and $\|\vec{\nabla} f\| = 0$, we can use the error $|f_i - f^*|$ for convergence analysis rather than $\|\vec{\nabla} f_i\|$, and $|f_i - f^*|$ is a better error measure because $\|\vec{\nabla} f_i\|$ is only a proxy for the quantity we really care about, which is $\|\mathbf{x}_i - \mathbf{x}^*\|$. Since $\mathbf{x}^*$ is not unique for a general convex function we cannot use $\|\mathbf{x}_i - \mathbf{x}^*\|$ directly; it would require a set distance $\min_{\mathbf{x} \in X^*} \|\mathbf{x}_i - \mathbf{x}\|$, where X^* is the set of all points at which f^* is obtained, which is more unwieldy than $|f_i - f^*|$. For a strictly convex function we finally have a unique $\mathbf{x}^*$, so we can define our error as $\|\mathbf{x}_i - \mathbf{x}^*\|$, which is our ultimate target as we are trying to find our optimal decision, which is in the domain.

5.5.2 Strong Convexity

In moving from vanilla convexity to strict convexity we updated the zeroth-order and first-order convexity conditions but argued that the second-order condition remains as $H(\mathbf{x}) \succeq 0$, including $H(\mathbf{x}^*) \succeq 0$. Recall from the Taylor series discussion that at $\mathbf{x}^*$ we look to $\vec{\nabla}^2 f(\mathbf{x}^*) = H(\mathbf{x}^*)$ in order to assess if it is a minimum or not, but if $\vec{\nabla}^2 f(\mathbf{x}^*) = \mathbf{0}$ then we have to look at higher-order derivative terms, which can be difficult to compute. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^4$ is an easy example of this problem: it is strictly convex, but to assess if we have gotten to the minimum point we have to assess $f^{(4)}(x) = 24$, because all earlier $f^{(i)}(x) = 0$ for $i = 1, 2, 3$. In higher dimensions this can be very costly to determine.

With this in mind, what if we consider restricting to convex functions with $H(\mathbf{x}) \succ 0$, i.e. positive definite Hessian everywhere. This is our strongest notion of convexity, which is aptly named *strong convexity*. The idea formalizes as $H(\mathbf{x}) \succeq mI \succ 0$ for all $\mathbf{x}$ in the domain and describe a function as *m-strongly convex* where m provides a uniform lower bound.

Definition 5.29 (Strongly convex function). Let $f : S \rightarrow \mathbb{R}$, $S \subseteq \mathbb{R}^n$, be a function, $\mathbf{x}, \mathbf{y} \in S$, $\lambda \in [0, 1]$, and $m > 0$. f is *m-strongly convex* if

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}) - \frac{m}{2} \lambda (1 - \lambda) \|\mathbf{x} - \mathbf{y}\|_2^2.$$

The easiest example of a strongly convex function is a parabola, i.e. $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$. We see it is 2-strongly convex by examining the Hessian: $\vec{\nabla}^2 f(x) = 2$, so $H(x) = 2 \succeq mI$ holds

with largest $m = 2$. We formalize the relationship with the Hessian shortly. To build intuition and highlight properties let us focus on the parabola $f(x) = x^2$. The inequality is

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y) - \lambda(1-\lambda) \|x - y\|_2^2 = \lambda f(x) + (1-\lambda)f(y) - \lambda(1-\lambda)(x-y)^2.$$

This inequality has the vanilla convexity convex combination of two points and corresponding convex combination of the function values (the line is above the curve). It also introduces a new term that is non-negative and subtracts it out. It may be useful to represent the inequality another way:

$$f(\lambda x + (1-\lambda)y) + \lambda(1-\lambda)(x-y)^2 \leq \lambda f(x) + (1-\lambda)f(y).$$

This tells us that the strongly convex function replicates the vanilla convex function's structure of having the line segment from the convex combination of the function values at two points providing an upper bound on the function value for all points in the convex combination of the two points, but it then adds an additional distance or "buffer" between that upper-bounding line and the function values.

Now consider the size of that buffer. We first pick the points that act as the end points of the convex combination, $x, y \in \mathbb{R}$. Once those are fixed the value of $(x-y)^2$ becomes fixed; of course, the further apart x and y are, the larger this value. Next we scale $(x-y)^2$ by the factor $\lambda(1-\lambda)$. This factor scales in the choice of $\lambda \in [0, 1]$ as in Figure 5.16.

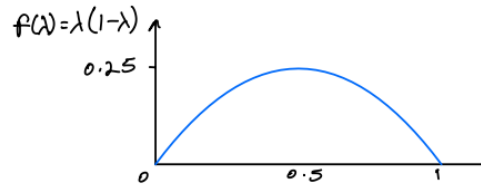


Figure 5.16: The factor $\lambda(1-\lambda)$ over $\lambda \in [0, 1]$: it is 0 at the endpoints and maximized at $\lambda = \frac{1}{2}$ with value $\frac{1}{4}$.

This tells us that the definition of a strongly convex function represents a vanilla convex function where the end points behave the same, but as we move away from either of the end points there is a guaranteed minimum distance between the upper-bounding line and the function value. Additionally, this guaranteed distance increases moving from an end point toward the midpoint and then decreases moving from the midpoint back to the other end point.

Theorem 5.30. *If f is a strongly convex function then f is also strictly convex.*

Exercise 5.31. Prove Theorem 5.30. This should be pretty clear from the definitions, subtracting a strictly positive buffer for $\lambda \in (0, 1)$.

We again want somewhat easy ways to characterize a strongly convex function. The zeroth-order condition is simple in that we just update the behavior of the function on lines in the domain to be strongly convex. The others are defined by the following equivalences.

Theorem 5.32. *Let $f : S \rightarrow \mathbb{R}$, $S \subseteq \mathbb{R}^n$ a convex domain, $\mathbf{x}, \mathbf{y} \in S$, and $m > 0$. The following are equivalent:*

- (i) f is m -strongly convex;

- (ii) (First order) $f(\mathbf{y}) \geq f(\mathbf{x}) + \vec{\nabla} f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{m}{2} \|\mathbf{y} - \mathbf{x}\|_2^2$ (if f is differentiable);
- (iii) (Strongly convex gradient monotonicity) $(\vec{\nabla} f(\mathbf{y}) - \vec{\nabla} f(\mathbf{x})) \cdot (\mathbf{y} - \mathbf{x}) \geq m \|\mathbf{y} - \mathbf{x}\|_2^2$ (if f is differentiable);
- (iv) (Second order) $\vec{\nabla}^2 f(\mathbf{x}) = H(\mathbf{x}) \succeq mI$ (if f is twice differentiable).

Proof. The key observation is that f is m -strongly convex if and only if the function $h(\mathbf{x}) = f(\mathbf{x}) - \frac{m}{2} \|\mathbf{x}\|_2^2$ is (vanilla) convex. This follows from the identity

$$\|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}\|_2^2 = \lambda \|\mathbf{x}\|_2^2 + (1 - \lambda) \|\mathbf{y}\|_2^2 - \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|_2^2,$$

so that subtracting $\frac{m}{2} \|\cdot\|_2^2$ from both sides of the m -strong convexity inequality turns the buffer term into exactly the vanilla convexity inequality for h . Now translate each characterization of convexity for h :

- First-order convexity of h (Theorem 5.16) reads $h(\mathbf{y}) \geq h(\mathbf{x}) + \vec{\nabla} h(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$. Substituting $h = f - \frac{m}{2} \|\cdot\|_2^2$ and $\vec{\nabla} h = \vec{\nabla} f - m\mathbf{x}$ and simplifying gives (ii).
- Gradient monotonicity of h (Theorem 5.24) reads $(\vec{\nabla} h(\mathbf{y}) - \vec{\nabla} h(\mathbf{x})) \cdot (\mathbf{y} - \mathbf{x}) \geq 0$; substituting $\vec{\nabla} h = \vec{\nabla} f - m\mathbf{x}$ gives $(\vec{\nabla} f(\mathbf{y}) - \vec{\nabla} f(\mathbf{x})) \cdot (\mathbf{y} - \mathbf{x}) - m \|\mathbf{y} - \mathbf{x}\|_2^2 \geq 0$, which is (iii).
- Second-order convexity of h (Theorem 5.17) reads $\vec{\nabla}^2 h(\mathbf{x}) = H(\mathbf{x}) - mI \succeq 0$, which is (iv).

Since all three are equivalent to convexity of h , and convexity of h is equivalent to (i), the four statements are equivalent. $\square$

Strongly Convex Sets

As with strictly convex functions, there is also a corresponding strongly convex set, but it is not connected via the epigraph. It is instead connected via the function's *sublevel sets*, and it is a one-directional relationship.

Definition 5.33 (Sublevel set). Let $f : S \rightarrow \mathbb{R}$, $S \subseteq \mathbb{R}^n$, be a function. The *sublevel sets* of f are the sets

$$S_\alpha = \{ \mathbf{x} \in S : f(\mathbf{x}) \leq \alpha \}.$$

These sets are subsets of the domain.

Definition 5.34 (Level set). Let $f : S \rightarrow \mathbb{R}$, $S \subseteq \mathbb{R}^n$, be a function. The *level sets* of f are the sets

$$L_\alpha = \{ \mathbf{x} \in S : f(\mathbf{x}) = \alpha \}.$$

Comparing to the definition of sublevel sets, level sets are the boundaries of the sublevel sets—the points where the function equals α .

Strongly convex sets are a bit more mathematically complex than strictly convex sets. The base definition unfortunately does not make use of an augmented idea of a convex combination like strictly convex sets do. Instead it uses an intersection of closed balls.

Definition 5.35 (Strongly convex set). A nonempty set $S \subseteq \mathbb{R}^n$ is m -strongly convex (for $m > 0$) if it is the intersection of a collection of closed balls of radius $\frac{1}{m}$; that is, there is a collection of closed balls $\bar{B}_{1/m}(\mathbf{c}_\alpha)$ such that

$$S = \bigcap_{\alpha} \bar{B}_{1/m}(\mathbf{c}_\alpha).$$

However, there is an equivalent definition which has the form of the convex combination, which we state as a theorem.

Theorem 5.36. Let $S \subseteq \mathbb{R}^n$ be nonempty, closed, and bounded. Then S is m -strongly convex if and only if for all $\mathbf{x}, \mathbf{y} \in S$, $\lambda \in [0, 1]$, and all $\mathbf{z} \in \mathbb{R}^n$ such that $\|\mathbf{z}\| \leq 1$,

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} + \frac{m}{2} \lambda (1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2 \mathbf{z} \in S.$$

Proof. ($\Rightarrow$) Suppose $S = \bigcap_{\alpha} \bar{B}_{1/m}(\mathbf{c}_\alpha)$, and write $r = \frac{1}{m}$. Fix $\mathbf{x}, \mathbf{y} \in S$, $\lambda \in [0, 1]$, and $\mathbf{z}$ with $\|\mathbf{z}\| \leq 1$; set $\mathbf{w} = \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$ and $s = \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2$. For each defining ball, using the identity for $\|\mathbf{w} - \mathbf{c}_\alpha\|^2$ and $\|\mathbf{x} - \mathbf{c}_\alpha\|, \|\mathbf{y} - \mathbf{c}_\alpha\| \leq r$,

$$\|\mathbf{w} - \mathbf{c}_\alpha\|^2 = \lambda \|\mathbf{x} - \mathbf{c}_\alpha\|^2 + (1 - \lambda) \|\mathbf{y} - \mathbf{c}_\alpha\|^2 - s \leq r^2 - s.$$

Since $\frac{m}{2}s = \frac{s}{2r}$ and $\sqrt{r^2 - s} \leq r - \frac{s}{2r}$ (square both sides), the triangle inequality gives

$$\|\mathbf{w} + \frac{m}{2}s\mathbf{z} - \mathbf{c}_\alpha\| \leq \sqrt{r^2 - s} + \frac{s}{2r} \leq r,$$

so the point lies in every $\bar{B}_{1/m}(\mathbf{c}_\alpha)$, hence in S .

($\Leftarrow$) Suppose S is nonempty, closed, and bounded and satisfies the convex-combination condition, and again write $r = \frac{1}{m}$. Taking $\mathbf{z} = \mathbf{0}$ shows $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in S$, so S is convex. We show S equals $\hat{S} := \bigcap \left\{ \bar{B}_r(\mathbf{c}) : S \subseteq \bar{B}_r(\mathbf{c}) \right\}$, the intersection of all closed balls of radius r that contain it; this family is nonempty (we exhibit a member below, using that S is bounded, hence $S \neq \mathbb{R}^n$). Clearly $S \subseteq \hat{S}$, so it remains to show $\hat{S} \subseteq S$.

Fix any $\mathbf{q} \notin S$. Since S is nonempty, closed, and convex, $\mathbf{q}$ has a unique nearest point $\mathbf{p} \in S$ (its projection onto S), and the projection satisfies the supporting inequality

$$\langle \mathbf{x} - \mathbf{p}, \mathbf{u} \rangle \leq 0 \quad \text{for all } \mathbf{x} \in S, \quad \mathbf{u} := \frac{\mathbf{q} - \mathbf{p}}{\|\mathbf{q} - \mathbf{p}\|}.$$

We claim the ball centered at $\mathbf{c} = \mathbf{p} - r\mathbf{u}$ contains S . Fix $\mathbf{x} \in S$ and apply the convex-combination condition to the pair $\mathbf{p}, \mathbf{x} \in S$ with $\mathbf{z} = \mathbf{u}$ and $\lambda \in [0, 1]$:

$$\mathbf{w}_\lambda := \lambda \mathbf{p} + (1 - \lambda) \mathbf{x} + \frac{m}{2} \lambda (1 - \lambda) \|\mathbf{p} - \mathbf{x}\|^2 \mathbf{u} \in S.$$

By the supporting inequality $\langle \mathbf{w}_\lambda - \mathbf{p}, \mathbf{u} \rangle \leq 0$, and since $\mathbf{w}_\lambda - \mathbf{p} = (1 - \lambda)(\mathbf{x} - \mathbf{p}) + \frac{m}{2} \lambda (1 - \lambda) \|\mathbf{p} - \mathbf{x}\|^2 \mathbf{u}$,

$$\langle \mathbf{w}_\lambda - \mathbf{p}, \mathbf{u} \rangle = (1 - \lambda) \left[\langle \mathbf{x} - \mathbf{p}, \mathbf{u} \rangle + \frac{m}{2} \lambda \|\mathbf{p} - \mathbf{x}\|^2 \right] \leq 0.$$

Dividing by $1 - \lambda > 0$ and letting $\lambda \rightarrow 1^-$,

$$\langle \mathbf{x} - \mathbf{p}, \mathbf{u} \rangle \leq -\frac{m}{2} \|\mathbf{x} - \mathbf{p}\|^2.$$

Therefore, using $rm = 1$,

$$\|\mathbf{x} - \mathbf{c}\|^2 = \|(\mathbf{x} - \mathbf{p}) + r\mathbf{u}\|^2 = \|\mathbf{x} - \mathbf{p}\|^2 + 2r \langle \mathbf{x} - \mathbf{p}, \mathbf{u} \rangle + r^2 \leq \|\mathbf{x} - \mathbf{p}\|^2 (1 - rm) + r^2 = r^2,$$

so $\mathbf{x} \in \bar{B}_r(\mathbf{c})$. As $\mathbf{x} \in S$ was arbitrary, $S \subseteq \bar{B}_r(\mathbf{c})$, so this ball belongs to our family. Finally, $\|\mathbf{q} - \mathbf{c}\| = \|(\mathbf{q} - \mathbf{p}) + r\mathbf{u}\| = \|\mathbf{q} - \mathbf{p}\| + r > r$, so $\mathbf{q} \notin \bar{B}_r(\mathbf{c})$ and hence $\mathbf{q} \notin \hat{S}$. Thus $\hat{S} \subseteq S$, giving $S = \hat{S}$, an intersection of closed balls of radius $r = \frac{1}{m}$. $\square$

This says that the ball of radius $\frac{m}{2}\lambda(1-\lambda)\|\mathbf{x} - \mathbf{y}\|^2$ around any convex combination $\lambda\mathbf{x} + (1-\lambda)\mathbf{y}$ of two points in S must also be in S . Strongly convex sets inherit the “curvature” of strongly convex functions.

Theorem 5.37. *If $f : S \rightarrow \mathbb{R}$, $S \subseteq \mathbb{R}^n$ convex, is an m -strongly convex function, then its sublevel sets $S_\alpha = \{\mathbf{x} \in S : f(\mathbf{x}) \leq \alpha\}$ are m_α -strongly convex.*

Proof. Omitted. It is a bit complex and in the author’s view does not add much insight. $\square$

To build some intuition consider $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ as $f(x) = x^2$ and $g(\mathbf{x}) = x_1^2 + x_2^2$. Both of these are strongly convex. Their sublevel sets are shown in Figure 5.17.

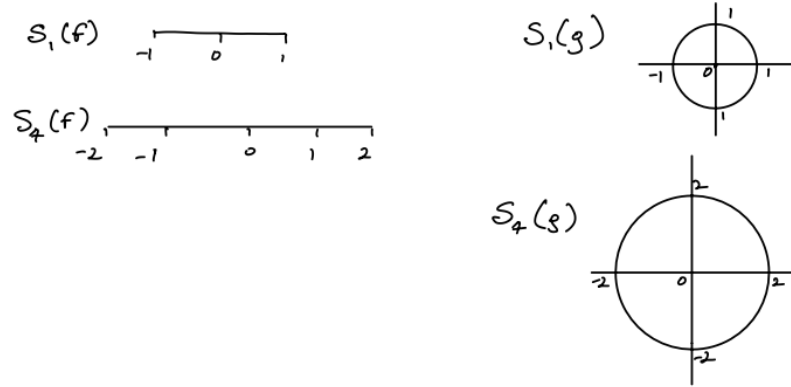


Figure 5.17: Sublevel sets S_1 and S_4 of $f(x) = x^2$ (intervals $[-1, 1]$ and $[-2, 2]$) and of $g(\mathbf{x}) = x_1^2 + x_2^2$ (disks of radius 1 and 2).

The “buffer” in the single-variable case is an interval $[a, b]$. In the two-dimensional case it is a more general closed ball that would appear as a circle. It is perhaps easier to notice in the two-dimensional case: at the boundary of the sublevel set, the curvature per unit distance on the boundary is different depending on α . When $\alpha = 1$ and considering a point d distance from the boundary, m_1 has to be smaller than m_2 for a point the same distance from the boundary, in order for the ball to be inside the set. As α gets bigger the local behavior of the boundary becomes more linear relative to a fixed move along the boundary so the strength of the needed curvature decreases. This allows m_α to decrease as α increases.

Theorem 5.38. *Strongly convex sets are bounded.*

Proof. By Definition 5.35, an m -strongly convex set S is an intersection of closed balls of radius $\frac{1}{m}$, so S is contained in any one of those balls. Hence S is bounded, with diameter at most $\frac{2}{m}$. $\square$

This result immediately makes clear why there is no relationship between a strongly convex function and its epigraph in terms of strongly convex sets: the epigraph of a convex function is unbounded.

Strongly convex functions and strongly convex sets show up most commonly in evaluating performance and convergence of algorithms, especially in constrained settings, i.e. optimizing over a specific set rather than all of $\mathbb{R}^n$. The reader is expected to have a general conceptual

understanding of strongly convex functions and sets, especially the characterization via first-/second-order convexity. A deeper understanding may be needed depending on research interests.

For strongly convex functions it can be somewhat easy to determine the value of $m > 0$ that defines its strong convexity by examining the Hessian: the second-order condition $H(\mathbf{x}) \succeq mI$ means m is the largest constant lower-bounding the eigenvalues of $H(\mathbf{x})$ over the domain. However, for strongly convex sets it is not as easy to determine the value of $m > 0$ which defines the strong convexity. It may however be possible to identify if a set is strongly convex by determining if the set coincides with the sublevel set of a strongly convex function. Consider the following example.

Proposition 5.39. *Let $X = \{ \mathbf{x} \in \mathbb{R}^n : \mathbf{x}^\top \mathbf{x} \leq 1 \}$. X is a 1-strongly convex set.*

Proof. X is closed and bounded, so we may verify the convex-combination characterization (Theorem 5.36) with $m = 1$. Let $\mathbf{x}, \mathbf{y} \in X$, $\lambda \in [0, 1]$, and $\mathbf{z}$ with $\|\mathbf{z}\| \leq 1$. Since $\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\mathbf{x}^\top \mathbf{y}$, we have $2\mathbf{x}^\top \mathbf{y} = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2$, so

$$\begin{aligned} \|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}\|^2 &= \lambda^2 \|\mathbf{x}\|^2 + (1 - \lambda)^2 \|\mathbf{y}\|^2 + 2\lambda(1 - \lambda) \mathbf{x}^\top \mathbf{y} \\ &= \lambda^2 \|\mathbf{x}\|^2 + (1 - \lambda)^2 \|\mathbf{y}\|^2 + \lambda(1 - \lambda) (\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2) \\ &= \lambda \|\mathbf{x}\|^2 + (1 - \lambda) \|\mathbf{y}\|^2 - \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2. \end{aligned}$$

Since $\mathbf{x}, \mathbf{y} \in X$ we have $\|\mathbf{x}\|^2, \|\mathbf{y}\|^2 \leq 1$, so

$$\|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}\|^2 \leq 1 - \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2.$$

Write $s = \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2$; since $\lambda(1 - \lambda) \leq \frac{1}{4}$ and $\|\mathbf{x} - \mathbf{y}\| \leq 2$, we have $s \in [0, 1]$. For $0 \leq s \leq 1$, $(1 - \frac{s}{2})^2 = 1 - s + \frac{s^2}{4} \geq 1 - s$, so

$$\|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}\|^2 \leq 1 - s \leq \left(1 - \frac{1}{2} \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2\right)^2 \Rightarrow \|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}\| \leq 1 - \frac{1}{2} \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2.$$

Therefore, by the triangle inequality and $\|\mathbf{z}\| \leq 1$,

$$\begin{aligned} \left\| \lambda \mathbf{x} + (1 - \lambda) \mathbf{y} + \frac{1}{2} \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2 \mathbf{z} \right\| &\leq \|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}\| + \frac{1}{2} \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2 \|\mathbf{z}\| \\ &\leq \left(1 - \frac{1}{2} \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2\right) + \frac{1}{2} \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2 = 1. \end{aligned}$$

Hence $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} + \frac{1}{2} \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2 \mathbf{z} \in X$, so X is 1-strongly convex. $\square$

This brings us to the end of the requisite results on convexity. Next we formally introduce gradient descent before connecting to optimization problems introduced by the linear setting called *linear programming*.

5.6 Gradient Descent

The notes thus far have essentially been in service of getting to gradient descent. Gradient descent is the foundational algorithm for optimization of high-dimensional and complex functions. One exception is that it is not used for linear programming, which instead makes use of the Simplex Algorithm, and we do not cover it in these notes but it may be covered in an introductory optimization course.

Gradient descent is the base form of algorithms generally classified as “gradient related.” This class of algorithms includes stochastic gradient descent and ADAM, which use the basic idea of gradient descent and build upon it. Now we introduce gradient descent.

Start by considering the multivariable Taylor series for a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^*$. We choose a point $\mathbf{x} \in \mathbb{R}^n$ and a direction to move in $\Delta \mathbf{x} \in \mathbb{R}^n$ to a new point $\mathbf{x} + \Delta \mathbf{x}$. The Taylor series expansion at the point $\mathbf{x}$ for the function value at the point $\mathbf{x} + \Delta \mathbf{x}$ is

$$f(\mathbf{x} + \Delta \mathbf{x}) = f(\mathbf{x}) + \vec{\nabla} f(\mathbf{x})^\top \Delta \mathbf{x} + \frac{1}{2} \Delta \mathbf{x}^\top \vec{\nabla}^2 f(\mathbf{x}) \Delta \mathbf{x} + o(\|\Delta \mathbf{x}\|^2).$$

In this formulation $\Delta \mathbf{x}$ includes both a direction and a magnitude. We can instead separate these into separate parameters as $\alpha \cdot \Delta \mathbf{x}$ where $\alpha \in \mathbb{R}$. The point we are moving to now becomes $\mathbf{x} + \alpha \Delta \mathbf{x}$, and the Taylor series is

$$f(\mathbf{x} + \alpha \Delta \mathbf{x}) = f(\mathbf{x}) + \alpha \vec{\nabla} f(\mathbf{x})^\top \Delta \mathbf{x} + \frac{\alpha^2}{2} \Delta \mathbf{x}^\top \vec{\nabla}^2 f(\mathbf{x}) \Delta \mathbf{x} + o(\alpha^2 \|\Delta \mathbf{x}\|^2).$$

Let us consider just the first-order term:

$$f(\mathbf{x} + \alpha \Delta \mathbf{x}) = f(\mathbf{x}) + \alpha \vec{\nabla} f(\mathbf{x})^\top \Delta \mathbf{x}.$$

The idea is simple: if we want to eventually get to a minimum point, let us just try to decrease the function value at the new point. An easy way to do this is to select $\Delta \mathbf{x} = -\vec{\nabla} f(\mathbf{x})$; then the first-order approximation gives

$$f(\mathbf{x} - \alpha \vec{\nabla} f(\mathbf{x})) = f(\mathbf{x}) + \alpha \vec{\nabla} f(\mathbf{x})^\top (-\vec{\nabla} f(\mathbf{x})) = f(\mathbf{x}) - \alpha \|\vec{\nabla} f(\mathbf{x})\|^2 \leq f(\mathbf{x}),$$

with equality if and only if $\|\vec{\nabla} f(\mathbf{x})\| = 0$, i.e. if and only if we are at a stationary point.

Therefore, starting at $\mathbf{x}$, we move to a new point $\mathbf{x}' = \mathbf{x} - \alpha \vec{\nabla} f(\mathbf{x})$ and we are in the same situation as when we were at the initial point $\mathbf{x}$. This sets up a sequence. We start at $\mathbf{x}_0$ then generate the sequence as

$$\boxed{\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \vec{\nabla} f(\mathbf{x}_k),}$$

where the subscript on the step size α indicates that we can vary the step size at each iteration.

This idea gives us the structure of the algorithm, but there are two details we need to resolve:

- (i) How do we choose the step size?
- (ii) Does this sequence actually converge?

It turns out these two are related. In particular, there are a few different commonly employed methods of finding α_k that give convergence. In these notes we will take a general approach that uses this problem to motivate broader mathematical ideas that can be applied here, but a Numerical Optimization course would likely focus on optimization-specific approaches which give guarantees under weaker conditions than we will use.

Let us define a map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ as

$$T(\mathbf{x}) = \mathbf{x} - \alpha \vec{\nabla} f(\mathbf{x}),$$

so that $\mathbf{x}_{k+1} = T(\mathbf{x}_k)$. Clearly if $\mathbf{x}^*$ is the (unique) minimum point of a strictly convex function then $T(\mathbf{x}^*) = \mathbf{x}^*$. We call $\mathbf{x}^*$ a *fixed point* of the map T . If T is a *contraction* then there are guarantees about a map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ converging to its fixed point.

5.6.1 Contractions and Fixed Points

Definition 5.40 (Transformation / operator). Let X be a metric space and let $T : X \rightarrow X$ be a map from X to X . We will call such a map T a *transformation* on X . If T is linear we refer to it as an *operator*.

Given a point $\mathbf{x} \in X$ we can apply a transformation T to obtain another point $T\mathbf{x} \in X$. In fact we can do this recursively, $T \circ T \circ \dots \circ T\mathbf{x} \in X$. This generates a sequence $\mathbf{x}_0 = \mathbf{x}$ and $\mathbf{x}_n = T^n \mathbf{x}$ for $n \geq 1$.

Definition 5.41 (Orbit). Let T be a transformation on a metric space X . Given an initial point $\mathbf{x} \in X$, the sequence generated by $\mathbf{x}_0 = \mathbf{x}$ and $\mathbf{x}_n = T^n \mathbf{x}$, $n \geq 1$, is called the *orbit* of $\mathbf{x}$ under T .

Definition 5.42 (Contraction). Let X be a metric space with distance d , and let $T : X \rightarrow X$ be a transformation on X . T is said to be a *contraction* if it is Lipschitz continuous with Lipschitz constant $\alpha < 1$. Formally, T is a contraction if there exists an $\alpha \in [0, 1)$ such that

$$d(T\mathbf{x}, T\mathbf{y}) \leq \alpha d(\mathbf{x}, \mathbf{y}) \quad \text{for all } \mathbf{x}, \mathbf{y} \in X.$$

It should be clear that the orbit of two points $\mathbf{x}, \mathbf{y} \in X$ under a contraction T on X will obey

$$d(T^n \mathbf{x}, T^n \mathbf{y}) \leq \alpha d(T^{n-1} \mathbf{x}, T^{n-1} \mathbf{y}) \leq \alpha (\alpha d(T^{n-2} \mathbf{x}, T^{n-2} \mathbf{y})) \leq \dots \leq \alpha^n d(\mathbf{x}, \mathbf{y}).$$

Thus if $\alpha \in [0, 1)$, as we need for a contraction, the orbits of any two points are getting closer and closer together. As a simple illustration of the idea, consider $X = \mathbb{R}$ and $T(x) = \pm \frac{1}{2}x$. For any $x, y \in \mathbb{R}$,

$$|Tx - Ty| = \left| \pm \frac{1}{2}x \mp \frac{1}{2}y \right| \leq \frac{1}{2} |x - y|.$$

Definition 5.43 (Fixed point). Let T be a transformation on a metric space X . A point $\mathbf{x}^* \in X$ is said to be a *fixed point* of T if $T\mathbf{x}^* = \mathbf{x}^*$.

Intuitively, if $\mathbf{x}^*$ is a fixed point of a contraction T on a metric space X and $\mathbf{x} \in X$ is any other point, then

$$d(T^n \mathbf{x}, T^n \mathbf{x}^*) \leq \alpha^n d(\mathbf{x}, \mathbf{x}^*), \quad \text{i.e.} \quad d(T^n \mathbf{x}, \mathbf{x}^*) \leq \alpha^n d(\mathbf{x}, \mathbf{x}^*),$$

and since $d(\mathbf{x}, \mathbf{x}^*)$ is a constant once $\mathbf{x}$ is selected, there is clearly convergence of the orbit of $\mathbf{x}$ under T toward such a fixed point, since $\alpha^n \rightarrow 0$ as $n \rightarrow \infty$. It turns out there is a slightly stronger result than this intuition, developed in the next theorem.

Theorem 5.44 (Banach's Fixed Point Theorem). *Let X be a complete metric space (so all Cauchy sequences converge), and let T be a contraction on X . T has a unique fixed point, and for all $\mathbf{x} \in X$ the orbit of $\mathbf{x}$ under T converges to this fixed point, $\mathbf{x}^*$.*

Proof. Fix any $\mathbf{x}_0 \in X$ and let $\mathbf{x}_n = T^n \mathbf{x}_0$. Consecutive terms satisfy $d(\mathbf{x}_{n+1}, \mathbf{x}_n) = d(T\mathbf{x}_n, T\mathbf{x}_{n-1}) \leq \alpha d(\mathbf{x}_n, \mathbf{x}_{n-1}) \leq \dots \leq \alpha^n d(\mathbf{x}_1, \mathbf{x}_0)$. For $m > n$, by the triangle inequality and the geometric series,

$$d(\mathbf{x}_m, \mathbf{x}_n) \leq \sum_{k=n}^{m-1} d(\mathbf{x}_{k+1}, \mathbf{x}_k) \leq d(\mathbf{x}_1, \mathbf{x}_0) \sum_{k=n}^{m-1} \alpha^k \leq d(\mathbf{x}_1, \mathbf{x}_0) \frac{\alpha^n}{1-\alpha} \xrightarrow{n \rightarrow \infty} 0,$$

so $\{\mathbf{x}_n\}$ is Cauchy. Since X is complete it converges to some $\mathbf{x}^*$. A contraction is continuous, so $T\mathbf{x}^* = T(\lim_n \mathbf{x}_n) = \lim_n T\mathbf{x}_n = \lim_n \mathbf{x}_{n+1} = \mathbf{x}^*$; hence $\mathbf{x}^*$ is a fixed point. If $\mathbf{y}^*$ were another fixed point, then $d(\mathbf{x}^*, \mathbf{y}^*) = d(T\mathbf{x}^*, T\mathbf{y}^*) \leq \alpha d(\mathbf{x}^*, \mathbf{y}^*)$ with $\alpha < 1$, forcing $d(\mathbf{x}^*, \mathbf{y}^*) = 0$, so the fixed point is unique. $\square$

This result is useful across a number of applied settings in mathematics and we will see one when we cover probability. For now, we can use this result to help us choose the step size in the gradient descent algorithm in order to guarantee convergence.

5.6.2 Convergence of Gradient Descent

For our discussion of convergence we will consider an m -strongly convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. It should be clear that to apply Banach's Fixed Point Theorem we need at least strict convexity, as the contraction has to be argued to have a unique fixed point. Strong convexity gives us this requirement but also additional properties we will highlight.

We need an additional requirement on f to make the gradient descent transformation a contraction: its gradient must be Lipschitz continuous. Recall that for a function $g : \mathbb{R}^n \rightarrow \mathbb{R}$, it is Lipschitz continuous (Definition 2.160) if there exists $L > 0$ such that $\|g(\mathbf{y}) - g(\mathbf{x})\| \leq L \|\mathbf{y} - \mathbf{x}\|$, so if f has a “Lipschitz gradient” it means

$$\|\vec{\nabla} f(\mathbf{y}) - \vec{\nabla} f(\mathbf{x})\| \leq L \|\mathbf{y} - \mathbf{x}\|.$$

This is also called L -smooth. Generally speaking, strong convexity gives us lower bounds on the behavior at a point, such as the first-order and second-order convexity conditions. These lower bounds give guarantees that gradient descent changes by at least a certain amount. We now show that a Lipschitz gradient gives us corresponding upper bounds, so that we can specify how big of a step we can allow in a gradient descent iteration.

The proofs in this subsection rely on integration, specifically the Fundamental Theorem of Calculus, which states that if f is a continuous function then $\int_a^b f'(t) dt = f(b) - f(a)$. We introduce integration formally when we cover probability; for most readers this should be familiar.

Theorem 5.45 (Descent Lemma). *If $f : S \rightarrow \mathbb{R}^*$, $S \subseteq \mathbb{R}^n$, has a Lipschitz continuous gradient with Lipschitz constant L , then for any points $\mathbf{x}, \mathbf{y} \in S$ where $\vec{\nabla} f(\mathbf{x})$ exists,*

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \vec{\nabla} f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|_2^2.$$

Proof. Let $\Delta \mathbf{x} = \mathbf{y} - \mathbf{x}$ and define $\phi : \mathbb{R} \rightarrow \mathbb{R}$ as

$$\phi(t) = f(\mathbf{x} + t\Delta \mathbf{x}) - (f(\mathbf{x}) + t\vec{\nabla} f(\mathbf{x})^\top \Delta \mathbf{x}).$$

Then ϕ is differentiable with $\phi'(t) = \vec{\nabla} f(\mathbf{x} + t\Delta \mathbf{x})^\top \Delta \mathbf{x} - \vec{\nabla} f(\mathbf{x})^\top \Delta \mathbf{x} = (\vec{\nabla} f(\mathbf{x} + t\Delta \mathbf{x}) - \vec{\nabla} f(\mathbf{x}))^\top \Delta \mathbf{x}$. By Cauchy–Schwarz and the Lipschitz gradient,

$$\phi'(t) \leq |(\vec{\nabla} f(\mathbf{x} + t\Delta \mathbf{x}) - \vec{\nabla} f(\mathbf{x}))^\top \Delta \mathbf{x}| \leq \|\vec{\nabla} f(\mathbf{x} + t\Delta \mathbf{x}) - \vec{\nabla} f(\mathbf{x})\|_2 \|\Delta \mathbf{x}\|_2 \leq L |t| \|\Delta \mathbf{x}\|_2^2.$$

By the Fundamental Theorem of Calculus, $\int_0^1 \phi'(t) dt = \phi(1) - \phi(0)$, where $\phi(1) = f(\mathbf{y}) - f(\mathbf{x}) - \vec{\nabla} f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$ and $\phi(0) = 0$. Thus

$$f(\mathbf{y}) - f(\mathbf{x}) - \vec{\nabla} f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) = \int_0^1 \phi'(t) dt \leq \int_0^1 Lt \|\Delta \mathbf{x}\|_2^2 dt = \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|_2^2,$$

which rearranges to the claim. $\square$

Theorem 5.46. *Let $f : S \rightarrow \mathbb{R}^*$, $S \subseteq \mathbb{R}^n$, and f be twice continuously differentiable on S . f has a Lipschitz gradient with Lipschitz parameter L if and only if for any $\mathbf{x} \in S$, $\|\vec{\nabla}^2 f(\mathbf{x})\|_2 \leq L$.*

Proof. ($\Leftarrow$) Suppose $\|\vec{\nabla}^2 f(\mathbf{x})\|_2 \leq L$ for all $\mathbf{x} \in S$. Let $\phi(t) = \vec{\nabla} f(\mathbf{x} + t\Delta \mathbf{x})$ with $\Delta \mathbf{x} = \mathbf{y} - \mathbf{x}$, so $\phi'(t) = \vec{\nabla}^2 f(\mathbf{x} + t\Delta \mathbf{x})\Delta \mathbf{x}$. By the Fundamental Theorem of Calculus,

$$\|\vec{\nabla} f(\mathbf{y}) - \vec{\nabla} f(\mathbf{x})\|_2 = \left\| \int_0^1 \vec{\nabla}^2 f(\mathbf{x} + t\Delta \mathbf{x})\Delta \mathbf{x} dt \right\|_2 \leq \int_0^1 \|\vec{\nabla}^2 f(\mathbf{x} + t\Delta \mathbf{x})\|_2 \|\Delta \mathbf{x}\|_2 dt \leq L \|\Delta \mathbf{x}\|_2,$$

so f has a Lipschitz gradient. ($\Rightarrow$) Suppose f has a Lipschitz gradient. Then $\|\vec{\nabla} f(\mathbf{x} + t\Delta \mathbf{x}) - \vec{\nabla} f(\mathbf{x})\|_2 \leq L \|t\Delta \mathbf{x}\|_2$; dividing by t and taking $t \rightarrow 0$, since norms are continuous we can move the limit inside to get $\|\vec{\nabla}^2 f(\mathbf{x})\Delta \mathbf{x}\|_2 \leq L \|\Delta \mathbf{x}\|_2$. Taking the supremum over $\|\Delta \mathbf{x}\|_2 \leq 1$ gives $\|\vec{\nabla}^2 f(\mathbf{x})\|_2 \leq L$. $\square$

Corollary 5.47. *Let $f : S \rightarrow \mathbb{R}^*$, $S \subseteq \mathbb{R}^n$, be twice continuously differentiable. f has a Lipschitz gradient if and only if all the eigenvalues of $\vec{\nabla}^2 f(\mathbf{x})$ are in $[-L, L]$ for all $\mathbf{x} \in S$.*

Proof. Let $\lambda_{\max}(A)$ for a square matrix A be defined as:

$$\lambda_{\max}(A) = \max_i |\lambda_i(A)|$$

so this is the maximum magnitude eigenvalue. The 2-norm of a matrix A is $\|A\|_2 = \sqrt{\lambda_{\max}(A^\top A)}$. Since $\vec{\nabla}^2 f(\mathbf{x})$ is a real symmetric matrix, its eigenvalue decomposition gives $\vec{\nabla}^2 f(\mathbf{x})^\top \vec{\nabla}^2 f(\mathbf{x}) = (U\Lambda_f U^\top)^\top U\Lambda_f U^\top = U\Lambda_f^2 U^\top$, so $\lambda_{\max}(U\Lambda_f^2 U^\top) = \lambda_{\max}^2$. Therefore $\|\vec{\nabla}^2 f(\mathbf{x})\|_2 = \sqrt{\lambda_{\max}^2} = |\lambda_{\max}| \leq L$, i.e. $-L \leq \lambda_{\max} \leq L$, and the same bound holds for every eigenvalue. $\square$

Now we can argue that for a function $f : S \rightarrow \mathbb{R}^*$, $S \subseteq \mathbb{R}^n$, that is both m -strongly convex and has an L -Lipschitz gradient, the gradient descent step-size transformation $T(\mathbf{x}) = \mathbf{x} - \alpha \vec{\nabla} f(\mathbf{x})$ is a contraction.

Proposition 5.48. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^*$, and let f be m -strongly convex and L -smooth. For such a function, if $0 < \alpha < \frac{2}{L}$ is a constant step size for gradient descent, gradient descent will converge to the minimum value of the function.*

Proof. Since f is both strongly convex and L -smooth, $mI \preceq \vec{\nabla}^2 f(\mathbf{x}) \preceq LI$ for all $\mathbf{x} \in \mathbb{R}^n$. Let $T(\mathbf{x}) = \mathbf{x} - \alpha \vec{\nabla} f(\mathbf{x})$. We want to find α such that $\|T(\mathbf{y}) - T(\mathbf{x})\|_2 \leq \rho \|\mathbf{y} - \mathbf{x}\|_2$ for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. First consider

$$T(\mathbf{y}) - T(\mathbf{x}) = (\mathbf{y} - \mathbf{x}) - \alpha(\vec{\nabla} f(\mathbf{y}) - \vec{\nabla} f(\mathbf{x})).$$

With $\Delta \mathbf{x} = \mathbf{y} - \mathbf{x}$, the Fundamental Theorem of Calculus gives $\vec{\nabla} f(\mathbf{y}) - \vec{\nabla} f(\mathbf{x}) = \int_0^1 \vec{\nabla}^2 f(\mathbf{x} + t\Delta \mathbf{x}) \Delta \mathbf{x} dt = H_{\Delta \mathbf{x}} \Delta \mathbf{x}$, where $H_{\Delta \mathbf{x}} = \int_0^1 \vec{\nabla}^2 f(\mathbf{x} + t\Delta \mathbf{x}) dt$. Substituting,

$$T(\mathbf{y}) - T(\mathbf{x}) = (I - \alpha H_{\Delta \mathbf{x}})(\mathbf{y} - \mathbf{x}), \quad \|T(\mathbf{y}) - T(\mathbf{x})\|_2 \leq \|I - \alpha H_{\Delta \mathbf{x}}\|_2 \|\mathbf{y} - \mathbf{x}\|_2.$$

Since $mI \preceq \vec{\nabla}^2 f(\mathbf{z}) \preceq LI$ for all $\mathbf{z}$, averaging preserves the eigenvalue bounds, so $mI \preceq H_{\Delta \mathbf{x}} \preceq LI$ and

$$\|I - \alpha H_{\Delta \mathbf{x}}\|_2 \leq \max\{|1 - \alpha m|, |1 - \alpha L|\}.$$

Choose $\rho = \max\{|1 - \alpha m|, |1 - \alpha L|\}$; then T is a contraction if $\rho < 1$, i.e. $\max\{|1 - \alpha m|, |1 - \alpha L|\} < 1$. The function $g(\alpha) = \max\{|1 - \alpha m|, |1 - \alpha L|\}$ is a mix of the two absolute value functions (Figure 5.18). We can see that if $0 < \alpha < \frac{2}{L}$ then $\rho < 1$ and we have a contraction. Thus when $\alpha \in (0, \frac{2}{L})$ we can guarantee convergence of gradient descent to the global minimum. $\square$

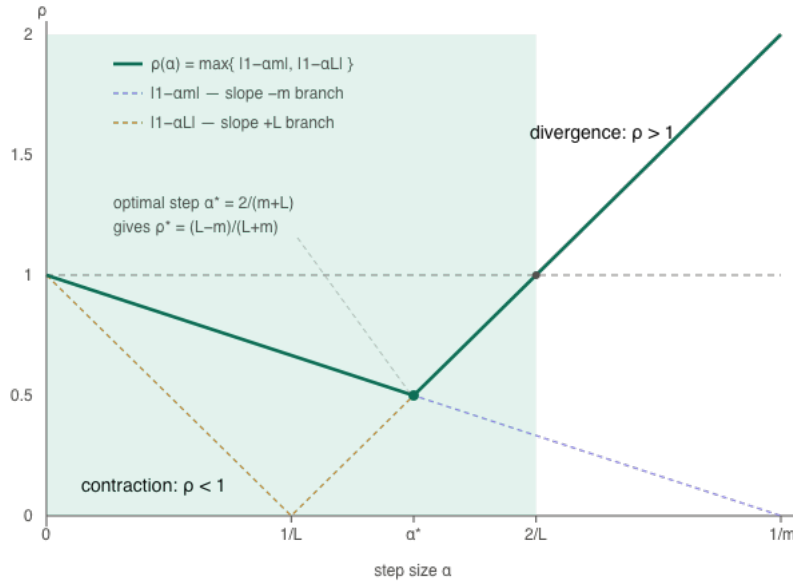


Figure 5.18: The contraction factor $\rho(\alpha) = \max\{|1 - \alpha m|, |1 - \alpha L|\}$ as a function of step size α . The envelope is below 1 (contraction) exactly for $0 < \alpha < \frac{2}{L}$, and is minimized at the optimal step $\alpha^* = \frac{2}{L+m}$, giving $\rho^* = \frac{L-m}{L+m}$; beyond $\frac{2}{L}$ the iteration loses the convergence guarantee.

Corollary 5.49. *The optimal step size is $\alpha^* = \frac{2}{L+m}$, which gives the smallest contraction factor $\rho^* = \frac{L-m}{L+m}$.*

Proof. The smaller ρ is, the stronger the contraction and the faster we expect to obtain convergence. By examining the graph of g (Figure 5.18), the two absolute-value branches cross where $1 - \alpha m = \alpha L - 1$, i.e. $\alpha^* = \arg \min_{\alpha} g(\alpha) = \frac{2}{L+m}$, giving $\rho^* = 1 - \alpha^* m = \frac{L-m}{L+m}$. $\square$

Definition 5.50 (Condition number). The *condition number* of a positive definite square matrix is the ratio of the largest eigenvalue to the smallest eigenvalue.

The condition number for this Hessian $\vec{\nabla}^2 f(\mathbf{x})$ is $\kappa = \frac{L}{m}$. We can rewrite

$$\rho^* = \frac{L - m}{L + m} = \frac{\kappa - 1}{\kappa + 1}.$$

The closer κ is to 1, the stronger the contraction property and thus the faster we expect convergence in terms of the number of iterations.

The results we developed above are for a function that is strongly convex and L -smooth. Typically we always require f be L -smooth for convergence, but similar results hold for vanilla convexity. One of the useful characteristics of strongly convex functions specifically is that as we move away from a point $\mathbf{x} \in S$ in any direction we have both a quadratic lower bound and a quadratic upper bound that “sandwiches” the function, as shown via the First-Order strong convexity condition and the Descent Lemma:

$$f(\mathbf{x}) + \vec{\nabla} f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{m}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 \leq f(\mathbf{y}) \leq f(\mathbf{x}) + \vec{\nabla} f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|_2^2.$$

Define $D_f(\mathbf{y} \mid \mathbf{x}) = f(\mathbf{y}) - (f(\mathbf{x}) + \vec{\nabla} f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}))$; then we can rewrite these inequalities as

$$\frac{m}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 \leq D_f(\mathbf{y} \mid \mathbf{x}) \leq \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|_2^2,$$

and illustrate in a single variable via Figure 5.19.

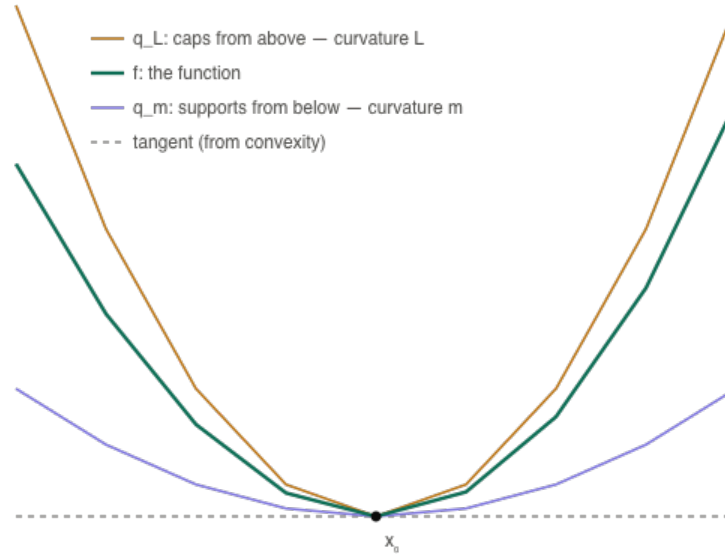


Figure 5.19: A strongly convex, L -smooth function f is sandwiched between the quadratic lower bound q_m with curvature m (supports from below) and the quadratic upper bound q_L with curvature L (caps from above), both agreeing with the tangent at $\mathbf{x}_0$.

This guarantees that the function behaves similarly to our gold standard of a parabola, i.e. the function is very well behaved for gradient descent. In fact when we have this kind of behavior on curvature guaranteed by the Hessian we can explore a higher-order Taylor series approximation for gradient descent.

Exercise 5.51. Consider the second-order Taylor series approximation

$$f(\mathbf{x} + \Delta \mathbf{x}) = f(\mathbf{x}) + \vec{\nabla} f(\mathbf{x})^\top \Delta \mathbf{x} + \frac{1}{2} \Delta \mathbf{x}^\top \vec{\nabla}^2 f(\mathbf{x}) \Delta \mathbf{x}.$$

What choice of $\Delta \mathbf{x}$ would lead to a gradient-descent-type algorithm? *Hint:* You may want to put a restriction similar to strong convexity on the Hessian. The algorithm you are likely to conceive is called *Newton's Method*.

This brings us to the end of discussing gradient descent in these notes. We next explore some fundamentals of an optimization problem as a mathematical object using linear programming as motivation.

5.7 Linear Programming

Recall that an optimization problem is specified as

$$\mathbf{x}^* = \arg \min_{\mathbf{x} \in X} f(\mathbf{x}).$$

We previously specified $X = \mathbb{R}^n$, but now we generalize the idea to an arbitrary set $X \subseteq \mathbb{R}^n$. While gradient descent almost always works for functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that are differentiable, including when $X \subseteq \mathbb{R}^n$ (although in this constrained case gradient descent serves as the base idea but uses augmented algorithms to address potentially leaving X via a gradient step), there is a more efficient algorithm available when the problem is linear, called the *Simplex Method*. We do not cover the Simplex Method in these notes, but we will use linear problems as motivation to develop core ideas employed in optimization. A linear problem is one where the objective function is linear and the set over which we are optimizing is defined by linear bounds.

Recall that when we previously discussed the definition of linear maps (Section 3.4) for functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ we required $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$ and $f(\alpha \mathbf{x}) = \alpha f(\mathbf{x})$ for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$. In our setting we want functions that output to the real line, $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Additionally, note that the definition of linearity we use above requires that $f(\mathbf{0}) = \mathbf{0}$. In a single-variable setting it means that every line defined by a linear function must pass through the origin. This is useful when mapping between vector spaces, but in our setting we want more flexibility to represent linear functions that have offsets from the origin.

Definition 5.52 (Affine function). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called *affine* if there is a linear map $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and a vector $\mathbf{c} \in \mathbb{R}^m$ such that $f(\mathbf{x}) = g(\mathbf{x}) + \mathbf{c}$.

There is an unfortunate notational inconsistency between functions/maps and polynomials, which are themselves really functions. A “linear” polynomial $f_p : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f_p(x) = ax + b$, $a, b \in \mathbb{R}$, but in the definitions we have adopted in these notes this is actually an *affine* function. It doesn’t affect the mathematics, but it is important to be aware of the terminology for effective communication. Colloquially “linear” and “affine” are used interchangeably in applied settings; we will use “linear” in the remainder of these notes to include affine unless we expressly designate the difference.

With the terminology covered, we can see very quickly that a function like $f : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$f(\mathbf{x}) = 5x_1 + 2x_2 - 1 = \mathbf{c}^\top \mathbf{x} - 1, \quad \mathbf{c} = \begin{bmatrix} 5 \\ 2 \end{bmatrix},$$

is linear, while a function $f(\mathbf{x}) = 5x_1 + 2x_2 + x_1x_2 - 1 = \mathbf{c}^\top \mathbf{x} + x_1x_2 - 1$ is not linear, because we cannot write x_1x_2 as a linear combination of x_1 and x_2 .

When $X \neq \mathbb{R}^n$ we often state the problem by explicitly writing out the set below the objective function as

$$\min_{\mathbf{x}} \mathbf{c}^\top \mathbf{x} \quad \text{subject to } \mathbf{x} \in X.$$

Of course this is not very informative about X without specifying the set.

5.7.1 Feasibility, Constraints, and Geometry

Let us say X is the unit square centered at the origin and we want to minimize the value of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ as $f(\mathbf{x}) = x_1 + x_2$ (Figure 5.20). We write this optimization problem as

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1 + x_2 \\ \text{s.t.} \quad & x_1 \leq 1, \quad x_1 \geq -1, \\ & x_2 \leq 1, \quad x_2 \geq -1. \end{aligned}$$

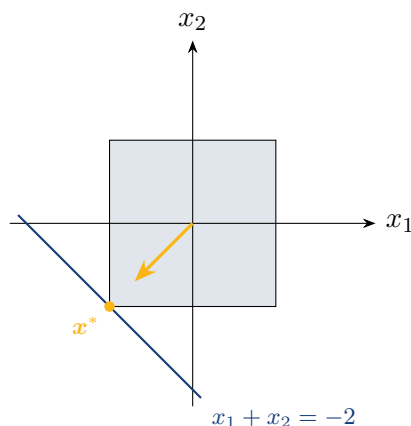


Figure 5.20: Minimizing $f(\mathbf{x}) = x_1 + x_2$ over the unit square. Moving in the negative-gradient direction $-\vec{\nabla}f = (-1, -1)$ pushes to the vertex $\mathbf{x}^* = (-1, -1)$, where the optimal value -2 is attained.

It should be clear that the optimal value of this problem is -2 , obtained at $\mathbf{x}^* = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. This is a very simple example where the solution is quite clear visually, but even in two dimensions it can sometimes be difficult to assess where the optimal point is by inspection. In such cases we can actually employ the gradient descent idea. Our function is $f(\mathbf{x}) = x_1 + x_2$, so the gradient is $\vec{\nabla}f(\mathbf{x}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

We start by finding any *feasible point*, which is any point $\mathbf{x} \in X$. Feasible points are an important concept because an optimization only has a possible solution if there exists a feasible point, i.e. X is not an empty set. This seems trivial but it is an important idea when trying to model real-world scenarios, because in trying to define the set over which a solution should be found it is not uncommon to find conflicting constraints such that X becomes empty. When this occurs we say the problem is *infeasible*.

Constraints are the limitations we place on the possible solution. In our example the constraint is that the solution must be in the unit square centered at the origin. If there are no constraints the problem is said to be *unconstrained*; if there are constraints it is a *constrained* problem.

Now we return to the example. Let us pick the origin as our initial feasible point. By gradient descent we want to move in the direction of the negative gradient at the point. Since the function is linear, the gradient is actually constant at every point, so we always move in direction $-\vec{\nabla}f(\mathbf{x}) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. By moving in the gradient descent direction we find the furthest point in the feasible set in that direction, and this is our solution. While this is not the method we use to solve these problems in general, it highlights useful ideas in linear programming.

Linear programming problems typically find a solution at a “corner” point, i.e. where two or more constraints intersect, which we call a *vertex*. It is also possible to obtain multiple solutions. Additionally, it is possible for the problem to be *unbounded*. These properties are actually connected to concepts we covered much earlier. Since we are simply connecting concepts we present the ideas as exercises to the reader.

Exercise 5.53. Consider the following linear program:

$$\begin{aligned} \max_{x_1, x_2} \quad & x_1 + x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4, \\ & x_1 \leq 3, \quad x_2 \leq 3, \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

- Is the feasible region convex? *Hint:* Consider the set defined by each constraint and observe the feasible region is an intersection.
- What is the optimal solution (x_1, x_2) to this problem?
- How does this solution relate to convexity? *Hint:* It may be worthwhile to do a tour of the vertices, i.e. calculating the function value at each vertex.
- Is it possible to determine the optimal value of the objective without doing any calculations at all? *Hint:* A gradient “ascent” approach can be useful here.

Exercise 5.54. Consider the following linear program:

$$\begin{aligned} \max_{x_1, x_2} \quad & x_1 + x_2 \\ \text{s.t.} \quad & -x_1 + x_2 \leq 1, \\ & x_1 - x_2 \leq 1, \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

- What is the optimal solution of this problem?
- Is the feasible region compact?
- Change the operation from maximization to minimization; what is the optimal solution to this new problem?
- Now consider the minimization problem we just introduced but set the constraints to be strict inequalities $(-x_1 + x_2 < 1, x_1 - x_2 < 1, x_1 > 0, x_2 > 0)$. Can you find the optimal solution to this new problem?

5.7.2 Standard Form

There are several typical conventions related to linear programming. First is the convention of referring to a linear programming problem as an *LP*. When we wrote our LP we implicitly assumed $x_1, x_2 \in \mathbb{R}$, and every point $(x_1, x_2) \in X$, X the unit square centered at the origin. We really should be explicit about this and refer to x_1 and x_2 as *free* variables, i.e. they can be positive or negative (or zero).

The Simplex Algorithm actually requires all variables to be non-negative, so when we write LPs we often want to translate them to *standard form*, which incorporates non-negativity and other conventions. A standard-form LP is given as

$$\min_x \mathbf{c}^\top \mathbf{x} \quad \text{s.t.} \quad A\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0},$$

where $\mathbf{c}, \mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$, and $A \in \mathbb{R}^{m \times n}$.

Let us illustrate the conventions using our original example problem, written with the constraints separated and x_1, x_2 free:

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1 + x_2 \\ \text{s.t.} \quad & x_1 \leq 1 \\ & x_1 \geq -1 \\ & x_2 \leq 1 \\ & x_2 \geq -1 \\ & x_1, x_2 \text{ free.} \end{aligned}$$

First we make all our variables non-negative. If a variable is already non-negative we are done; if not, define the variable as consisting of a positive part and a negative part:

$$x_1 = x_1^+ - x_1^-, \quad x_2 = x_2^+ - x_2^-, \quad x_1^+, x_1^-, x_2^+, x_2^- \geq 0.$$

We are not losing or introducing anything by this representation; the problems are equivalent. Our LP now becomes, omitting the subscript on the “min” operation,

$$\begin{aligned} \min \quad & (x_1^+ - x_1^-) + (x_2^+ - x_2^-) \\ \text{s.t.} \quad & x_1^+ - x_1^- \leq 1 \\ & x_1^+ - x_1^- \geq -1 \\ & x_2^+ - x_2^- \leq 1 \\ & x_2^+ - x_2^- \geq -1 \\ & x_1^+, x_1^-, x_2^+, x_2^- \geq 0. \end{aligned}$$

Next we convert the inequalities to equalities. For any inequality with a variable z and constant a as $z \leq a$, we introduce a *slack* variable $s = a - z \geq 0$ so that $z + s = a$; when $s = 0$ the constraint “binds.” Applying these, our LP becomes

$$\begin{aligned} \min \quad & (x_1^+ - x_1^-) + (x_2^+ - x_2^-) \\ \text{s.t.} \quad & x_1^+ - x_1^- + s_1 = 1, \\ & x_1^+ - x_1^- - s_2 = -1, \\ & x_2^+ - x_2^- + s_3 = 1, \\ & x_2^+ - x_2^- - s_4 = -1, \\ & x_1^+, x_1^-, x_2^+, x_2^-, s_1, s_2, s_3, s_4 \geq 0, \end{aligned}$$

where the slack variables are moved to the LHS of the equality constraints, hence the sign differences. In matrix and vector notation, with $\mathbf{x} = [x_1^+, x_1^-, x_2^+, x_2^-, s_1, s_2, s_3, s_4]^\top$,

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

so the LP is $\min_{\mathbf{x}} \mathbf{c}^\top \mathbf{x}$ s.t. $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$. This standard form is the typical representation we use when discussing LPs, and the reader is encouraged to become comfortable with it.

Remark 5.55. There are other useful strategies for getting to standard form, such as problems defined with absolute values, maximums, minimums, etc. in the objective or constraints, but we do not explore those here. Our goal is to introduce the core mathematical structure. Specific modeling strategies should be covered in a class that introduces linear programming or from any standard text on linear programming.

5.7.3 Duality

In order to motivate our next concept we return to representing an LP by individual constraints and variables. Consider an LP in standard form but represented individually:

$$\begin{aligned} y^* = \min \quad & 5x_1 + x_2 + 5x_3 \\ \text{s.t.} \quad & 2x_1 - x_2 + x_3 = 1, \\ & x_1 + x_2 + 2x_3 = 5, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Consider the point $\mathbf{x} = (0, 1, 2)$. This is clearly feasible since $2(0) - 1 + 2 = 1$ and $0 + 1 + 2(2) = 5$. The objective value is $5(0) + 1 + 5(2) = 11$. Since this is a feasible point and we are minimizing, the objective value at a feasible point provides an *upper bound* on the optimal objective value, i.e. $y^* \leq 11$. Is this optimal?

Let us consider the two constraints and multiply them by scalars λ_1, λ_2 (which *preserve the feasible set*, since they are equalities) and add:

$$\lambda_1(2x_1 - x_2 + x_3) + \lambda_2(x_1 + x_2 + 2x_3) = \lambda_1(1) + \lambda_2(5).$$

Rearranging by variable,

$$x_1(2\lambda_1 + \lambda_2) + x_2(-\lambda_1 + \lambda_2) + x_3(\lambda_1 + 2\lambda_2) = \lambda_1 + 5\lambda_2.$$

If we can pick appropriate values for the multipliers λ_1, λ_2 , then we can get a *lower bound* relationship for the objective function. In equation form,

$$\lambda_1 + 5\lambda_2 = x_1(2\lambda_1 + \lambda_2) + x_2(-\lambda_1 + \lambda_2) + x_3(\lambda_1 + 2\lambda_2) \stackrel{?}{\leq} 5x_1 + x_2 + 5x_3.$$

Since $x_1, x_2, x_3 \geq 0$, in order to get this inequality we must have each coefficient no larger than the corresponding objective coefficient:

$$\begin{aligned} 2\lambda_1 + \lambda_2 &\leq 5, \\ -\lambda_1 + \lambda_2 &\leq 1, \\ \lambda_1 + 2\lambda_2 &\leq 5. \end{aligned}$$

Taking $(\lambda_1, \lambda_2) = (0, 0)$ satisfies these and gives the lower bound 0, so $0 \leq y^* \leq 11$. We can improve. Taking $(\lambda_1, \lambda_2) = (1, 2)$ is feasible ($2 + 2 = 4 \leq 5$, $-1 + 2 = 1 \leq 1$, $1 + 4 = 5 \leq 5$) and gives the lower bound $\lambda_1 + 5\lambda_2 = 1 + 10 = 11$. Thus $11 \leq y^* \leq 11$, which means $\mathbf{x} = (0, 1, 2)$ must be an optimal solution.

The reader may have noticed that the inequalities we developed for the multipliers λ look like constraints, and in fact we were implicitly solving an LP:

$$\begin{aligned} \max_{\lambda_1, \lambda_2} \quad & \lambda_1 + 5\lambda_2 \\ \text{s.t.} \quad & 2\lambda_1 + \lambda_2 \leq 5, \\ & -\lambda_1 + \lambda_2 \leq 1, \\ & \lambda_1 + 2\lambda_2 \leq 5, \\ & \lambda_1, \lambda_2 \text{ free.} \end{aligned}$$

This is referred to as the *dual linear program*, often just “dual,” to the original LP. The original LP is referred to as the *primal linear program*, often just “primal.” In constrained optimization the notion of a primal–dual pair is called *duality*, alluding to the two-sided nature of a constrained optimization problem.

The idea of duality is present across all of optimization, not just in linear programs. The general idea is that for an objective function being minimized, and thus approaching the optimal value via a descent, there is a corresponding problem which approaches the value of the objective function via ascent in pursuit of maximizing the objective, and vice versa for a (primal) maximization problem.

Constructing the Dual

This is in fact a general result. To formalize it we generalize the construction of the dual, starting from a primal in standard form. Let

$$(P) : \quad \min_{\mathbf{x}} \quad \mathbf{c}^\top \mathbf{x} \quad \text{s.t.} \quad A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}.$$

Consider a relaxed problem which allows deviation from the restriction $A\mathbf{x} = \mathbf{b}$ but puts a penalty on that restriction:

$$(P') : \quad \min_{\mathbf{x} \geq \mathbf{0}} \quad \mathbf{c}^\top \mathbf{x} + \lambda^\top (\mathbf{b} - A\mathbf{x}).$$

Readers familiar with the concept of the *Lagrangian* will recognize this structure, where λ is called a *Lagrange multiplier*; here λ is our dual (decision) variable. Define a function of λ representing the result of (P') after minimization:

$$\begin{aligned} g(\lambda) &= \min_{\mathbf{x} \geq \mathbf{0}} [\mathbf{c}^\top \mathbf{x} + \lambda^\top (\mathbf{b} - A\mathbf{x})] \\ &= \min_{\mathbf{x} \geq \mathbf{0}} [\mathbf{c}^\top \mathbf{x} - \lambda^\top A\mathbf{x} + \lambda^\top \mathbf{b}] \\ &= \lambda^\top \mathbf{b} + \min_{\mathbf{x} \geq \mathbf{0}} [\mathbf{c}^\top \mathbf{x} - \lambda^\top A\mathbf{x}], \end{aligned}$$

since $\lambda^\top \mathbf{b}$ is unrelated to the minimization over $\mathbf{x}$. Let $\mathbf{x}^*$ be the solution to (P) ; since $\mathbf{x}^*$ satisfies $A\mathbf{x}^* = \mathbf{b}$ and is feasible for the relaxed problem,

$$\begin{aligned} g(\lambda) &= \min_{\mathbf{x} \geq \mathbf{0}} [\mathbf{c}^\top \mathbf{x} + \lambda^\top (\mathbf{b} - A\mathbf{x})] \\ &\leq \mathbf{c}^\top \mathbf{x}^* + \lambda^\top (\mathbf{b} - A\mathbf{x}^*) = \mathbf{c}^\top \mathbf{x}^*, \end{aligned}$$

so $g(\lambda)$ is a lower bound on the primal optimal value for every $\lambda \in \mathbb{R}^m$. In particular this holds for the λ that maximizes the lower bound. Now consider

$$g(\lambda) = \lambda^\top \mathbf{b} + \min_{x \geq 0} (\mathbf{c}^\top - \lambda^\top A)x.$$

If $\mathbf{d} = \mathbf{c} - A^\top \lambda$ has any negative entry d_i , we can set $x_j = 0$ for $j \neq i$ and $x_i \rightarrow +\infty$ so that $\min_{x \geq 0} \mathbf{d}^\top \mathbf{x} \rightarrow -\infty$, i.e. the term is unbounded below. When an optimization problem is unbounded we do not return a solution, so to keep the lower bound non-trivial we require $\mathbf{c}^\top - \lambda^\top A \geq \mathbf{0}^\top$, i.e. $\lambda^\top A \leq \mathbf{c}^\top$. Thus our dual problem becomes

$$(D): \quad \max_{\lambda} \lambda^\top \mathbf{b} \quad \text{s.t. } \lambda^\top A \leq \mathbf{c}^\top, \lambda \text{ free.}$$

For a primal problem in standard form, this is the standard corresponding dual problem.

Weak and Strong Duality

Our derivation of the dual illustrates a few important observations. The first is *weak duality*, the dual objective acting as a lower bound for the primal objective. The second concerns *feasibility*: if there exists an $\mathbf{x} \in \mathbb{R}^n$ with $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$, we say the problem is *primal feasible*; if there exists $\lambda \in \mathbb{R}^m$ with $\lambda^\top A \leq \mathbf{c}^\top$, we say it is *dual feasible*. From the derivation, dual feasibility relates to primal boundedness and primal feasibility to dual boundedness. Note, however, that this does not imply that dual infeasibility guarantees primal unboundedness; we can have both dual infeasibility and primal infeasibility.

Theorem 5.56 (Weak Duality). *Let $X = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$ and $D = \{\lambda \in \mathbb{R}^m : \lambda^\top A \leq \mathbf{c}^\top\}$. Let $y^* = \min_{\mathbf{x} \in X} \mathbf{c}^\top \mathbf{x}$ and $z^* = \max_{\lambda \in D} \lambda^\top \mathbf{b}$. Then $z^* \leq y^*$. Moreover, if $\mathbf{x} \in X$ and $\lambda \in D$ then $\lambda^\top \mathbf{b} \leq \mathbf{c}^\top \mathbf{x}$.*

Proof. For any $\mathbf{x} \in X$ and $\lambda \in D$, $\lambda^\top A \leq \mathbf{c}^\top$ and $\mathbf{x} \geq \mathbf{0}$ give $\lambda^\top A\mathbf{x} \leq \mathbf{c}^\top \mathbf{x}$; and $A\mathbf{x} = \mathbf{b}$ gives $\lambda^\top \mathbf{b} = \lambda^\top A\mathbf{x} \leq \mathbf{c}^\top \mathbf{x}$. Taking the maximum over $\lambda \in D$ on the left and the minimum over $\mathbf{x} \in X$ on the right gives $z^* \leq y^*$. $\square$

Corollary 5.57. *Let $X \neq \emptyset$ and $D \neq \emptyset$. If there exist $\mathbf{x} \in X$ and $\lambda \in D$ such that $\lambda^\top \mathbf{b} = \mathbf{c}^\top \mathbf{x}$, then λ is the optimal solution to the dual LP and $\mathbf{x}$ is the optimal solution to the primal LP.*

Proof. By Weak Duality (Theorem 5.56), every dual objective value is $\leq z^* \leq y^* \leq$ every primal objective value. The given pair achieves $\lambda^\top \mathbf{b} = \mathbf{c}^\top \mathbf{x}$, so both equal the common value $z^* = y^*$; a feasible point achieving the optimal value is optimal, so $\mathbf{x}$ and λ are optimal for the primal and dual respectively. $\square$

If the primal and dual objectives are in fact equal at optimality, we call it *strong duality*. This was the case in our example, where $11 \leq y^* \leq 11$.

Remark 5.58. We adopt the convention that $\inf \emptyset = +\infty$ and $\sup \emptyset = -\infty$. If there is a minimization problem $y^* = \min_{\mathbf{x} \in S} f(\mathbf{x})$ and $S = \emptyset$, then $y^* = +\infty$; similarly for a maximization problem $z^* = \max_{\lambda \in S} g(\lambda)$ with $S = \emptyset$, $z^* = -\infty$. Feasibility of either problem guarantees that there is an optimal objective value in $\mathbb{R}^* = \mathbb{R} \cup \{-\infty, +\infty\}$; if both are feasible then, by weak duality bounding each problem by the other, both optimal values are finite.

Theorem 5.59 (Strong Duality). *If a (primal) linear program has an optimal solution, so does its dual, and the objective function values for both primal and dual are equal:*

$$\boldsymbol{\lambda}^{*\top} \mathbf{b} = \mathbf{c}^\top \mathbf{x}^*.$$

Proof. Omitted in this iteration of these notes. It can be proved using supporting hyperplanes via convex sets. $\square$

Strong duality is a much “stronger” condition and does not hold in general. In most well-studied problems, particularly convex problems with inequality constraints, strong duality has some guarantees for existence. One such guarantee is *Slater’s condition*, which states that for a convex problem with inequality constraints, if there exists a “strictly feasible” point, i.e. one in the interior of the set defined by the inequality constraints, then strong duality holds. We mention this just to contextualize strong duality more generally.

The Canonical Form and the Dual of the Dual

When we consider weak duality we only need to connect a primal–dual pair by one being maximization and one being minimization. In our construction of the dual we considered the minimization problem to be the primal, but what if we considered the maximization problem to be the primal? First we establish the primal–dual pair for the *canonical form*.

Proposition 5.60. *For an LP given in canonical form, the primal–dual pair is*

$$(P) : \min_{\mathbf{x}} \mathbf{c}^\top \mathbf{x} \text{ s.t. } A\mathbf{x} \geq \mathbf{b}, \mathbf{x} \geq \mathbf{0}; \quad (D) : \max_{\boldsymbol{\lambda}} \boldsymbol{\lambda}^\top \mathbf{b} \text{ s.t. } \boldsymbol{\lambda}^\top A \leq \mathbf{c}^\top, \boldsymbol{\lambda} \geq \mathbf{0}.$$

Exercise 5.61. Prove Proposition 5.60. *Hint:* Consider converting canonical form to standard form using slack variables and then using the standard-form primal–dual as an intermediary step.

Theorem 5.62. *The dual of the dual is the primal.*

Proof. Start from the standard-form primal $(P) : \min \mathbf{c}^\top \mathbf{x}$ s.t. $A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}$, with dual $(D) : \max \boldsymbol{\lambda}^\top \mathbf{b}$ s.t. $\boldsymbol{\lambda}^\top A \leq \mathbf{c}^\top, \boldsymbol{\lambda}$ free. Rewrite (D) as a minimization by negating: $\min(-\mathbf{b})^\top \boldsymbol{\lambda}$ s.t. $A^\top \boldsymbol{\lambda} \leq \mathbf{c}, \boldsymbol{\lambda}$ free. Put this in standard form (splitting the free $\boldsymbol{\lambda}$ into positive and negative parts and adding slack variables), form its dual by the standard construction, and simplify. The resulting problem is $\min \mathbf{c}^\top \mathbf{x}$ s.t. $A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}$, which is exactly (P) . $\square$

This is an incredibly useful result, as it allows us to move generally between primal and dual, minimization and maximization, much more freely.

It is worth noting that weak duality almost always holds wherever we have a primal–dual pair for an optimization problem. In iterative approaches like gradient descent, this can be used as a measure of convergence called the *primal–dual gap*, which is measured as $f(\mathbf{x}) - g(\boldsymbol{\lambda})$, when a primal problem is paired with its dual either in the objective function or when calculated as a check on convergence. When this gap decreases we can conclude we are approaching convergence, and we use a tolerance criterion on this gap as an exit criterion.

5.7.4 Feasibility and Farkas' Lemma

The reader should have observed that the notion of feasibility plays an important role in our analysis of linear programs. One easy way to augment a (primal) LP from one trying to find both an optimal objective value and corresponding solution, if it exists, into an LP that simply returns a truth value as to whether the problem is (primal) feasible is to use the following LP:

$$(P): \quad \min_{\mathbf{x}} \quad \mathbf{0}_n^\top \mathbf{x} \quad \text{s.t.} \quad A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}.$$

This LP will return 0 if feasible and $+\infty$ otherwise. We can therefore assign a truth value for feasibility of TRUE if it returns 0 and FALSE if it does not. We call this a *feasibility LP*. The dual to this feasibility LP is

$$(D): \quad \max_{\boldsymbol{\lambda}} \quad \boldsymbol{\lambda}^\top \mathbf{b} \quad \text{s.t.} \quad \boldsymbol{\lambda}^\top A \leq \mathbf{0}_n^\top, \boldsymbol{\lambda} \text{ free.}$$

Based on our discussion about the relationship between primal feasibility and dual boundedness, we know that if the primal is feasible then the dual is bounded. The contrapositive is: if the dual is unbounded then the primal is infeasible. One way to write this unboundedness condition is: given $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, the set

$$Z = \left\{ \boldsymbol{\lambda} \in \mathbb{R}^m : \boldsymbol{\lambda}^\top A \leq \mathbf{0}_n^\top, \boldsymbol{\lambda}^\top \mathbf{b} > 0 \right\}$$

is not empty. This set describes a $\boldsymbol{\lambda}$ satisfying dual feasibility for which the dual objective is positive; if it is non-empty then Z also contains $\alpha \boldsymbol{\lambda}$ for all $\alpha > 0$ (scaling preserves both conditions), so (D) is unbounded. Thus $Z \neq \emptyset$ makes (D) unbounded, which means the primal is infeasible: $Z \neq \emptyset \Rightarrow X = \emptyset$, where $X = \{\mathbf{x} : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$.

The converse also holds: if the primal is infeasible then its dual is unbounded.

Proposition 5.63. *If the primal of the feasibility LP is infeasible, then its dual is unbounded.*

Proof. Assume for contradiction that the dual is bounded. For a bounded LP there exists a solution which obtains this bounded objective value; for (D) let this be $\boldsymbol{\lambda}^*$. Since the dual of the dual is the primal (Theorem 5.62), (P) is the dual of (D) . By Strong Duality (Theorem 5.59), if (D) has an optimal solution then its dual (P) must also have an optimal solution $\mathbf{x}^*$ with $\mathbf{0}_n^\top \mathbf{x}^* = \boldsymbol{\lambda}^{*\top} \mathbf{b}$, which implies $X \neq \emptyset$ and contradicts infeasibility of the primal. So the dual of the feasibility LP must be unbounded, i.e. $X = \emptyset \Rightarrow Z \neq \emptyset$. $\square$

Combining the two directions gives Farkas' Lemma.

Theorem 5.64 (Farkas' Lemma). *For a given $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, define the sets $X = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$ and $Z = \{\boldsymbol{\lambda} \in \mathbb{R}^m : \boldsymbol{\lambda}^\top A \leq \mathbf{0}_n^\top, \boldsymbol{\lambda}^\top \mathbf{b} > 0\}$. Then*

$$X = \emptyset \quad \text{if and only if} \quad Z \neq \emptyset.$$

Proof. Both directions were established above: $Z \neq \emptyset \Rightarrow X = \emptyset$ from the boundedness/scaling argument, and $X = \emptyset \Rightarrow Z \neq \emptyset$ from Proposition 5.63. (It is possible to prove Farkas' Lemma without appealing to the feasibility LP and its primal-dual pair, but that proof is algebraic and tedious, using induction, and offers little insight.) $\square$

It is worth drawing to the reader's attention that Farkas' Lemma is equivalently $X \neq \emptyset$ if and only if $Z = \emptyset$, i.e. only one set can be non-empty for a given pair $(A, \mathbf{b})$. This gives us a way to assess primal feasibility through Z .

It may be natural to wonder whether a similar result exists for dual feasibility, and the answer is yes. The dual feasibility LP, given $A \in \mathbb{R}^{m \times n}$ and $\mathbf{c} \in \mathbb{R}^n$, has primal–dual pair

$$(P_D) : \max_{\lambda} \lambda^\top \mathbf{0}_m \text{ s.t. } \lambda^\top A \leq \mathbf{c}^\top; \quad (D_D) : \min_{\mathbf{x}} \mathbf{c}^\top \mathbf{x} \text{ s.t. } A\mathbf{x} = \mathbf{0}_m, \mathbf{x} \geq \mathbf{0}_n.$$

Define the sets $Y = \{\lambda \in \mathbb{R}^m : \lambda^\top A \leq \mathbf{c}^\top\}$ and $W = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}_m, \mathbf{x} \geq \mathbf{0}_n, \mathbf{c}^\top \mathbf{x} < 0\}$. Here W plays the analogous role to Z .

Exercise 5.65. Argue via a set argument that $W \neq \emptyset$ represents (D_D) being unbounded.

Theorem 5.66. For a given $A \in \mathbb{R}^{m \times n}$ and $\mathbf{c} \in \mathbb{R}^n$, define the sets $Y = \{\lambda \in \mathbb{R}^m : \lambda^\top A \leq \mathbf{c}^\top\}$ and $W = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}_m, \mathbf{x} \geq \mathbf{0}_n, \mathbf{c}^\top \mathbf{x} < 0\}$. Then

$$Y = \emptyset \text{ if and only if } W \neq \emptyset.$$

Exercise 5.67. Prove Theorem 5.66. *Hint:* Use the same strategy as for Farkas' Lemma—weak duality gives one direction, and the dual of the dual combined with strong duality gives the other.

There are versions of Farkas' Lemma and its dual feasibility analog, but in keeping with the scope of these notes we only mention this to give the reader awareness of how these generalize.

Conclusion of Optimization

We conclude our coverage of optimization here. We covered two major parts of optimization needed for an introductory graduate-level course in optimization. Specifically, the topics related to gradient descent provide a strong foundation for coursework in computational and algorithmic aspects of optimization, while the topics we covered via linear programming introduce the important foundational ideas for the theoretical treatment of optimization problems and support further study of topics like semi-definite programming or quadratic programming.

The following sections of these notes introduce the complementary topics needed for a first-year Operations Research (or similar) program, which are generally stochastic processes. We again develop the fundamental mathematical ideas with a focus toward applications.

To be continued...

The probability and stochastic processes material is not yet written and will be added in a future revision of these notes.